

FRM Part II Exam

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Questions with Answers - Operational Risk and Resiliency

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Reading 103: Introduction to Operational Risk and Resilience

Q.5044 Which of the following risks falls within the scope of an operational risk management (ORM) framework?

- A. Credit risk associated with a company's investments in the stock market.
- B. Political risk resulting from changes in government policies.
- C. Legal risk arising from breach of contracts, laws and legislations.
- D. Interest rate risk associated with fluctuations in interest rates.

The correct answer is **C**.

Operational risk management (ORM) framework encompasses a wide range of risks that can affect an organization's operations. The operational risk definition provided by the BCBS includes legal risk but excludes strategic and reputational risk. Legal risk refers to the potential for losses or liabilities arising from contractual breaches, legal relevance, laws and regulations, and the risk of financial harm in the event of errors or breaches. Legal losses can be associated with various operational event types, but they are particularly linked to event types 3 (Employment practices and workplace safety) and 7 (Execution, delivery, and process management). While credit risk, political risk, and interest rate risk are important risks to manage, they fall outside the scope of operational risk management and are typically managed separately.

A is incorrect. Credit risk is the risk of loss arising from a borrower's failure to repay a debt or meet its obligations falls under the category of credit risk management, which is a separate type of risk management framework.

B is incorrect. While political risk can affect an organization's operations, it falls outside the scope of operational risk management. Political risk is typically managed through a separate type of risk management framework known as geopolitical risk management.

D is incorrect. Interest rate risk is the risk of loss resulting from fluctuations in interest rates. This type of risk falls under the category of market risk management, which is another type of risk management framework.

Things to Remember

- Operational Risk Management (ORM) framework encompasses a wide range of risks that can affect an organization's operations.
- Legal risk refers to the potential for losses or liabilities arising from contractual breaches, legal relevance, laws and regulations.
- Legal risk is particularly linked to event types such as Employment practices and

workplace safety, and Execution, delivery, and process management.

- Credit risk, political risk, and interest rate risk are important risks to manage but fall outside the scope of operational risk management.
 - Credit risk is managed through credit risk management framework, political risk through geopolitical risk management, and interest rate risk through market risk management.
-

Q.5045 Mr. Jonathan Howard, FRM, is presenting on the ORM framework. Jonathan highlights several points regarding the ORM framework. Which of the following statements made by Jonathan is incorrect?

- A. Companies should develop a holistic picture of their risk management practices to understand the relationships between actions, tools, and techniques
- B. A good ORM framework should include governance and conduct risk as the umbrella of all other risk management actions
- C. Regulated financial service companies are required to define their risk appetite and tolerance as a means of achieving their objectives
- D. Risk monitoring focuses on the definition, discovery, selection, and categorization, of the risks faced by a business or in a given activity

The correct answer is **D**.

The statement that 'Risk monitoring focuses on the definition, discovery, selection, and categorization, of the risks faced by a business or in a given activity' is incorrect. In the context of Operational Risk Management (ORM), risk identification, not risk monitoring, is the process that focuses on the definition, discovery, selection, and categorization of risks. This process aims to identify as many risks as possible that a business or a given activity might face. On the other hand, risk monitoring is a continuous process that tracks and reports on identified risks and the effectiveness of risk mitigation strategies. It involves the regular review of risk management activities and the updating of risk information to ensure that the risk management strategy remains effective.

Choice A is incorrect. This statement is accurate as it emphasizes the importance of a holistic approach in risk management practices. Understanding the relationships between actions, tools, and techniques helps companies to identify potential risks and develop effective strategies to mitigate them.

Choice B is incorrect. This statement correctly highlights that a good ORM framework should include governance and conduct risk as an overarching element of all other risk management

actions. Governance ensures that risks are managed according to the company's policies and procedures, while conduct risk relates to the risks associated with unethical or inappropriate behavior by employees.

Choice C is incorrect. The statement accurately reflects that regulated financial service companies are required to define their risk appetite and tolerance as part of their strategic objectives. Risk appetite refers to the level of risk a company is willing to accept in pursuit of its objectives, while risk tolerance refers to the degree of variability in investment returns that an organization can withstand.

Things to Remember

- Operational Risk Management (ORM) is a framework for identifying, assessing, monitoring, and mitigating operational risks within an organization.
 - Risk identification is the process of defining, discovering, selecting, and categorizing risks faced by a business or in a given activity.
 - Risk monitoring is a continuous process that tracks and reports on identified risks and the effectiveness of risk mitigation strategies.
 - Risk appetite refers to the level of risk a company is willing to accept in pursuit of its objectives, while risk tolerance refers to the degree of variability in investment returns that an organization can withstand.
 - Governance and conduct risk are important components of an effective ORM framework, ensuring that risks are managed according to policies and procedures and addressing risks associated with unethical behavior.
-

Q.5047 Which of the following statements best illustrates the evolution of compliance risk management in the financial industry?

- A. Compliance risk has gradually become less significant, as evidenced by the reduced number of regulatory fines in recent years.
- B. Compliance risk was initially captured by event types 3 (EPWS) and 7 (EDPM), but has since evolved to be managed separately.
- C. Regulatory fines for compliance risk breaches have increased over time, prompting banks to establish dedicated departments for compliance risk management.
- D. Operational risk management departments now oversee compliance departments in

most large banks, reflecting the growing importance of operational risk.

The correct answer is **C**.

The significance of compliance risk in the financial industry has indeed escalated over the years. This is evident from the increasing regulatory fines imposed on financial institutions for breaches in compliance. For instance, HSBC was fined \$1.9 billion in 2012, and Lloyds Banking Group was fined £21.9 billion. These hefty penalties have prompted many large banks to establish dedicated departments specifically for managing compliance risk. In some banks, such as JPMC and BNPP, the operational risk management department is situated within the compliance department, indicating the growing importance of compliance risk. Therefore, statement C accurately illustrates the evolution of compliance risk management in the financial industry.

Choice A is incorrect. The significance of compliance risk in the financial industry has not diminished over time. In fact, it has become more prominent due to increased regulatory scrutiny and the potential for significant fines and reputational damage resulting from non-compliance.

Choice B is incorrect. While event types 3 (EPWS) and 7 (EDPM) may have initially captured some aspects of compliance risk, it's not accurate to say that compliance risk management evolved from these categories. Compliance risk has always been a distinct area requiring specialized knowledge and strategies.

Choice D is incorrect. Although operational risk management is an important function within banks, it does not typically oversee the compliance department. These are separate functions with different responsibilities - operational risk focuses on risks arising from failed processes or systems, while compliance ensures adherence to laws and regulations.

Things to Remember

- Compliance risk management involves identifying, assessing, and mitigating risks associated with non-compliance with laws, regulations, and internal policies.
- Regulatory fines for compliance breaches can have significant financial and reputational impacts on financial institutions.
- Compliance risk management requires specialized knowledge and expertise to navigate the complex regulatory environment.
- Effective compliance risk management programs include policies, procedures, training, monitoring, and reporting mechanisms.
- Compliance risk is a key component of overall risk management frameworks in financial institutions.

Q.5048 BCBS categorizes operational risk into seven broad categories, commonly known as "Basel types level 1." These types are further divided into regulatory types (level 2) and examples (level 3). An FRM Part II candidate highlights several points regarding the Basel event risk type categories. Which of the following statements is **correct**?

- A. Internal fraud and external fraud are under the same event risk category
- B. Acts that go against laws put in place to safeguard the health, safety, and general well-being of employees and customers fall under clients, products, and business practices
- C. Issues such as data entry errors and unfinished legal documents fall under execution, delivery, and process management
- D. Losses due to theft and hacking are examples of event risks under the damage to physical assets category

The correct answer is **C**.

The category of 'execution, delivery, and process management' under the BCBS's operational risk framework encompasses issues related to the failure to execute transactions and manage processes correctly. This includes data entry errors and unfinished legal documents. These issues can lead to operational losses and are therefore considered as operational risks. The BCBS's framework is designed to help financial institutions identify, assess, monitor, and mitigate these risks. By categorizing these risks, institutions can develop targeted strategies to manage them and reduce potential losses.

Choice A is incorrect. Internal fraud and external fraud are not under the same event risk category. Internal fraud falls under the 'Internal Fraud' category, which includes acts of a type intended to defraud, misappropriate property or circumvent regulations, the law or company policy, excluding diversity & discrimination events, involving at least one internal party. On the other hand, external fraud falls under 'External Fraud' category which includes acts of a type intended to defraud, misappropriate property or circumvent the law by a third party.

Choice B is incorrect. Acts that go against laws put in place to safeguard health and safety do not fall under clients, products and business practices but rather they fall under 'Employment Practices and Workplace Safety'. This category includes acts inconsistent with employment, health or safety laws or agreements from payment of personal injury claims or diversity & discrimination events.

Choice D is incorrect. Losses due to theft and hacking do not fall under damage to physical assets but rather they are categorized as 'External Fraud'. The damage to physical assets refers more towards natural disaster-induced losses such as those from earthquakes or floods.

Things to Remember

- Operational risk is the risk of loss resulting from inadequate or failed internal processes, people, and systems, or from external events.

- Basel Committee on Banking Supervision (BCBS) provides a framework for categorizing operational risk into seven broad categories known as "Basel types level 1."
 - These Basel types are further divided into regulatory types (level 2) and examples (level 3) to provide a detailed classification of operational risks.
 - Internal fraud and external fraud are distinct categories under operational risk, each with its own characteristics and implications for financial institutions.
 - Acts that violate laws related to health, safety, and well-being of employees and customers are typically categorized under 'Employment Practices and Workplace Safety' in operational risk frameworks.
 - Execution, delivery, and process management category in operational risk covers issues related to the failure to execute transactions and manage processes correctly, such as data entry errors and incomplete legal documents.
 - Damage to physical assets category in operational risk typically includes losses resulting from physical damage to assets due to events like theft, natural disasters, or vandalism.
-

Q.5049 Which of the following is **most likely** an example of execution, delivery, and process management (EDPM) type of operational risk of the "Basel types level 1"?

- A. Destruction of equipment
- B. Employment contract termination
- C. Vendor disputes
- D. Unauthorized activities by employees

The correct answer is **C**.

Vendor disputes fall under the category of Execution, Delivery, and Process Management (EDPM) type of operational risk according to the Basel types level 1 classification. EDPM risks are associated with failed transaction processing or process management, or from relations with

trade counterparties and vendors. Vendor disputes can arise due to disagreements over the terms of contracts, quality of goods or services provided, payment issues, or other aspects of the vendor-client relationship. These disputes can disrupt the normal operations of a business, leading to financial losses and reputational damage. Therefore, managing vendor disputes effectively is a crucial aspect of operational risk management.

Choice A is incorrect. Destruction of equipment falls under the category of Physical Environment and Safety Risks, not Execution, Delivery, and Process Management (EDPM). This type of risk involves damage to physical assets or inability to use them due to natural or man-made disasters.

Choice B is incorrect. Employment contract termination is more likely to be classified as an Employment Practices and Workplace Safety Risk rather than EDPM. This type of risk arises from acts inconsistent with employment, health or safety laws or agreements, from payment of personal injury claims, or from diversity/discrimination events.

Choice D is incorrect. Unauthorized activities by employees are typically categorized as Internal Fraud under Basel's classification system for operational risks. These activities involve acts intended to defraud, misappropriate property or circumvent regulations, the law or company policy that are perpetrated by an entity's own staff.

Things to Remember

- **Operational Risk:** It is the risk of loss resulting from inadequate or failed internal processes, people, and systems, or from external events. It includes risks such as fraud, legal risks, physical or environmental risks, and more.
- **Basel Types Level 1:** Basel Committee on Banking Supervision has classified operational risks into eight categories, including Execution, Delivery, and Process Management (EDPM), Internal Fraud, Employment Practices and Workplace Safety, Clients, Products, and Business Practices, Damage to Physical Assets, Business Disruption and System Failures, and external fraud.
- **Vendor Management:** It involves managing relationships with third-party vendors who provide goods or services to a company. Effective vendor management is crucial for mitigating operational risks related to vendor disputes, non-performance, or breaches of contract.
- **Reputational Damage:** Operational risks, such as vendor disputes, can lead to reputational damage for a company. Maintaining a good reputation is essential for attracting customers, investors, and business partners.

Q.5051 An investment firm has contracted a risk professional and wishes to discuss the characteristics of operational loss events and challenges that may arise in managing operational risk. Which of the following characteristics **correctly** matches its description?

- A. Heterogenous – Operational risk is highly varied because it encompasses diverse risks, such as fraud in retail transactions
- B. Idiosyncratic and diffuse – The distribution of operational risk is highly skewed, with a higher concentration of the density being in the lowest part of the distribution
- C. Interconnected – Operational risk arises from each person and process within the firm; therefore, everyone has to take part in managing operational risk
- D. Heavy-Tailed – Operational risk is evolving in nature

The correct answer is **A**.

Operational risk is indeed heterogeneous. This means that it is highly varied and encompasses a wide range of diverse risks. For example, operational risk can include risks such as fraud in retail transactions, system failures, process inefficiencies, and human errors. Each of these risks has different causes, consequences, and distributions of losses. This diversity in operational risk makes it challenging for risk professionals to manage, as they need to understand and mitigate a wide range of potential risks. Therefore, a comprehensive and flexible risk management strategy is required to effectively manage operational risk.

Choice B is incorrect. While operational risk can indeed be idiosyncratic and diffuse, the statement that the distribution of operational risk is highly skewed with a higher concentration of density in the lowest part of the distribution is not necessarily accurate. Operational risks can have severe impacts, and their distribution may not always be skewed towards lower severity events. The severity and frequency of operational risk events can vary greatly depending on numerous factors such as the nature of operations, controls in place, etc.

Choice C is incorrect. Although it's true that everyone within a firm has a role to play in managing operational risk because it arises from each person and process within the firm, this does not accurately describe a characteristic of operational risk events themselves. Rather, it describes an approach to managing such risks.

Choice D is incorrect. The statement that "Operational risk is evolving in nature" does not specifically characterize operational risk events or their implications for managing operational risks. While it's true that these risks evolve over time due to changes in processes, systems or external environment etc., this choice doesn't provide specific insight into characteristics or implications for management strategies related to these types of risks.

Things to Remember

- Operational risk encompasses a wide range of diverse risks, such as fraud, system failures, process inefficiencies, and human errors.

- Operational risk events can have varying causes, consequences, and distributions of losses, making them heterogeneous in nature.
 - A comprehensive and flexible risk management strategy is essential to effectively manage operational risk due to its diverse and evolving nature.
 - Operational risk events can have severe impacts, and their distribution may not always be skewed towards lower severity events.
 - Operational risk management involves the participation of everyone within a firm, as these risks arise from each person and process.
 - Operational risks evolve over time due to changes in processes, systems, or the external environment.
-

Q.5052 Bank ABC wishes to strengthen its operational resilience. The bank invites a consultant to give more insights into this area. In his definition of resilience, the consultant clarifies that according to BCBS, resilience cannot be defined in a single sentence but rather comprises four components. Which of the following components is **correctly** defined?

- A. Continuity of business services: To contribute to the stability of the system, firms should respond to disruptions, maintain trust among key stakeholders, and provide clarity of communication during a crisis
- B. Important business services: From a process-based view of continuity, the regulator moved to a service-based view to ensure continuity
- C. Management of disruption: It protects vital business services from disruption
- D. Lessons learned: Firms should learn from past events and cover predictable shocks only

The correct answer is **B**.

The term 'Important business services' refers to the shift in focus from a process-based view of continuity to a service-based view. This shift is crucial to ensure the continuity of operations. In the process-based view, the emphasis is on maintaining the continuity of specific processes within the organization. However, this approach may not necessarily ensure the continuity of the overall business services that the organization provides. Therefore, the regulator moved to a service-based view. In this approach, the focus is on ensuring the continuity of the services that the organization provides, regardless of the specific processes involved. This approach is more

holistic and is better suited to ensuring operational resilience in the face of disruptions.

Choice A is incorrect. Responding to disruptions, maintaining trust among key stakeholders, and provide clarity of communication during a crisis falls under the management of disruption.

Choice C is incorrect. The management of disruption involves more than just protecting vital business services from disruption. It also includes identifying potential sources of disruption, assessing their impact, and developing strategies to mitigate their effects.

Choice D is incorrect. Lessons learned should not only cover predictable shocks but also unpredictable ones. Firms should learn from all past events and use this knowledge to improve their resilience against future disruptions, regardless of whether these disruptions are predictable or not.

Things to Remember

- Operational Resilience: The ability of a firm to prevent, respond to, recover from, and learn from operational disruptions.
- Business Continuity Management (BCM): The process of creating systems of prevention and recovery to deal with potential threats to a company.
- Regulatory Requirements: Various regulatory bodies, such as BCBS, set guidelines and requirements for operational resilience to ensure the stability of the financial system.
- Risk Management: Identifying, assessing, and prioritizing risks followed by coordinated and economical application of resources to minimize, monitor, and control the probability and impact of unfortunate events.
- Disruption Management: Strategies and plans put in place to manage and mitigate disruptions to business operations.

Q.5053 The Federal Reserve's Sound Practices for Strengthening Operational Resilience, published in 2020 along similar business lines and tolerance levels, illustrates that operational resilience is an important element in an Operational Risk Management Framework. Which of the following is **not** a regulatory expectation for operational resilience in line with the BCBS?

- A. Effective coordination of ORM relies on a solid foundation of governance and assigning roles and responsibilities to each party.
- B. Firms are required to monitor and report the coordination and maintenance of Business Continuity Management (BCM) and IT systems resilience.

C. A strong ORM framework is necessary in order to achieve operational resilience.

D. Firms are required to reduce their reliance on third parties.

The correct answer is **D**.

The statement that firms are required to reduce their reliance on third parties is not a regulatory expectation for operational resilience in line with the BCBS. The Federal Reserve's Sound Practices for Strengthening Operational Resilience does not encourage firms to reduce their use of third parties. Instead, it encourages firms to properly manage third parties as they are among typical areas that can expose firms to significant risks. This includes ensuring that third parties adhere to the same standards of operational resilience as the firm itself, and that they have robust contingency plans in place to manage potential disruptions. This approach recognizes the interconnected nature of modern financial systems, where third-party service providers often play a critical role in a firm's operations. Reducing reliance on third parties is not a practical or effective strategy for enhancing operational resilience. Instead, firms should focus on managing and mitigating the risks associated with third-party relationships.

Choice A is incorrect. The Basel Committee on Banking Supervision (BCBS) indeed expects effective coordination of ORM to rely on a solid foundation of governance and assigning roles and responsibilities to each party. This is a key aspect of operational resilience as it ensures that all parties involved in the ORM process understand their roles and responsibilities, thereby reducing the risk of errors or oversights.

Choice B is incorrect. According to BCBS, firms are indeed required to monitor and report the coordination and maintenance of Business Continuity Management (BCM) and IT systems resilience. This requirement ensures that firms are actively managing their operational risks, including those related to business continuity and IT systems.

Choice C is incorrect. As per BCBS guidelines, a strong ORM framework is necessary in order to achieve operational resilience. Without an effective ORM framework in place, it would be difficult for a firm to manage its operational risks effectively.

Things to Remember

- Operational Resilience: The ability of an organization to continue operating and delivering services in the face of disruptive events.
- Business Continuity Management (BCM): The process of creating systems of prevention and recovery to deal with potential threats to a company.
- Third-Party Risk Management: The process of managing risks associated with third-party vendors, suppliers, and service providers.
- Basel Committee on Banking Supervision (BCBS): An international standard-setting body that provides guidelines and recommendations for the banking industry.

- Operational Risk Management (ORM): The process of identifying, assessing, and controlling risks arising from operational factors within an organization.
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Reading 104: Risk Governance

Q.5055 In an FRM presentation on the Basel regulatory expectations for the governance of an operational risk management framework, an FRM candidate wishes to know how one can examine whether the ORM framework is being implemented at a firm. Which of the following questions should **not** be used to examine the above case?

- A. Is there evidence that all material events are captured in event reports?
- B. Does the value of each risk indicator come from an independent source?
- C. Are the presented data sufficient for decision-making?
- D. Does the information pertain to the senior management?

The correct answer is **D**.

'Does the information pertain to the senior management?' is not an appropriate question to evaluate the implementation of an ORM framework. The question is not relevant because the information should pertain to the level of management it is intended for, not specifically to the senior management. The ORM framework is a comprehensive approach that involves all levels of management, not just the senior management. Therefore, the information should be relevant and useful to all involved parties, not just the senior management. The focus should be on whether the information is sufficient for decision-making at all levels, not just at the senior level.

Choice A is incorrect. Asking for evidence that all material events are captured in event reports is a valid inquiry. This helps to ensure that the ORM framework is comprehensive and captures all significant operational risks.

Choice B is incorrect. Checking if the value of each risk indicator comes from an independent source is also a relevant question. Independent sources can provide unbiased data, which contributes to the accuracy and reliability of risk indicators.

Choice C is incorrect. Assessing whether the presented data are sufficient for decision-making forms an integral part of evaluating an ORM framework's effectiveness. If the data provided are not adequate or relevant, it could lead to poor decision-making and ineffective risk management.

Things to Remember

- Operational Risk Management (ORM) framework is a structured approach to identifying, assessing, monitoring, and managing operational risks within an organization.
- Event reporting is a crucial component of ORM, as it helps in capturing and documenting all material events that could impact the organization's operations.

- Risk indicators are quantitative or qualitative measures used to assess and monitor operational risks. It is important for these indicators to come from reliable and independent sources to ensure accuracy.
 - Data quality and sufficiency are key factors in effective decision-making within an ORM framework. Insufficient or irrelevant data can lead to poor risk management decisions.
 - An effective ORM framework involves all levels of management within an organization, not just senior management. Each level of management plays a role in implementing and adhering to the ORM framework.
-

Q.5056 A company's operational risk is managed through several committees that make collegial decisions based on information provided by different levels of the firm's decision-making hierarchy and information escalated by those committees. Which of the following is the **correct** function of the operational risk committee?

- A. Overseeing, managing, and reporting to the executive risk committee
- B. Overseeing the activities of a specific business line or function
- C. Overseeing all operational risks
- D. Reviewing and monitoring the investigation of large incidents

The correct answer is **A**.

The primary function of an operational risk committee is to oversee, manage, and report a comprehensive picture of the company's operational risks to the executive risk committee, management committee, and board risk committee. This involves coordinating with various levels of the firm's decision-making hierarchy and ensuring that relevant information is escalated to the appropriate committees. The operational risk committee plays a crucial role in the company's risk management structure by providing a holistic view of operational risks, which enables the executive risk committee and the board risk committee to make informed decisions about risk mitigation strategies and policies.

Choice B is incorrect. While an operational risk committee may oversee the activities of a specific business line or function, this is not its primary function. The main role of the operational risk committee is to oversee, manage and report on all operational risks to the executive risk committee.

Choice C is incorrect. Although overseeing all operational risks might seem like a plausible answer, it's not entirely accurate. The primary responsibility of an operational risk committee isn't just oversight but also managing and reporting these risks to the executive risk committee.

Choice D is incorrect. Reviewing and monitoring investigations of large incidents are part of the responsibilities of an operational risk committee but it does not constitute its primary function. Its main role involves overseeing, managing and reporting on all aspects related to operational risks.

Things to Remember

- **Operational Risk Management:** The process of identifying, assessing, and prioritizing operational risks within a company to minimize, monitor, and control the probability or impact of these risks.
 - **Risk Committees:** Various committees within a company that are responsible for overseeing different aspects of risk management, such as operational risk committee, executive risk committee, management committee, and board risk committee.
 - **Decision-Making Hierarchy:** The structure within a company that outlines the levels of decision-making authority and responsibility, from operational levels to executive levels.
 - **Risk Escalation:** The process of raising important risk-related information to higher levels of management or committees for decision-making and action.
 - **Risk Mitigation Strategies:** Actions taken to reduce the likelihood or impact of risks, including risk avoidance, risk reduction, risk transfer, or risk acceptance.
-

Q.5057 The 3rd principle of operational risk management outlines the roles of the board of directors in operational risk governance. Which of the following roles of the board is in line with principle 3?

- A. Identify the types and levels of operational risks the bank is willing to assume, as well as approve risk appetite and risk tolerance statements
- B. Regularly review the bank's risk appetite and tolerance statements' appropriateness
- C. Ensure the ORM framework is subject to independent review by sufficiently skilled personnel
- D. Ensure that they consider all risks when approving the bank's risk appetite and tolerance statements which provide details on risk limits and thresholds.

The correct answer is **C**.

The third principle of operational risk management (ORM) emphasizes the board of directors' responsibility to ensure that the ORM framework is subject to independent review by sufficiently skilled personnel. This principle underscores the importance of independent review in identifying potential weaknesses in the ORM framework and ensuring its effectiveness. The board of directors, as the governing body of the bank, is responsible for ensuring that the ORM framework is robust and capable of managing operational risks effectively. This includes ensuring that the framework is reviewed independently by personnel with the necessary skills and expertise. Independent review provides an unbiased assessment of the ORM framework, which can help identify areas for improvement and ensure that the framework is aligned with the bank's operational risk profile.

Choice A is incorrect. While the board of directors plays a crucial role in operational risk management, it is not their responsibility to identify the types and levels of operational risks the bank is willing to assume. This task typically falls under the purview of risk management professionals within the organization who have specialized knowledge and expertise in identifying and assessing various types of risks.

Choice B is incorrect. Although reviewing the appropriateness of risk appetite and tolerance statements regularly is an important aspect of ORM, this responsibility does not specifically fall under the third principle which addresses the board's role in governing operational risk. The third principle primarily focuses on ensuring that ORM framework undergoes independent review by sufficiently skilled personnel.

Choice D is incorrect. Ensuring consideration for all risks when approving risk appetite and tolerance statements, which provide details on risk limits and thresholds, does not fall under the third principle's purview concerning governing operational risk by board members. This aspect may be part of other principles but not specifically addressed in Principle 3.

Things to Remember

- Operational Risk Management (ORM) is the process of identifying, assessing, and managing the risks associated with the day-to-day operations of an organization.
- The board of directors plays a crucial role in overseeing and governing the operational risk management framework of a bank or financial institution.
- Risk appetite refers to the level of risk that an organization is willing to accept in pursuit of its objectives, while risk tolerance defines the acceptable level of variation around the risk appetite.
- Independent review of the ORM framework by skilled personnel helps in identifying weaknesses, improving effectiveness, and aligning the framework with the organization's risk profile.

- Operational risk governance involves establishing clear roles and responsibilities for managing operational risks within an organization.
-

Q.5058 The Bank of India wishes to get a deeper understanding of the three lines of defense. To achieve this, the bank has invited an operational risk specialist to shed more light on this topic. Which of the following roles did the specialist highlight under the first line of defense?

- A. Monitoring and reporting the operational risk profiles of business units
- B. The development and maintenance of operational risk management and measurement policies, standards, and guidelines, as well as the design and delivery of operational risk
- C. Reviewing other lines of business
- D. Reviewing and taking part in the monitoring and reporting of the operational risk profile

The correct answer is **A**.

The first line of defense in the three lines of defense model in operational risk management is primarily responsible for managing operational risk on a day-to-day basis. This includes identifying, assessing, controlling, and mitigating operational risks. One of the key responsibilities of the first line of defense is to keep track of the operational risk profiles of the business units and report them. This involves monitoring the operational risks inherent in the business, developing appropriate controls to manage these risks, and evaluating the effectiveness and design of these controls. By keeping track of the operational risk profiles, the first line of defense can provide timely and accurate information to the second line of defense, which is responsible for overseeing the operational risk management framework and policies.

Choice B is incorrect. The development and maintenance of operational risk management and measurement policies, standards, and guidelines, as well as the design and delivery of operational risk, are typically responsibilities associated with the second line of defense. The first line of defense primarily involves managing risks on a day-to-day basis.

Choice C is incorrect. Reviewing other lines of business is not a primary responsibility associated with any specific line of defense in the three-line model. Each line has its own set of responsibilities related to risk management.

Choice D is incorrect. This task falls under both the second and third lines, which provide oversight functions.

Things to Remember

- Three lines of defense model in operational risk management

- Responsibilities of the first line of defense
 - Responsibilities of the second line of defense
 - Responsibilities of the third line of defense
 - Operational risk profiles of business units
 - Monitoring and reporting of operational risk profiles
-

Q.5061 According to the 4th principle of operational risk management, the board must identify the types and levels of operational risks the bank is willing to assume, as well as approve risk appetite and risk tolerance statements. Which of the following is **not** a correct feature of these statements?

- A. Be easy to communicate and understand
- B. Provide reasons for taking or avoiding certain operational risks
- C. Be forward-looking and subject to scenario and stress testing
- D. Perform scenario analysis retrospectively

The correct answer is **D**.

'Perform scenario analysis retrospectively' is not a correct feature of risk appetite and risk tolerance statements according to the 4th principle of operational risk management. These statements are meant to be forward-looking and subject to scenario and stress testing. This means they should anticipate future scenarios and risks, rather than analyzing past scenarios retrospectively. The purpose of these statements is to guide the bank's future actions and decisions regarding operational risks, so they need to be proactive and forward-thinking, not retrospective.

Choice A is incorrect. The risk appetite and tolerance statements should indeed be easy to communicate and understand. This ensures that all stakeholders, including the board, management, employees, and even external parties such as regulators or investors have a clear understanding of the bank's approach to operational risk.

Choice B is incorrect. Providing reasons for taking or avoiding certain operational risks is an important feature of these statements. It helps in aligning the bank's strategic objectives with its operational risk management framework.

Choice C is incorrect. Being forward-looking and subject to scenario and stress testing are key characteristics of these statements according to the 4th principle of operational risk management. This allows banks to anticipate potential future risks and prepare accordingly.

Things to Remember

- Risk Appetite: The level of risk that an organization is willing to accept in pursuit of its objectives. It reflects the organization's overall attitude towards risk-taking.
 - Risk Tolerance: The acceptable level of variation around the organization's risk appetite. It helps in determining the boundaries within which risks can be taken.
 - Scenario Analysis: A technique for analyzing possible future events by considering alternative outcomes. It helps in assessing the impact of different scenarios on the organization.
 - Stress Testing: A risk management technique used to evaluate the resilience of financial institutions and their portfolios under adverse conditions. It involves testing the organization's ability to withstand severe shocks.
 - Operational Risk Management: The process of identifying, assessing, controlling, and mitigating operational risks within an organization. It involves understanding the potential risks associated with people, processes, systems, and external events.
-

Reading 105: Risk Identification

Q.5065 Which of the following is most likely a document that includes all operational risks of a firm, the likelihood of the risks and the controls applied to each risk?

- A. Risk universe
- B. Top-ten risks
- C. Risk register
- D. Shock scenarios

The correct answer is **C**.

The risk register is the central repository of all operational risks in financial firms. It is a document that includes all operational risks of a firm, the likelihood of the risks and the controls applied to each risk. The risk register is a critical tool in risk management as it provides a comprehensive view of all the risks a firm is exposed to, their potential impact, and the measures taken to mitigate them. It is a dynamic document that is updated regularly to reflect changes in the risk landscape. The risk register helps in prioritizing risks based on their likelihood and impact, and in developing appropriate risk response strategies. It also serves as a communication tool to inform all stakeholders about the risks and the actions taken to manage them.

Choice A is incorrect. A Risk universe is a tool used to identify potential risks in different areas of an organization. While it does list out all possible risks, it does not necessarily include their respective likelihoods or the specific controls implemented to mitigate each risk.

Choice B is incorrect. The Top-ten risks document typically lists only the most significant risks faced by an organization, rather than a comprehensive list of all operational risks. Furthermore, this document may not always detail the likelihoods of these risks or the specific controls implemented for mitigation.

Choice D is incorrect. Shock scenarios are hypothetical extreme events used for stress testing and do not provide a comprehensive list of all operational risks, their respective likelihoods, and mitigation strategies.

Things to Remember

- Operational risk: The risk of loss resulting from inadequate or failed internal processes, people, and systems, or from external events. It includes legal risk but excludes strategic and reputational risks.
- Likelihood of risks: The probability or chance that a specific risk event will occur.
- Controls: Measures or actions taken to mitigate or manage risks effectively.

- Risk management: The process of identifying, assessing, and prioritizing risks followed by coordinated and economical application of resources to minimize, monitor, and control the probability and impact of unfortunate events.
 - Risk mitigation: The process of reducing the impact or likelihood of a risk event by implementing specific actions or controls.
-

Q.5066 Which of the following four main categories of controls are implemented to reduce the likelihood of risks materializing by mitigating their possible causes?

- A. Detective controls
- B. Corrective controls
- C. Preventative controls
- D. Directive controls

The correct answer is **C**.

Preventative controls are designed to reduce the likelihood of risks materializing by mitigating their possible causes. These controls are proactive measures that are put in place to prevent an undesirable event from occurring. They are designed to manage risks before they have an impact on the organization. Preventative controls may include a wide range of activities such as system and process design, training, and supervision. The goal of these controls is to identify potential risks and take action to eliminate or reduce them before they can affect the organization. This is achieved by identifying the potential causes of risks and implementing measures to mitigate these causes. Examples of preventative controls include access controls, segregation of duties, and approval authorities.

Choice A is incorrect. Detective controls are not designed to mitigate the possible causes of risks, but rather to identify and react to instances where a risk has materialized. They are reactive in nature and come into play after a risk event has occurred.

Choice B is incorrect. Corrective controls, similar to detective controls, are also reactive in nature. They aim at correcting or rectifying the situation after a risk event has occurred rather than mitigating the possible causes of risks.

Choice D is incorrect. Directive controls guide actions towards compliance with policies and procedures but they do not specifically target the mitigation of potential causes of risks.

Things to Remember

- Detective controls are reactive in nature and are designed to identify and react to

instances where a risk has materialized.

- Corrective controls are also reactive and aim at correcting or rectifying the situation after a risk event has occurred.
 - Directive controls guide actions towards compliance with policies and procedures but do not specifically target the mitigation of potential causes of risks.
 - Preventative controls are proactive measures put in place to prevent an undesirable event from occurring by mitigating their possible causes.
 - Examples of preventative controls include access controls, segregation of duties, and approval authorities.
-

Q.5067 Which of the following risk identification processes begins at the executive level, then to the business units and finally to individual business processes?

- A. Bottom-up risk identification
- B. Event and loss data analysis
- C. Top down risk identification
- D. Risk and control self-assessment

The correct answer is **C**.

This process begins at the highest level of the organization, typically the board or executive level. The purpose of this approach is to identify risks that could potentially impact the strategic objectives of the organization. From the executive level, the process moves down to the various business units, where risks related to specific operational activities are identified. Finally, the process reaches the individual business processes, where risks associated with specific tasks or procedures are identified. This approach ensures that risks are identified and assessed in the context of the organization's overall strategic objectives, and that risk management activities are aligned with these objectives.

Choice A is incorrect. Bottom-up risk identification starts at the individual business processes and moves up to the various business units, and finally reaches the executive or board level. This is opposite to what is described in the question.

Choice B is incorrect. Event and loss data analysis involves studying past events and losses to identify potential risks. It does not necessarily follow a hierarchical process from top (executive or board level) to bottom (individual business processes).

Choice D is incorrect. Risk and control self-assessment (RCSA) involves individuals assessing their own risks within their respective roles or departments, rather than a process that initiates at the executive or board level.

Things to Remember

- Risk Identification: The process of identifying, assessing, and prioritizing risks that could potentially affect the achievement of an organization's objectives.
 - Enterprise Risk Management (ERM): ERM is a holistic approach to managing all types of risks faced by an organization, integrating risk management practices across the entire organization.
 - Risk Appetite: The amount and type of risk that an organization is willing to take in order to meet its strategic objectives.
 - Risk Tolerance: The acceptable level of variation that an organization is willing to take in achieving its objectives.
 - Risk Management Framework: A structured approach to identifying, assessing, and managing risks within an organization.
-

Q.5068 Which of the following is most likely risks that a firm has identified as being on the horizon, relatively small but on the rise with the potential for significant impact in the future?

- A. Emerging risks
- B. Risk universe
- C. Taxonomies
- D. Risk register

The correct answer is **A**.

Emerging risks are risks that a firm has identified as being on the horizon, relatively small but on the rise with the potential for significant impact in the future. These risks are not yet fully understood, making them difficult to quantify and manage. They could be related to a variety of factors, including changes in the market, technological advancements, regulatory changes, or societal shifts. The identification and management of emerging risks is a critical aspect of a firm's overall risk management strategy, as it allows the firm to anticipate and prepare for potential future threats.

Choice B is incorrect. The term "Risk Universe" refers to the set of all possible risks that a firm might face. It does not specifically refer to risks that are currently small but have the potential to significantly impact the firm in the future.

Choice C is incorrect. Taxonomies in risk management are used for classification of risks into different categories or levels, they do not represent a type of risk that is small now but could have significant impact in future.

Choice D is incorrect. A Risk Register is a document used by firms to identify, assess and track risks over time, it does not denote a specific type of risk like emerging ones which are currently small but can potentially have significant impacts.

Things to Remember

- Emerging risks are often associated with new technologies, changing market dynamics, regulatory changes, or shifts in societal behavior.
- Identifying and managing emerging risks requires a proactive approach and continuous monitoring of the external environment.
- Scenario analysis and stress testing are common techniques used to assess the potential impact of emerging risks on a firm.
- Effective communication and collaboration across different functions within a firm are essential for addressing emerging risks.

Q.5069 Which of the following is not a top-down risk identification tool?

- A. Exposures and vulnerabilities
- B. Risk wheel
- C. Process mapping
- D. Horizon scanning

The correct answer is **C**.

Process mapping is not a top-down risk identification tool. Instead, it is a bottom-up risk identification technique. This method involves outlining the steps of a process in a detailed manner, considering the risks associated with each specific set of actions, and asking what could potentially go wrong at each stage. The focus here is on individual tasks or processes, and the risks are identified at this granular level. These identified risks are then aggregated to form the overall risk profile. This approach is in contrast to the top-down approach, where the focus is on the overall risk profile, and then specific risks are identified by drilling down into this profile.

Choice A is incorrect. Exposures and vulnerabilities are indeed a part of top-down risk identification tools. They involve identifying the overall risk profile by examining the potential exposures and vulnerabilities that an organization might face.

Choice B is incorrect. The Risk wheel is also a top-down approach to risk identification. It starts with the overall risk profile, then drills down into specific risks by considering various factors such as external environment, internal environment, strategy and operations.

Choice D is incorrect. Horizon scanning falls under the category of top-down risk identification tools as well. It involves looking at the broader picture or horizon to identify potential risks that could impact an organization's overall risk profile.

Things to Remember

- Top-down risk identification tools focus on starting with the overall risk profile and then drilling down into specific risks.
- Bottom-up risk identification techniques involve starting with individual tasks or processes and identifying risks at a granular level.
- Exposures and vulnerabilities are key components of top-down risk identification tools.
- The Risk wheel is a top-down approach that considers various factors like external environment, internal environment, strategy, and operations.

- Horizon scanning involves looking at the broader picture to identify potential risks that could impact the organization's risk profile.
-

Q.5070 Which of the following is not classified as an exposure under top-down risk identification tools?

- A. Critical third parties
- B. Key distribution channels
- C. Systems overdue for updates
- D. Main drivers of revenues

The correct answer is **C**.

Exposures and vulnerabilities are top-down risk identification tools. Business risk exposure is inherent in every financial firm while vulnerabilities are the weakest links in business activities.

The key benefit of using a list of exposures and vulnerabilities as a brainstorming technique for risk identification is that it is business specific.

Examples of vulnerabilities are issues in control systems, systems overdue for updates, overdue resolutions of issues, stand-alone systems, unmonitored operations or people, blind spots among others.

A, B & D are incorrect. Examples of exposures are critical third parties, key persons, key distribution channels, main drivers of revenue, sources of goodwill among others.

Things to Remember

- Exposures and vulnerabilities are key components of top-down risk identification tools.
 - Business risk exposure is inherent in every financial firm.
 - Vulnerabilities represent the weakest links in business activities.
 - Using a list of exposures and vulnerabilities as a brainstorming technique for risk identification is business-specific.
 - Examples of vulnerabilities include issues in control systems, systems overdue for updates, overdue resolutions of issues, stand-alone systems, unmonitored operations or people, and blind spots.
 - Examples of exposures include critical third parties, key persons, key distribution channels, main drivers of revenue, and sources of goodwill.
-

Q.5071 Which of the following bottom-up risk identification tools relates to incidents that could have resulted in operational losses but did not due to interventions outside normal controls?

- A. Internal losses
- B. External losses
- C. Near misses
- D. Process mapping

The correct answer is **C**.

Near misses are incidents that could have resulted in an operational loss but did not because of good luck or intervention outside of the normal course of controls. An example would be sending funds to the wrong person but having the funds reversed before the funds could be withdrawn. This concept is crucial in operational risk management as it helps organizations identify potential risks and take preventive measures. By analyzing near misses, organizations can understand the weaknesses in their operational processes and controls, and take corrective actions to prevent actual losses in the future. This proactive approach to risk management can significantly enhance an organization's resilience to operational risks.

Choice A is incorrect. Internal losses refer to the actual losses that a firm incurs due to operational risk events within the organization. They do not account for incidents that did not result in losses due to external interventions.

Choice B is incorrect. External losses are those incurred by other firms in the same industry and are used as a benchmark or reference point for assessing potential operational risks. They do not specifically relate to incidents that were prevented from causing loss through interventions outside of normal controls.

Choice D is incorrect. Process mapping is a tool used in operational risk management to identify potential risks by visually representing the steps involved in a process, their sequence, and decision points. It does not directly deal with incidents that had the potential for loss but were prevented due to external interventions.

Things to Remember

- **Operational Risk Management:** Involves identifying, assessing, and managing risks arising from operational failures, including processes, people, systems, and external events.
- **Loss Events:** Actual losses incurred by a firm due to operational risk events, which can be used to analyze trends and patterns in risk exposure.
- **Risk Mitigation:** Strategies and actions taken to reduce the likelihood or impact of

potential risks, including preventive measures and controls.

- Key Risk Indicators (KRIs): Quantitative or qualitative metrics used to monitor and assess the likelihood of potential risk events before they occur.
 - Scenario Analysis: A technique used to assess the potential impact of specific risk events by modeling different scenarios and outcomes.
-

Q.5072 Which of the following is not one of the six components of PESTLE that are used for scanning horizon risks?

- A. Political component
- B. Economic component
- C. Labor market component
- D. Environmental component

The correct answer is C.

PESTLE is an acronym that stands for Political, Economic, Social, Technological, Legal, and Environmental. These are the six components of the PESTLE analysis, a tool used by organizations to scan their external macro-environment. Each component represents a type of factor that can influence an organization's operations and performance.

The Labor market component is not a part of the PESTLE analysis. While labor market conditions can certainly impact an organization, they are typically considered under the Economic or Social components of the PESTLE analysis. Economic factors include economic growth, exchange rates, inflation rate, and labor market conditions. Social factors include health consciousness, population growth rate, age distribution, career attitudes, and emphasis on safety.

Therefore, the Labor market component is not one of the six components of the PESTLE analysis used for scanning horizon risks.

Choice A is incorrect. The Political component is a part of the PESTLE analysis. It involves understanding the impact of government policies, regulations, and political stability on an organization's operations.

Choice B is incorrect. The Economic component is also a part of the PESTLE analysis. It examines economic factors such as inflation rates, interest rates, foreign exchange rates, economic growth patterns etc., that affect an organization's operations and decision-making process.

Choice D is incorrect. The Environmental component forms a part of the PESTLE analysis too.

This aspect looks at environmental considerations that might affect an organization like climate change policies or natural disaster risks.

Things to Remember

- PESTLE analysis is a strategic tool used by organizations to understand and analyze the external macro-environmental factors that can impact their operations.
 - Political factors in PESTLE analysis include government stability, trade regulations, tax policies, and political trends.
 - Economic factors in PESTLE analysis encompass economic growth, interest rates, exchange rates, inflation, and labor market conditions.
 - Social factors in PESTLE analysis cover demographics, cultural trends, lifestyle changes, and social attitudes.
 - Technological factors in PESTLE analysis focus on technological advancements, innovation, automation, and research and development.
 - Legal factors in PESTLE analysis involve laws, regulations, compliance requirements, and litigation risks.
 - Environmental factors in PESTLE analysis consider environmental regulations, sustainability practices, climate change, and natural disaster risks.
-

Q.5073 Which of the following is most likely a bias that an external expert can help mitigate during scenario analysis?

- A. An excessive focus on scenarios driven by internal causes
- B. Myopia
- C. Initiation of discussions
- D. External loss data

The correct answer is **B**.

Myopia, also known as nearsightedness, is a cognitive bias that causes individuals to place too

much emphasis on recent events while underestimating the likelihood or impact of events in the distant future. This bias can significantly impact the results of scenario analysis, as it may lead to an overestimation of the likelihood or impact of recent events and an underestimation of the likelihood or impact of future events. An external expert can help mitigate this bias by providing an objective perspective and helping to ensure that the analysis takes into account a broad range of potential future scenarios, not just those that are influenced by recent events.

Choice A is incorrect. An excessive focus on scenarios driven by internal causes is a bias that can occur during scenario analysis, but it's not necessarily mitigated by an external expert. This bias refers to the tendency of individuals within an organization to focus more on scenarios that are driven by factors within the organization, rather than considering external factors or events. While an external expert may provide a fresh perspective and help broaden the scope of the analysis, they may not necessarily mitigate this particular bias.

Choice C is incorrect. Initiation of discussions is not a recognized bias in risk management or scenario analysis. It refers to starting conversations or debates about potential risks or scenarios, which is generally considered a positive action in risk management as it encourages open communication and collaboration.

Choice D is incorrect. External loss data isn't a type of bias; instead, it's a source of information used in risk management and scenario analysis to understand potential losses from past events that have occurred outside the organization. An external expert might bring additional insights into interpreting this data but wouldn't mitigate any biases associated with it.

Things to Remember

- Confirmation bias: This is the tendency to search for, interpret, favor, and recall information in a way that confirms one's preexisting beliefs or hypotheses.
- Groupthink: This occurs within a group of people when the desire for harmony or conformity in the group results in an irrational or dysfunctional decision-making outcome.
- Availability heuristic: This is a mental shortcut that relies on immediate examples that come to mind when evaluating a specific topic, concept, method, or decision.
- Overconfidence bias: This bias involves an individual's subjective confidence in their judgments being reliably greater than their objective accuracy, especially when confidence is relatively high.

Q.5074 Which of the following Basel Category level 1 event relates to losses arising from acts inconsistent with employment, health, or safety laws or from diversity / discrimination events?

- A. Client, products & business practices
- B. Employment practices and workplaces safety
- C. Damage to physical assets
- D. Business disruption and system failures

The correct answer is **B**.

Employment practices and workplace safety encompasses losses that arise from violations of employment, health, or safety regulations or agreements. It also includes losses that result from having to pay for personal injury claims, or from incidents involving diversity or discrimination. These losses can occur in a variety of ways, such as through lawsuits, fines, or settlements. The category is designed to capture a wide range of potential losses that can occur in the workplace, and is a critical component of the Basel framework's approach to risk management.

Choice A is incorrect. The category of "Client, products & business practices" is associated with losses that occur due to the failure of fiduciary responsibilities, improper business or market practices, product flaws, and advisory activities. It does not cover losses related to employment regulations or diversity/discrimination incidents.

Choice C is incorrect. "Damage to physical assets" refers to losses resulting from the damage or destruction of a company's physical assets due to natural disasters or other events such as vandalism. This category does not include losses related to non-compliance with employment, health and safety regulations.

Choice D is incorrect. "Business disruption and system failures" pertains to losses caused by hardware/software failures, power outages or other disruptions in normal business operations. It does not encompass incidents related to violations of employment laws or discrimination issues.

Things to Remember

- Basel Category level 1 events are used in the Basel framework to categorize different types of operational risk events that can impact a financial institution.
- Operational risk is the risk of loss resulting from inadequate or failed internal processes, people, and systems, or from external events.
- Employment practices and workplace safety is one of the key categories under Basel Category level 1 events, focusing on losses related to violations of employment, health, or safety regulations.
- Diversity and discrimination events can lead to significant financial and reputational losses for a firm, making it important to manage and mitigate such risks effectively.

- Effective risk management practices include implementing policies and procedures to ensure compliance with employment laws, promoting a diverse and inclusive workplace culture, and providing training to employees on discrimination and harassment prevention.
-

Reading 106: Risk Measurement and Assessment

Q.5157 Joel and Mark, FRM Part II candidates, are discussing BCBS' guidelines on the need to report comprehensive data regarding operational risk events. During the discussion, the following statements are made. Which statement is most likely correct?

- A. While the Basel Committee has set a minimum threshold for loss reporting at €20,000 (\$22,000), setting reporting thresholds at zero is considered best practice so as to capture every operational loss or simplify instructions to the business units that do not need to estimate a loss before deciding to report incidents.
- B. Regulatory guidelines dictate that firms must report any incidents causing them both financial losses and non-financial impacts.
- C. Both direct and indirect losses must be reported.
- D. Grouped losses are distinct operational risk events connected through a common loss amount.

The correct answer is **C**.

Both direct and indirect losses must be reported. Direct losses refer to the immediate financial impact following an operational risk event. This could include costs associated with remediation efforts, financial consequences arising from erroneous transactions, or compensation paid to clients. On the other hand, indirect losses are more challenging to identify as they represent the subsequent consequences stemming from an operational risk event. These could include reputational damage, loss of business, or increased regulatory scrutiny, which may not have immediate financial implications but can significantly affect the firm's operations and profitability in the long run. Therefore, both types of losses are crucial to capture a comprehensive picture of the operational risk landscape and inform effective risk management strategies.

Choice A is incorrect. While it is true that the Basel Committee has set a minimum threshold for loss reporting at €20,000 (\$22,000), setting reporting thresholds at zero is not considered best practice. This could lead to an overload of information and make it difficult for firms to identify significant operational risk events. Furthermore, business units do need to estimate a loss before deciding to report incidents as this helps in prioritizing and managing risks effectively.

Choice B is incorrect. The BCBS guidelines do not dictate that firms must report any incidents causing them both financial losses and non-financial impacts. While both types of impacts are important, the guidelines primarily focus on financial losses as these can be quantified and measured more accurately.

Choice D is incorrect. Grouped losses are not distinct operational risk events connected through a common loss amount. Instead, they refer to multiple losses arising from the same root cause or event type which are grouped together for reporting purposes.

Things to Remember

- Operational Risk: The risk of loss resulting from inadequate or failed internal processes, people, and systems, or from external events.
 - Direct Losses: Immediate financial impacts following an operational risk event.
 - Indirect Losses: Subsequent consequences stemming from an operational risk event that may not have immediate financial implications.
 - Basel Committee on Banking Supervision (BCBS): Sets global standards for the regulation and supervision of banks and financial institutions.
 - Reporting Thresholds: Minimum levels of loss or impact that trigger the reporting of operational risk events.
 - Regulatory Guidelines: Rules and requirements set by regulatory bodies to ensure compliance and risk management in financial institutions.
-

Q.5159 What is the primary use of the Swiss Cheese model?

- A. A method to assess the impact of an attack on an asset.
- B. A tool for generating scenarios with FAIR methodology.
- C. A framework for analyzing and identifying security.
- D. A concept that uses layers of defense to prevent hazards.

The correct answer is **D**.

The Swiss Cheese model is primarily used as a concept that employs layers of defense to prevent hazards. This model metaphorically represents how multiple defenses, or 'layers of cheese,' are necessary to create effective safety systems. Each layer serves as a defense against hazards, and each has its own weaknesses, creating holes in the protective barrier. The idea is that all defenses need to be in place and working properly in order to protect against potential hazards. The model was first proposed by James Reason, and it has become widely used in risk management as a way to identify vulnerabilities and increase safety protocols.

Choice A is incorrect. The Swiss Cheese model is not primarily used to assess the impact of an attack on an asset. While it can be used to understand how different layers of defense might mitigate the impact of a potential hazard, its primary use is not in assessing the impact itself.

Choice B is incorrect. The Swiss Cheese model does not directly relate to FAIR (Factor

Analysis of Information Risk) methodology, which focuses on quantifying risk and uncertainty in digital environments. The Swiss Cheese model instead provides a visual representation for understanding how multiple layers of defense work together to prevent hazards.

Choice C is incorrect. Although the Swiss Cheese model can be used as part of a broader security analysis framework, its primary purpose isn't for analyzing and identifying security measures but rather illustrating how multiple defenses work together to prevent hazards from materializing.

Things to Remember

- The Swiss Cheese model was first proposed by James Reason as a way to understand how multiple layers of defense can work together to prevent hazards.
 - Each layer in the Swiss Cheese model represents a defense mechanism, and the holes in the cheese represent potential weaknesses in the system.
 - It is important to ensure that all layers of defense are in place and functioning properly to effectively prevent hazards.
 - The Swiss Cheese model is widely used in industries such as healthcare, aviation, and risk management to improve safety protocols.
 - Understanding the concept of "cumulative effect" is crucial in the Swiss Cheese model, as it highlights how multiple failures or weaknesses in different layers can align to allow a hazard to pass through.
-

Q.5160 What is the purpose of Monte Carlo simulations in the FAIR model of managing operational risk?

- A. To estimate the frequency and magnitude of a potential loss event.
- B. To generate scenarios with an asset at risk, a threat community, a threat type and an effect.
- C. To provide the distribution of simulated scenario losses as output.
- D. To determine the best course of action to prevent a potential loss event.

The correct answer is **C**.

Monte Carlo simulations are used in the FAIR model to provide the distribution of simulated scenario losses based on factor estimates expressed as distributions. The factor estimates come from business experts who estimate the frequency and probable loss magnitude for each scenario. The Monte Carlo simulations then use these factor estimates as inputs to generate outputs in the form of distributions of simulated scenario losses.

Option A is incorrect because Monte Carlo simulations do not estimate the frequency and magnitude - this is done by business experts.

Option B is incorrect because Monte Carlo simulations do not generate scenarios with an asset at risk, a threat community, a threat type and an effect - this is done through risk assessment and analysis.

Option D is incorrect because Monte Carlo simulations are not used to determine the best course of action to prevent a potential loss event - this is determined through other methods such as cost/benefit analysis or decision trees.

Things to Remember

- Monte Carlo simulations are a computational technique that relies on repeated random sampling to obtain numerical results.
 - FAIR (Factor Analysis of Information Risk) is a framework for understanding, analyzing, and measuring information risk.
 - Operational risk is the risk of loss resulting from inadequate or failed internal processes, people, and systems, or from external events.
 - Scenario analysis is a technique for analyzing possible future events by considering alternative possible outcomes.
 - Factor estimates in the FAIR model are expressed as distributions to capture the uncertainty and variability in risk factors.
-

Reading 107: Risk Mitigation

Q.5075 According to the international standards of enterprise risk management ISO 31000, there are four ways to address risks. Which of the following is **correct** in this context?

- A. All risks can be transferred to a third party
- B. Termination should be the first response action in case of an operational risk event
- C. Risk can be transferred through external insurance and outsourcing
- D. Tolerance involves all types of risk mitigations, especially internal controls aimed at reducing the probability

The correct answer is **C**.

Risk transfer is one of the ways to address risks according to the ISO 31000 standards. This method involves shifting the risk to another party. This can be achieved through various means, including external insurance and outsourcing. External insurance allows an organization to transfer the financial risk associated with a particular event to an insurance company. On the other hand, outsourcing involves delegating certain business operations to third-party entities, thereby transferring the associated risks as well. This method is particularly useful when the third party has better capabilities or resources to manage the risk. However, it's important to note that risk transfer doesn't eliminate the risk entirely; it merely shifts the responsibility of managing the risk.

Choice A is incorrect. While risk transfer is a valid method of addressing risks, it's not accurate to say that all risks can be transferred to a third party. Some risks are inherent and cannot be completely transferred.

Choice B is incorrect. Termination should not necessarily be the first response action in case of an operational risk event according to ISO 31000 standards. The response should depend on the nature and severity of the risk, and could involve other strategies such as mitigation or acceptance.

Choice D is incorrect. Tolerance does not involve all types of risk mitigations, especially internal controls aimed at reducing the probability. Tolerance refers to accepting the existence of a particular risk and deciding to live with it rather than taking actions to remove or mitigate it.

Things to Remember

- Enterprise Risk Management (ERM) is a comprehensive approach to identifying, assessing, and managing risks across an organization.
- ISO 31000 is an international standard that provides guidelines and principles for effective risk management.

- Risk transfer is a risk management strategy where the risk is shifted to another party, such as through insurance or outsourcing.
 - Risk mitigation involves taking actions to reduce the probability or impact of a risk.
 - Risk acceptance is a strategy where the organization acknowledges the existence of a risk and decides not to take any action to mitigate it.
-

Q.5076 Christian Grey, an FRM Part II candidate, wishes to present on different types of internal controls, the process of internal control design, and control testing in operational risk management. Which of the following statement made by Grey is **correct**?

- A. According to the Institute of Internal Auditors, controls can be of four types, i.e., preventive, detective, corrective, and directive controls
- B. Directive controls aim to alert the firm if an incident occurs to accelerate its resolution and limit the impact of the incident on the firm or its stakeholders
- C. Examples of preventive controls include smoke alarms and credit card notifications of potentially fraudulent transactions
- D. Directive controls are always part of control taxonomies

The correct answer is **A**.

The Institute of Internal Auditors (IIA) classifies internal controls into four types: preventive, detective, corrective, and directive. Preventive controls are designed to prevent errors or irregularities from occurring. They are proactive controls that help to ensure departmental directives are carried out and that the organization's objectives are achieved. Detective controls, on the other hand, are designed to find errors or irregularities that have already occurred. Corrective controls aim to correct errors that have been detected, while directive controls guide operations towards achieving the organization's objectives. Grey's statement accurately reflects this classification.

Choice B is incorrect. Directive controls do not aim to alert the firm if an incident occurs. Instead, they are designed to guide operations towards achieving set objectives and ensuring compliance with laws and regulations. Alerting the firm of incidents is typically a function of detective controls, which identify and report on incidents that have already occurred.

Choice C is incorrect. The examples provided are not preventive controls but rather detective controls. Preventive controls aim to prevent an incident from occurring in the first place, such as segregation of duties or authorization requirements for certain transactions. Smoke alarms and credit card notifications of potentially fraudulent transactions are examples of detective controls as they identify and report on incidents after they have occurred.

Choice D is incorrect. While directive controls can be part of control taxonomies, it's not always the case as it depends on how a particular organization structures its internal control framework. Some organizations may choose to include directive controls in their taxonomy while others may not.

Things to Remember

- Internal controls are processes implemented by an organization to help achieve its objectives related to financial reporting, operations, compliance, and safeguarding assets.
 - Preventive controls are designed to prevent errors or irregularities from occurring in the first place.
 - Detective controls are designed to find errors or irregularities that have already occurred.
 - Corrective controls aim to correct errors that have been detected.
 - Directive controls guide operations towards achieving the organization's objectives and ensuring compliance with laws and regulations.
 - Control taxonomies are hierarchical structures that categorize and organize different types of controls within an organization.
-

Q.5077 Among the four ways to address risk, treatment is the most common risk response, which involves risk mitigation through various control plans. Controls can be of different classes. In this chapter, however, we have adopted the classification used by the Institute of Internal Auditors. Which of the following types of control fall under this classification?

- A. Preventive controls
- B. Key controls
- C. Manual controls
- D. Automated controls

The correct answer is **A**.

Preventive controls are one of the four types of controls classified by the Institute of Internal

Auditors (IIA). These controls are designed to reduce the likelihood of an incident occurring. They are proactive measures taken to prevent a risk from materializing. This could include actions such as implementing security measures to prevent unauthorized access to data, or establishing procedures and policies that ensure compliance with regulations. The goal of preventive controls is to deter undesirable events from happening in the first place. They are considered the most effective type of control as they help in avoiding potential risks altogether.

Choice B is incorrect. Key controls are not a type of control as per the classification provided by the Institute of Internal Auditors (IIA). They are rather an important subset of controls that are crucial for the effective functioning of a system or process, but they do not form a separate category in themselves.

Choice C is incorrect. Manual controls refer to those risk management procedures that require human intervention and oversight. While they are indeed a form of control, they do not constitute a distinct category according to IIA's classification.

Choice D is incorrect. Automated controls, similar to manual ones, involve mechanisms that operate without human intervention. However, like manual controls, they too do not represent a separate class within IIA's categorization scheme.

Things to Remember

- Risk treatment is a crucial step in the risk management process, involving the implementation of strategies to address and mitigate identified risks.
- Controls play a vital role in risk management by helping organizations monitor, evaluate, and mitigate risks effectively.
- Preventive controls focus on reducing the likelihood of risks materializing, making them a proactive risk management approach.
- Key controls are essential controls that are critical for the success and integrity of a system or process.
- Manual controls require human intervention and oversight, while automated controls operate without direct human involvement.
- The classification of controls by the Institute of Internal Auditors helps in organizing and understanding different types of controls for effective risk management.

Q.5078 David Hans, FRM, works as the risk manager at ABC Bank. In one of his presentations, David states that a firm's internal controls are its foundation for risk mitigation. He further goes ahead to state several issues concerning internal controls. Which of the following statements is

incorrect in this regard?

- A. A key control is a control that can sufficiently mitigate risk on its own.
- B. Controls can either be manual or automated in nature.
- C. Control automation can transform human error risk into technology and model risk
- D. Automated data back-up is an essential component of control testing.

The correct answer is **D**.

The statement that 'Automated data back-up is an essential component of control testing' is incorrect. Control testing is a process that involves assessing the effectiveness of a control system in mitigating risks. It does not necessarily involve automated data backup. Automated data backup, on the other hand, is a component of control automation. Control automation refers to the use of technology to automate certain control processes, thereby reducing the risk of human error. Automated data backup is a part of this process as it ensures that data is automatically saved and can be recovered in case of any data loss. Therefore, it is a component of control automation and not control testing.

Choice A is incorrect. A key control is indeed a control that can sufficiently mitigate risk on its own. It is a critical part of the internal control system and has a significant impact on the achievement of an entity's objectives.

Choice B is incorrect. Controls can either be manual or automated in nature. Manual controls are performed by individuals, while automated controls are performed by systems or applications.

Choice C is incorrect. Control automation can transform human error risk into technology and model risk, as it reduces the chance of human errors but introduces risks associated with technology failures or model inaccuracies.

Things to Remember

- Internal controls are processes implemented by an organization to help ensure the achievement of its objectives, including the reliability of financial reporting, compliance with laws and regulations, and the effectiveness and efficiency of operations.
- Key controls are controls that are essential in mitigating significant risks to an acceptable level.
- Control testing involves evaluating the design and operating effectiveness of internal controls to ensure they are functioning as intended.
- Control automation uses technology to streamline and automate control processes,

reducing the reliance on manual intervention and minimizing the risk of human error.

- Automated data backup is a critical component of control automation as it helps in ensuring data integrity, availability, and recoverability in case of data loss.
-

Q.5079 An FRM candidate is preparing for May exam. In one of the open discussion forums, the candidate states that a control should be effectively designed so as to be applied effectively and hence be able to mitigate risk effectively. Ineffectively designed controls waste resources and may give unrealistic expectations resulting in vulnerabilities. He goes ahead to state the types of weakly designed controls. Which of the following is a least likely a type of a weakly designed control highlighted by the candidate?

- A. "Optimistic control."
- B. "More of the same."
- C. "Collective controls."
- D. "System-based data validation."

The correct answer is **D**.

'System-based data validation.' This type of control is an example of automated controls. Automated controls, such as system-based data validation checks in data collection tools, are designed to ensure the accuracy and reliability of data. These controls are typically embedded in the systems and processes of an organization and operate automatically, without the need for human intervention. They are designed to prevent errors and fraud, ensure compliance with regulations, and enhance the overall effectiveness and efficiency of operations. Automated controls like system-based data validation are generally considered to be well-designed controls as they are effective, efficient, and reliable. They do not waste resources and do not create unrealistic expectations or vulnerabilities. Therefore, 'System-based data validation' is least likely to represent a type of poorly designed control.

Choice A is incorrect. "Optimistic control" refers to a type of control that assumes the best-case scenario and does not adequately prepare for potential risks or adverse events. This type of control can be poorly designed as it may not provide sufficient protection against risks.

Choice B is incorrect. "More of the same" refers to a situation where an organization continues to implement the same controls, even when they have proven ineffective in mitigating risk. This approach can lead to resource wastage and does not address underlying vulnerabilities, making it a representation of poor design.

Choice C is incorrect. "Collective controls" refer to controls that are implemented across an entire organization or system, without considering specific risk profiles or needs of different departments or units. While collective controls can be effective in some cases, they may also

overlook unique risks associated with specific areas within an organization, leading to potential vulnerabilities.

Things to Remember

- Controls should be effectively designed to mitigate risks effectively and efficiently.
 - Ineffectively designed controls can waste resources and create vulnerabilities.
 - Automated controls, such as system-based data validation, are designed to ensure accuracy and reliability of data.
 - Optimistic controls assume the best-case scenario and may not adequately prepare for risks.
 - Implementing the same controls that have proven ineffective can lead to resource wastage.
 - Collective controls implemented across an organization may overlook specific risks in different areas.
-

Q.5080 The Bank of India is in the process of implementing an effective control system. Its risk management unit has clarified that control designs should be assessed, and if satisfactory, they can be tested to check whether they are operationally effective. Which of the following is **not** a type of control testing?

- A. Examination
- B. Observation
- C. Self-certification
- D. Independence

The correct answer is **D**.

Independence of the testing party is one of the factors that influence the effectiveness of control testing.

We have four primary types of control testing, presented in their level of scrutiny. The greater the inherent risk, the more rigorous the control testing must be.

The following are the main types of control testing:

- Self-certification or inquiry. Given the lack of evidence, it is reasonable to limit this assessment to secondary controls or controls related to environments with low inherent risk.
- Examination. Written documentation of the process, as well as written evidence of the results, is needed to support this claim. The quality and relevance of documentation determine the effectiveness of this testing method. In addition, it is more suitable for automated checks and sampling of manual checks since it provides moderate assurance.
- Observation. It involves observing the execution of the control process in real time so that its design and effectiveness can be judged. This testing control is suitable for key controls.
- Reperformance (reproduction or parallel testing). This is the strongest form of testing, which involves the tester reproducing the control process on a sample of transactions and comparing the results with those previously obtained by the process.

Things to Remember

- Control testing is a crucial part of the risk management process to ensure that controls are effective in mitigating risks.
- Independence of the testing party is important to maintain objectivity and credibility in the control testing process.
- Control testing should be tailored based on the inherent risk of the process being tested.
- Regular review and updates of control testing procedures are necessary to adapt to changing risks and environments.

Q.5081 Joseph Bolts, FRM, is a risk manager at the Bank of Baroda. In his recent presentation to the board of directors, Joseph highlights that while the firm strives to establish effective control

testing, we have several factors that determine the level of this effectiveness. Which of the following factors **does not** influence the effectiveness of control testing?

- A. The independence of the testing party
- B. The frequency of testing
- C. Scope and sample
- D. Reperformance

The correct answer is **D**.

Reperformance is not a factor that influences the effectiveness of control testing. Instead, it is one of the four types of control testing. Repformance involves independently executing the controls to verify whether they are functioning as intended. While it is a method of control testing, it does not influence the effectiveness of the testing process itself. The effectiveness of control testing is determined by factors such as the independence of the testing party, the frequency of testing, and the scope and sample size of the test. Therefore, reperformance does not influence the effectiveness of control testing.

Choice A is incorrect. The independence of the testing party can significantly impact the effectiveness of control testing. If the party conducting the test has a vested interest in the outcome, it may lead to biased results and thus affect its effectiveness.

Choice B is incorrect. The frequency of testing also plays a crucial role in determining its effectiveness. Infrequent tests may not capture all potential risks or control failures, thereby reducing their efficacy.

Choice C is incorrect. The scope and sample size used for control testing can greatly influence its effectiveness as well. A narrow scope or small sample size might not provide an accurate representation of all possible scenarios, leading to ineffective controls.

Things to Remember

- Control testing is a crucial component of risk management that helps ensure the effectiveness of internal controls within an organization.
- The independence of the testing party is essential to maintain objectivity and credibility in the control testing process.
- The frequency of testing determines how often controls are assessed, with more frequent testing generally leading to better risk mitigation.
- The scope and sample size of control testing refer to the extent of coverage and the number of items tested, which can impact the reliability of the results.

- Repformance, as mentioned in the question, is a type of control testing method that involves independently executing controls to verify their functionality.
-

Q.5082 To effectively mitigate human errors, we should first categorize these errors accordingly. Identifying slips and mistakes is the first step in categorizing human error. Which of the following categories of human errors is correctly described?

- A. Slips – These are wrong choices made when someone faces a new situation due to a lack of familiarity with a process
- B. Rule-based mistakes – The perpetrator understands the right thing to do but decides to act against the rules
- C. Knowledge-based mistakes – These are the wrong choices made when someone faces a new situation due to a lack of familiarity with a process or a lack of training and guidance
- D. Violation – These are involuntary errors caused by inattention, distraction, and fatigue

The correct answer is C.

Knowledge-based mistakes refer to incorrect decisions made when an individual encounters a new situation and lacks familiarity with the process or lacks adequate training and guidance. These mistakes occur when a person does not have the necessary knowledge to handle a situation correctly. For instance, a new employee might make a knowledge-based mistake when faced with a task they have not been trained to perform. This type of error can be mitigated through proper training and guidance, ensuring that individuals are adequately prepared to handle new situations. It's also important to create an environment where individuals feel comfortable asking for help when they encounter unfamiliar situations.

Choice A is incorrect. Slips are not wrong choices made when someone faces a new situation due to a lack of familiarity with a process. Instead, slips are typically associated with automatic behavior and occur when attention is diverted from the task at hand.

Choice B is incorrect. Rule-based mistakes do not involve an individual deciding to act against the rules. Rather, they occur when an individual makes an error in the application or interpretation of a rule that they understand correctly.

Choice D is incorrect. Violations are not involuntary errors caused by inattention, distraction, and fatigue. They are deliberate deviations from procedures, rules or standards and often result from conscious decisions by individuals who believe their actions will achieve a desired outcome.

Things to Remember

- Human error can be categorized into different types such as slips, mistakes, rule-based mistakes, knowledge-based mistakes, and violations.

- Slips are errors that occur during automatic behavior when attention is diverted from the task at hand.
 - Mistakes can be rule-based or knowledge-based, depending on the understanding and application of rules or knowledge in a given situation.
 - Rule-based mistakes involve errors in the application or interpretation of rules that are understood correctly.
 - Knowledge-based mistakes occur when individuals lack the necessary knowledge or training to handle a new situation correctly.
 - Violations are deliberate deviations from procedures, rules, or standards, often resulting from conscious decisions by individuals.
 - Proper training, guidance, and creating an environment where individuals feel comfortable seeking help can help mitigate human errors.
-

Q.5083 To improve the quality of an operational process and reduce the potential for human error, the risk management unit should first identify these errors and then apply several methods to assess and mitigate risks related to these errors. Which of the following statements is **incorrect** in light of this statement?

- A. The Lean Six Sigma is applied to remove and reduce waste and variation by analyzing processes and collaborative tasks hence minimizing variations
- B. Six Sigma improves quality by identifying and eliminating causes of errors or defects and minimizing variability in industrial processes
- C. Quality improvement follows the 'plan', 'do', 'study', 'act' (PDSA) cycle.
- D. Under the "Dr. Deming cycle", 'do' refers to analyzing the collected data, comparing the set targets, and evaluating opportunities for improvement.

The correct answer is **D**.

The statement in choice D is incorrect because it misrepresents the 'Do' phase of the 'Plan', 'Do', 'Study', 'Act' (PDSA) cycle, also known as the 'Dr. Deming cycle'. In this cycle, 'Do' refers to the implementation of the plan and the recording of its progress. It does not involve the analysis of collected data, comparison of set targets, or evaluation of opportunities for improvement. These

activities are part of the 'Study' phase. Therefore, the statement in choice D is incorrect, making it the correct answer to the question.

Choice A is incorrect. The Lean Six Sigma methodology indeed focuses on reducing waste and variation in processes. It does this by identifying and removing the causes of defects, minimizing variability, and improving the quality of process outputs.

Choice B is incorrect. Six Sigma is a set of techniques used for process improvement by eliminating defects and ensuring quality. It aims to reduce process variability, thereby enhancing product or service consistency.

Choice C is incorrect. The 'plan', 'do', 'study', 'act' (PDSA) cycle accurately describes a common approach to continuous quality improvement in operational risk management methodologies.

Things to Remember

- Operational Risk Management: The process of identifying, assessing, and prioritizing risks in operational processes to minimize potential losses.
 - Human Error: Mistakes or oversights made by individuals that can lead to operational failures or losses.
 - Risk Assessment: The process of evaluating potential risks, their likelihood, and impact on operations.
 - Risk Mitigation: Strategies and actions taken to reduce the impact or likelihood of identified risks.
 - Continuous Improvement: The ongoing effort to improve processes, products, or services by making small, incremental changes.
-

Q.5084 Businesses face significant operational risks when they embark on new projects, products, and initiatives that are unfamiliar or unfamiliar to them. Which of the following statements is **correct** in this context?

- A. As a best practice, the owner of each new initiative should present a business case to show the allocation of resources
- B. When acquiring new assets, it is easier to assess operational risk than credit risk
- C. The acquired firm should not provide any information as this makes operational risk assessment even more difficult

D. When projects are merged, the risks of the acquired assets remain with the original firm

The correct answer is **A**.

The owner of each new initiative should present a business case to show the allocation of resources. This is considered a best practice in business operations. A well-structured business case provides a comprehensive overview of the initiative, including its objectives, alternatives, expected benefits, commercial aspects, and risks. By presenting a business case, the owner of the initiative can effectively communicate the potential value and risks of the initiative to stakeholders. This can facilitate informed decision-making and ensure that resources are allocated appropriately. Furthermore, a business case can serve as a reference point for monitoring and controlling the initiative as it progresses.

Choice B is incorrect. Operational risk assessment can be more complex than credit risk assessment when acquiring new assets. This is because operational risks involve a wide range of uncertainties, including those related to the integration of new assets into existing operations, changes in market conditions, and potential regulatory issues.

Choice C is incorrect. The acquired firm should provide as much information as possible to facilitate the operational risk assessment process. Withholding information can lead to incomplete or inaccurate risk assessments, which could potentially expose the business to unforeseen risks and liabilities.

Choice D is incorrect. When projects are merged, the risks associated with the acquired assets do not necessarily remain with the original firm. Instead, these risks are typically transferred to or shared with the acquiring entity depending on the terms of acquisition agreement.

Things to Remember

- **Operational Risk:** Involves the risk of loss resulting from inadequate or failed internal processes, people, and systems, or from external events.
- **Credit Risk:** The risk of loss resulting from a borrower's failure to repay a loan or meet contractual obligations.
- **Business Case:** A document that outlines the justification for a project or initiative by detailing its objectives, benefits, costs, and risks.
- **Stakeholders:** Individuals or groups who have an interest in the success of a project or initiative and can influence or be influenced by its outcomes.
- **Integration Risks:** Risks associated with combining new assets or operations with existing ones, including challenges in systems, processes, and culture.

- Regulatory Risks: Risks arising from non-compliance with laws, regulations, or industry standards that can lead to financial penalties or reputational damage.
-

Q.5085 Bank A wishes to acquire all the assets of Bank B. The risk unit of Bank A is therefore concerned about the possible operational risks that may arise if they go ahead to acquire assets of Bank B. Which of the following is **not** a correct way in which the risk function of Bank A will involve in the acquisition of assets of Bank B?

- A. Doing a thorough assessment of the operational risk related to the assets of Bank B
- B. The risk unit of Bank A should ask Bank B to present information on payrolls, customers, payroll and management systems, and its communication with other companies
- C. The board of directors can create a risk profile to familiarize the management with potential operational risks related to these new business initiatives.
- D. Bank B should provide Bank A with data on collateral, obligors, and terms and conditions in order for them to assess credit risk

The correct answer is **C**.

The statement that the board of directors can create a risk profile to familiarize the management with potential operational risks related to these new business initiatives is incorrect. In the context of risk management, it is the Operational Risk Management (ORM) function that is responsible for creating a risk profile. The ORM function is designed to identify, assess, monitor, and control operational risk. It is not the responsibility of the board of directors to create a risk profile. The board of directors is responsible for overseeing the risk management framework and ensuring that it is effectively implemented. However, the day-to-day management of risk, including the creation of risk profiles, is typically delegated to the ORM function or other risk management professionals within the organization.

Choice A is incorrect. The risk unit of Bank A should indeed conduct a thorough assessment of the operational risks related to the assets of Bank B. This is an integral part of their role in the asset acquisition process.

Choice B is incorrect. It's not wrong for the risk unit of Bank A to ask for information on payrolls, customers, payroll and management systems, and its communication with other companies from Bank B. This information can help them understand potential operational risks better.

Choice D is incorrect. While it's true that data on collateral, obligors, and terms and conditions are important for assessing credit risk, this doesn't negate the fact that it's also part of the role of Bank A's risk function in an asset acquisition process.

Things to Remember

- Operational Risk Management (ORM) function is responsible for creating a risk profile to identify, assess, monitor, and control operational risk.
 - Board of directors oversee the risk management framework and ensure its effective implementation.
 - Day-to-day management of risk, including the creation of risk profiles, is typically delegated to the ORM function or other risk management professionals within the organization.
 - Thorough assessment of operational risks related to acquired assets is crucial for the risk unit of Bank A.
 - Information on payrolls, customers, management systems, and communication with other companies can help in understanding potential operational risks.
 - Data on collateral, obligors, and terms and conditions are important for assessing credit risk in asset acquisition processes.
-

Q.5086 Paul Schering, FRM, works as a risk manager at ABC Bank. Paul wishes to present to the bank approaches firms should use to mitigate the impact of operational risk events. Which of the following statements highlighted by Paul is **correct**?

- A. A contingency plan is simply a "Plan B" or an alternative action if the result of a future event does not go as expected
- B. The first step in business continuity management (BCM) is identifying threats and risks and linking these risks to the firm's key operational risks
- C. In case of a crisis, a firm should have at least one response team: the technical team
- D. In case of a crisis, a communications team to assess the risk event and restore normal processes

The correct answer is **A**.

A contingency plan is indeed a 'Plan B' or an alternative course of action if the outcome of a future event does not proceed as anticipated. Contingency planning is a component of business continuity management (BCM), disaster recovery plans (DRP), and corrective risk management. It should clearly specify who is responsible for what and when in the event of a crisis. In broader

terms, contingency planning involves providing alternatives in systems, people, and processes. This ensures that the organization can continue to function and recover quickly in the event of a disruption or crisis.

Choice B is incorrect. The first step in business continuity management (BCM) is not identifying threats and risks and linking these risks to the firm's key operational risks. Rather, it involves understanding the organization, which includes identifying critical products and services, defining the organization's risk appetite, and understanding legal and regulatory requirements.

Choice C is incorrect. While having a technical team in place during a crisis can be beneficial, it should not be the only response team a firm has. A comprehensive crisis management plan should include multiple teams with different areas of expertise such as operations, communications, legal etc., each tasked with specific responsibilities to ensure an effective response.

Choice D is incorrect. A communications team alone cannot assess the risk event and restore normal processes during a crisis situation. This requires a coordinated effort from various teams including risk assessment team for evaluating the severity of risk event , operations team for restoring normal processes along with communication team for managing internal & external communication.

Things to Remember

- Business Continuity Management (BCM) involves developing strategies to ensure that essential business functions can continue during and after a disaster or crisis.
- Disaster Recovery Plans (DRP) focus on the recovery of IT systems and infrastructure after a disaster.
- Corrective Risk Management involves identifying and addressing risks before they materialize into issues.
- Contingency planning should clearly define roles and responsibilities in the event of a crisis.
- A comprehensive crisis management plan should include multiple response teams with different areas of expertise.
- Communication is a critical aspect of crisis management, but it is not the sole responsibility of a communications team to assess and restore normal processes.

Q.5088 Operational risk can be transferred through external insurance and outsourcing. Which

of the following statements is incorrect regarding risk transfer?

- A. There is a trade-off decision between the insurance premium versus the volatility
- B. In external insurance, the risk is not necessarily fully transferred, as the amount of compensation depends on the premiums paid
- C. Outsourcing may result in third-party risk
- D. It is hard to transfer both risk exposure and consequences

The correct answer is **D**.

The statement 'It is hard to transfer both risk exposure and consequences' is incorrect. In the context of operational risk, external insurance policies are particularly suitable for risks that are predictable and easy to transfer in terms of both risk exposure and consequences. This makes risk mitigation effective for those who take out insurance. The predictability of the risk allows for proper underwriting and pricing by the insurer, ensuring that the risk transfer is beneficial for both parties. Therefore, it is not necessarily hard to transfer both risk exposure and consequences, as the statement suggests.

Choice A is incorrect. The statement accurately represents a consideration in risk transfer. There is indeed a trade-off decision between the insurance premium and the volatility of the risk being insured. Higher volatility risks would typically require higher premiums for coverage.

Choice B is incorrect. This statement correctly highlights that in external insurance, the risk isn't necessarily fully transferred as the amount of compensation depends on the premiums paid. If an event occurs that exceeds the coverage limit set by the premium, then some of that risk remains with the company.

Choice C is incorrect. This statement correctly identifies another potential issue with outsourcing as a strategy for operational risk mitigation - it may result in third-party risks such as vendor or supply chain disruptions, which are outside of direct control by your company.

Things to Remember

- Risk transfer is a key component of risk management strategies, allowing organizations to mitigate potential losses by shifting the risk to another party.
- External insurance involves purchasing insurance policies from third-party insurers to protect against specific risks, such as operational, liability, or property risks.
- Outsourcing refers to the practice of contracting out certain business functions or processes to external vendors or service providers.
- Third-party risk is the potential risk that arises from engaging with external parties,

such as vendors, suppliers, or service providers, and can impact the organization's operations and reputation.

- Effective risk transfer requires careful consideration of the terms of the insurance policy or outsourcing agreement to ensure that the organization is adequately protected.
-

Reading 108: Risk Reporting

Q.5091 Which of the following is **not** a type of information critical in the operational risk requirements?

- A. Qualitative information on operational risk management
- B. Historical losses
- C. Business indicator and subcomponents
- D. Risk appetite metrics

The correct answer is **D**.

Risk appetite metrics, while important in the broader context of risk management, are not considered a type of information that is critical in operational risk requirements. Risk appetite metrics are used to measure the level of risk that an organization is willing to accept in pursuit of its objectives. They are typically defined by the organization's board of directors and senior management and are used to guide the organization's risk-taking activities. However, in the context of operational risk requirements, the focus is more on specific types of information that can help identify, assess, and manage operational risks. These include qualitative information on operational risk management, historical losses, and business indicator and subcomponents. Therefore, risk appetite metrics do not fall into this category.

Choice A is incorrect. Qualitative information on operational risk management is indeed considered critical in operational risk requirements. This type of information provides insights into the organization's culture, governance, and other non-quantifiable aspects that can significantly impact the level of operational risks.

Choice B is incorrect. Historical losses are also deemed critical in operational risk requirements as they provide a historical perspective on the frequency and severity of losses due to operational risks. They serve as a basis for estimating potential future losses.

Choice C is incorrect. Business indicators and subcomponents are crucial pieces of information in understanding and managing operational risks. They help identify areas within an organization that may be vulnerable to significant operational risks.

Things to Remember

- Risk appetite metrics are used to measure the level of risk that an organization is willing to accept in pursuit of its objectives.
- Qualitative information on operational risk management provides insights into the organization's culture, governance, and other non-quantifiable aspects.
- Historical losses help in understanding the frequency and severity of losses due to

operational risks.

- Business indicators and subcomponents are crucial in identifying areas vulnerable to operational risks within an organization.
-

Q.5092 Which of the following types of information pertains to the presentation of an entity's governance and risk management structures that are established to manage and mitigate risk?

- A. Qualitative information on operational risk management
- B. Historical losses
- C. Business indicators and subcomponents
- D. Incidents and near misses

The correct answer is **A**.

Qualitative information is crucial for presenting an organization's governance and risk management structures. Operational risk management involves the identification, assessment, monitoring, and mitigation of risks that arise from operational failures such as system failures, procedural errors, or disruptions. Qualitative information in this context could include details about the organization's risk appetite, risk management strategies, risk mitigation measures, and the roles and responsibilities of different entities within the organization in managing risk. This information provides a comprehensive view of the organization's approach to managing operational risk, thereby helping stakeholders understand how the organization is structured to manage and mitigate risk.

Choice B is incorrect. Historical losses are important for understanding the past performance of an organization and can be used to predict future risks. However, they do not specifically relate to the presentation of governance and risk management structures.

Choice C is incorrect. Business indicators and subcomponents can provide insights into the operational efficiency and financial health of an organization, but they do not directly depict the governance and risk management structures.

Choice D is incorrect. Incidents and near misses are crucial for identifying potential risks in an organization's operations, but they do not specifically represent the governance or risk management structures in place.

Things to Remember

- **Operational Risk Management:** Involves the identification, assessment, monitoring, and mitigation of risks that arise from operational failures.

- Risk Appetite: Refers to the level of risk that an organization is willing to accept in pursuit of its objectives.
 - Risk Management Strategies: Plans and actions implemented by an organization to identify, assess, and mitigate risks.
 - Risk Mitigation Measures: Steps taken to reduce the impact or likelihood of risks occurring.
 - Roles and Responsibilities: Clearly defined duties and accountabilities of individuals or entities within an organization in managing risk.
-

Q.5094 Which of the following components of operational risk reporting involves reporting a list of the top overall risks?

- A. Heatmap and risk register.
- B. Risk appetite metrics.
- C. Top-10 risks and risk outlook.
- D. Incidents and near misses.

The correct answer is **C**.

The 'Top-10 risks and risk outlook' is indeed the component of operational risk reporting that involves reporting a list of the top overall risks. This component is crucial as it provides a snapshot of the most significant risks that an organization faces at a given point in time. It is often derived from the risk register or risk inventory, which is a comprehensive list of all the risks identified by the organization. The 'Top-10 risks and risk outlook' not only lists these risks but also provides an outlook on how these risks might evolve in the future. This information is invaluable for decision-makers as it helps them prioritize their risk mitigation efforts and allocate resources effectively.

Choice A is incorrect. While a heatmap and risk register are components of operational risk reporting, they do not specifically report the top-10 overall risks. A heatmap visually represents the severity and likelihood of risks, while a risk register records details about identified risks.

Choice B is incorrect. Risk appetite metrics are used to measure an organization's willingness to take on risk, but they do not provide a list of the top-10 overall risks.

Choice D is incorrect. 'Incidents and near misses' refers to actual events or situations that almost led to an undesired outcome such as financial loss or legal issues. This component does

not necessarily report on the top-10 overall risks in an organization.

Things to Remember

- Operational risk reporting is essential for organizations to identify, assess, monitor, and mitigate risks that could impact their operations and objectives.
 - Heatmaps are visual tools that help organizations prioritize risks based on their severity and likelihood of occurrence.
 - Risk appetite metrics help organizations understand and quantify the level of risk they are willing to accept in pursuit of their objectives.
 - Incidents and near misses are valuable sources of information for organizations to learn from past events and improve their risk management processes.
 - Top-10 risks and risk outlook provide a focused view of the most critical risks facing an organization, helping decision-makers allocate resources effectively.
-

Q.5095 Why are near-miss occurrences included in the reporting of incidents in organizations with strong risk cultures?

- A. To assess the cost of close calls.
- B. To determine the importance of close calls.
- C. To analyze the potential consequence that was unintentionally avoided.
- D. To evaluate the frequency of close calls.

The correct answer is **C**.

Organizations with strong risk cultures include near-miss occurrences in the reporting of incidents primarily to analyze the potential consequence that was unintentionally avoided. This practice allows them to understand the severity of the risk that was narrowly missed and to take necessary precautions to prevent such incidents in the future. It helps in identifying the vulnerabilities in the system and provides an opportunity to improve the risk management strategies. By analyzing the potential consequences, organizations can learn from these near-miss incidents without having to experience the actual loss or damage. This proactive approach to risk management is a hallmark of organizations with strong risk cultures.

Choice A is incorrect. While assessing the cost of close calls can be a part of risk management,

it is not the primary reason for including near-miss occurrences in incident reports. The main purpose is to analyze what could have happened had the event not been avoided, which helps in understanding and mitigating future risks.

Choice B is incorrect. Determining the importance of close calls may be a part of risk analysis but it's not the primary reason for their inclusion in incident reports. The key objective is to understand potential consequences that were unintentionally avoided, thereby helping organizations prepare better for similar situations in future.

Choice D is incorrect. Evaluating the frequency of close calls can provide insights into how often such incidents occur, but this does not directly contribute to understanding or mitigating potential risks that were unintentionally avoided - which remains as the main purpose behind including near-miss occurrences in incident reports.

Things to Remember

- Organizations with strong risk cultures prioritize learning from near-miss incidents to improve risk management strategies.
 - Near-miss occurrences are valuable in identifying vulnerabilities in the system and enhancing risk mitigation measures.
 - Understanding the potential consequences of near-miss incidents helps in preparing for similar situations in the future.
 - Proactive risk management involves analyzing near-miss occurrences to prevent actual losses or damages.
 - Assessing the cost, determining importance, and evaluating frequency are secondary considerations in comparison to learning from potential consequences of near-miss incidents.
-

Q.5096 Which of the following is not one of the three options worth considering when aggregating qualitative data?

- A. Conversion and addition
- B. Categorization
- C. Horizon scanning
- D. Worst-case reporting

The correct answer is **C**.

Choice A is incorrect. Conversion and addition is indeed a method used for aggregating qualitative data. This method involves converting the qualitative data into numerical form and then adding it up to get a total score or value.

Choice B is incorrect. Categorization is also a common method used in the aggregation of qualitative data. It involves grouping similar types of data together into categories, which can then be analyzed collectively.

Choice D is incorrect. Worst-case reporting, while not as commonly used as the other methods, can still be considered a form of aggregating qualitative data. This approach focuses on identifying and presenting the worst possible outcomes or scenarios based on the available data.

Things to Remember

- Qualitative data: Data that is descriptive and non-numeric in nature, often obtained through interviews, observations, or open-ended survey questions.
 - Quantitative data: Data that is numerical and can be quantified, often obtained through structured surveys, experiments, or measurements.
 - Data aggregation: The process of combining and summarizing data from multiple sources or variables to provide a more comprehensive view or analysis.
 - Weighting: Assigning different levels of importance or influence to different data points or variables when aggregating data.
 - Scenario analysis: A technique used to analyze possible future events by considering multiple scenarios or outcomes based on different assumptions.
-

Q.5097 Which of the following stakeholder groups is authorized by the board to monitor the effectiveness of the firm's risk management framework?

- A. The audit committee
- B. The risk committee
- C. Executive committee
- D. Business line managers

The correct answer is **B**.

The risk committee is a specialized group within the board of directors that is specifically authorized to monitor the effectiveness of the firm's risk management framework. This committee is typically composed of members who have a deep understanding of the various risks that the firm might face, including operational, financial, and strategic risks. The risk committee's primary role is to ensure that the firm has robust risk management policies and procedures in place and that these are being effectively implemented. The committee also reviews and assesses the firm's risk profile and risk appetite, and ensures that the firm's risk management activities align with its strategic objectives. The risk committee plays a crucial role in promoting a strong risk culture within the firm, and in ensuring that risk considerations are integrated into decision-making processes at all levels of the firm.

Choice A is incorrect. The audit committee is primarily responsible for overseeing the integrity of the company's financial statements, internal controls over financial reporting, and the performance of internal and external audits. While they may review risk management policies and procedures as part of their duties, they are not specifically authorized to oversee and evaluate the effectiveness of the firm's risk management framework.

Choice C is incorrect. The executive committee typically handles a range of issues including strategic planning, resource allocation, and operational decisions. Although they might be involved in discussions about risk management strategies or initiatives, their primary role does not include overseeing or evaluating the effectiveness of these frameworks.

Choice D is incorrect. Business line managers are responsible for managing risks within their specific business units or departments but do not have a firm-wide oversight role in evaluating the effectiveness of overall risk management framework.

Things to Remember

- The risk committee is a specialized group within the board of directors specifically authorized to monitor the effectiveness of the firm's risk management framework.
- The audit committee is responsible for overseeing the integrity of the company's financial statements and internal controls over financial reporting.
- The executive committee handles strategic planning, resource allocation, and

operational decisions.

- Business line managers are responsible for managing risks within their specific business units or departments.
 - Effective risk management involves identifying, assessing, and mitigating risks that could impact the firm's objectives and performance.
-

Q.5098 Which of the following stakeholder groups is responsible for collecting all relevant operational risk information from the business lines to produce aggregated, synthesized reporting and provide feedback to the business lines?

- A. The risk champions
- B. The operational risk committee
- C. The audit committee
- D. The executive committee

The correct answer is **B**.

The operational risk committee is the group responsible for collecting all relevant operational risk information from the business lines. This committee is tasked with producing aggregated, synthesized reporting based on the collected data. The operational risk committee plays a crucial role in operational risk management as it provides a holistic view of the operational risks faced by the organization. By aggregating and synthesizing the risk information, the committee can identify patterns, trends, and areas of concern that may not be apparent at the individual business line level. This comprehensive view allows the committee to provide valuable feedback to the business lines, helping them to manage their operational risks more effectively.

Choice A is incorrect. The risk champions are typically responsible for promoting a risk-aware culture within their respective business lines, not for the collection and consolidation of operational risk data.

Choice C is incorrect. The audit committee's role primarily involves overseeing the organization's internal control systems, financial reporting processes, and audits of financial statements. They do not typically handle the collection and consolidation of operational risk data.

Choice D is incorrect. The executive committee usually focuses on strategic decision-making based on the information provided by other groups such as the operational risk committee. They do not directly involve themselves in collecting or consolidating operational risk data.

Things to Remember

- Operational Risk Management: The process of identifying, assessing, and controlling operational risks within an organization.
 - Business Lines: Different divisions or units within an organization that are responsible for specific business activities.
 - Aggregated Reporting: Combining data from multiple sources to provide a comprehensive view of risks or performance.
 - Synthesized Reporting: Analyzing and summarizing data to extract key insights and trends.
 - Risk Champions: Individuals within business lines who promote risk awareness and best practices.
 - Audit Committee: A subcommittee of the board of directors responsible for overseeing financial reporting and internal controls.
 - Executive Committee: A group of top executives responsible for strategic decision-making within an organization.
-

Q.5099 Which of the following is a challenge of non-financial risk data reporting?

- A. Risk appetite metrics
- B. Action plans and follow up
- C. Asymmetry of operational risk event data
- D. Incidents and near misses

The correct answer is **C**.

Asymmetry of operational risk event data is indeed a challenge in non-financial risk data reporting. This is because operational risk event data is often characterized by a small number of low-frequency, high-severity loss occurrences that account for the majority of operational loss severity. This asymmetry can make it difficult to accurately assess and manage operational risk, as the majority of losses are caused by relatively rare events. Therefore, addressing this asymmetry is a key challenge in non-financial risk data reporting.

Choice A is incorrect. Risk appetite metrics are not a challenge in non-financial risk data reporting but rather a tool used to measure and manage the level of risk an organization is willing to accept. They help in decision-making and strategic planning, but do not pose a challenge in the reporting process itself.

Choice B is incorrect. Action plans and follow-ups are part of the risk management process, which includes identifying risks, assessing their potential impact, developing strategies to mitigate them, and monitoring progress. While they may present challenges in terms of execution or compliance, they do not represent a specific challenge related to non-financial risk data reporting.

Choice D is incorrect. Incidents and near misses are sources of operational risk data that need to be reported for effective management of non-financial risks. However, they themselves do not constitute a challenge in the reporting process; instead, how these incidents are recorded, analyzed and communicated could potentially pose challenges.

Things to Remember

- **Operational risk:** Operational risk is the risk of loss resulting from inadequate or failed internal processes, people, and systems, or from external events. It includes risks such as fraud, legal risks, and cyber risks.
 - **Loss severity:** Loss severity refers to the magnitude of a loss that occurs as a result of an operational risk event. Understanding the severity of losses is crucial for effective risk management.
 - **Risk management process:** The risk management process involves identifying, assessing, prioritizing, and mitigating risks to minimize their impact on an organization's objectives. It is a continuous process that helps organizations proactively manage risks.
 - **Reporting and communication:** Effective reporting and communication of non-financial risk data are essential for decision-making and strategic planning. Clear and accurate reporting helps stakeholders understand the risks facing the organization and take appropriate actions.
-

Reading 109: Integrated Risk Management

Q.5100 Which of the following statements is correct regarding Integrated Risk Management?

- A. It is a framework that focuses on identifying and managing individual risks separately
- B. It involves the centralization of risk management activities within a single department
- C. It considers the potential impact of multiple risks occurring simultaneously
- D. It is primarily focused on compliance and regulatory requirements

The correct answer is **C**.

Integrated Risk Management (IRM) indeed considers the potential impact of multiple risks occurring simultaneously. This is a key characteristic of the IRM approach. Rather than treating risks as isolated events, IRM recognizes that risks are often interconnected and that the occurrence of one risk can trigger or exacerbate other risks. This understanding of the interrelationships between risks allows organizations to better anticipate and prepare for potential risk events. By considering the potential impact of multiple risks occurring simultaneously, organizations can develop more robust risk management strategies that take into account the complexity and interconnectedness of risks.

Choice A is incorrect. Integrated Risk Management (IRM) does not focus on identifying and managing individual risks separately. Instead, it takes a comprehensive view of all risks across an organization, considering their potential interrelationships and the possibility of multiple risks occurring simultaneously.

Choice B is incorrect. While IRM may involve some degree of centralization in risk management activities, its primary characteristic is not the centralization within a single department. Rather, it emphasizes on a holistic approach to risk management that spans across different departments and functions within an organization.

Choice D is incorrect. Although compliance and regulatory requirements are important aspects of any risk management strategy, they are not the primary focus of Integrated Risk Management. The main aim of IRM is to identify, prioritize, and manage risks in a way that considers their potential interrelationships and simultaneous occurrence.

Things to Remember

- Integrated Risk Management (IRM) is a holistic approach to risk management that considers the interconnectedness of risks.
- IRM focuses on identifying, prioritizing, and managing risks in a way that takes into account the potential impact of multiple risks occurring simultaneously.
- Risks are often interrelated, and the occurrence of one risk can trigger or exacerbate

other risks, highlighting the importance of an integrated approach to risk management.

- IRM involves collaboration and coordination across different departments and functions within an organization to effectively manage risks.
 - While compliance and regulatory requirements are important, they are not the primary focus of Integrated Risk Management.
-

Q.5103 A newly hired risk manager is preparing to present to the risk committee on the role of ERM in financial services in ensuring the solvency and sustainability of an institution through appropriate capital funding that covers any unexpected losses relating to any of the main risk classes. Which of the following points highlighted by the risk manager is **correct**?

- A. An enterprise risk management framework and activities consist of regulatory capital and economic capital only
- B. Regulatory capital is the internal capital that firms estimate, reflecting both their risk profile and potential needs to cover unexpected losses
- C. Pillar 2, introduced under Basel II, is about market discipline
- D. Basel regulations bear no legal grounds

The correct answer is **D**.

Basel regulations do not have any legal standing. Instead, individual countries decide whether to incorporate the Basel standards into their domestic laws and regulations. The Basel Accords, developed by the Basel Committee on Banking Supervision (BCBS), are a set of recommendations for regulations in the banking industry. While these accords are not legally binding, they have been widely adopted worldwide due to their comprehensive approach to banking supervision. The accords aim to ensure that financial institutions have enough capital on account to meet obligations and absorb unexpected losses. However, the implementation of these standards is at the discretion of individual countries, and they are not legally enforceable unless enacted into law by the respective governments.

Choice A is incorrect. An enterprise risk management framework and activities do not consist of regulatory capital and economic capital only. It also includes operational risk, credit risk, market risk, liquidity risk, strategic risk among others.

Choice B is incorrect. Regulatory capital is not the internal capital that firms estimate to cover unexpected losses. Instead, it's the minimum amount of equity a bank must hold to reduce the risk of insolvency set by banking regulators.

Choice C is incorrect. Pillar 2 under Basel II does not pertain to market discipline; rather it

deals with supervisory review process which allows regulators to review an institution's internal assessment process and ensure that they hold sufficient capital for their level of risks.

Things to Remember

- Enterprise Risk Management (ERM) framework includes regulatory capital, economic capital, operational risk, credit risk, market risk, liquidity risk, strategic risk, among others.
 - Regulatory capital is the minimum amount of equity a bank must hold to reduce the risk of insolvency, set by banking regulators.
 - Pillar 2 under Basel II pertains to the supervisory review process, allowing regulators to review an institution's internal assessment process and ensure sufficient capital for their risks.
 - Basel regulations are recommendations by the Basel Committee on Banking Supervision (BCBS) and are not legally binding unless adopted into domestic laws by individual countries.
-

Q.5105 A risk manager of a large bank recommends that the bank should consider not only regulatory and economic capital requirements but also assess aggregate capital needs. Which of the following statements is **correct** regarding capital aggregation and diversification in the ERM context?

- A. Diversification can only be achieved across different risk classes
- B. To determine the risk capital for a particular business unit within a larger firm, each unit is viewed individually.
- C. We have diversification benefits whenever we have a correlation of exactly +1
- D. We can have large diversification benefits when operational risk is aggregated with other risks

The correct answer is **D**.

The statement that we can have large diversification benefits when operational risk is aggregated with other risks is accurate. This is because operational risk tends to behave independently, unlike credit and market risks, which often show increased correlations during a crisis. Therefore, by aggregating operational risk with other types of risks, a firm can achieve

significant diversification benefits. This is a key principle in Enterprise Risk Management (ERM), where the goal is to manage and mitigate various types of risks within a firm in a holistic manner.

Choice A is incorrect. Diversification can be achieved not only across different risk classes but also within the same risk class. For example, a bank can diversify its credit risk by lending to borrowers in different industries or geographical locations.

Choice B is incorrect. This statement does not accurately reflect the principles of capital aggregation and diversification. Capital aggregation involves summing up individual risks to get a total risk measure, while diversification involves spreading risks across various assets or business units to reduce exposure.

Choice C is incorrect. Diversification benefits do not occur when we have a correlation of exactly +1 between two risks. In fact, if two risks are perfectly positively correlated (+1), there would be no diversification benefit at all because both risks would move in the same direction at the same time.

Things to Remember

- Capital aggregation involves summing up individual risks to get a total risk measure
 - Diversification benefits can be achieved not only across different risk classes but also within the same risk class
 - Correlation of exactly +1 between two risks eliminates diversification benefits
 - Operational risk tends to behave independently, making it a good candidate for diversification benefits when aggregated with other risks
 - ERM focuses on managing and mitigating various types of risks within a firm in a holistic manner
-

Q.5108 The operational risk manager of a bank has asked a junior analyst to model total operational risk losses and the frequency and severity of operational risk losses. Which of the following method would the junior analyst apply?

- A. Reverse stress testing
- B. Loss distribution approach
- C. Monte Carlo simulation
- D. AMA approach

The correct answer is **B**.

The LDA is a commonly used method for modeling operational risk losses in banks. It involves the use of statistical techniques to estimate the distribution of losses that could arise from operational risk. The LDA models the frequency and severity of losses separately and then combines them to estimate the total loss distribution. Some LDA models, such as frequency and severity models, project losses based on Monte Carlo simulations. This approach allows for a comprehensive analysis of potential losses, taking into account both the likelihood and impact of operational risk events. Therefore, the LDA would be the most appropriate method for the junior analyst to use in this scenario.

Choice A is incorrect. Reverse stress testing is a risk management tool used to evaluate the potential impact of severe events or market conditions on a bank's financial condition. It does not specifically model the total operational risk losses, nor the frequency and severity of these losses.

Choice C is incorrect. Monte Carlo simulation is a computational algorithm that relies on repeated random sampling to obtain numerical results; however, it does not specifically focus on modeling operational risk losses in banking.

Choice D is incorrect. The Advanced Measurement Approach (AMA) for operational risk allows banks to develop their own empirical model to quantify required capital for operational risk, but it doesn't directly involve modeling the total operational risk losses as well as their frequency and severity.

Things to Remember

- **Operational Risk:** The risk of loss resulting from inadequate or failed internal processes, people, and systems, or from external events.
- **Loss Distribution Approach (LDA):** A method used to model operational risk losses by estimating the distribution of potential losses based on frequency and severity models.
- **Frequency and Severity Models:** Models used in LDA to separately estimate the frequency of operational risk events and the severity of the losses that could result from these events.
- **Monte Carlo Simulation:** A computational algorithm that uses random sampling to model the probability of different outcomes in a process that involves uncertainty.
- **Reverse Stress Testing:** A risk management tool used to assess the impact of extreme events or adverse market conditions on a bank's financial stability.
- **Advanced Measurement Approach (AMA):** An approach that allows banks to develop

their own models to quantify required capital for operational risk based on their internal data and risk management practices.

Q.5109 A newly hired risk manager of a bank wishes to implement a robust operational risk stress testing framework at the bank. Which of the following is a potential challenge the manager is **unlikely** to face when developing and implementing models used in stress testing Operational risk?

- A. Legal risk is characterized by the delay between adverse macroeconomic conditions and legal losses suffered by banks
- B. It is challenging for Conditional LDA to justify the severity percentile choice
- C. Some LDA assumptions do not align with stress testing objectives
- D. Quantitative-Qualitative Approach Dimension

The correct answer is **D**.

Quantitative-Qualitative Approach Dimension is not a challenge that the risk manager is likely to face when developing and implementing models used in stress testing Operational risk. This term refers to one of the two dimensions of stress testing. A stress testing taxonomy, which includes the Quantitative-Qualitative Approach Dimension, is a tool that aids in understanding the evolution of stress testing and the variety of stress testing practices. Therefore, it is not a challenge but rather a methodological approach that can be used in the process of stress testing.

Choice A is incorrect. Legal risk is indeed characterized by the delay between adverse macroeconomic conditions and legal losses suffered by banks. This delay can make it difficult to accurately model and predict future legal risks, which can be a significant challenge in stress testing operational risk.

Choice B is incorrect. Justifying the severity percentile choice in Conditional Loss Distribution Approach (LDA) can indeed be challenging. The choice of severity percentile has a significant impact on the results of the stress test, making it a critical decision that requires careful consideration and justification.

Choice C is incorrect. Some assumptions made in Loss Distribution Approach (LDA) may not align with stress testing objectives, posing another challenge for operational risk stress testing. For example, LDA assumes that loss events are independent and identically distributed, which may not hold true under stressed conditions.

Things to Remember

- Legal risk is characterized by the delay between adverse macroeconomic conditions and legal losses suffered by banks

- Conditional Loss Distribution Approach (LDA) is a common method used in operational risk modeling
- Loss Distribution Approach (LDA) assumptions may not always align with stress testing objectives
- Quantitative-Qualitative Approach Dimension is a methodological approach in stress testing

When developing and implementing models for stress testing operational risk, it is important to consider the various challenges and assumptions that come with different modeling approaches. Understanding the nuances of legal risk, the complexities of severity percentile choices in LDA, and the alignment of assumptions with stress testing objectives are all crucial aspects to consider in building a robust stress testing framework.

Reading 110: Cyber-resilience: Range of Practices

Q.4263 Assume you are the chief systems manager at your local bank. How best would you approach the issue of cyber security in line with the Basel Committee Report on cyber-resilience among regulated institutions?

- A. To identify all instances of cyber warfare and establish the severity and potential damage of attacks, and ensure that findings are publicized and acted upon immediately.
- B. To single out all potentially crippling cyber-related vulnerabilities that expose the bank to large-scale monetary or nonmonetary loss
- C. Accept that there can be no absolute security and instead work on developing a robust IT system and build local and international cooperation and information exchange in order to reduce threat and build resilience
- D. To identify all instances of cyber warfare and potential vulnerabilities with an eye on complete eradication of threats

The correct answer is **C**.

The reality of cyber security is that there is no such thing as 'absolute security'. Cyber threats are constantly evolving and new ones are emerging every day. It is virtually impossible to identify and eliminate all potential attack points. Therefore, the most effective approach is to accept this reality and focus on developing a robust IT system that is resilient to these threats. This involves building both local and international cooperation and information exchange networks to reduce the threat and protect critical information infrastructures. This approach aligns with the Basel Committee's emphasis on cyber-resilience among regulated institutions.

Choice A is incorrect. While identifying instances of cyber warfare and assessing their severity and potential damage is important, publicizing these findings immediately may not be the best approach. This could potentially expose the bank to further attacks by revealing its vulnerabilities. Moreover, this option does not consider building resilience or cooperation which are key aspects of a comprehensive cybersecurity strategy.

Choice B is incorrect. Singling out all potentially crippling cyber-related vulnerabilities that expose the bank to large-scale monetary or nonmonetary loss is a part of addressing cybersecurity issues but it's not sufficient on its own. It lacks consideration for developing robust IT systems and fostering local and international cooperation for information exchange, which are crucial in reducing threats and building resilience.

Choice D is incorrect. Identifying all instances of cyber warfare and potential vulnerabilities with an aim towards complete eradication of threats might be unrealistic as new threats can emerge constantly due to technological advancements. Instead, focusing on reducing threat levels through robust IT systems development, local and international cooperation for information exchange would be more effective in enhancing cyber-resilience.

Things to Remember

- Basel Committee on Banking Supervision (BCBS) sets global standards for the regulation and supervision of banks and financial institutions.
 - Cyber resilience refers to an organization's ability to resist, respond to, and recover from cyber attacks.
 - Robust IT systems are essential for protecting critical information infrastructures from cyber threats.
 - Building local and international cooperation for information exchange helps in sharing threat intelligence and enhancing cyber security.
 - Cyber security involves not only technical measures but also organizational and human factors to effectively mitigate risks.
-

Q.4264 In response to the increasing number of threats in the cyber space, the Basel committee has come up with a report aimed at inculcating cyber resilience across the banking industry. The cyber risk resilience framework encompasses all of the following EXCEPT:

- A. Threat anticipation
- B. Adapting to changes in the cyber space
- C. Rapid recovery from cyber incidents
- D. None - All of the above form part of the cyber risk resilience framework

The correct answer is **D**.

The Basel Committee's report on cyber resilience includes all of the elements listed in the options. The Financial Stability Board (FSB) defines cyber resilience as the 'ability of an organization to continue to carry out its mission by anticipating and adapting to cyber threats and other relevant changes in the environment and by withstanding, containing and rapidly recovering from cyber incidents.' Therefore, all the options - threat anticipation, adapting to changes in the cyber space, and rapid recovery from cyber incidents - are part of the cyber risk resilience framework.

Choice A is incorrect. Threat anticipation is indeed a part of the cyber risk resilience framework. It involves identifying potential cyber threats and preparing for them in advance to minimize their impact.

Choice B is incorrect. Adapting to changes in the cyber space is also included in the

framework. This element emphasizes on the need for banks to continuously update and adapt their cybersecurity measures as per evolving cyber threats.

Choice C is incorrect. Rapid recovery from cyber incidents forms an integral part of the framework as well, highlighting the importance of quick response and recovery mechanisms post a cybersecurity breach.

Things to Remember

- Cyber risk resilience framework is essential for banks to mitigate the increasing threats in the cyber space.
 - Threat anticipation involves identifying potential cyber threats and preparing for them in advance.
 - Adapting to changes in the cyber space requires continuous updating and adapting of cybersecurity measures.
 - Rapid recovery from cyber incidents is crucial for banks to quickly respond and recover from cybersecurity breaches.
 - Basel Committee's report on cyber resilience aims to enhance the ability of organizations to withstand, contain, and recover from cyber incidents.
-

Q.4265 Capital Bank just went through a serious system breach that resulted in massive loss of sensitive customer data. The information security department is attempting to restore the system as well as located critical data backups. Unfortunately, it appears no one knows exactly what they are supposed to do. The bank also has a rapid recovery plan in place but the relevant personnel do not know what protocol to follow in the execution process. What's more, the recovery team is struggling to put in a well-coordinated effort to carry out specific tasks. Which of the following vulnerabilities is most likely to blame for this scenario?

- A. Lack of a business impact analysis
- B. Failure to back up sensitive data adequately
- C. Failure to set up an alternate system
- D. Failure to test the disaster recovery strategy

The correct answer is **D**.

The primary issue in this scenario is the failure to test the disaster recovery strategy. This is highlighted in the Basel Committee Report on Cyber-Resilience Practices, which emphasizes the importance of sufficient business continuity testing. Despite having a contingency and recovery plan in place, many regulated entities do not conduct adequate testing to ensure the effectiveness of these plans. A disaster recovery test is crucial as it ensures that all team members are aware of their roles and responsibilities and are familiar with the steps to be followed during the recovery period. In the case of Capital Bank, the lack of such testing has resulted in confusion and inefficiency among the recovery team, thereby exacerbating the impact of the system breach.

Choice A is incorrect. While a business impact analysis is crucial for understanding the potential effects of system interruptions, it does not directly address the issues of role confusion and lack of protocol understanding among the recovery team. These problems are more related to disaster recovery strategy testing, which ensures that all personnel understand their roles and responsibilities during a crisis.

Choice B is incorrect. The failure to back up sensitive data adequately could indeed result in data loss during a system breach. However, this issue does not explain the confusion about roles and responsibilities or the ineffective execution of the rapid recovery plan among relevant personnel.

Choice C is incorrect. Setting up an alternate system can be part of a disaster recovery strategy, but it doesn't necessarily ensure that team members understand their roles or how to execute recovery plans effectively. This choice doesn't address the main problem described in this scenario - lack of clarity on roles and ineffective execution due to misunderstanding protocols.

Things to Remember

- Business Impact Analysis (BIA) is a critical component of risk management that helps organizations identify and prioritize their business functions and processes, understand the potential impact of disruptions, and establish recovery time objectives.
- Disaster Recovery Testing involves simulating potential disaster scenarios to ensure that the recovery plan is effective, all team members are familiar with their roles, and the organization can recover critical systems and data in a timely manner.
- Alternate System Setup is a part of a comprehensive disaster recovery strategy that involves having backup systems or infrastructure in place to ensure business continuity in case of a system failure or breach.
- Role Clarity and Protocol Understanding are essential for effective crisis management and recovery. Clear communication, defined roles, and regular training are key to ensuring that all team members know what to do in case of an emergency.

Q.4266 Considering commerce and marketing, which of the following is a significant obstacle to developing cyber resilience among regulated institutions around the globe?

- A. Use of technology, including high-level automation and integration with third parties
- B. Large-scale use of third party services
- C. Cloud computing and related services
- D. All of the above

The correct answer is **D**.

The biggest stumbling block toward inculcating cyber resilience among regulated institutions has been high-level automation and use of systems that are heavily integrated with third-party service providers and customers. This has resulted in an attack surface that is growing by the day and has only served to increase accessibility from potential adversaries. Increased third party integration implies that the perimeter of interest to financial sector regulators has gotten bigger, and cloud computing means the perimeter is shared.

Options B, and C all come up as a result of high-level automation and integration.

Things to Remember

- Cyber resilience refers to an organization's ability to prevent, detect, respond to, and recover from cyber attacks or security breaches.
 - Regulated institutions in the financial sector are subject to strict cybersecurity regulations and guidelines to protect sensitive data and maintain trust with customers.
 - Third-party risk management is crucial for regulated institutions as they often rely on external vendors for various services, increasing the potential attack surface.
 - Cloud computing offers scalability and cost-efficiency benefits but also introduces new security challenges related to data protection and access control.
-

Q.4267 According to the Basel Committee report on cyber resiliency among institutions, which of the following jurisdictions tend to have the least robust regulatory information sharing frameworks?

- A. Those with minimum freewill information sharing arrangements
- B. Those with observable practices for information-sharing among banks

C. Jurisdictions that have recorded the smallest number of cyber-related incidences

D. Jurisdictions with the highest number of systematically important banks

The correct answer is **B**.

The Basel Committee's report indicates that jurisdictions with observable practices for information-sharing among banks tend to have less robust regulatory information sharing frameworks. This is because the regulators in these jurisdictions may not feel the need to enforce stringent information-sharing policies if voluntary peer sharing practices are well established and effective. Therefore, these jurisdictions may rely more on the voluntary information-sharing practices among banks rather than on robust regulatory frameworks.

Choice A is incorrect. While it might seem intuitive that jurisdictions with minimum freewill information sharing arrangements would have the least robust regulatory information sharing frameworks, this is not necessarily the case. The term "freewill" implies that these jurisdictions allow banks to decide whether or not to share information, which could potentially lead to more robust frameworks if banks choose to share extensively. Therefore, this choice does not definitively indicate a lack of robustness in regulatory information sharing frameworks.

Choice C is incorrect. The number of cyber-related incidences recorded in a jurisdiction does not directly correlate with the robustness of its regulatory information sharing framework. A low number of incidents could be due to effective cybersecurity measures rather than a strong framework for sharing regulatory information.

Choice D is incorrect. The presence of systematically important banks within a jurisdiction does not necessarily imply weak or strong regulatory information-sharing frameworks. These banks may have advanced internal systems for managing cyber risk but it doesn't provide any direct indication about the quality or effectiveness of their external communication and cooperation with regulators and other institutions.

Things to Remember

- Cyber resiliency is the ability of an organization to prepare for, respond to, and recover from cyber attacks or incidents.
- Regulatory information sharing frameworks are essential for promoting collaboration and coordination among financial institutions and regulators to address cyber threats.
- Systemically important banks are institutions whose failure could potentially have a significant impact on the stability of the financial system.
- Effective cyber resiliency requires not only robust regulatory frameworks but also active information sharing and cooperation among institutions and regulators.
- Voluntary information sharing practices among banks can complement regulatory

frameworks but may not always be sufficient to ensure comprehensive cyber resiliency.

Q.4269 With respect to cyber security strategy as outlined in the Basel Committee report on cyber-resilience, all of the following statements are correct EXCEPT?

- A. All regulators expect regulated entities to have a board approved information security strategy
- B. Most jurisdictions have included cyber-risk within their broader risk management frameworks
- C. Most supervisors review regulated entities' information security strategies, but very few require or evaluate those entities' standalone cyber-security strategies.
- D. In most jurisdictions the development of a cyber-security strategy is a mandatory requirement anchored in law

The correct answer is **D**.

The Basel Committee report on cyber-resilience does not mandate the development of a cyber-security strategy in most jurisdictions. While it is true that cyber-security is a critical aspect of information security, the report does not require regulated entities to have a standalone cyber-security strategy. Instead, the focus is on having a board-approved information security strategy, policy, and procedures that effectively oversee technology. This includes, but is not limited to, cyber-security. Therefore, the statement that 'In most jurisdictions the development of a cyber-security strategy is a mandatory requirement anchored in law' is incorrect.

Choice A is incorrect. The Basel Committee report does indeed state that all regulators expect regulated entities to have a board approved information security strategy. This is part of the broader framework for managing cyber risk and ensuring cyber resilience.

Choice B is incorrect. According to the Basel Committee report, most jurisdictions have included cyber-risk within their broader risk management frameworks. This integration allows for a more comprehensive approach to managing and mitigating risks associated with cybersecurity.

Choice C is incorrect. It's true that most supervisors review regulated entities' information security strategies, but very few require or evaluate those entities' standalone cyber-security strategies as per the Basel Committee report.

Things to Remember

- Basel Committee on Banking Supervision (BCBS) plays a key role in setting global standards for the regulation and supervision of banks and financial institutions.

- Cyber resilience refers to an organization's ability to prepare for, respond to, and recover from cyber attacks or security breaches.
 - Information security strategy encompasses a range of measures and practices designed to protect an organization's information assets from unauthorized access, use, disclosure, disruption, modification, or destruction.
 - Regulators and supervisors play a crucial role in overseeing the implementation of cyber security measures in regulated entities to ensure the stability and security of the financial system.
 - Effective cyber security strategies often involve a combination of technical controls, policies and procedures, employee training, incident response plans, and continuous monitoring.
-

Q.4270 Which of the following regulatory approaches has been adopted by jurisdictions as a way of enforcing cyber-security strategy requirements among regulated entities?

- I. Mandatory sector specific or cross-sector cyber-security requirements
 - II. A requirement to develop internal cyber-security strategies by financial institutions
 - III. Examining whether institutions have an active IT strategy and accompanying security provisions
- A. I and II only
 - B. II only
 - C. III only
 - D. All three

The correct answer is **D**.

The Basel Committee report on Cyber-resilience notes that jurisdictions enforce cyber-security strategy requirements using one or a combination of the following:

- I. Regulator-developed cyber security strategy requirements that must be observed by all financial institutions. This is by far the most common approach especially among

emerging market economies

- II. Financial institutions may be required to develop their own cyber-security strategies that are in compliance with existing risk management principles
- III. The regulators actively examine whether financial entities have an IT strategy and security provisions. This is especially common in Europe

Things to Remember

- Cybersecurity risk management is a critical aspect of overall risk management for financial institutions
 - Regulators may also conduct regular cybersecurity assessments and audits to ensure compliance
 - Regulatory requirements for cybersecurity are constantly evolving to keep up with the changing threat landscape
 - Cybersecurity incidents can have significant financial and reputational impacts on regulated entities
-

Q.4271 John Henderson, FRM, is the newly appointed chief officer in charge of information systems and security at Capital Bank. Upon scrutinizing the bank's cyber-security strategy, he has found that the bank lacks a well thought out business continuity plan that can be adopted in the event of an exceptional event or crisis. With the help of other executives, he proceeds to conduct a business impact assessment and singles out the most critical activities, resources, and services that would be in need of rapid restoration in the event of a cyber-attack. Which of the following activities would be **most critical** before finalizing and implementing the newly developed plan?

- A. Consultations with other banks in the same jurisdiction
- B. Continuity tests
- C. A detailed review of past cyber-related incidences
- D. Data recovery tests

The correct answer is **B**.

Continuity tests are the most critical activity before finalizing and implementing a newly developed business continuity plan. The Basel committee, a global standard setter for the

prudential regulation of banks, emphasizes the importance of business continuity tests. These tests are designed to validate the effectiveness of the business continuity and crisis response plans. For instance, the tests can assess whether it is feasible to restore services within the stipulated timelines. This is crucial because, in the event of a cyber-attack or any other crisis, the ability to quickly restore critical services can significantly mitigate the impact on the bank's operations and reputation. Therefore, conducting continuity tests is a key step in ensuring that the business continuity plan is robust and effective.

Choice A is incorrect. While consultations with other banks in the same jurisdiction can provide valuable insights and best practices, it is not the most critical activity before finalizing and implementing a business continuity plan. The primary focus should be on ensuring that the plan works effectively within the specific context of Capital Bank.

Choice C is incorrect. A detailed review of past cyber-related incidences can help to identify potential vulnerabilities and threats, but it does not directly test or validate the effectiveness of a business continuity plan. Therefore, while useful, this activity is not as critical as conducting continuity tests.

Choice D is incorrect. Data recovery tests are an important part of any business continuity plan, especially in relation to cyber-attacks where data loss can occur. However, these tests are only one component of a comprehensive business continuity strategy which also includes aspects like restoring operations and services, communication plans etc., hence they do not hold precedence over overall continuity tests.

Things to Remember

- Business Continuity Planning (BCP) is a proactive approach to creating a plan that ensures an organization can continue operating during and after a disaster or any disruptive event.
- Business Impact Analysis (BIA) is a key component of BCP that identifies critical business functions and the impact of a disruption to those functions.
- Risk Assessment is another important step in BCP that involves identifying potential risks and their likelihood of occurrence.
- Incident Response Plan (IRP) is a predefined set of instructions for how an organization will respond to potential security incidents.
- Disaster Recovery Plan (DRP) focuses on the recovery of IT systems and data after a disaster or disruption.

Q.4272 According to the Basel Committee Report on Cyber-Resilience Practices, which of the

following is the “least observed practice across jurisdictions” with respect to information sharing?

- A. Information sharing among banks
- B. Information sharing among regulators
- C. Information sharing by banks with regulators
- D. Information sharing by banks with security agencies

The correct answer is **B**.

The Basel Committee Report on Cyber-Resilience Practices has identified that the least observed practice in terms of information sharing is among regulators. This is a concerning situation, particularly considering the increasing sophistication and global nature of cyber-fraud. The lack of information sharing among regulators can lead to a lack of awareness about emerging risks and can hinder the development of a comprehensive, coordinated response. This could potentially leave certain industry sectors vulnerable to cyber threats. Therefore, there is a pressing need to enhance information sharing among regulators to ensure a robust and effective response to cyber threats.

Choice A is incorrect. Information sharing among banks is not the least prevalent practice. Banks often share information with each other to mitigate risks and enhance their cyber-resilience practices. This collaboration allows them to learn from each other's experiences and implement effective strategies.

Choice C is incorrect. Information sharing by banks with regulators is also not the least prevalent practice. Banks are required to report certain types of information to regulators, such as incidents of cyber-attacks or breaches, as part of their regulatory obligations.

Choice D is incorrect. Information sharing by banks with security agencies isn't the least prevalent either. In fact, it's quite common for banks to collaborate with security agencies in order to strengthen their defenses against potential cyber threats.

Things to Remember

- **Information Sharing:** Sharing of information among banks, regulators, and security agencies is crucial for enhancing cyber-resilience practices and combating cyber threats.
- **Cyber-Resilience:** The ability of an organization to prepare for, respond to, and recover from cyber attacks or incidents.
- **Regulatory Obligations:** Banks are required to comply with regulations that mandate reporting of cyber incidents to regulators.

- Cyber-Fraud: Criminal activities conducted through the use of computers or the internet, often involving financial fraud or data breaches.
 - Global Nature of Cyber Threats: Cyber threats can originate from anywhere in the world and can target organizations across borders, highlighting the need for international cooperation and information sharing.
-

Q.4273 Exim Bank has just completed a risk assessment and business impact analysis (BIA) with respect to cyber-attacks and the latest emerging threats and vulnerabilities in the cyber space. However, the bank's information security manager and business department manager don't seem to agree on who will ultimately be responsible for detailed evaluation of the results and risk analysis. Which of the following would be the best cause of action in these circumstances?

- A. Acceptance and implementation of the information security manager's decision on the risk to the bank
- B. Acceptance and implementation of the business department manager's decision on the risk to the bank
- C. Creation of a new risk assessment and BIA plan to iron out the differences
- D. Review the report with senior management for final input

The correct answer is **D**.

The senior management and executives of an organization play a pivotal role in the evaluation and management of cyber risk. This is highlighted in the Basel Committee Report on Cyber Resilience. Just like with other types of risks, the senior management is ultimately responsible for promoting and maintaining cyber resilience within their institutions. They have the authority and responsibility to streamline and resolve any issues that might arise during the process of implementing a solution against cyber risk. In the given scenario, the disagreement between the information security manager and the business department manager can be resolved by involving the senior management. They can review the report, provide their inputs, and make the final decision. This approach ensures that the decision is made at the highest level, taking into consideration the overall strategic objectives and risk appetite of the organization.

Choice A is incorrect. While the information security manager's decision on the risk to the bank is important, it should not be accepted and implemented without considering other perspectives. The information security manager may have a deep understanding of cyber threats and vulnerabilities, but they might lack a comprehensive view of business operations and strategic objectives.

Choice B is incorrect. Similarly, accepting and implementing the business department manager's decision on the risk to the bank would also be inappropriate. Although this individual

likely has a strong understanding of business operations, they may not fully comprehend or appreciate all aspects of cyber risks.

Choice C is incorrect. Creating a new risk assessment and BIA plan to iron out differences could be time-consuming and costly without necessarily resolving disagreements between different stakeholders in an organization. It would be more efficient to review existing assessments with senior management who can provide balanced input based on their overall understanding of both business operations and potential risks.

Things to Remember

- Risk assessment and business impact analysis (BIA) are crucial processes in identifying, evaluating, and prioritizing risks that could potentially impact an organization's operations.
 - Cyber risk management involves assessing and mitigating risks related to information security, data breaches, and cyber-attacks.
 - Senior management's involvement in decision-making related to cyber risk is essential to ensure alignment with strategic objectives and risk appetite.
 - Collaboration between different departments, such as information security and business operations, is key to effectively managing cyber risks.
 - Regular review and updates of risk assessment plans are necessary to adapt to evolving cyber threats and vulnerabilities.
-

Q.4478 In the context of cyber-resilience practices, which of the following is/are considered (a) third-party(ies)?

- A. Cloud computing services
- B. Computer hardware
- C. Trading platforms
- D. All of the above

The correct answer is **D**.

All of the options listed, i.e., cloud computing services, computer hardware, and trading platforms, are considered third-parties in the context of cyber-resilience practices. Cloud

computing services are often outsourced by organizations to manage and store their data. Computer hardware, such as servers and workstations, are typically purchased from external suppliers. Trading platforms, which are used to conduct financial transactions, are also considered third-parties as they are usually operated by external entities. Therefore, all of these entities fall under the category of third-parties as they are external to the organization and play a crucial role in its operations.

Choice A is incorrect. While cloud computing services are indeed a third-party entity, this choice alone does not encompass all the possible third-party entities that an organization might rely on for its operations in cyber-resilience.

Choice B is incorrect. Computer hardware can be considered as a third-party entity if it's provided by an external vendor. However, similar to Choice A, this option alone does not cover all the potential third-party entities involved in cyber-resilience.

Choice C is incorrect. Trading platforms can also be classified as a third-party entity if they are externally provided and managed. But again, this choice doesn't include all other possible external entities that contribute to an organization's cyber-resilience.

Things to Remember

- Third-party risk management is a critical aspect of cyber-resilience practices, as organizations need to assess and mitigate the risks associated with external entities.
 - Vendor management is essential in ensuring the security and reliability of third-party services and products used by an organization.
 - Supply chain security is another important consideration in cyber-resilience, as vulnerabilities in the supply chain can impact the overall security posture of an organization.
 - Regulatory compliance requirements often mandate organizations to have robust third-party risk management processes in place to protect sensitive data and systems.
-

Q.4481 What is cyber resilience?

- A. The ability of an entity to continue to execute its purpose by anticipating and adapting to cyber threats
- B. The ability of an entity to rapidly recover from the cyber occurrence
- C. All of the above
- D. None of the above

The correct answer is **C**.

Cyber resilience encompasses both the ability of an entity to continue to execute its purpose by anticipating and adapting to cyber threats (Choice A) and the ability of an entity to rapidly recover from the cyber occurrence (Choice B). Cyber resilience is not just about being able to resist and withstand cyber threats, but also about being able to quickly recover and adapt in the face of these threats. This involves having robust security measures in place, as well as effective incident response and recovery plans. It's about being prepared for any potential cyber threats, and being able to respond and adapt quickly when they occur.

Choice A is incorrect. While it is true that cyber resilience involves the ability of an entity to continue its operations by anticipating and adapting to cyber threats, this definition alone does not fully encompass the concept of cyber resilience. Cyber resilience also includes the ability to rapidly recover from a cyber occurrence, which is not mentioned in this option.

Choice B is incorrect. This choice only focuses on one aspect of cyber resilience - the ability to rapidly recover from a cyber occurrence. However, it misses out on another crucial aspect - the ability to anticipate and adapt to potential threats, which makes this choice incomplete in describing the full concept of cyber resilience.

Choice D is incorrect. The statement that none of these options describe the concept of cyber resilience is false as both choices A and B partially describe aspects of it but neither fully encapsulates all elements involved in achieving complete cybersecurity resiliency.

Things to Remember

- Cyber resilience is a critical component of cybersecurity and involves both anticipating and adapting to cyber threats as well as rapidly recovering from cyber occurrences.
- It is important for entities to have robust security measures in place, effective incident response and recovery plans, and the ability to quickly adapt to changing cyber threats.
- Cyber resilience is not just about withstanding cyber threats, but also about being able to recover and adapt quickly in the face of these threats.
- Regular testing and updating of cyber resilience strategies is essential to ensure they remain effective against evolving cyber threats.

Q.4482 Assume that you are a cyber risk manager for a regulated company in a country where cybersecurity regulations are absent. What is the best course of action you should take to ensure that your company is secured against cyber threats?

A. Implement the international standard and use prescribed guidance and supervisory

practices

- B. Develop new regulations to govern cyber risk in your organization
- C. Develop a sound cybersecurity regulation according to existing country regulations
- D. Develop cyber risk awareness culture in your company according to existing country regulations

The correct answer is **A**.

In the absence of specific cybersecurity regulations in a country, the best course of action for a cyber risk manager would be to adhere to international standards and utilize the guidance and supervisory practices prescribed by these standards. These international standards are developed by experts in the field and are widely recognized and accepted. They provide a comprehensive framework for managing cyber risks and include best practices for identifying, assessing, and mitigating these risks. Implementing these standards would ensure that the company is adequately protected against potential cyber threats, even in the absence of local regulations. Furthermore, these standards are often used as a benchmark by regulators and stakeholders to assess a company's cybersecurity posture. Therefore, adhering to these standards would not only ensure the company's security but also enhance its reputation and credibility in the eyes of stakeholders.

Choice B is incorrect. While developing new regulations to govern cyber risk in your organization might seem like a good idea, it may not be the most effective strategy. This is because creating new regulations can be time-consuming and costly, and there's no guarantee that these regulations will cover all potential cyber threats. Furthermore, without expertise in cybersecurity, the developed regulations may not be comprehensive or up-to-date with current threats.

Choice C is incorrect. Developing a sound cybersecurity regulation according to existing regulations would not be possible in this scenario as it was stated that the company operates in a country where there are no specific regulations governing cybersecurity.

Choice D is incorrect. While developing a cyber risk awareness culture according to existing regulations could potentially help mitigate some risks, it would not ensure adequate protection against all potential cyber threats. This approach relies heavily on individual employees' understanding and adherence to security practices which can vary greatly and leave gaps in protection.

Things to Remember

- International standards for cybersecurity, such as ISO 27001, provide a comprehensive framework for managing cyber risks.
- Adhering to international standards can enhance a company's cybersecurity posture and reputation.

- Developing new regulations for cyber risk management can be time-consuming and costly.
 - Creating a cyber risk awareness culture is important but may not be sufficient to protect against all cyber threats.
 - Regulatory compliance is crucial in cybersecurity to ensure data protection and risk management.
-

Q.4484 According to the Basel Committee's report on the regulated institutions, cyber risk awareness and risk culture is enhanced through:

- A. Cyber training, incorporated in all phases of employment-recruitment to the termination in a regulated institution
- B. Having effective processes and controls that ensure that employees, contractors, and third-party dealers understand their roles and responsibilities in the quest to reduce the risk of theft, fraud, or misuse of the institution's facilities
- C. Establishing a common risk culture to ensure effective cyber-risk management
- D. All of the above

The correct answer is **D**.

The Basel Committee's report on regulated institutions emphasizes the importance of cyber risk awareness and a robust risk culture. It suggests several methods to enhance these aspects, all of which are represented in the choices provided. Choice A refers to the incorporation of cyber training in all phases of employment, from recruitment to termination. This is a crucial aspect of creating a culture of cyber risk awareness as it ensures that all employees, regardless of their role or tenure, are equipped with the necessary knowledge and skills to identify and mitigate cyber risks. Choice B refers to the establishment of effective processes and controls that ensure that all stakeholders, including employees, contractors, and third-party dealers, understand their roles and responsibilities in reducing the risk of theft, fraud, or misuse of the institution's facilities. This is an essential component of a robust risk culture as it ensures that everyone involved in the institution's operations is accountable for managing cyber risks. Finally, Choice C refers to the establishment of a common risk culture to ensure effective cyber-risk management. This is a fundamental aspect of a robust risk culture as it ensures that all stakeholders share the same understanding and approach to managing cyber risks.

Choice A is incorrect. While cyber training is indeed a crucial part of enhancing cyber risk awareness, it is not the only method recommended by the Basel Committee. The report also emphasizes on having effective processes and controls in place and establishing a common risk

culture.

Choice B is incorrect. Although having effective processes and controls that ensure employees, contractors, and third-party dealers understand their roles in reducing risks of theft, fraud or misuse of facilities is important, this alone does not fully represent the recommendations made in the Basel Committee's report. Other methods such as cyber training and establishing a common risk culture are also suggested.

Choice C is incorrect. Establishing a common risk culture to ensure effective cyber-risk management is one of the recommendations made by the Basel Committee but it does not encompass all suggestions made in the report. Cyber training for all phases of employment and having effective processes are also recommended.

Things to Remember

- Cyber risk awareness is crucial for regulated institutions to mitigate potential threats and vulnerabilities in their operations.
 - Risk culture plays a significant role in shaping the behavior and decision-making processes of employees towards managing cyber risks effectively.
 - Basel Committee's recommendations are aimed at promoting a comprehensive approach to cyber risk management within regulated institutions.
 - Effective communication and training programs are essential for fostering a culture of cyber risk awareness and accountability among all stakeholders.
 - Regular assessments and updates of cyber risk management practices are necessary to adapt to evolving cyber threats and vulnerabilities.
-

Reading 111: Case Study: Cyberthreats and Information Security Risks

Q.5110 Which of the following is not an example of an involuntary disclosure under the taxonomy of information security risks?

- A. Database loss
- B. Virus infection
- C. System disruptions
- D. Loss of printed documents

The correct answer is **B**.

Virus infection is not an example of an involuntary disclosure under the taxonomy of information security risks. Instead, it falls under the category of data theft or corruption caused by external factors. In this scenario, the disclosure of information is not unintentional or involuntary. Rather, it is a deliberate act by an external entity (the virus) to access, steal, or corrupt the data. This is different from involuntary disclosure, where the information is unintentionally exposed due to various reasons such as system disruptions, database loss, loss of devices by staff members, or accidental mentions of confidential information when communicating to outsiders.

Choice A is incorrect. Database loss can be considered as an 'involuntary disclosure' because it involves the unintentional exposure of sensitive information. This could occur due to various reasons such as system failures, human errors, or cyber attacks.

Choice C is incorrect. System disruptions can also lead to 'involuntary disclosure'. For instance, during a system disruption, sensitive data might become accessible to unauthorized individuals or entities.

Choice D is incorrect. Loss of printed documents falls under the category of 'involuntary disclosure'. If these documents contain sensitive information and are lost or misplaced, they could potentially be found and accessed by unauthorized individuals.

Things to Remember

- Information Security Risks: Involuntary disclosure is just one category of information security risks. Other categories include data breaches, cyber attacks, insider threats, and social engineering.
- Data Theft: Virus infections can lead to data theft, where sensitive information is intentionally accessed or stolen by malicious software or attackers.
- Data Corruption: Virus infections can also cause data corruption, where the integrity or usability of data is compromised.

- Preventive Measures: To mitigate information security risks, organizations can implement measures such as encryption, access controls, regular backups, and employee training on cybersecurity best practices.
 - Compliance Requirements: Many industries have specific regulations and compliance requirements related to information security, such as GDPR, HIPAA, and PCI DSS.
-

Q.5111 Which of the following is not one of the five guidelines offered by The National Institute of Standards and Technology (NIST) on cybersecurity standards?

- A. Identify
- B. Protect
- C. Mitigate
- D. Recover

The correct answer is C.

Mitigate is not one of the five guidelines offered by The National Institute of Standards and Technology (NIST) on cybersecurity standards. While mitigation is a common term used in risk management and information security, it is not specifically listed as one of the five key steps in the NIST's cybersecurity framework. Mitigation generally refers to the process of reducing the severity, seriousness, or painfulness of something. In the context of cybersecurity, this could involve actions taken to reduce the impact of a security breach or to prevent future breaches. However, the NIST's framework focuses on the steps of Identify, Protect, Detect, Respond, and Recover.

Choice A is incorrect. "Identify" is indeed one of the five key guidelines established by NIST for cybersecurity standards. It involves understanding the business context, resources that support critical functions, and related cybersecurity risks to an organization's information systems.

Choice B is incorrect. "Protect" is also a part of NIST's five guidelines for cybersecurity standards. This guideline focuses on developing and implementing appropriate safeguards to ensure delivery of critical infrastructure services.

Choice D is incorrect. "Recover" too falls under the NIST's five guidelines for cybersecurity standards. It emphasizes on developing and implementing activities necessary to restore any capabilities or services that were impaired due to a cybersecurity event.

Things to Remember

- The National Institute of Standards and Technology (NIST) is a non-regulatory federal

agency within the U.S. Department of Commerce that develops and promotes measurement, standards, and technology to enhance economic security and improve quality of life.

- Cybersecurity standards are guidelines and best practices designed to protect computer systems, networks, and data from cyber threats.
 - Identify: In the context of cybersecurity, this involves understanding and documenting the assets, business processes, and potential threats to an organization's information systems.
 - Protect: This guideline focuses on implementing safeguards to ensure the security and integrity of critical infrastructure services.
 - Detect: This step involves continuous monitoring and detection of cybersecurity events to provide timely response and mitigation.
 - Respond: In the event of a cybersecurity incident, organizations need to have response plans in place to contain the impact and recover from the incident.
 - Recover: This guideline emphasizes the importance of restoring capabilities and services that were affected by a cybersecurity event.
-

Q.5112 Which of the following is an external cause of information security risk?

- A. Database loss
- B. Loss of printed documents
- C. Systems disruptions
- D. Departing employees taking proprietary information

The correct answer is **C**.

Systems disruptions are an example of an external cause of information security risk. External causes of information security risks are factors that originate outside of the organization but can still impact the confidentiality, integrity, and availability of information and systems. Examples of

external causes include system disruptions, hacking, phishing, theft, or transfer of digital/physical information. System disruptions can occur due to various reasons such as natural disasters, power outages, or cyber-attacks. These disruptions can lead to downtime, loss of data, and can severely impact the operations of an organization. Therefore, it is crucial for organizations to have robust disaster recovery and business continuity plans in place to mitigate the impact of such disruptions.

Choice A is incorrect. Database loss is typically an internal risk, often resulting from technical failures or human errors within the organization. It could be due to hardware failure, software bugs, or even accidental deletion by an employee.

Choice B is incorrect. Loss of printed documents is also an internal risk as it usually occurs due to mishandling of physical documents within the organization's premises. This could be due to negligence, lack of proper storage facilities or inadequate document management procedures.

Choice D is incorrect. Departing employees taking proprietary information represents an internal threat rather than external. This can occur when employees who have access to sensitive information leave the company and take this information with them for personal gain or use at a new employer.

Things to Remember

- Internal causes of information security risks originate from within the organization and can include factors such as employee negligence, inadequate security measures, or technical failures.
- External causes of information security risks come from outside the organization and can include factors such as hacking, phishing, natural disasters, or theft.
- System disruptions can lead to downtime, loss of data, and impact the operations of an organization, highlighting the importance of disaster recovery and business continuity planning.
- Having robust access control measures, encryption protocols, and regular security audits can help mitigate information security risks.
- Employee training and awareness programs are essential to educate staff about the importance of information security and how to prevent security breaches.

Q.5113 Which of the following five guidelines offered by The National Institute of Standards and Technology (NIST) on cybersecurity standards is related to reporting an attack to law enforcement and other authorities?

- A. Recover
- B. Respond
- C. Protect
- D. Detect

The correct answer is **B**.

The 'Respond' guideline provided by the National Institute of Standards and Technology (NIST) is specifically designed to address the process of reporting a cyber attack to law enforcement and other relevant authorities. This guideline emphasizes the importance of creating and regularly testing a plan for reporting such incidents. The goal is to ensure that organizations are prepared to take immediate and effective action in the event of a cyber attack. This includes not only identifying and mitigating the attack but also communicating the incident to the appropriate authorities. This is crucial for several reasons. Firstly, it allows law enforcement agencies to investigate the incident and potentially take action against the perpetrators. Secondly, it can help other organizations to be aware of the threat and take necessary precautions. Lastly, it contributes to the overall body of knowledge about cyber threats, helping to improve cybersecurity measures and strategies.

Choice A is incorrect. The "Recover" guideline from NIST focuses on developing and implementing the appropriate activities to maintain plans for resilience and to restore any capabilities or services that were impaired due to a cybersecurity event. It does not specifically address the process of reporting a cyber attack.

Choice C is incorrect. The "Protect" guideline from NIST involves developing and implementing the appropriate safeguards to ensure delivery of critical infrastructure services. This includes access control, awareness and training, data security, information protection processes and procedures, maintenance, protective technology etc., but it does not involve reporting a cyber attack.

Choice D is incorrect. The "Detect" guideline from NIST refers to developing and implementing the appropriate activities to identify the occurrence of a cybersecurity event in a timely manner which includes anomalies detection, security continuous monitoring etc., but it doesn't cover reporting an incident.

Things to Remember

- Reporting cyber attacks to law enforcement and other authorities is crucial for investigation, potential legal action, and raising awareness.
- Having a well-defined and tested incident response plan is essential for effective reporting and handling of cyber attacks.
- Timely and accurate reporting of cyber attacks can help in mitigating the impact and

preventing future incidents.

- Organizations should also consider legal and regulatory requirements when reporting cyber attacks to authorities.
-

Q.5114 Which of the following is **not** one of the actions under the respond guideline of the National Institute of Standards and Technology (NIST) on cybersecurity standards?

- A. Notifying customers, employees, and others whose data may be at risk
- B. Keeping business operations up and running
- C. Investigating any unusual activities on your network or by your staff
- D. Reporting the attack to law enforcement and other authorities

The correct answer is **C**.

The action of 'Investigating any unusual activities on your network or by your staff' is not part of the 'respond' guideline of the NIST cybersecurity standards. Instead, this action falls under the 'detect' guideline. The 'detect' guideline focuses on the identification of potential cybersecurity events and the subsequent assessment of their impact. This includes monitoring and analyzing the organization's networks and systems to identify any unusual or suspicious activities that could indicate a cybersecurity threat. Therefore, investigating unusual activities is a proactive measure aimed at detecting potential threats before they can cause significant damage, rather than a reactive measure in response to a confirmed cybersecurity incident.

Choice A is incorrect. Notifying customers, employees, and others whose data may be at risk is indeed a part of the 'respond' guideline of the NIST cybersecurity standards. This action ensures that those potentially affected by a cyber attack are aware of the situation and can take necessary precautions to protect their information.

Choice B is incorrect. Keeping business operations up and running during a cyber attack is also included in the 'respond' guideline. The aim here is to minimize disruption to services while dealing with the incident.

Choice D is incorrect. Reporting an attack to law enforcement and other authorities falls under the 'respond' guideline as well. This step helps in investigating the incident further and possibly preventing similar attacks in future.

Things to Remember

- NIST Cybersecurity Framework: The NIST Cybersecurity Framework provides a policy framework of computer security guidance for how private sector organizations in the

United States can assess and improve their ability to prevent, detect, and respond to cyber attacks.

- **Incident Response Plan:** An incident response plan is a set of instructions to help an organization detect, respond to, and recover from cybersecurity incidents. It outlines the steps to take when a security breach occurs.
 - **Incident Response Team:** An incident response team is a group of individuals within an organization responsible for preparing for and responding to cybersecurity incidents. This team is crucial in implementing the incident response plan effectively.
 - **Forensic Analysis:** Forensic analysis involves collecting, preserving, and analyzing digital evidence in a way that maintains its integrity for potential use in legal proceedings or investigations.
-

Q.5115 Which of the following is a category of informational controls that address actions taken by people when it comes to handling and protecting information?

- A. Technical controls
- B. Behavioral controls
- C. Prevention controls
- D. Detection controls

The correct answer is **B**.

Behavioral controls are a category of informational controls that address actions taken by people when it comes to handling and protecting information. They are a type of administrative control that focuses on influencing the behavior of people within an organization to reduce information security risks. Examples of behavioral controls include security awareness training, policies and procedures, background checks, and security clearances. These controls are designed to ensure that individuals within an organization are aware of the potential risks associated with information handling and are equipped with the necessary knowledge and skills to mitigate these risks. They are crucial in creating a security-conscious culture within an organization, thereby enhancing the overall effectiveness of information security management.

Choice A is incorrect. Technical controls, also known as logical controls, primarily involve the use of software and hardware to protect information systems. They do not directly influence the behavior of individuals within an organization.

Choice C is incorrect. Prevention controls are measures taken to prevent security incidents from occurring in the first place. While they may indirectly influence behavior by setting up barriers or restrictions, their primary focus is not on influencing individual behavior.

Choice D is incorrect. Detection controls are designed to identify and respond to security incidents after they have occurred. They do not primarily focus on influencing individual behavior but rather on identifying breaches in security.

Things to Remember

- Information security controls can be categorized into three main types: technical controls, administrative controls, and physical controls.
- Technical controls involve the use of technology to protect information systems, such as firewalls, encryption, and access controls.
- Administrative controls, also known as organizational controls, focus on the policies, procedures, and practices that govern how an organization manages its information security program.
- Physical controls involve the use of physical measures to protect information systems, such as locks, security cameras, and biometric access controls.
- Security awareness training is a key component of behavioral controls, as it helps educate employees about security best practices and their role in protecting sensitive information.
- Background checks and security clearances are examples of behavioral controls that help ensure individuals with access to sensitive information are trustworthy and have the necessary qualifications.

Q.5116 Which of the following is not a requirement for a company to be certified as ISO27001 compliant?

- A. Have an Information Security Management System (ISMS) that manages its information security risks
- B. Design and implement information security, including effective and comprehensive controls

C. Adopt an ongoing risk management process

D. Guidance in response and recovery from cybersecurity incidents

The correct answer is **D**.

The requirement for guidance in response and recovery from cybersecurity incidents is not a part of ISO27001 compliance. This is a guideline provided by the NIST Framework for Improving Critical Infrastructure Cybersecurity. ISO27001 focuses on the establishment and maintenance of an Information Security Management System (ISMS), risk management, and the implementation of effective controls. While it does require a procedure for responding to and managing information security incidents, it does not specifically require guidance in response and recovery from cybersecurity incidents. Therefore, this is not a requirement for a company to be certified as ISO27001 compliant.

Choice A is incorrect. Having an Information Security Management System (ISMS) that manages its information security risks is indeed a requirement for ISO27001 certification. The ISMS should be designed to ensure the selection of adequate and proportionate security controls that protect information assets and give assurance to interested parties.

Choice B is incorrect. Designing and implementing effective and comprehensive controls for information security is also a requirement for ISO27001 certification. These controls are necessary to manage or reduce the risks identified through the risk assessment process.

Choice C is incorrect. Adopting an ongoing risk management process is another requirement for ISO27001 certification. This involves conducting regular reviews and audits of the ISMS to ensure its continual improvement in line with changes in the threat environment, business circumstances, legal requirements, etc.

Things to Remember

- ISO27001 is an international standard for Information Security Management Systems (ISMS) that provides a framework for organizations to manage and protect their information assets.
- ISMS helps organizations identify, manage, and minimize risks to information security, ensuring the confidentiality, integrity, and availability of information.
- Risk management is a key component of ISO27001, requiring organizations to assess risks, implement controls to mitigate those risks, and regularly review and update their security measures.
- ISO27001 certification demonstrates to stakeholders, customers, and partners that an organization has implemented best practices for information security management.

- Continuous improvement is a fundamental principle of ISO27001, requiring organizations to regularly review and enhance their ISMS to adapt to changing threats and business environments.
-

Q.5117 A risk manager at a large bank claims that when talking about information control, it is important to consider the different types or categories of control that exist. These categories can provide a framework for understanding how information is being controlled, who has control over it, and what the implications of that control may be. Which of the following is a correct category of information control?

- A. Protect
- B. Recover
- C. Behavioral
- D. Detect

The correct answer is **C**.

Information control can be broadly classified into two categories: Behavioral controls and Technical controls. Behavioral controls involve the implementation of policies, procedures, and training programs that aim to influence the behavior of individuals who handle sensitive information. The primary objective of this category of information control is to minimize the potential for human error or deliberate misconduct that could compromise the confidentiality, integrity, or availability of information. This is achieved by addressing human behaviors related to the handling and protection of information. Behavioral controls are crucial in any information security framework as they directly deal with the human element, which is often considered the weakest link in the security chain.

Choice A is incorrect. "Protect" is not a category of information control but rather an action or measure taken within the framework of information control. It refers to the steps taken to safeguard sensitive data from unauthorized access, use, disclosure, disruption, modification, or destruction.

Choice B is incorrect. "Recover" also does not represent a category of information control. Instead, it's a part of disaster recovery planning which involves restoring operations critical for the resumption of business after a disaster or disruption in services.

Choice D is incorrect. "Detect" like protect and recover does not represent a category but rather an action within the framework of information control. It refers to identifying potential threats and vulnerabilities that could compromise data security.

Things to Remember

- Technical controls are another category of information control that involve the use of technology to protect, detect, and respond to security threats.
 - Information control frameworks such as ISO 27001 provide guidelines and best practices for implementing effective controls to protect information assets.
 - Access controls, encryption, and intrusion detection systems are examples of technical controls used to secure information.
 - Regular monitoring and auditing of information controls are essential to ensure their effectiveness and compliance with regulations.
-

Q.5118 Which of the following five guidelines offered by The National Institute of Standards and Technology (NIST) on cybersecurity standards involves creating and sharing a company cybersecurity policy that covers roles and responsibilities of employees?

- A. Recover
- B. Protect
- C. Identify
- D. Detect

The correct answer is **C**.

The 'Identify' guideline provided by NIST is primarily concerned with the identification of all the resources that a company uses. This includes equipment, software, and data. In addition to this, the 'Identify' guideline also involves the creation and sharing of a company's cybersecurity policy. This policy is crucial as it outlines the roles and responsibilities of the employees in relation to cybersecurity. It helps in ensuring that all employees are aware of their duties and responsibilities in maintaining the security of the company's digital assets. This guideline is the first step in a company's cybersecurity strategy as it helps in identifying the resources that need to be protected and the roles that employees will play in this protection.

Choice A is incorrect. The 'Recover' guideline by NIST refers to the development and implementation of appropriate activities necessary to restore any capabilities or services that were impaired due to a cybersecurity event. It does not involve the creation and dissemination of a company's cybersecurity policy.

Choice B is incorrect. The 'Protect' guideline by NIST involves developing and implementing safeguards to ensure delivery of critical infrastructure services, rather than outlining roles and responsibilities in a cybersecurity policy.

Choice D is incorrect. The 'Detect' guideline by NIST refers to the development and implementation of appropriate activities that identify the occurrence of a cybersecurity event, not about creating or disseminating a company's cybersecurity policy.

Things to Remember

- NIST (National Institute of Standards and Technology) provides guidelines and standards for cybersecurity to help organizations improve their security posture.
 - Cybersecurity policy is a crucial document that outlines the rules, responsibilities, and practices employees must follow to protect the organization's digital assets.
 - Identifying resources is the first step in developing a cybersecurity strategy as it helps in understanding what needs to be protected.
 - Protecting, detecting, and recovering are also important aspects of cybersecurity, but they focus on different stages of the cybersecurity process.
-

Q.5119 Which of the following actions does **not** fall under the 'protect' step of the National Institute of Standards and Technology (NIST) guidelines?

- A. Controlling who logs onto a company's network
- B. Updating security software regularly
- C. Creating a company cybersecurity policy that covers roles and responsibilities of employees
- D. Having formal policies for safely disposing of electronic files

The correct answer is **C**.

Creating a company cybersecurity policy that covers roles and responsibilities of employees does not fall under the 'protect' step of the National Institute of Standards and Technology (NIST) guidelines. This action is actually a part of the 'identify' step of the NIST framework. The 'identify' step involves understanding the business context, the resources that support critical functions, and the related cybersecurity risks. This enables an organization to focus and prioritize its efforts, consistent with its risk management strategy and business needs. The creation of a cybersecurity policy that outlines the roles and responsibilities of employees is a crucial part of this step as it helps to clearly define who is responsible for what in the context of cybersecurity.

Choice A is incorrect. Controlling who logs onto a company's network is indeed part of the

'protect' step in the NIST guidelines. This action helps to ensure that only authorized individuals have access to sensitive information, thereby reducing the risk of cyber threats.

Choice B is incorrect. Regularly updating security software aligns with the 'protect' step as it helps to safeguard digital assets by ensuring that any potential vulnerabilities are addressed promptly and effectively.

Choice D is incorrect. Having formal policies for safely disposing of electronic files also falls under the 'protect' step in NIST's cybersecurity framework. Proper disposal prevents unauthorized access or recovery of sensitive data from discarded devices or files.

Things to Remember

- The NIST cybersecurity framework consists of five core functions: Identify, Protect, Detect, Respond, and Recover.
 - The 'protect' step focuses on implementing safeguards to ensure delivery of critical infrastructure services.
 - Controlling access to systems and data, updating security software, and securely disposing of electronic files are all key actions under the 'protect' step.
 - The 'identify' step involves understanding the business context, resources, and cybersecurity risks to prioritize efforts effectively.
 - A cybersecurity policy outlining roles and responsibilities of employees is crucial for defining accountability and ensuring compliance with security measures.
-

Q.6531 Which of the following lessons from the Equifax case study is most likely applicable to financial institutions seeking to enhance their cybersecurity risk management framework?

- A. A well-developed incident response plan can significantly reduce reputational and financial damage from cyber breaches.
- B. Outsourcing cybersecurity functions reduces the risk exposure of financial institutions.
- C. Prioritizing cost reduction over cybersecurity investments leads to stronger operational resilience.
- D. Delaying disclosure of a cybersecurity breach allows financial institutions to manage public relations better and minimize regulatory penalties.

The correct answer is **A**.

The Equifax breach demonstrated that delayed and ineffective responses can exacerbate financial losses and regulatory scrutiny. An effective incident response plan should include rapid detection, containment, communication, and remediation protocols to minimize damage.

B is incorrect. While outsourcing may provide specialized expertise, it introduces third-party risk, requiring robust vendor risk management practices. The Equifax breach exposed vulnerabilities in external software that were not properly patched, underscoring the importance of internal oversight of outsourced functions.

C is incorrect. The Equifax case highlights the dangers of underfunding cybersecurity. Insufficient investment in security infrastructure and poor risk prioritization contributed to the breach, demonstrating that cost-cutting in security leads to higher long-term risks.

D is incorrect. Delayed breach disclosure increased regulatory scrutiny and public backlash in the Equifax case. Regulators emphasize timely reporting of breaches to protect consumers and maintain trust. Failure to do so can result in heavier penalties and reputational damage.

Things to Remember:

- Effective incident response plans minimize breach impact and regulatory penalties.
 - Cybersecurity investment is critical—cost-cutting in security can lead to catastrophic breaches.
 - Third-party vendors introduce risks that require robust oversight.
 - Timely breach disclosure is essential to maintain trust and comply with regulations.
-

Q.6532 Which of the following best explains how poor vulnerability management contributed to the Equifax breach?

- A. Equifax failed to identify and patch a known vulnerability in a timely manner, leaving its systems exposed to cyberattacks.
- B. Equifax's security team deliberately ignored critical security patches to reduce operational disruptions.
- C. Equifax lacked an incident response plan, which prevented it from addressing the breach efficiently.
- D. Equifax experienced a zero-day attack, meaning no patch was available to mitigate the risk.

The correct answer is **A**.

The Equifax breach was caused by a failure to patch a known vulnerability in the Apache Struts framework, which was publicly disclosed and had an available patch. Equifax's security team did not apply the patch in a timely manner, leaving sensitive consumer data exposed to hackers.

B is incorrect. There is no evidence that Equifax's security team deliberately ignored security patches. However, the failure to patch critical vulnerabilities reflects weak internal controls and oversight in vulnerability management.

C is incorrect. Equifax did have an incident response plan, but its poor execution and lack of preparedness worsened the breach's impact. The main cause of the breach was unpatched software, not the absence of an incident response strategy.

D is incorrect. The breach did not involve a zero-day attack (a vulnerability exploited before a patch is available). The Apache Struts vulnerability was known, and a patch had been released, but Equifax failed to apply it.

Things to Remember:

- Regular vulnerability management is critical—organizations must apply patches promptly.
 - Unpatched software vulnerabilities are among the leading causes of cyber breaches.
 - Delays in patching known threats increase exposure to cyber risks.
 - Strong internal controls ensure that security updates are applied in a timely manner.
-

Q.6533 Which of the following best describes how ineffective internal controls at Equifax contributed to the breach?

- A. Equifax lacked a cybersecurity framework, making it impossible to detect threats.
- B. Equifax's monitoring system failed to detect and prevent unauthorized access to sensitive data.
- C. Equifax employees intentionally disabled security alerts, allowing hackers to remain undetected.
- D. Equifax lacked sufficient financial resources to invest in cybersecurity controls.

The correct answer is **B**.

Equifax's internal controls failed to detect and block unauthorized access to sensitive data for over two months. The company's intrusion detection system (IDS) was misconfigured, and expired encryption certificates further weakened security monitoring. As a result, critical alerts went unnoticed, allowing attackers to exfiltrate data without immediate detection.

A is incorrect. Equifax had a cybersecurity framework in place, but it was not effectively implemented. The breach resulted from weak security monitoring, not a complete absence of cybersecurity policies.

C is incorrect. There is no evidence that Equifax employees intentionally disabled security alerts. Instead, poor security oversight and expired encryption certificates contributed to the failure in detecting the breach.

D is incorrect. Equifax, as a large financial institution, had the financial capacity to invest in cybersecurity. The breach was not caused by a lack of funds but by poor risk management and weak internal controls.

Things to Remember:

- Effective monitoring systems are essential for detecting cyber threats in real time.
- Expired security certificates weaken network defenses and prevent proper alerting.
- Weak internal controls allow breaches to go undetected for extended periods.
- Organizations must continuously update and test security controls to ensure effectiveness.

Q.6534 Which of the following best explains how poor breach disclosure practices worsened the impact of the Equifax data breach?

- A. Equifax immediately informed the public and regulators, minimizing reputational damage
- B. Equifax delayed disclosing the breach, leading to increased regulatory scrutiny and public backlash.
- C. Equifax disclosed the breach before understanding its full impact, which helped reduce penalties.
- D. Equifax fully complied with disclosure regulations, avoiding any negative consequences.

The correct answer is **B**.

Equifax delayed notifying the public and regulators about the breach for several weeks, despite knowing that sensitive consumer data had been compromised. This delay increased regulatory scrutiny, led to significant reputational damage, and resulted in legal and financial penalties.

A is incorrect. Equifax did not immediately disclose the breach. The company waited several weeks before informing the public, which worsened consumer trust and regulatory consequences.

C is incorrect. Equifax did not disclose the breach before understanding its impact. Instead, it delayed notification, which was viewed as a failure in transparency and accountability.

D is incorrect. Equifax did not fully comply with disclosure regulations. Many regulators criticized the company for failing to act quickly, and it faced legal repercussions for its slow response.

Things to Remember:

- Timely breach disclosure is critical for maintaining consumer trust and regulatory compliance.
- Delays in notifying stakeholders can lead to higher legal, financial, and reputational consequences.
- Transparency in cybersecurity incidents is essential for reducing regulatory penalties.
- Failure to disclose breaches promptly can result in lawsuits, fines, and long-term brand damage.

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Reading 112: Sound Management of Risks related to Money Laundering and Financing of Terrorism

Q.2988 In the context of a banking institution, the anti-money laundering representative plays a crucial role in ensuring the organization's compliance with anti-money laundering regulations. This representative is responsible for regularly updating the senior management about the progress and effectiveness of the anti-money laundering program. Which report would be the most beneficial for the representative to use in order to provide a comprehensive update to the senior management about the progress of the anti-money laundering program?

- A. Law enforcement inquiry details
- B. Credit exposure report
- C. Management changes notification
- D. Report on audits and examinations results

The correct answer is **D**.

Reports on audit and examination results would be the most useful report for informing senior management about the organization's anti-money laundering program's progress. This report would provide an in-depth look at the organization's anti-money laundering program, highlighting its strengths and weaknesses. It would contain information on any audits or examinations that have taken place, as well as any findings or recommendations made by auditors or examiners. The report would also detail any actions taken in response to these findings, as well as any improvements or enhancements made to the program over time.

A is incorrect. These details are typically related to specific cases and investigations, rather than the overall progress of the program.

B is incorrect. Credit exposure report: A credit exposure report provides information about the amount of credit risk the bank is exposed to, but it is not necessarily related to the progress of the anti-money laundering program.

C is incorrect. While changes in management can impact the program's effectiveness, this information does not provide an overview of the program's progress or its strengths and weaknesses.

Things to Remember

- **Anti-Money Laundering (AML) Regulations:** Regulations put in place to prevent money laundering activities within financial institutions.
- **Senior Management Reporting:** Providing regular updates to senior management is crucial for ensuring transparency and accountability in compliance programs.

- Audit and Examination Processes: Regular audits and examinations help identify gaps and areas of improvement in the AML program.
 - Compliance Program Effectiveness: Assessing the effectiveness of the AML program is essential for mitigating risks and maintaining regulatory compliance.
-

Q.2991 Blackest Bank wants to promote an anti-money laundering culture. Which of the following is an appropriate action by the senior management to enable them to achieve the said task?

- A. As an employment condition, compliance with anti-money laundering procedures should be included.
- B. The management should have close ties with the anti-money laundering program's independent auditors.
- C. Employee's compensation should be based on the how many suspicious activities they engage in.
- D. Back-end employees must attend training sessions with frontline employees.

The correct answer is **A**.

Things to Remember

1. Anti-money laundering measures are crucial for financial institutions to prevent and detect the use of their services for money laundering or terrorist financing activities.
 2. Effective anti-money laundering measures include establishing robust internal controls, providing regular training to employees, conducting regular audits and reviews, and ensuring compliance with all relevant laws and regulations.
 3. It is important for organizations to foster a culture of compliance where all employees understand their roles and responsibilities in relation to anti-money laundering procedures and are committed to adhering to these procedures.
 4. Any measure that could potentially incentivize suspicious activities or compromise the independence and objectivity of audits and reviews should be avoided as it could undermine the effectiveness of the anti-money laundering program.
-

Q.3122 Under what circumstances may a bank rely on a third party for customer due diligence (CDD)?

- A. When the third party has an established business relationship with the customer.

- B. When the third party is a bank or financial institution, regardless of the nature of the relationship with the customer.
- C. When the third party is subject to different levels of supervision and regulation than the bank, but is able to demonstrate a strict AML/CFT program.
- D. When the bank conducts periodic checks to ensure the third party's CDD process is more comprehensive than its own.

The correct answer is **A**.

A bank may rely on a third party for customer due diligence (CDD) when the third party has an established business relationship with the customer. This is because the third party, having an established relationship, would have a better understanding of the customer's financial behavior and risk profile. The bank can leverage this understanding to conduct a more effective CDD. However, the bank must establish a written document acknowledging the reliance on the other party's CDD processes. This is to ensure that the bank has a clear understanding of the third party's CDD processes and can hold them accountable for any lapses in the CDD.

Choice B is incorrect. While it might seem logical for a bank to rely on another financial institution for CDD, the nature of the relationship with the customer is crucial. Simply being a bank or financial institution does not automatically qualify a third party to conduct CDD on behalf of another bank.

Choice C is incorrect. The level of supervision and regulation that a third party is subject to, even if different from that of the bank, does not necessarily make them suitable for conducting CDD. They must also have an established business relationship with the customer and be able to demonstrate strict adherence to AML/CFT programs.

Choice D is incorrect. Conducting periodic checks on the third party's CDD process does not justify relying on them for this task unless they have an established business relationship with the customer. The comprehensiveness of their process alone cannot be used as a criterion for selection.

Things to Remember

- Customer Due Diligence (CDD) is a critical process for banks to assess the risk associated with their customers and prevent money laundering and terrorist financing.
- Third-party reliance for CDD can help banks leverage existing relationships and expertise to enhance their risk assessment processes.
- It is important for banks to have clear written agreements with third parties outlining the responsibilities and expectations regarding CDD processes.
- Regulatory authorities often require banks to have robust oversight and monitoring

mechanisms in place when relying on third parties for CDD.

- Compliance with Anti-Money Laundering (AML) and Combating the Financing of Terrorism (CFT) regulations is a key aspect of effective CDD processes.
-

Q.3123 A bank in Italy holds a business account for an Italian company that sells suits throughout Europe and North America. Information provided during the account opening process states that the purpose of this account is to receive payment for sales. A year-long review of the account shows a pattern of wire transfers coming from pass-through accounts. There are also significant transactions involving purchases of garment and cotton from China and India. The MOST important factor in assessing whether money laundering is a threat is that:

- A. The account is apparently used for both sales and purchases.
- B. Payments originate from third party accounts.
- C. Account holder maintains raw materials rather than finished pieces of clothing.
- D. Most transactions involve wire transfers rather than cash deposits.

The correct answer is **B**.

The fact that payments are originating from third-party accounts, specifically pass-through accounts, is the most significant factor when assessing the threat of money laundering. Pass-through accounts are those through which banking institutions extend money transfer privileges to the customers of other institutions, often foreign banks. These accounts can be more susceptible to higher risk because banks do not subject the foreign customers to the same level of due diligence as domestic customers who want to open checking and other accounts. Therefore, there is a possibility that the money wired into the account comes from illicit activities. This makes it a critical factor in assessing the potential threat of money laundering.

Choice A is incorrect. The account being used for both sales and purchases is a normal business practice and does not necessarily indicate money laundering. It's common for businesses to have accounts that handle both inflows (sales) and outflows (purchases).

Choice C is incorrect. The fact that the account holder maintains raw materials rather than finished pieces of clothing doesn't necessarily suggest money laundering. This could simply be indicative of the nature of their business operations, which involves manufacturing suits.

Choice D is incorrect. While cash transactions can often be a red flag in terms of potential money laundering, the use of wire transfers isn't inherently suspicious or indicative of illicit activity. Many legitimate businesses prefer wire transfers due to their speed, convenience, and security.

Things to Remember

- Money laundering is the process of making illegally obtained money appear legal by passing it through a complex sequence of banking transfers or commercial transactions.
 - Pass-through accounts are often used in money laundering schemes to obfuscate the original source of funds.
 - Due diligence is the process of investigating and verifying the background, reputation, and financial stability of individuals or businesses before entering into a financial transaction.
 - Wire transfers are electronic funds transfers from one person or entity to another, often used for large or international transactions.
 - Cash deposits are often scrutinized more closely in anti-money laundering efforts due to the difficulty in tracing the original source of the funds.
-

Q.3124 Which of the following is the main role of supervisors in banks under the Anti-Money Laundering (AML) and Countering Financing of Terrorism (CFT) framework put forth by the Basel Committee?

- A. Approval and oversight of AML/CFT risk management policies and procedures.
- B. Advising banks on the best risk management strategies.
- C. Evaluating whether the reporting entity has an appropriate and reasonable risk assessment, and an AML/CFT programme that reflects inherent risks.
- D. Helping banks to develop a sound AML/CFT risk management system that can keep track of all customer transactions.

The correct answer is **C**.

The role of supervisors is to independently scrutinize and verify AML/CFT policies and procedures. They have a mandate to ensure that banks in their jurisdiction maintain sound ML/FT risk management to protect the integrity of both the banks and the financial system as a whole.

A is incorrect. Approval and oversight is the responsibility of the board.

B and D are duties of the chief AML/CFT officer.

Things to Remember

- Supervisors play a crucial role in ensuring that banks comply with AML/CFT regulations to prevent money laundering and terrorist financing.
- They assess the effectiveness of the AML/CFT programs implemented by banks to mitigate risks associated with financial crimes.
- Supervisors conduct on-site examinations and off-site monitoring to evaluate the adequacy of banks' AML/CFT controls.
- Regular communication and cooperation between supervisors and banks are essential for maintaining a robust AML/CFT framework.
- Supervisors have the authority to impose sanctions and penalties on banks that fail to meet AML/CFT requirements.

Q.3125 What is the role of the AML/CFT chief officer in the second layer of defense in anti-money

laundering and countering the financing of terrorism?

- A. Conducting customer due diligence checks.
- B. Continuously monitoring the bank's compliance with AML/CFT duties.
- C. Conducting forensic investigations into suspicious transactions.
- D. Reviewing and approving high-risk transactions.

The correct answer is **B**.

The AML/CFT chief officer is responsible for continuously monitoring the bank's compliance with AML/CFT duties as part of the second layer of defense in anti-money laundering and countering the financing of terrorism. This includes conducting sample testing to ensure compliance and reviewing exception reports to alert senior management or the board of directors if there are concerns that AML/CFT procedures are not being addressed in a responsible manner.

A is incorrect. Customer due diligence checks are typically conducted by the first line of defense, which includes customer-facing employees such as relationship managers.

C is incorrect. Forensic investigations into suspicious transactions are typically conducted by specialized units within the bank's compliance function.

D is incorrect. Reviewing and approving high-risk transactions is typically the responsibility of the bank's compliance function or risk management function.

Things to Remember

- AML (Anti-Money Laundering) refers to a set of regulations, laws, and procedures designed to prevent criminals from disguising illegally obtained funds as legitimate income.
- CFT (Countering the Financing of Terrorism) involves measures to combat the financing of terrorism and ensure that financial systems are not used for terrorist activities.
- The second layer of defense in AML/CFT typically involves oversight and monitoring functions to ensure compliance with regulations and policies.
- Sample testing is a common method used to assess the effectiveness of AML/CFT controls and procedures.
- Exception reports are generated when transactions or activities deviate from

established norms or raise suspicions, requiring further investigation.

Q.3126 Paul Khan, a risk manager at the bank of India, is presenting to the board of directors on important AML/CFT considerations including responsibilities of various components of AML/CFT governance. What is the responsibility of internal audit in the bank's AML/CFT policies and procedures?

- A. Monitoring customer transactions.
- B. Approving new customer accounts.
- C. Evaluating the effectiveness of risk management and controls.
- D. Developing AML/CFT policies and procedures.

The correct answer is **C**.

The primary responsibility of the internal audit department in a bank's AML/CFT policies and procedures is to evaluate the effectiveness of risk management and controls. The internal audit department is considered the third line of defense in a bank's risk management structure. They are responsible for providing an independent and objective review of the bank's operations and controls, including those related to AML/CFT. They assess the adequacy and effectiveness of the bank's risk management system and internal controls, identify areas of risk and non-compliance, and recommend improvements. Their findings are reported to the board of directors' audit committee or a similar oversight body, which then takes appropriate action based on these findings.

Choice A is incorrect. Monitoring customer transactions is not the primary responsibility of the internal audit department. This task typically falls under the purview of the compliance or operations department, which has direct access to transaction data and can monitor for suspicious activity on a real-time basis.

Choice B is incorrect. Approving new customer accounts is also not a primary function of the internal audit department. This responsibility usually lies with the client relationship management or business development teams, who are in charge of onboarding new clients and ensuring they meet all necessary regulatory requirements.

Choice D is incorrect. Developing AML/CFT policies and procedures isn't primarily done by the internal audit department but rather by a specialized team within compliance or risk management departments that have expertise in regulatory requirements and best practices for AML/CFT governance.

Things to Remember

- AML (Anti-Money Laundering): AML refers to a set of procedures, laws, and regulations designed to stop the practice of generating income through illegal actions.

- CFT (Combating the Financing of Terrorism): CFT involves efforts to prevent and deter the financing of terrorism and terrorist organizations.
 - Three Lines of Defense Model: The three lines of defense model is a widely accepted framework for organizing risk management responsibilities in a financial institution. The first line of defense includes business units managing risks, the second line consists of risk management and compliance functions, and the third line is internal audit providing independent assurance.
 - Board of Directors: The board of directors is responsible for overseeing the bank's operations, including risk management and compliance. They rely on reports from internal audit to ensure that the bank's AML/CFT policies and procedures are effective.
-

Q.3127 The following are lines of defense in the context of AML/CFT EXCEPT:

- A. The supervisor
- B. The internal audit function
- C. The chief AML/CFT officer and the compliance department
- D. Customer-facing activities

The correct answer is **A**.

The supervisor is not considered a line of defense in the context of AML/CFT. The lines of defense in AML/CFT are typically internal mechanisms within an organization designed to prevent, detect, and mitigate money laundering and terrorist financing risks. These include the business units or customer-facing activities, the Chief AML/CFT Officer and the compliance department, and the internal audit function. The supervisor, while playing a crucial role in overseeing and enforcing compliance with AML/CFT regulations, is not a line of defense in the same sense as these internal mechanisms. The supervisor's role is more of an external oversight function, ensuring that the organization's internal lines of defense are functioning effectively and in compliance with relevant regulations.

Choice B is incorrect. The internal audit function is indeed a line of defense in the AML/CFT context. It provides an independent, objective assurance and consulting activity designed to add value and improve an organization's operations. It helps an organization accomplish its objectives by bringing a systematic, disciplined approach to evaluate and improve the effectiveness of risk management, control, and governance processes.

Choice C is incorrect. The chief AML/CFT officer and the compliance department are also

considered as a line of defense in the AML/CFT context. They are responsible for developing policies and procedures to detect, prevent, monitor, report on money laundering/terrorist financing activities.

Choice D is incorrect. Customer-facing activities are also considered as a line of defense in the AML/CFT context because they involve direct interaction with customers which can help identify suspicious transactions or behaviors that may indicate money laundering or terrorist financing activities.

Things to Remember

- AML (Anti-Money Laundering): AML refers to a set of regulations, laws, and procedures designed to prevent criminals from disguising illegally obtained funds as legitimate income.
 - CFT (Combating the Financing of Terrorism): CFT involves efforts to prevent and deter the financing of terrorism and terrorist organizations.
 - Know Your Customer (KYC): KYC is a process that financial institutions use to verify the identity of their clients to assess potential risks of illegal intentions.
 - Suspicious Activity Reporting (SAR): SAR is a requirement for financial institutions to report any suspicious transactions that may indicate money laundering or terrorist financing activities.
 - Risk-Based Approach: A risk-based approach involves assessing and managing risks associated with money laundering and terrorist financing based on the institution's risk profile.
-

Q.3128 What is the reasoning behind implementing a “risk-based anti-money laundering and combating financial terrorism approach”?

- A. It allows banks to focus on selling products that surpass a specified “hurdle” rate of return.
- B. A qualitative approach would yield better results than a quantitative approach.
- C. Banks can best detect instances of money laundering by customers where the money laundering risks are high.
- D. It allows banks to best monitor their profits.

The correct answer is **C**.

The primary reason for implementing a 'risk-based anti-money laundering and combating financial terrorism approach' is to enable banks to better detect instances of money laundering by customers where the money laundering risks are high. This approach allows banks to consider all relevant inherent and residual risk factors at various levels, including country, sector, bank, and business relationship. By doing so, banks can determine their risk profile and create customer risk profiles based on the nature and amount of their transactions. This enables them to group accounts based on the level of risk posed, making it easier to identify and flag suspicious activity.

Choice A is incorrect because the 'risk-based anti-money laundering and combating financial terrorism approach' is not primarily designed to allow banks to focus on selling products that surpass a specified 'hurdle' rate of return. While banks do aim to sell products that yield high returns, the primary purpose of this approach is to detect and prevent money laundering and financial terrorism, not to maximize profits.

Choice B is incorrect because the 'risk-based anti-money laundering and combating financial terrorism approach' is not about choosing a qualitative approach over a quantitative one. Both qualitative and quantitative methods can be used in this approach, depending on the specific circumstances and requirements. The main objective of this approach is to identify and manage money laundering risks effectively, regardless of the methods used.

Choice D is incorrect because the 'risk-based anti-money laundering and combating financial terrorism approach' is not primarily aimed at allowing banks to monitor their profits. While profit monitoring is an important aspect of banking operations, the main purpose of this approach is to detect and prevent money laundering and financial terrorism. Profit monitoring and risk management are two distinct aspects of banking operations.

Things to Remember

- Money Laundering: The process of making illegally-gained proceeds appear legal and legitimate.
- Financial Terrorism: The use of financial means to fund terrorist activities.
- Risk-Based Approach: A method of identifying, assessing, and managing risks based on the level of risk posed.
- Customer Risk Profiles: Profiles created by banks to assess the risk level associated with each customer based on their transactions and activities.
- Suspicious Activity: Transactions or behavior that may indicate money laundering or terrorist financing.

Q.3129 Simon works as the chief risk officer at XYZ Bank. He is looking at the transactions of one of the bank's customers, Mr. Lincoln, a commercial account holder and owner of a check cashing company. Over the last eight months, Mr. Lincoln has made multiple check deposits but not a single withdrawal of cash against those deposits. Mr. Lincoln also deposited two checks for US\$10,000 each that were issued by an infamous casino in town. When checking the account's details, Simon finds out that during account opening, Mr. Lincoln went to great lengths to establish the various fees and commissions attached to his account. Mr. Lincoln also has a savings account at the bank, but it has had little activity over the same period. What should arouse Simon's suspicion the most? Mr. Lincoln:

- A. Showed an untypical level of curiosity about fees.
- B. Made significant deposits from casinos.
- C. Has multiple accounts at the institution.
- D. Did not make withdrawals of cash against check deposits.

The correct answer is **D**.

Check cashing companies, also known as money services businesses, provide customers with an easy way to turn their checks into cash without having to rely on a bank account. As such, one would expect to see deposit activity that's commensurate with cash withdrawals as the money is released to the relevant persons. That this did not happen for a prolonged period raises questions as to the source of the check deposits. This is unusual and suspicious behavior, especially for a check cashing company. The lack of cash withdrawals against check deposits could indicate potential money laundering or other illicit activities. Therefore, this should arouse Simon's suspicion the most.

Choice A is incorrect. While it may be unusual for a customer to show an atypical level of curiosity about fees, this behavior alone does not necessarily indicate suspicious activity. It could simply mean that Mr. Lincoln is a cautious and informed customer who wants to understand all the costs associated with his account.

Choice B is incorrect. Although the source of funds can sometimes be a red flag for potential money laundering or other illicit activities, depositing checks from casinos in itself isn't necessarily suspicious. Casinos are legitimate businesses and it's possible that Mr. Lincoln received these checks as winnings or payments for services rendered.

Choice C is incorrect. Having multiple accounts at the same institution isn't inherently suspicious either, especially if one account is used for business transactions (like check cashing) and another for personal savings.

Things to Remember

- Money Services Businesses (MSBs) like check cashing companies provide customers with the ability to convert checks into cash without using a traditional bank account.

- Unusual patterns of deposit activity without corresponding withdrawals could indicate potential money laundering or illicit activities.
 - Understanding the source of funds, especially from high-risk sources like casinos, is important in assessing potential risks of money laundering.
 - Customer behavior, such as showing curiosity about fees, may not always indicate suspicious activity and could simply reflect a cautious approach.
 - Having multiple accounts at the same institution is not inherently suspicious and can be for legitimate reasons like separating business and personal finances.
-

Q.3130 A large banking group has an AML compliance program that addresses procedures for filing Suspicious Transaction Reports and includes policies, procedures and internal controls for customer identification, information sharing, account monitoring, and identifying money laundering red flags. Each of the bank's 12 branches undergoes mandatory AML/CFT trainings in April and November each year, all conducted as online conferences via a video link. The board does not take the Internet training. Instead, the chief risk officer organizes a luncheon at the head office where an outsider comes in and trains them. The program provides for the appointment of a chief ALM/CFT officer, and twice a year the chief ALM/CFT officer conducts an audit of the ALM/CFT framework. In what respect does the program need improvement?

- A. The AML program should be tested more than twice per year.
- B. Employees should be trained in a classroom, not via the internet because physical training is better.
- C. The group should consolidate the training sessions across its subsidiaries into a single event.
- D. The AML/CFT program should be tested by an independent party, not the chief ALM/CFT officer.

The correct answer is **D**.

The AML/CFT program should be tested by an independent party, not the chief AML/CFT officer. The role of internal audit, which is considered the third line of defense, is crucial in independently evaluating a bank's risk management and controls. The office conducting the audit should have sufficient independence to assess adherence to various policies and procedures without any compromise. In this case, since the chief risk officer, who also serves as the developer and advisor on AML and CFT matters, conducts the audit, there could be a potential conflict of interest. This is because they are assessing the same policies and procedures

that they have developed. Therefore, to ensure the effectiveness and integrity of the audit, it should be conducted by an independent party.

Choice A is incorrect. The frequency of testing the AML program twice a year is generally considered adequate in most banking institutions. Increasing the frequency may not necessarily lead to improved compliance or detection of money laundering activities.

Choice B is incorrect. The mode of training, whether online or in-person, does not inherently determine its effectiveness. As long as the content and delivery are effective and comprehensive, online training can be just as beneficial as classroom training.

Choice C is incorrect. Consolidating all training sessions into a single event may not be practical due to logistical constraints and could potentially overwhelm employees with information overload. Spreading out the trainings allows for better absorption and understanding of the material.

The AML/CFT program should indeed be tested by an independent party rather than by someone who has been involved in its implementation (the chief AML/CFT officer). This ensures objectivity and impartiality in assessing the effectiveness of the program.

Things to Remember

- AML (Anti-Money Laundering) Compliance: A set of regulations and procedures designed to prevent the generation of income through illegal activities.
- CFT (Combating the Financing of Terrorism): Measures to prevent and combat the financing of terrorism activities.
- Suspicious Transaction Reports: Reports filed by financial institutions to regulatory authorities when they suspect a transaction may be related to money laundering or terrorist financing.
- Customer Identification: Procedures to verify the identity of customers to prevent fraud and money laundering.
- Internal Controls: Policies and procedures implemented within an organization to ensure compliance with regulations and safeguard assets.
- Red Flags: Warning signs or indicators of potential money laundering activities that employees should be trained to recognize.
- Chief AML/CFT Officer: Responsible for overseeing and implementing the AML/CFT program within an organization.

- Independent Party Testing: Having an external party assess the effectiveness of compliance programs to ensure objectivity and impartiality.
 - Internal Audit: The third line of defense in the Three Lines of Defense model, responsible for independently evaluating risk management and controls.
-

Q.5452 Global Bank, a leading financial institution, is in the process of reviewing its risk management procedures. The bank's Chief Risk Officer, Mr. Anderson, has been appointed to lead this initiative. As part of the review, he is keen on incorporating the best practices as recommended by the Basel committee for the assessment, management, mitigation, and monitoring of money laundering and financing of terrorism (ML/FT) risks. Mr. Anderson convened a meeting with his team to discuss key aspects that should be emphasized in the revised policies and procedures. Which of the following actions, based on the Basel committee's best practices, should Mr. Anderson incorporate for effective management of ML/FT risks?

- A. Integrate international risk assessments and country reports to augment the bank's internal monitoring process.
- B. Prioritize customer due diligence (CDD) rules as the primary tool for risk identification.
- C. Mainly focus on the bank's own risk assessment of ML/FT risks.
- D. Delegate the responsibility of ML/FT risk management to the board of directors.

The correct answer is **A**.

To ensure a thorough understanding and monitoring of ML/FT risks, banks should utilize both internal and external sources. By incorporating international risk assessments and country reports, banks can have a more comprehensive view of potential risks, especially in a global context.

B is incorrect. Although CDD rules are crucial, they shouldn't be the sole focus. It's vital to adopt a comprehensive approach by incorporating other measures and tools in addition to CDD.

C is incorrect. Relying mainly on the bank's own risk assessment might miss out on global trends and evolving risks. External guidelines and international standards can offer insights that the bank's internal assessment might overlook.

D is incorrect. While the board of directors should oversee the risk management process, day-to-day management and operational decision-making should involve specialized officers, such as a qualified chief AML/CFT officer. This ensures that the bank has dedicated expertise managing and mitigating ML/FT risks.

Things to Remember

- Money Laundering (ML) and Financing of Terrorism (FT) risks are significant concerns for financial institutions due to regulatory requirements and potential reputational damage.
 - The Basel Committee on Banking Supervision provides guidelines and best practices for banks to effectively manage ML/FT risks.
 - Customer Due Diligence (CDD) is a key component of anti-money laundering (AML) efforts, but it should be complemented with other risk management tools.
 - International risk assessments and country reports can provide valuable insights into global trends and risks that may impact a bank's operations.
 - Delegating ML/FT risk management to specialized officers, such as a Chief AML/CFT Officer, ensures dedicated expertise in managing these risks.
-

Q.5453 Global Trust Bank is a rapidly growing financial institution with an expanding international clientele. To ensure its continued growth and maintain its reputation, the bank's senior management has stressed the importance of strict adherence to global best practices in managing risks. As part of this strategy, they are revising their guidelines on combating money laundering and financing of terrorism (ML/FT). Ms. Carter, the bank's Head of Risk and Compliance, is overseeing the revision process and has sought input from her team on how best to align their strategies with the Basel committee's recommendations. Which of the following strategies, based on the Basel committee's best practices, would best enhance Global Trust Bank's approach to ML/FT risk management?

- A. Regularly update bank protocols based on core principles for effective banking supervision.
- B. Limit the bank's reliance on external sources, focusing on internal transaction data for ML/FT risk assessment.
- C. Develop a singular mitigation strategy for all ML/FT risks, ensuring uniformity across all operations.
- D. Designate the ML/FT risk management tasks to a specific department without cross-collaboration with other bank units.

The correct answer is **A**.

For a bank to stay compliant and mitigate risks effectively, it's essential to be up-to-date with the core principles for effective banking supervision. Regularly updating the bank's protocols ensures that they remain relevant and effective as global standards and risks evolve.

B is incorrect. A comprehensive approach to ML/FT risk assessment requires the integration of both internal transaction data and external sources such as international risk assessments and country reports. Limiting reliance on external sources may result in an incomplete understanding of global ML/FT risks.

C is incorrect. ML/FT risks can be diverse and context-specific. Implementing a one-size-fits-all mitigation strategy can lead to gaps in risk management and may not address specific risks effectively.

D is incorrect. ML/FT risk management is a multi-faceted challenge that often requires cross-collaboration between different bank units. Designating tasks to a specific department without collaboration might limit the bank's ability to gain a comprehensive view and manage risks holistically.

Things to Remember

- Money Laundering (ML) and Financing of Terrorism (FT) are significant risks that financial institutions need to manage effectively to maintain their reputation and comply with regulations.
- The Basel Committee on Banking Supervision provides global standards and best practices for risk management in the banking sector, including guidelines on combating ML/FT risks.
- Effective ML/FT risk management involves a comprehensive approach that integrates internal transaction data with external sources for a holistic view of risks.
- Risk management strategies should be tailored to the specific nature of ML/FT risks, as they can vary in complexity and context.
- Collaboration and coordination between different departments within a bank are crucial for addressing ML/FT risks effectively and gaining a comprehensive view of the risks.

Q.5454 Regent Bank, a globally renowned financial institution, is aiming to bolster its Anti-Money Laundering and Countering Financing of Terrorism (AML/CFT) protocols. Mrs. Fernandez, the bank's newly appointed Chief AML/CFT officer, is tasked with evaluating and improving the existing frameworks in accordance with global best practices. To achieve this,

Mrs. Fernandez considers the importance of the three lines of defense and the role of the chief AML/CFT officer in effective risk management. Based on global best practices, which of the following measures should Mrs. Fernandez prioritize to enhance Regent Bank's AML/CFT risk management?

- A. Limit the responsibilities of the Chief AML/CFT officer to solely the operational aspects, excluding any form of executive oversight.
- B. Ensure that the Chief AML/CFT officer has a direct reporting line to senior management or the board, clarifying the relationship between the chief officer and other officers.
- C. Standardize the AML/CFT training content for all departments to maintain uniformity in knowledge dissemination across the bank.
- D. Allocate the Chief AML/CFT officer as the sole authority to modify and adjust bank AML/CFT policies without consulting the board or senior management.

The correct answer is **B**.

Establishing a direct reporting line for the Chief AML/CFT officer to senior management or the board is pivotal. It not only underscores the significance of the AML/CFT function within the organization but also ensures that concerns, suggestions, and findings of the officer are accorded the attention and urgency they deserve at the highest echelons of the bank. Furthermore, clear reporting lines prevent potential conflicts of interest and facilitate swift decision-making processes.

A is incorrect. The Chief AML/CFT officer's role should not be limited. Their responsibilities should encompass both operational and executive oversight to ensure comprehensive risk management.

C is incorrect. While standardization ensures uniformity, AML/CFT training should be adapted to the specific roles and potential risks encountered by different departments. Implementing a blanket training approach may lead to gaps in knowledge where specialized training is necessary.

D is incorrect. While the Chief AML/CFT officer plays a central role in shaping AML/CFT policies, it is essential that they collaborate with the board and senior management. This collective decision-making process ensures holistic policy development that considers multiple perspectives and expertise.

Things to Remember

- **Three Lines of Defense Model:** This model is a widely accepted framework for effective risk management in organizations. It delineates responsibilities among operational management, risk management and compliance functions, and internal audit to ensure robust risk oversight and control.

- AML/CFT Compliance: Anti-Money Laundering (AML) and Countering Financing of Terrorism (CFT) are critical components of regulatory compliance in the financial industry. They aim to prevent illicit activities such as money laundering and terrorist financing.
 - Direct Reporting Lines: Establishing clear reporting lines for key roles, such as the Chief AML/CFT officer, ensures accountability, transparency, and effective communication within the organization.
 - Policy Development: Collaborative policy development involving key stakeholders like the board, senior management, and compliance officers is essential for creating robust and comprehensive AML/CFT frameworks.
-

Q.5455 Falcon International Bank, known for its expansive global operations, is in the process of overhauling its Anti-Money Laundering and Countering Financing of Terrorism (AML/CFT) protocols. As a part of this initiative, Mr. Thompson, the bank's Chief Risk Officer, is collaborating with external consultants to devise a strategy that aligns with international best practices. Their discussions mainly revolve around the roles of the different lines of defense and the importance of human resources in this context. Considering global best practices, which of the following strategies should Mr. Thompson adopt to fortify Falcon International Bank's AML/CFT risk management procedures?

- A. Establish a centralized AML/CFT training program that provides identical content to all employees to ensure consistent knowledge across departments.
- B. Facilitate robust communication channels between the three lines of defense, ensuring that the first line - business units, are proactive in identifying, assessing, and controlling AML/CFT risks.
- C. Assign the internal audit function as the primary decision-maker for amending AML/CFT policies, thereby ensuring an unbiased review and implementation process.
- D. Empower the compliance function with advanced technology tools, relegating human intervention to the background, for efficient and error-free monitoring.

The correct answer is **B**.

Effective AML/CFT risk management necessitates a well-coordinated approach among the three lines of defense. Particularly, the first line - the business units, must be empowered and educated to actively identify, assess, and control risks. This proactive stance ensures that potential threats

are mitigated at the onset, reducing the burden on subsequent defense lines and promoting a risk-aware culture.

A is incorrect. While uniformity in training ensures a consistent baseline of knowledge, AML/CFT training should be tailored to the specific responsibilities and potential risks faced by individual departments. A one-size-fits-all approach might lead to knowledge gaps in specialized areas, making it less effective.

C is incorrect. While the internal audit function plays a pivotal role in reviewing AML/CFT policies, entrusting them with the sole authority to amend these policies could lead to a lack of holistic input. Collaborative decision-making, involving various stakeholders, is crucial for comprehensive and effective policy formulation.

D is incorrect. While technology tools enhance efficiency in monitoring, human judgment remains indispensable in AML/CFT risk management. Relegating human intervention to the background could result in overlooking nuanced risks and threats that technology might miss. The optimal approach combines the strengths of both technology and human expertise.

Things to Remember

- Three lines of defense model in risk management
 - Importance of collaboration between internal and external stakeholders in risk management
 - Role of technology in enhancing risk management processes
 - Importance of tailored training programs for specific departments
 - Need for a risk-aware culture within an organization
-

Q.5456 Global United Bank is conducting an internal review of its governance structures. The Board of Directors has specifically requested an evaluation of the Internal Audit department's role in the context of Anti-Money Laundering and Countering Financing of Terrorism (AML/CFT) compliance. Mr. Kingston, the Senior Internal Auditor, has been given the task to ensure that the internal audit's capabilities align with industry best practices and regulatory requirements. What measure should Mr. Kingston prioritize to ensure that the Internal Audit function effectively contributes to the AML/CFT framework of Global United Bank, in line with best practices?

- A. Streamline the Internal Audit function to focus primarily on post-incident reporting to avoid interfering with the proactive risk management activities of the compliance department.
- B. Delegate the authority to revise AML/CFT policies exclusively to the Internal Audit department to maintain an element of independence and objectivity in policy formulation.

C. Institute a regular rotation of audit personnel within the AML/CFT function to different departments to broaden their understanding of various business operations and risks.

D. Introduce a continuous auditing process for AML/CFT activities, enabling real-time monitoring and evaluation, thus providing immediate feedback for risk mitigation.

The correct answer is **D**.

Incorporating a continuous auditing approach allows the Internal Audit function to provide ongoing assurance and consultative feedback, which is essential in the dynamic field of AML/CFT. This helps the bank to promptly identify and address risks, ensuring that AML/CFT measures are effective and updated in real time to align with the ever-changing regulatory landscape and patterns of financial crime.

A is incorrect. The Internal Audit function should not be limited to post-incident reporting as this diminishes its role in preventive control and oversight. Best practices suggest a proactive engagement in reviewing and advising on AML/CFT controls before incidents occur.

B is incorrect. While the Internal Audit function should evaluate and provide recommendations on AML/CFT policies, the exclusive authority to revise policies should not reside with Internal Audit. Such responsibilities should involve a collaborative approach with the AML/CFT compliance team and senior management to ensure balanced and comprehensive policy development.

C is incorrect. Although rotating personnel can be beneficial for broadening experience, regularly moving auditors may disrupt the continuity and depth of expertise required for effective AML/CFT auditing. Auditors need to develop and maintain specialized knowledge in AML/CFT to conduct thorough and insightful audits.

Things to Remember

- Anti-Money Laundering (AML): A process designed to prevent individuals from disguising illegally obtained funds as legitimate income.
- Countering Financing of Terrorism (CFT): Measures to combat the financing of terrorist activities.
- Continuous Auditing: A method used to perform audit-related activities on a more frequent basis, often leveraging technology for real-time monitoring.
- Internal Audit Function: An independent, objective assurance and consulting activity designed to add value and improve an organization's operations.
- Regulatory Requirements: Rules and guidelines set by regulatory bodies that

organizations must adhere to in order to ensure compliance and mitigate risks.

Q.5457 During the development of its Customer Acceptance Policy (CAP), MetroBank is focusing on aligning with best practices for the acceptance of new clients while also adhering to the principles of financial inclusion. Based on the recommendations by the Basel committee, which of the following approaches should MetroBank prioritize to ensure both compliance and inclusivity in its CAP?

- A. Implement a universal policy that restricts the opening of accounts to only those customers who can provide a comprehensive financial profile, thereby minimizing the risk of money laundering and terrorist financing.
- B. Enforce a stringent CAP that mandates enhanced due diligence for all new customers, irrespective of their risk profile, to maintain a consistent level of scrutiny across all customer interactions.
- C. Develop a CAP that incorporates differentiated due diligence measures, considering factors such as the customer's geographical location, type of product desired, and the level of risk associated with their profile, to balance risk management with non-discrimination.
- D. Establish a CAP that exclusively targets high-risk individuals, such as politically exposed persons (PEPs) and individuals from high-risk countries, with enhanced due diligence, while simplifying the process for all other customers.

The correct answer is **C**.

The correct approach for MetroBank is to develop a Customer Acceptance Policy that allows for the application of risk-based due diligence measures. This means differentiating the level of due diligence based on various risk factors associated with the customer, such as their geographical location, the type of products they are interested in, their transactional profiles, and any other relevant risk indicators. This risk-based approach enables the bank to effectively manage potential risks without being overly restrictive, thereby promoting financial inclusion and avoiding unfair discrimination against individuals who may be financially or socially disadvantaged.

A is incorrect. A universal policy that heavily restricts account opening can be overly prohibitive and could potentially deny access to legitimate customers who are financially or socially disadvantaged, which goes against the principles of financial inclusion.

B is incorrect. Mandating enhanced due diligence for all new customers would result in an inefficient allocation of resources and could lead to unnecessary barriers for low-risk individuals, which is not in line with recommended practices that call for proportionality based on risk.

D is incorrect. While focusing on high-risk individuals for enhanced due diligence is important, it should not be done to the exclusion of applying a risk-based approach across all customers.

Simplifying the process for all other customers without any due diligence could expose the bank to unforeseen risks.

Things to Remember

- Customer Acceptance Policy (CAP) is a key component of a bank's anti-money laundering (AML) and counter-terrorist financing (CTF) framework.
 - Risk-based approach in AML/CFT involves assessing the risk associated with each customer and applying appropriate due diligence measures accordingly.
 - Financial inclusion aims to provide access to financial services to all individuals, including those who are traditionally underserved or excluded from the financial system.
 - Politically Exposed Persons (PEPs) are individuals who are entrusted with prominent public functions and may pose a higher risk of corruption or money laundering.
 - Enhanced Due Diligence (EDD) involves conducting additional checks and monitoring for high-risk customers to mitigate the increased risk associated with their accounts.
-

Reading 113: Case Study: Financial Crime and Fraud

Q.5120 An operational risk manager at the bank is presenting on financial crimes and fraud. He starts his presentation by defining financial crime. Which of the following is the correct definition of a financial crime?

- A. Any criminal conduct relating to money or to, financial services, or markets
- B. Losses due to acts intended to defraud, misappropriate property or circumvent regulations, the law, or company policy excluding diversity/discrimination events involving at least one internal party
- C. Losses due to acts of a type intended to defraud, misappropriate property or circumvent the law by a third party
- D. Misappropriation of assets, such as extortion, embezzlement, malicious destruction of assets, bribery, and tax evasion

The correct answer is **A**.

Financial crime, as defined by the Financial Conduct Authority's (FCA) Handbook of the UK, encompasses any kind of criminal conduct relating to money or to financial services or markets. This includes any offence involving fraud or dishonesty, misconduct in, or misuse of information relating to, a financial market, handling the proceeds of crime, or the financing of terrorism. This definition is broad and encompasses a wide range of activities, reflecting the diverse nature of financial crimes. Financial crimes can include everything from money laundering and fraud to bribery and corruption. They can be committed by individuals, organizations, or even states, and can have serious consequences, including financial loss, reputational damage, and legal penalties.

Choice B is incorrect. While this option does cover some aspects of financial crime, such as fraud and misappropriation of property, it unnecessarily excludes diversity/discrimination events. Financial crimes can involve a wide range of illegal activities, not just those intended to defraud or misappropriate property. Furthermore, the involvement of an internal party is not a necessary condition for an act to be considered a financial crime.

Choice C is incorrect. This choice incorrectly limits the definition of financial crimes to acts committed by third parties only. Financial crimes can be committed by anyone involved in the handling or management of money or financial services, including employees and executives within an organization.

Choice D is incorrect. Although this option includes several types of misconduct that could be classified as financial crimes (e.g., extortion, embezzlement), it fails to capture the full breadth and complexity of what constitutes a 'financial crime'. For instance, it does not mention fraudulent activities related to financial markets or services which are also part of 'financial crime'.

Things to Remember

- Financial crimes can have serious consequences, including financial loss, reputational damage, and legal penalties.
 - Financial crimes can be committed by individuals, organizations, or even states.
 - Financial crimes encompass a wide range of activities, reflecting the diverse nature of financial crimes.
 - Handling the proceeds of crime or the financing of terrorism are also considered financial crimes.
 - Internal parties are not a necessary condition for an act to be considered a financial crime.
 - Financial crimes can involve illegal activities beyond just those intended to defraud or misappropriate property.
 - Fraudulent activities related to financial markets or services are also part of financial crimes.
-

Q.5122 A risk manager at a large bank states that the bank has zero tolerance for internal fraud. He goes ahead to highlight that the bank has a robust framework of controls and measures to mitigate internal fraud risks. Which of the following is a component of such a framework?

- A. Inspections
- B. Selection
- C. Placement
- D. Layering

The correct answer is **B**.

Selection is a crucial component of a framework designed to control and mitigate internal fraud risks. This process involves the careful screening of employees and associated third parties. The organization's culture is also taken into account during this step. When firms hire individuals who align with their values and ethical standards, managing these employees becomes significantly easier. Selection also plays a vital role in Anti-Money Laundering (AML) and third-party risk management, serving as an important mitigation mechanism.

Other components of a robust fraud risk management framework include Prevention, Detection, and Deterrents. Prevention involves clearly defining the rights, authority, and access of each function to effectively manage fraud risk. Detection is critical in limiting the impact of an operational risk event, with detective controls acting as a deterrent as well. Fraud is least likely to occur if the consequences are severe. Effective supervision and monitoring help limit internal fraud. Deterrents, which are sanctions and actions announced following any act of fraud, also disincentivize employees to commit fraud, thus promoting the risk-reward balance.

Choice A is incorrect. While inspections can be a part of a comprehensive system to mitigate risks, they are not specifically targeted towards internal fraud. Inspections are more general in nature and aim to ensure compliance with all policies and procedures, not just those related to fraud.

Choice C is incorrect. Placement refers to the process of assigning employees to specific roles within the organization. Although it's important for risk management, it doesn't directly contribute to mitigating the risks associated with internal fraud.

Choice D is incorrect. Layering is a term used in money laundering processes where illegal funds are made difficult to trace back by creating complex layers of financial transactions. It does not relate directly with measures taken against internal fraud in an organization.

Things to Remember

- Internal fraud: Refers to fraudulent activities committed by individuals within an organization, such as employees or managers, for personal gain or to harm the organization.
- Controls and measures: These are policies, procedures, and practices put in place to prevent, detect, and respond to risks, including internal fraud.
- Employee screening: Involves background checks, reference checks, and other processes to ensure that individuals hired by the organization are trustworthy and aligned with its values.
- Anti-Money Laundering (AML): A set of regulations and practices designed to prevent the illegal generation of income through financial systems.
- Third-party risk management: Involves assessing and mitigating risks associated with vendors, suppliers, and other external parties that interact with the organization.
- Prevention, Detection, and Deterrents: Key components of a fraud risk management framework aimed at reducing the likelihood and impact of internal fraud incidents.
- Supervision and monitoring: Continuous oversight of activities and transactions to

identify suspicious behavior or anomalies that could indicate internal fraud.

- Risk-reward balance: Refers to the concept that individuals are less likely to engage in fraudulent activities if the potential consequences outweigh the benefits.
-

Q.5123 An operational risk manager at a bank has asked a junior analyst to prepare a presentation on AML risk management to be presented to the board's risk committee. Which of the following controls falls under the "deterrents" step of AML controls?

- A. Transaction monitoring system
- B. Staff information and training
- C. Ethos and values
- D. Legal pursuits

The correct answer is **D**.

Deterrents are measures taken to discourage or prevent fraudulent activities. They are typically punitive actions that are announced following any act of fraud. The purpose of deterrents is to disincentivize employees from committing fraud by promoting a balance between risk and reward. Examples of deterrents in the context of Anti-Money Laundering (AML) controls include escalation to the relevant financial intelligence unit (FIU), legal pursuits, and closure of accounts. Legal pursuits, in particular, serve as a strong deterrent as they involve the potential for legal action and penalties against those involved in fraudulent activities. This can include fines, imprisonment, and other legal consequences. Therefore, legal pursuits are an appropriate control to include under the 'deterrents' step of AML controls.

Choice A is incorrect. A transaction monitoring system is not a deterrent but rather a detection tool used in the process of Anti-Money Laundering (AML) controls. It helps in identifying suspicious transactions that may indicate money laundering activities, but it does not deter individuals from engaging in such activities.

Choice B is incorrect. Staff information and training are crucial for ensuring that employees understand their roles and responsibilities in preventing money laundering. However, this falls under the 'prevention' step of AML controls rather than 'deterrents'. While well-informed staff can help deter money laundering by being vigilant and proactive, this control itself does not directly act as a deterrent.

Choice C is incorrect. Ethos and values form part of an organization's culture which can influence behavior towards compliance with laws and regulations including those related to AML. However, they do not directly serve as deterrents to money laundering activities.

Things to Remember

- Anti-Money Laundering (AML) controls are designed to prevent, detect, and deter money laundering activities within financial institutions.
 - Deterrents in AML controls are punitive measures aimed at discouraging individuals from engaging in fraudulent activities.
 - Examples of deterrents in AML controls include legal pursuits, escalation to financial intelligence units, and closure of accounts.
 - Transaction monitoring systems are detection tools used to identify suspicious transactions, not deterrents.
 - Staff information and training are important for prevention of money laundering, but they fall under the prevention step rather than deterrents.
 - Ethos and values contribute to an organization's culture of compliance but do not directly act as deterrents to money laundering.
-

Q.5124 A newly hired money laundering risk officer is presenting on AML risk management. He highlights that it is common for criminals to disguise the proceeds of their criminal activities into legitimate sources of funds in two or three phases. Which of the following is a phase of money laundering?

- A. Placement
- B. Deterrent
- C. Detection
- D. Protection

The correct answer is **A**.

Placement is indeed a phase of money laundering. It involves all methods intended to disguise the origins of the funds. This could include cash transfer to businesses, false invoicing, use of trusts and offshore companies, 'smurfing' (keeping a bank account or credit card under the Anti-Money Laundering (AML) reporting threshold by making a series of small transactions rather than a single large transaction), using foreign bank accounts, and so on. The goal of this phase is to introduce the illicit money into the financial system in such a way that it appears to be legitimate.

Choice B is incorrect. Deterrent is not a stage in the money laundering process. It refers to measures taken to prevent or discourage certain behaviors, including illegal activities such as money laundering. However, it does not represent a stage in the process of money laundering itself.

Choice C is incorrect. Detection is also not a stage in the money laundering process. Detection refers to identifying and recognizing suspicious activities that may indicate money laundering, but it does not represent an actual step that criminals undertake when they launder their illicit funds.

Choice D is incorrect. Protection does not represent a stage in the money laundering process either. While criminals involved in this illegal activity do seek ways to protect their ill-gotten gains and avoid detection by authorities, protection itself isn't considered as one of the distinct stages of converting illicit proceeds into seemingly legitimate funds.

Things to Remember

- Money laundering is the process of disguising the origins of illegally obtained money, typically by means of transfers involving foreign banks or legitimate businesses.
 - There are generally three main stages of money laundering: Placement, Layering, and Integration.
 - Layering involves separating the illicit funds from their original source through a complex series of financial transactions.
 - Integration is the final stage where the laundered money is reintroduced into the economy and appears to be legitimate.
 - Anti-Money Laundering (AML) regulations and compliance programs are designed to detect and prevent money laundering activities.
-

Q.5125 The CEO of a bank highlights that the bank is practicing comprehensive AML risk management. Which of the following statements would justify the CEO's claim that the bank is practicing comprehensive Anti-Money Laundering (AML) risk management?

- A. The bank has established robust customer due diligence procedures.
- B. The bank has never had a customer involved in a money laundering scheme.
- C. The bank has hired a new CEO with extensive experience in AML risk management.
- D. The bank has reported suspicious transactions to the relevant authorities.

The correct answer is **A**.

Comprehensive AML risk management involves a proactive approach to mitigating risks associated with money laundering and terrorism financing. A key component of this approach is Customer Due Diligence (CDD). CDD procedures require banks to perform background checks and verify the identities of their customers. This helps to ensure that the bank is not inadvertently facilitating illicit activities. By establishing robust CDD procedures, the bank is demonstrating a commitment to preventing money laundering and terrorism financing, which is a key aspect of comprehensive AML risk management. Therefore, if the bank has indeed established robust CDD procedures, this would provide strong evidence to support the CEO's claim that the bank is practicing comprehensive AML risk management.

Choice B is incorrect. The fact that a bank has never had a customer involved in a money laundering scheme does not necessarily indicate comprehensive AML risk management. It could simply mean that the bank has been fortunate or unaware of such activities. Effective AML risk management involves proactive measures, such as robust customer due diligence procedures, rather than relying on the absence of past incidents.

Choice C is incorrect. Hiring a new CEO with extensive experience in AML risk management can be beneficial for the bank's overall approach to managing this type of risk, but it does not provide evidence of comprehensive AML practices currently in place at the bank. The CEO's experience alone cannot guarantee effective implementation and operation of robust policies, procedures, and controls.

Choice D is incorrect. Reporting suspicious transactions to relevant authorities is an important part of AML compliance but it doesn't necessarily indicate comprehensive AML risk management by itself. Comprehensive approach would involve preventive measures like establishing robust customer due diligence procedures which can help prevent illicit activities from occurring in the first place.

Things to Remember

- Customer Due Diligence (CDD) is a key component of comprehensive AML risk management, involving background checks and identity verification of customers.
- Proactive measures, such as robust CDD procedures, are essential in preventing money laundering and terrorism financing in banks.
- Effective AML risk management requires more than just the absence of past incidents; it involves proactive risk mitigation strategies.
- Experience of a new CEO in AML risk management can be beneficial, but it does not guarantee comprehensive AML practices in place.
- Reporting suspicious transactions to relevant authorities is important for AML

compliance, but preventive measures like CDD are crucial for comprehensive risk management.

Q.5126 In its 2022 report, the FCA examines financial crime controls at challenger banks, which are fully digital and offer customers the ability to open accounts very quickly. Which of the following is a key finding highlighted by UK regulators in their examination of financial crime controls at challenger banks in their 2022 report?

- A. Challenger banks tend to perform better than traditional banks in identifying higher-risk customers.
- B. Challenger banks need to improve their systems for identifying and verifying customer information.
- C. Challenger banks are not required to follow AML regulations because they operate fully digitally.
- D. Traditional banks are more susceptible to financial crime than challenger banks.

The correct answer is **B**.

The 2022 report by the Financial Conduct Authority (FCA) on the examination of financial crime controls at challenger banks highlighted that these banks need to enhance their systems for identifying and verifying customer information. The report underscored the risk associated with the rapid account opening process offered by these banks. The information collected during this process may not be sufficient to identify customers who pose a higher risk, thereby making it challenging for these banks to effectively manage their Anti-Money Laundering (AML) risks. As a result, the regulators have recommended that challenger banks bolster their systems for identifying and verifying customer information to better manage their AML risks and prevent financial crimes.

Choice A is incorrect. The report did not conclude that challenger banks perform better than traditional banks in identifying higher-risk customers. In fact, the FCA's examination found that these digital banks need to improve their systems for identifying and verifying customer information, which includes the identification of higher-risk customers.

Choice C is incorrect. Regardless of whether a bank operates digitally or traditionally, it is required to follow Anti-Money Laundering (AML) regulations. The mode of operation does not exempt any financial institution from adhering to these regulations.

Choice D is incorrect. The FCA's report did not suggest that traditional banks are more susceptible to financial crime than challenger banks. The susceptibility to financial crime depends on the effectiveness of a bank's financial crime controls and not its mode of operation.

Things to Remember

- Challenger banks: These are smaller, newer banks that compete with traditional banks, often by specializing in areas such as online banking, mobile apps, and customer experience.
 - Financial Conduct Authority (FCA): The FCA is a regulatory body in the UK that oversees financial firms and markets to ensure they operate in a way that protects consumers and maintains the integrity of the financial system.
 - Anti-Money Laundering (AML) regulations: These regulations are designed to prevent money laundering and the financing of terrorism by requiring financial institutions to implement policies and procedures to detect and report suspicious activities.
 - Customer due diligence (CDD): This is a process that financial institutions must perform to verify the identity of their customers and assess the risks associated with doing business with them.
 - Know Your Customer (KYC): KYC is a key component of AML regulations that requires financial institutions to verify the identity of their customers before providing services to them.
-

Q.5127 Which of the following is a lesson learned from the USAA Federal Savings Bank (FSB) case study, where it was fined \$140 million by the Financial Crimes Enforcement Network (FinCEN) and the Office of the Comptroller of the Currency (OCC) for failing to implement and maintain a BSA/AML compliance program?

- A. Banks should prioritize customer service over compliance to remain competitive.
- B. Banks should implement robust BSA/AML compliance programs to avoid penalties.
- C. Banks should minimize their reporting of suspicious activities to avoid regulatory scrutiny.
- D. Banks should shift their compliance focus away from AML to other areas such as cybersecurity.

The correct answer is **B**.

The USAA FSB case study underscores the critical importance of having a robust Bank Secrecy

Act/Anti-Money Laundering (BSA/AML) compliance program in place. The bank was penalized with a hefty fine of \$140 million for its failure to implement and maintain such a program, thereby exposing itself to the risks of money laundering and terrorist financing activities. This case serves as a stark reminder for all banks about the potential consequences of non-compliance with BSA/AML regulations. Therefore, the primary lesson to be learned from this case is the necessity for banks to prioritize BSA/AML compliance in order to avoid penalties and regulatory scrutiny.

Choice A is incorrect. Prioritizing customer service over compliance can lead to severe penalties and reputational damage, as seen in the USAA FSB case. Compliance with regulatory requirements such as BSA/AML should not be compromised for competitiveness.

Choice C is incorrect. Minimizing the reporting of suspicious activities can result in non-compliance with BSA/AML regulations, leading to hefty fines and sanctions from regulatory bodies like FinCEN and OCC. Banks are required to report any suspicious activities promptly to avoid such consequences.

Choice D is incorrect. While cybersecurity is an important area of focus for banks, shifting compliance focus away from AML could lead to non-compliance with BSA/AML regulations, resulting in significant penalties as demonstrated by the USAA FSB case study.

Things to Remember

- Bank Secrecy Act/Anti-Money Laundering (BSA/AML) compliance programs are essential for banks to detect and prevent money laundering and terrorist financing activities.
- Regulatory bodies like the Financial Crimes Enforcement Network (FinCEN) and the Office of the Comptroller of the Currency (OCC) oversee and enforce compliance with BSA/AML regulations.
- Non-compliance with BSA/AML regulations can result in hefty fines, sanctions, and reputational damage for banks.
- Reporting of suspicious activities is a crucial part of BSA/AML compliance to prevent financial crimes.
- Compliance with BSA/AML regulations should be a top priority for banks to avoid penalties and regulatory scrutiny.

Reading 114: Guidance on Managing Outsourcing Risk

Q.2318 Tummers Bank from New York, USA, is considering outsourcing some of its activities to a third party. Which of the following risks (among others) should the bank consider before making a final decision?

- A. Credit, market, and operational risks
- B. Reputational, operational, and compliance risks
- C. Portfolio, counterparty, and market risks
- D. Country, legal, and counterparty risks

The correct answer is **B**.

When a financial institution like Tummers Bank considers outsourcing some of its activities, it must consider a variety of risks. These risks include reputational, operational, and compliance risks. Reputational risks refer to the potential damage to the bank's reputation that could occur if the third party fails to meet the bank's standards or if there is a breach of data or security. Operational risks refer to the potential for loss resulting from inadequate or failed internal processes, people, and systems, or from external events. This includes the risk that the third party may not be able to carry out the outsourced activities to the required standard. Compliance risks involve the potential for legal penalties, financial forfeiture, and material loss an institution might suffer as a result of its failure to comply with laws, regulations, rules, related self-regulatory organization standards, and codes of conduct applicable to its banking activities. Therefore, these three risks are crucial for Tummers Bank to consider before making a final decision on outsourcing.

Choice A is incorrect. While credit, market, and operational risks are important for any financial institution to consider, they are not the most relevant in the context of outsourcing. Credit and market risks primarily relate to investment decisions rather than operational decisions such as outsourcing.

Choice C is incorrect. Portfolio, counterparty, and market risks are more related to investment management and trading activities. These types of risk do not directly arise from outsourcing functions of a bank.

Choice D is incorrect. Country risk could be relevant if the bank was considering outsourcing functions overseas; however, it's not necessarily a primary concern in all cases of outsourcing. Legal risk would be a consideration but it's typically addressed through contracts with service providers rather than being an ongoing risk associated with the decision itself. Counterparty risk can be considered part of credit risk which as explained above isn't directly related to an outsourcing decision.

Things to Remember

- Outsourcing is the practice of hiring a third-party company or individual to perform

tasks, handle operations, or provide services that are traditionally done by the bank internally.

- Reputational risk is the potential for negative public perception or damage to the bank's brand and image.
 - Operational risk involves the potential for loss due to internal processes, systems, or external events.
 - Compliance risk refers to the potential legal and financial consequences of failing to comply with laws, regulations, and industry standards.
 - Credit risk is the risk of loss resulting from a borrower's failure to repay a loan or meet contractual obligations.
 - Market risk is the risk of losses in financial instruments due to changes in market conditions such as interest rates, exchange rates, and asset prices.
 - Portfolio risk is the risk associated with the overall investment portfolio of the bank.
 - Counterparty risk is the risk that the other party in a financial transaction will not fulfill its obligations.
 - Country risk is the risk associated with investing or operating in a particular country due to political, economic, or social factors.
 - Legal risk is the risk of legal actions, penalties, or losses due to non-compliance with laws and regulations.
-

Q.2322 New Savings Bank from Texas, USA, recently outsourced its IT services to Novel IT Service company (NIS) from Los Angeles, USA. NIS has a rich history spanning several decades but has recently been the subject of public criticism for various legal violations, as well as poor service delivery, punctuated by costly delays. NIS has most likely exposed the bank to:

- A. Outsourcing risk
- B. Credit risk

C. Reputational risk

D. Concentration risk

The correct answer is **C**.

Reputational risk is the potential for negative publicity, public perception, or uncontrollable events to adversely impact a company's reputation, thereby affecting its revenue. In the context of the question, New Savings Bank has outsourced its IT services to Novel IT Service company (NIS), which has been under public scrutiny for legal violations and poor service delivery. These issues have the potential to negatively impact the bank's reputation if they become associated with NIS's negative public image. This could lead to loss of customer trust, which could in turn lead to loss of customers and revenue. Therefore, the bank is most likely exposed to reputational risk.

Choice A is incorrect. Outsourcing risk refers to the potential negative impacts that can arise from an organization's decision to outsource certain tasks or services. While New Savings Bank is indeed outsourcing its IT services, the question specifically asks about the type of risk associated with NIS's legal violations and poor service delivery, which falls under reputational risk rather than outsourcing risk.

Choice B is incorrect. Credit risk pertains to the possibility of a loss resulting from a borrower's failure to repay a loan or meet contractual obligations. In this scenario, there is no mention of any loans or credit agreements between New Savings Bank and NIS, hence credit risk does not apply here.

Choice D is incorrect. Concentration risk refers to the potential for losses to occur due to heavy investment in a particular asset class, sector, region or other area of exposure without adequate diversification. In this case scenario, there isn't any indication that New Savings Bank has heavily invested in one particular area without sufficient diversification; thus concentration risk does not apply here.

Things to Remember

- Reputational risk can have a significant impact on a company's brand value and can lead to long-term consequences such as loss of customers, revenue, and trust.
- Outsourcing risk involves the potential negative impacts that can arise from an organization's decision to outsource certain tasks or services, such as loss of control, quality issues, and dependency on third-party vendors.
- Credit risk is the risk of financial loss resulting from a borrower's failure to repay a loan or meet contractual obligations, and it is a key consideration in lending and investment activities.

- Concentration risk arises when a company has a significant exposure to a single asset class, sector, or region, which can increase vulnerability to adverse events in that specific area.
 - When outsourcing services, it is crucial for companies to conduct thorough due diligence on potential service providers to assess their reputation, track record, and compliance with regulations to mitigate reputational risks.
-

Q.2323 Which of the following statements is correct?

- A. After the outsourcing of an activity, all responsibility with regard to outsourced activities is transferred to the third party.
- B. After the outsourcing of an activity, the third party and senior management have partial responsibility.
- C. After the outsourcing of an activity, senior management is still responsible for normal functioning of the bank.
- D. It is not possible to outsource accounting services.

The correct answer is **C**.

After the outsourcing of an activity, senior management is still responsible for normal functioning of the bank. This is because the use of service providers does not relieve a financial institution's board of directors and senior management of their responsibility to ensure that outsourced activities are conducted in a safe-and-sound manner and in compliance with applicable laws and regulations. Policies governing the use of service providers should be established and approved by the board of directors, or an executive committee of the board. This means that even though the activity has been outsourced, the senior management still has the responsibility to ensure that the activity is being conducted properly and in accordance with all relevant laws and regulations.

Choice A is incorrect. Even though an activity is outsourced, the responsibility for ensuring that the outsourced activities are conducted in a safe-and-sound manner and in compliance with applicable laws and regulations still lies with the senior management of the financial institution. The third party does not assume all responsibilities.

Choice B is incorrect. While it's true that both parties have some level of responsibility, it's not accurate to say they share equal or partial responsibility. The ultimate accountability for ensuring compliance and safety remains with senior management of the financial institution, regardless of any outsourcing arrangements.

Choice D is incorrect. It's certainly possible to outsource accounting services within a financial

institution; however, this doesn't change who retains overall responsibility for ensuring these services are conducted safely and compliantly - which remains with senior management.

Things to Remember

- Outsourcing: The practice of contracting out certain business functions or activities to external service providers.
 - Senior Management Responsibility: Senior management of a financial institution is ultimately responsible for the overall functioning and compliance of the institution.
 - Board of Directors: The governing body of a financial institution responsible for overseeing the institution's activities and ensuring compliance with regulations.
 - Service Provider Governance: Policies and procedures should be in place to govern the use of service providers and ensure compliance with regulations.
 - Compliance: Ensuring that all activities, including outsourced ones, are conducted in accordance with relevant laws and regulations.
-

Q.2324 WPC performs an audit on financial statements of Anderson Bank. After performing really well, the bank decides to offer the company an internal audit role in addition to the existing role. The move exposes the bank to:

- A. Country risk
- B. Operational risk
- C. Reputational risk
- D. Compliance risk

The correct answer is **D**.

Compliance risk is the potential for losses and legal penalties due to failure to comply with laws or regulations. In this case, the Sarbanes-Oxley Act of 2002, a U.S. law, specifically prohibits a registered public accounting firm from performing certain non-audit services for a public company client for whom it performs financial statement audits. This is to ensure the independence and objectivity of the audit. If WPC were to accept the offer to perform both external and internal audits for Anderson Bank, it could potentially violate this law, thereby exposing the bank to compliance risk. This could result in legal penalties for the bank, and could also damage its reputation.

Choice A is incorrect. Country risk refers to the potential for losses due to a nation's political, legal, economic or social conditions. It does not apply in this scenario as the situation described does not involve any changes in these conditions.

Choice B is incorrect. Operational risk refers to the risk of loss resulting from inadequate or failed internal processes, people and systems or from external events. While there may be some operational risks involved in WPC performing both internal and external audits for Anderson Bank, such as potential conflicts of interest or lack of independence, these are more appropriately classified under compliance risk.

Choice C is incorrect. Reputational risk involves damage to a company's reputation that could lead to financial loss or difficulties in maintaining relationships with customers, shareholders and other stakeholders. Although reputational damage could potentially occur if WPC were found to be conducting its audits improperly due to its dual role at Anderson Bank, this would be a consequence rather than a direct type of risk associated with the arrangement.

Things to Remember

- Compliance risk is the potential for losses and legal penalties due to failure to comply with laws or regulations.
 - The Sarbanes-Oxley Act of 2002 prohibits a registered public accounting firm from performing certain non-audit services for a public company client for whom it performs financial statement audits to ensure independence and objectivity.
 - Operational risk refers to the risk of loss resulting from inadequate or failed internal processes, people, and systems or from external events.
 - Reputational risk involves damage to a company's reputation that could lead to financial loss or difficulties in maintaining relationships with stakeholders.
-

Q.2325 An American bank is considering outsourcing its IT operations to an Indian IT provider. Is it in order for the bank's risk management team to analyze the provider's financial condition before making a final decision?

- A. No, because the bank should only be concerned with country risk.
- B. No, because it's illegal to outsource IT services in the first place.
- C. Yes, in order to access the financial stability and integrity of the service provider.
- D. Yes, because it's a requirement under Basel II regulation.

The correct answer is **C**.

Financial institutions should review the financial condition of the service provider and its closely-related affiliates. A bad financial condition may be an indicator of potential problems in the future which could result in interruption of service providing etc.

Things to Remember

1. Outsourcing IT operations can offer several benefits to banks, including cost savings, access to specialized expertise, and the ability to focus on core business activities. However, it also comes with risks, including operational, reputational, and financial risks.
 2. To manage these risks, banks should conduct thorough due diligence on potential service providers. This includes analyzing the provider's financial condition, as well as other factors such as its technical capabilities, track record, and compliance with relevant laws and regulations.
 3. While country risk is an important factor to consider when outsourcing operations to a foreign service provider, it is not the only factor. Banks should also consider the specific risks associated with the service provider itself, including its financial stability.
 4. Basel II regulation encourages banks to maintain a robust risk management framework, but it does not specifically mandate that banks analyze the financial condition of their service providers. Therefore, banks should not rely solely on regulatory requirements when deciding whether to analyze a service provider's financial condition.
-

Q.2326 Sandero bank from Carrington, North Dakota, is considering outsourcing part of its IT services to a third party. Such a move will most likely involve sharing of some nonpublic personal information about the bank's customers with the third party. Should the bank go ahead with its plan?

- A. No, all outsourcing activities that can reasonably be expected to expose nonpublic personal information are forbidden.
- B. Yes, but the bank should refrain from sharing all nonpublic personal information.

C. Yes, the bank could outsource an activity which requires usage of nonpublic personal information, but the service provider must comply with applicable privacy laws and regulation.

D. No, because such a move would open doors to possible lawsuits by aggrieved customers.

The correct answer is **C**.

Yes, the bank could outsource an activity which requires usage of nonpublic personal information, but the service provider must comply with applicable privacy laws and regulation. This is because the privacy laws and regulations are designed to protect the privacy and security of nonpublic personal information (NPPI). These laws and regulations apply not only to financial institutions like Sandero bank, but also to their service providers. Therefore, if Sandero bank decides to outsource part of its IT services to a third party, the third party must comply with these laws and regulations. This means that the third party must implement appropriate measures to protect the NPPI from unauthorized access, use, disclosure, alteration, and destruction. These measures may include, for example, the use of encryption technologies, firewalls, intrusion detection systems, and access controls. In addition, the third party must provide adequate training to its employees about the importance of protecting NPPI and the consequences of non-compliance. Furthermore, the third party must regularly monitor and audit its privacy and security practices to ensure their effectiveness and compliance with the laws and regulations. Finally, the third party must promptly report any privacy or security incidents to Sandero bank and cooperate with the bank in the investigation and resolution of such incidents.

Choice A is incorrect. While privacy laws and regulations do place restrictions on the sharing of nonpublic personal information, they do not outright forbid all outsourcing activities that could potentially expose such information. Instead, these laws typically require that appropriate safeguards be put in place to protect this information.

Choice B is incorrect. This choice suggests that the bank should refrain from sharing all nonpublic personal information during outsourcing. However, this may not be practical or even possible in some cases where the outsourced service requires access to such data for operational purposes. The key is ensuring that any shared data is protected and used in compliance with applicable privacy laws and regulations.

Choice D is incorrect. While it's true that mishandling of customer data could potentially lead to lawsuits, simply outsourcing IT services does not automatically open doors to legal action by customers. As long as the bank and its service provider comply with relevant privacy laws and regulations, including those pertaining to data protection and consent for data use, they can mitigate this risk.

Things to Remember

- Nonpublic personal information (NPPI) includes sensitive data such as social security numbers, financial account information, and personal identification details.
- Privacy laws and regulations, such as the Gramm-Leach-Bliley Act (GLBA) in the United

States, mandate the protection of NPPI by financial institutions and their service providers.

- Outsourcing IT services can provide cost savings and specialized expertise, but it also introduces risks related to data security and privacy.
 - Service level agreements (SLAs) between the bank and the third-party service provider should clearly outline data protection requirements and compliance measures.
 - Data encryption, access controls, regular security audits, and incident response plans are essential components of a robust data protection strategy.
-

Q.2327 Fort Worth Bank from Texas, USA, is considering outsourcing its retail loans collection process to ICAP, a service provider from Mexico. What should its risk management team do?

- A. Carry out a risk assessment of the move.
- B. Analyze contract provisions and considerations.
- C. Perform a due diligence of the service provider.
- D. All of the above.

The correct answer is **D**.

The risk management team of Fort Worth Bank should indeed carry out all the actions listed in the options. When a financial institution is considering outsourcing a significant process such as retail loans collection, it is crucial to conduct a comprehensive risk assessment. This involves identifying potential risks and evaluating their potential impact on the bank's operations. The risk assessment should consider various factors, including the service provider's capabilities, the regulatory environment in the service provider's country, and the potential impact on the bank's customers.

Furthermore, the risk management team should also analyze the contract provisions and considerations. This includes understanding the terms and conditions of the contract, the responsibilities of each party, and the mechanisms for dispute resolution. The contract should also include provisions for the bank to monitor the service provider's performance and ensure compliance with regulatory requirements.

Finally, performing due diligence of the service provider is another critical step. The due diligence process involves verifying the service provider's credentials, assessing its financial stability, and evaluating its track record in providing similar services. The due diligence process also includes reviewing the service provider's internal controls, data security measures, and

business continuity plans.

Choice A is incorrect. While carrying out a risk assessment of the move is an important step, it alone is not sufficient to mitigate potential risks. The bank's risk management team should also consider other factors such as contract provisions and due diligence of the service provider.

Choice B is incorrect. Analyzing contract provisions and considerations is crucial, but this alone does not cover all aspects of risk mitigation. It needs to be complemented with a comprehensive risk assessment and due diligence process.

Choice C is incorrect. Performing a due diligence of the service provider can help identify potential issues that might arise in future, but it should be accompanied by a thorough risk assessment and analysis of contract provisions for complete mitigation of risks.

Things to Remember

- Outsourcing risk management
 - Regulatory considerations in outsourcing
 - Vendor risk management
 - Operational risk in outsourcing
 - Contract management in outsourcing
-

Q.2329 A bank based in Palermo, Italy, is considering outsourcing its IT services and is preparing a review of financial condition of IAM Systems – the most popular service provider in town. Which of the following would not form part of the financial review process?

- A. The adequacy of the service provider's insurance coverage.
- B. The adequacy of the service provider's review of the financial condition of any subcontractors.
- C. The potential financial impact of the bank's business on the provider.
- D. The potential impact of the provider's past clients on the bank's financial condition.

The correct answer is **D**.

Financial institutions should review the financial condition of the service provider and its closely-related affiliates. The financial review may include:

- The service provider's most recent financial statements and annual report with regard

to outstanding commitments, capital strength, liquidity and operating results

- The service provider's sustainability, including factors such as the length of time that the service provider has been in business and the service provider's growth of market share for a given service
- **The potential impact of the financial institution's business relationship on the service provider's financial condition**
- The service provider's commitment (both in terms of financial and staff resources) to provide the contracted services to the financial institution for the duration of the contract
- **The adequacy of the service provider's insurance coverage**
- **The adequacy of the service provider's review of the financial condition of any subcontractors**
- Other current issues the service provider may be facing that could affect future financial performance

Things to Remember

- Outsourcing: The practice of hiring a third-party service provider to perform tasks, handle operations, or provide services that are traditionally done in-house.
- Financial Due Diligence: The process of investigating, analyzing, and assessing the financial health and stability of a potential business partner or service provider before entering into a business relationship.
- Subcontractors: Third-party entities hired by the main service provider to perform specific tasks or services as part of the overall service delivery.
- Business Impact Analysis: Evaluating the potential financial consequences and operational disruptions that may result from a specific event, such as a business relationship with a service provider.
- Contractual Obligations: The terms and conditions outlined in a contract that specify

the responsibilities, commitments, and expectations of both parties involved in the business relationship.

Q.2330 Stroud Bank from Gloucester, UK, is in the process of executing a major merger. The bank will be outsourcing the activities related to the relocation of facilities, as required by the merger, which should take considerable efforts. The bank is reviewing operations and internal controls of several service providers in order to make an informed decision. Which of the following should not be included in the review?

- A. Training, including compliance training for staff.
- B. Compliance with the environmental sustainability of the business.
- C. Business resumption and contingency planning.
- D. Service support and delivery.

The correct answer is **B**.

While environmental sustainability is an important aspect of any business, it is not typically a factor that a financial institution would consider when reviewing the operations and internal controls of a service provider for outsourcing activities related to a merger. The primary focus of such a review would be on the service provider's ability to effectively and efficiently carry out the tasks required by the merger, including relocation of facilities, staff training, business resumption and contingency planning, and service support and delivery. Environmental sustainability, while important, is not directly related to these tasks and therefore would not typically be included in the review.

Choice A is incorrect. Training, including compliance training for staff, is a crucial element to consider during the review process. It ensures that the service provider's employees are well-equipped with the necessary skills and knowledge to perform their tasks effectively and in accordance with regulatory requirements.

Choice C is incorrect. Business resumption and contingency planning should be considered as part of this review process. This would provide Stroud Bank with an understanding of how the service provider plans to resume operations in case of any disruptions or unforeseen circumstances, ensuring continuity of services.

Choice D is incorrect. Service support and delivery are essential elements to consider during this review process as they directly impact the quality of services provided by potential providers. This includes factors such as response times, availability of support personnel, and efficiency in delivering services.

Things to Remember

- Outsourcing: The practice of contracting out certain business activities or functions to

external service providers.

- Internal Controls: Processes put in place by an organization to ensure the integrity of financial and accounting information, compliance with laws and regulations, and the effectiveness and efficiency of operations.
 - Merger: A corporate strategy involving the combination of two or more companies into a single company.
 - Business Continuity Planning: The process of creating systems of prevention and recovery to deal with potential threats to a company.
 - Service Level Agreements (SLAs): Contracts that define the level of service expected from a service provider.
-

Q.5257 Which of the following **best** describes the key elements of contracts and agreements related to the cost and compensation of service providers?

- A. Contracts and agreements should only describe the compensation to be paid to the service provider without addressing any other related expenses.
- B. Contracts and agreements should only address the payment of legal, audit, and examination fees related to the activity performed by the service provider.
- C. Contracts and agreements should only address the responsibility for the maintenance of equipment, hardware, and software related to the activity performed by the service provider.
- D. Contracts and agreements should describe the compensation, variable charges, and any fees to be paid for non-recurring items and special requests.

The correct answer is **D**.

Contracts and agreements with service providers should be comprehensive and cover all aspects related to cost and compensation. This includes the basic compensation to be paid to the service provider, any variable charges that may arise due to changes in the scope of work or other factors, and any fees that may be incurred for non-recurring items and special requests. This ensures that both parties have a clear understanding of the financial obligations involved, reducing the potential for disputes and misunderstandings. It also allows for better budgeting and financial planning by both parties. This approach aligns with best practices in contract management and is recommended by experts in the field.

Choice A is incorrect. While compensation to the service provider is an essential component, it should not be the only aspect addressed in contracts and agreements. Other related expenses such as variable charges, fees for non-recurring items and special requests should also be included to ensure a comprehensive cost structure.

Choice B is incorrect. Although legal, audit, and examination fees are important considerations in a service provider agreement, they do not encapsulate all the essential components of cost and compensation. The agreement should also cover aspects like compensation to the service provider, variable charges and any fees for non-recurring items or special requests.

Choice C is incorrect. Responsibility for maintenance of equipment, hardware, and software are indeed important considerations but they do not represent all essential components that need to be addressed in terms of cost and compensation in a service provider agreement. Other elements such as direct compensation to the service provider along with variable charges or any other additional costs must also be included.

Things to Remember

- Contracts and agreements with service providers should be comprehensive and cover all aspects related to cost and compensation
 - Variable charges may arise due to changes in the scope of work or other factors
 - Fees may be incurred for non-recurring items and special requests
 - Clear understanding of financial obligations reduces disputes and misunderstandings
 - Comprehensive agreements allow for better budgeting and financial planning
-

Reading 115: Case Study: Third-Party Risk Management

Q.5129 Which of the following is not a step in the Third-Party Risk Management life cycle?

- A. Remediation
- B. Risk rating & evaluation
- C. Shared assessments
- D. Continuous monitoring

The correct answer is C.

Shared assessment is a US based certifying organization specializing in Third-Party Risk Management and delivering the professional certification of third-party risk management professionals.

The five stages of the professional certification of third-party risk management professionals are:

- i. Business model decision
- ii. Evaluation, risk rating, due diligence
- iii. RFPs (requests for proposal) and contracts
- iv. Monitoring (continuous and ongoing)
- v. Remediation or termination

Things to Remember

- Third-Party Risk Management (TPRM) involves identifying, assessing, and controlling risks associated with outsourcing to third-party vendors.
- Remediation involves taking corrective actions to address identified risks or issues in the third-party relationship.
- Risk rating & evaluation involves assessing the level of risk posed by a third-party vendor and evaluating their performance against predefined criteria.
- Continuous monitoring is an ongoing process of monitoring the activities and performance of third-party vendors to ensure compliance with established standards and requirements.
- Shared assessments refer to the practice of sharing assessment results and information about third-party vendors among organizations to streamline the evaluation process.

Q.5130 Which of the five steps in the Third-Party Risk Management cycle involves choosing a third-party service provider after evaluating the risk appetite of the firm?

- A. Evaluation, risk rating, due diligence
- B. Business model decision
- C. Contracts and contract management
- D. Continuous monitoring

The correct answer is **B**.

The 'Business model decision' step in the Third-Party Risk Management cycle is primarily concerned with the selection of a third-party service provider. This step involves making strategic decisions about whether to outsource certain activities or keep them in-house. The choice of a provider's quality and price are important considerations in this step. These decisions are closely related to the risk appetite of the firm, as the firm must balance the potential benefits of outsourcing with the potential risks. The risk appetite of the firm can influence the level of risk it is willing to accept in its business model, and therefore, in its choice of third-party service providers.

Choice A is incorrect. While evaluation, risk rating, and due diligence are important steps in the third-party risk management cycle, they primarily involve assessing the potential risks associated with a third-party service provider rather than selecting one based on the firm's risk appetite. These steps help to identify and quantify the risks but do not directly involve making a decision about whether or not to engage with a particular service provider.

Choice C is incorrect. Contracts and contract management are concerned with formalizing the relationship between the firm and its chosen third-party service provider once that selection has been made. This step involves setting out terms of engagement, responsibilities, performance metrics etc., but it does not directly deal with selecting a service provider based on risk appetite.

Choice D is incorrect. Continuous monitoring refers to ongoing oversight of the relationship with a third-party service provider after it has been established. It involves tracking performance against agreed-upon metrics and managing any emerging risks or issues that arise during execution of services by third party providers but does not involve selection of these providers.

Things to Remember

- Third-Party Risk Management (TPRM) involves identifying, assessing, and controlling risks associated with outsourcing activities to third-party service providers.
- Risk appetite refers to the level of risk that an organization is willing to accept in pursuit of its objectives. It is influenced by various factors, including the organization's strategic goals, regulatory environment, and tolerance for risk.

- Due diligence is the process of investigating and evaluating a potential business decision or partner before entering into a contract or agreement. It helps in assessing the risks and benefits associated with the decision.
 - Contract management involves the administration of contracts to ensure that both parties fulfill their obligations and that the relationship is managed effectively. It includes monitoring performance, resolving disputes, and ensuring compliance with terms and conditions.
 - Continuous monitoring is an essential part of TPRM to ensure that the risks associated with third-party relationships are continuously assessed and managed over time. It helps in identifying and addressing any emerging risks or issues promptly.
-

Q.5131 Which of the following is a good risk management practice of the remediation or termination step of the Third-Party Risk Management life cycle?

- A. Defining trigger events for reassessment
- B. Establish limits on the outsourcing by third parties
- C. Having an exit strategy
- D. Use of standard assessment questionnaires

The correct answer is **C**.

Having an exit strategy is a good risk management practice under the remediation and termination step of the Third-Party Risk Management life cycle. This step involves taking necessary actions to address identified risks and, if necessary, terminating the relationship with the third party. An exit strategy or termination clause provides a clear plan for ending the relationship with the third party if the situation deteriorates beyond repair. This could be due to a variety of reasons such as the third party's failure to meet contractual obligations, significant changes in the third party's business condition, or the identification of unacceptable risks during the continuous monitoring process. Having an exit strategy ensures that the organization is prepared for such scenarios and can minimize potential losses and disruptions to its operations.

Choice A is incorrect. Defining trigger events for reassessment is an important part of the risk management process, but it is not specifically associated with the remediation or termination step. This practice typically falls under the monitoring and review stage of the Third-Party Risk Management life cycle, where ongoing assessment of third-party performance and risk exposure takes place.

Choice B is incorrect. Establishing limits on outsourcing by third parties can be a good risk management practice, but it does not directly relate to remediation or termination step. This action usually pertains to the initial stages of third-party relationship establishment where organizations define their outsourcing policies and set boundaries for third-party involvement.

Choice D is incorrect. The use of standard assessment questionnaires can be a useful tool in evaluating potential risks associated with third parties; however, this practice generally applies to earlier stages in the Third-Party Risk Management life cycle such as selection and due diligence phase rather than remediation or termination step.

Things to Remember

- Third-Party Risk Management involves identifying, assessing, monitoring, and managing risks associated with outsourcing to third parties.
 - Exit strategies are crucial in risk management to mitigate potential losses and disruptions in case of deteriorating relationships with third parties.
 - Continuous monitoring is essential in the Third-Party Risk Management life cycle to ensure ongoing assessment of third-party performance and risk exposure.
 - Standard assessment questionnaires are commonly used in the selection and due diligence phase of Third-Party Risk Management to evaluate potential risks.
-

Q.5132 Which of the following is **not** an example of an event that can trigger the reassessment of contracts with third parties?

- A. Data breaches
- B. A merger or acquisition
- C. Regulatory change
- D. Risk appetite

The correct answer is **D**.

Risk appetite is not an event that can trigger the reassessment of contracts with third parties. Risk appetite refers to the level of risk that an organization is willing to accept in pursuit of its objectives, before action is deemed necessary to reduce the risk. It is a strategic decision made by the organization and is not an event or incident that would necessitate a reassessment of contracts. A company's risk appetite will influence its decision-making process, including whether to outsource certain activities or keep them in-house. However, it does not directly trigger a reassessment of existing contracts with third parties.

Choice A is incorrect. Data breaches are a significant event that can necessitate the reassessment of contracts with third parties. If a data breach occurs, it may indicate that the third party's security measures are inadequate, which could lead to potential legal and financial risks for the business.

Choice B is incorrect. A merger or acquisition is another event that typically triggers a reassessment of contracts with third parties. This is because such events often result in changes to business operations and strategies, which may affect existing contractual relationships.

Choice C is incorrect. Regulatory changes can also trigger a reassessment of contracts with third parties. Changes in laws or regulations can impact the terms and conditions of existing contracts, making it necessary for businesses to review these agreements to ensure compliance.

Things to Remember

- Contractual Risk Management
- Third-party Risk Management
- Risk Appetite Framework
- Outsourcing Risk Management
- Legal and Regulatory Compliance

Q.5133 Which of the five steps in the Third-Party Risk Management cycle requires sound due diligence and verification of third-party service providers?

- A. Remediation or termination
- B. Continuous monitoring
- C. Evaluation and risk rating
- D. Business model decision

The correct answer is **C**.

This is the second stage of the Third-Party Risk Management cycle. It involves a thorough due diligence process and verification of third-party service providers. The aim of this step is to assess the potential risks associated with engaging a third-party service provider. The level of due diligence required may vary depending on the nature of the third-party service provider. For instance, a third-party service provider that will have access to sensitive information may require more extensive due diligence compared to one that will not. This approach, known as proportionality of approach, is considered a good risk management practice.

Choice A is incorrect. Remediation or termination is a step in the Third-Party Risk Management cycle that occurs after a risk has been identified and evaluated. It involves taking corrective action to mitigate the risk or terminating the relationship with the third-party service provider if necessary. This step does not involve due diligence process and verification of third-party service providers.

Choice B is incorrect. Continuous monitoring refers to ongoing oversight of third-party relationships to ensure they continue to meet contractual obligations and compliance requirements, as well as manage any changes in their risk profile over time. While it's an important part of managing third-party risks, it doesn't necessitate a thorough due diligence process and verification which are typically conducted before engaging with a third party.

Choice D is incorrect. The business model decision stage involves deciding whether to outsource certain functions or processes based on strategic considerations such as cost, efficiency, expertise, etc., rather than conducting due diligence on potential service providers.

Things to Remember

- Third-Party Risk Management (TPRM) involves identifying, assessing, and controlling risks associated with outsourcing to third-party service providers.
- Due diligence is the process of investigating and verifying the information provided by a third-party service provider to assess their reliability, reputation, and ability to meet contractual obligations.
- Proportionality of approach in risk management refers to tailoring the level of due

diligence and risk assessment to the specific characteristics and risks associated with each third-party service provider.

- Continuous monitoring is essential in TPRM to ensure that third-party relationships remain compliant with regulations and contractual agreements and to promptly address any emerging risks.
 - Remediation or termination is a critical step in TPRM that involves taking corrective actions to address identified risks or terminating the relationship with a third-party service provider if the risks are too high.
 - Business model decision involves strategic considerations such as cost, efficiency, expertise, and competitive advantage when deciding whether to outsource certain functions or processes.
-

Q.5134 Which of the following is **not** a common third-party risk?

- A. Service disruption
- B. Third parties
- C. Accidental data privacy breach
- D. Compliance breaches

The correct answer is **B**.

Third parties are not necessarily a common third-party risk. It is the use of third parties that increases a firm's exposure to third-party risks. Third parties are providers of goods and services that are not internal to the firm.

Common third-party risks include service disruption, failings in service quality, fraud, accidental data privacy breach or intentional information leak, compliance breaches, espionage and IP theft, and reputational damage.

Things to Remember

- Third-party risk management involves identifying, assessing, and mitigating risks associated with outsourcing to third parties.
 - Service disruption can occur when a third-party provider fails to deliver services as agreed, leading to operational disruptions for the firm.
 - Accidental data privacy breaches can result from mishandling of sensitive information by third-party vendors, potentially leading to regulatory fines and reputational damage.
 - Compliance breaches by third parties can expose the firm to legal and regulatory risks, especially if the third party fails to adhere to industry standards or regulations.
-

Q.5135 Which of the following is a good practice when addressing fourth-party risk?

- A. Establish standards on outsourcing
- B. Define trigger events for reassessment
- C. Have an exit strategy
- D. Having a termination clause

The correct answer is **A**.

Establishing standards on outsourcing is a good practice when addressing fourth-party risk. This is because it allows the firm to set certain standards or limits on the outsourcing activities carried out by third-party vendors. These standards can be a replication of the rules that the firm applies to its own vendors, thereby ensuring that the vendors also apply them to their own vendors and contractors. This practice is beneficial as it helps in maintaining control over the outsourcing activities and mitigating any potential risks associated with them.

Choice B is incorrect. While defining trigger events for reassessment can be a good practice in managing third-party risk, it does not directly address the issue of fourth-party risk. Fourth-party risk arises when a third party outsources some of its tasks or services to another entity (the fourth party). Therefore, establishing standards on outsourcing would be more beneficial in managing this specific type of risk.

Choice C is incorrect. Having an exit strategy is indeed important in any contract management process, but it does not specifically target the management of fourth-party risks. An exit strategy would come into play when there are significant issues with the third party that cannot be resolved, and does not necessarily prevent or manage risks associated with the fourth parties they may engage.

Choice D is incorrect. Similar to choice C, having a termination clause can be beneficial in general contract management but it doesn't directly help manage fourth-party risks. A termination clause allows for ending the contract under certain conditions but doesn't set standards or limits on outsourcing which is crucial for managing fourth-party risks.

Things to Remember

- Third-party risk management: Involves assessing and managing risks associated with third-party vendors or service providers.
- Fourth-party risk: Arises when a third party outsources some of its tasks or services to another entity, introducing additional layers of risk.
- Outsourcing standards: Setting standards on outsourcing activities helps in controlling risks associated with third and fourth parties.
- Contract management: Involves overseeing contracts, including aspects like exit strategies and termination clauses.

Q.5136 Which of the following is **not** an action that should be undertaken during the wind-down process of contracts?

- A. The transmission of intellectual property
- B. Plan to transition to in-house services
- C. Have audit rights on their vendors
- D. Provide evidence of data transfer or destruction

The correct answer is **C**.

Having audit rights on vendors is not typically part of the wind-down process. Audit rights are a necessity for firms whose third parties also outsource services to other parties. It is necessary for the firms to verify by themselves the application of rules the vendors use. However, this is not a step that is typically undertaken during the wind-down process. Instead, audit rights are usually exercised during the course of the relationship with the vendor, to ensure compliance with contractual obligations and regulatory requirements.

Choice A is incorrect. The transmission of intellectual property is typically part of the wind-down process. This ensures that any proprietary information or technology that was shared with the vendor is returned to the company, protecting its intellectual assets.

Choice B is incorrect. Planning to transition to in-house services is also a common step in the wind-down process. This involves preparing internal resources and personnel to take over the tasks previously handled by the third-party vendor, ensuring business continuity.

Choice D is incorrect. Providing evidence of data transfer or destruction is another crucial step in winding down contracts with third-party vendors. This helps confirm that all sensitive data has been appropriately handled, either transferred back to the company or securely destroyed.

Things to Remember

- Wind-down process of contracts involves various steps to smoothly transition services back in-house or to another vendor.
- Intellectual property transmission is crucial to protect proprietary information during the wind-down process.
- Planning for transition to in-house services ensures business continuity and a seamless handover of responsibilities.
- Providing evidence of data transfer or destruction is essential to safeguard sensitive information.
- Audit rights on vendors are typically exercised during the course of the relationship, not specifically during the wind-down process.

Q.5137 Which of the following steps of the Third-Party Risk Management life cycle involves keeping track of service provision, quality SLAs, and compliance with regulation?

- A. Remediation or termination
- B. Business model decision
- C. Evaluation and risk rating
- D. Continuous monitoring

The correct answer is **D**.

Continuous monitoring is the step in the Third-Party Risk Management life cycle that involves keeping track of service provision, quality SLAs, and compliance with regulations. This step is crucial as it ensures that the third-party service provider is meeting the agreed-upon standards and adhering to the regulations. Continuous monitoring also involves setting trigger events for reassessment, not just at the end-of-contract. This means that the performance of the third-party service provider is constantly being evaluated and any deviations from the agreed-upon standards or regulations are quickly identified and addressed. This continuous monitoring and evaluation help in mitigating risks and ensuring that the third-party service provider is delivering as per the expectations.

Choice A is incorrect. Remediation or termination is a step in the third-party risk management process, but it typically comes into play when there are significant issues with the service provision or compliance breaches. It does not primarily focus on continuous monitoring of SLAs and regulatory compliance.

Choice B is incorrect. The business model decision step involves deciding whether to outsource a particular function or keep it in-house based on various factors such as cost, expertise, and strategic importance. This step does not involve continuous monitoring of service provision and compliance with regulations.

Choice C is incorrect. Evaluation and risk rating involves assessing the potential risks associated with outsourcing a particular function to a third party. While this step may include setting trigger events for reassessment, its primary focus is not on ongoing monitoring of service provision and regulatory compliance.

Things to Remember

- Third-Party Risk Management: The process of evaluating and managing the risks associated with outsourcing certain functions to third-party service providers.
- Service Level Agreements (SLAs): Agreements that define the level of service expected from a service provider, including quality, performance, and responsibilities.

- Regulatory Compliance: Ensuring that the third-party service provider complies with relevant laws, regulations, and industry standards.
 - Risk Mitigation: Strategies and actions taken to reduce the impact of potential risks on an organization.
 - Trigger Events: Specific events or indicators that signal the need for reassessment or action in the risk management process.
-

Reading 116: Case Study: Investor Protection and Compliance Risks in Investment Activities

Q.5139 Which of the following is not one of the activities addressed by the Markets in Financial Instruments Regulation (MIFIR)?

- A. Investment advisory definition and objectivity
- B. Best deal execution for the clients
- C. Protections for whistleblowers
- D. Transactions with qualified counterparties

The correct answer is C.

Protections for whistleblowers is not one of the activities addressed by the Markets in Financial Instruments Regulation (MIFIR). This is because the protection of whistleblowers is typically covered under different legislation. In the United States, for example, the Dodd-Frank Wall Street Reform and Consumer Protection Act provides significant protections for whistleblowers. These protections include confidentiality, protection from retaliation, and potential financial rewards for providing information that leads to successful enforcement action. Therefore, while whistleblower protection is a critical aspect of financial regulation, it is not a focus of MIFIR.

Choice A is incorrect. The Markets in Financial Instruments Regulation (MIFIR) does cover the definition and objectivity of investment advisory. It provides guidelines on how investment advice should be given, ensuring that it is objective and in the best interest of the client.

Choice B is incorrect. MIFIR also addresses best deal execution for clients. It mandates that financial institutions must take all necessary steps to obtain the best possible result for their clients when executing orders.

Choice D is incorrect. Transactions with qualified counterparties are indeed covered by MIFIR. The regulation outlines specific rules for these transactions to ensure transparency and fairness in financial markets.

Things to Remember

- Markets in Financial Instruments Regulation (MIFIR) is a European Union regulation that aims to increase transparency and investor protection in financial markets.
- MIFIR covers a wide range of activities related to trading, reporting, and transparency in financial instruments.
- Best execution is a key concept in MIFIR, requiring firms to take all sufficient steps to obtain the best possible result for their clients when executing orders.

- Qualified counterparties are entities such as banks, investment firms, and insurance companies that are considered to have the knowledge and experience to understand the risks involved in financial transactions.
 - Whistleblower protections are important in financial regulation to encourage individuals to report misconduct or violations of regulations without fear of retaliation.
-

Q.5144 Which of the following is an effect of the Volcker rule?

- A. Enhanced witness protection
- B. Buying and selling of securities through clearing houses
- C. Stopping commercial banks from speculation and proprietary trading
- D. Establish the Consumer Financial Protection Bureau

The correct answer is **C**.

The Volcker Rule was designed to prevent commercial banks from engaging in speculative activities and proprietary trading for profit. This rule was a response to the 2008 financial crisis, during which risky trading activities by banks contributed to the collapse of financial markets. By prohibiting these activities, the Volcker Rule aims to protect depositors' funds and maintain the stability of the financial system. The rule specifically limits banks' investments in hedge funds and private equity funds, which are often associated with high levels of risk. Therefore, the primary effect of the Volcker Rule is to stop commercial banks from speculation and proprietary trading.

Choice A is incorrect. Enhanced witness protection is not a direct effect of the Volcker Rule. The Volcker Rule primarily focuses on limiting risky financial activities by banks, such as proprietary trading and certain investments in hedge funds and private equity funds, rather than enhancing witness protection.

Choice B is incorrect. While the Dodd-Frank Act does include provisions related to clearing houses, these are not directly related to the Volcker Rule. The rule's main purpose is to limit speculative trading activities by commercial banks, not necessarily promoting buying and selling of securities through clearing houses.

Choice D is incorrect. The establishment of the Consumer Financial Protection Bureau (CFPB) was indeed a part of the Dodd-Frank Wall Street Reform and Consumer Protection Act but it was not a direct result of implementing the Volcker Rule. The CFPB was created to protect consumers from deceptive financial practices whereas the Volcker rule specifically targets risk-taking activities by banks.

Things to Remember

- The Volcker Rule is a regulation that prohibits banks from engaging in certain types of speculative trading activities, such as proprietary trading and investments in hedge funds and private equity funds.
 - It was implemented as part of the Dodd-Frank Wall Street Reform and Consumer Protection Act in response to the 2008 financial crisis.
 - The rule is named after Paul Volcker, a former chairman of the Federal Reserve who advocated for restrictions on risky trading activities by banks.
 - The primary goal of the Volcker Rule is to protect depositors' funds and prevent banks from taking excessive risks that could lead to financial instability.
 - Compliance with the Volcker Rule requires banks to establish robust compliance programs and reporting mechanisms to ensure they are not engaging in prohibited activities.
-

Q.5146 Which of the following is the key issue addressed by the creation of the Markets in Financial Instruments Directive II (MIFID II)?

- A. The requirements for regulatory reporting and transaction transparency
- B. The disclosure of transaction data to supervisors and regulators
- C. The asymmetry in information between buyers and sellers
- D. The oversight and supervision of workers and trades

The correct answer is **A**.

MiFID II, the Markets in Financial Instruments Directive II, is a regulatory framework created by the European Union to improve transparency and investor protection in financial markets. The main focus of MiFID II is on **regulatory reporting and transaction transparency**. This means that financial firms must report transaction details to regulators and disclose trade information to the public, ensuring that all market participants have a clearer view of market activities.

This transparency requirement goes beyond simply sharing information with regulators. By making transaction data available, MiFID II aims to reduce unfair practices and create a more open market where prices and trades are visible to all.

While MiFID II also addresses **information asymmetry**—helping ensure buyers and sellers have similar information—it achieves this as part of its larger goal to increase transparency. Therefore, the directive's key issue is primarily about regulatory reporting and transaction transparency, making **Option A** the correct answer.

B is incorrect because MiFID II's main goal extends beyond just the disclosure of data to supervisors and regulators. While data disclosure is part of MiFID II, the directive's broader focus is on creating transparency and establishing detailed reporting requirements across the market, not solely on internal disclosures.

C is incorrect because although MiFID II helps reduce information asymmetry between buyers and sellers, this is a secondary effect. The primary focus is on enforcing transparency and comprehensive reporting standards for fairer market practices overall, not specifically targeting information asymmetry.

D is incorrect because MiFID II does not primarily focus on the direct supervision or oversight of individual workers. Its main aim is to ensure transparent and well-regulated financial transactions in the market, rather than oversight of personnel within firms.

Things to Remember

- Markets in Financial Instruments Directive II (MIFID II) is a regulatory framework in the European Union that aims to increase transparency and investor protection in financial markets.
- MIFID II requires the disclosure of transaction data to supervisors and regulators to ensure market integrity and protect investors.
- Regulatory reporting and transaction transparency are important components of MIFID II, but the primary focus is on disclosure of transaction data to regulators.
- Information asymmetry between buyers and sellers is a common issue in financial markets, but MIFID II's main goal is to address transparency and investor protection.
- Oversight and supervision of workers and trades are important for market integrity, but they were not the key issues addressed by the creation of MIFID II.

Q.5147 Which of the following is a protection provided to investors through the Investor Protection Act – Dodd-Frank?

- A. Employee education
- B. Best deal execution for the clients

C. Establishing the Volcker Rule

D. Fair and non-misleading communication with customers

The correct answer is **C**.

The Investor Protection Act – Dodd-Frank established the Volcker Rule, which is named after former Federal Reserve Chairman Paul Volcker. The rule seeks to stop commercial banks from engaging in profit-driven speculation and proprietary trading, as well as limiting banks' investments in hedge funds and private equity funds. The Volcker Rule is designed to reduce the risks that banks pose to the financial system and to protect depositors. By prohibiting banks from engaging in certain types of risky activities, the rule aims to prevent the kind of financial instability that led to the 2008 financial crisis. The Volcker Rule is a key component of the Dodd-Frank Act's efforts to reform Wall Street and protect investors.

Choice A is incorrect. Employee education is not a protection provided to investors through the Investor Protection Act – Dodd-Frank. While employee education can be beneficial for improving the overall quality of service provided by financial institutions, it does not directly protect investors or their investments.

Choice B is incorrect. Best deal execution for clients, while an important aspect of fiduciary duty in finance, is not specifically addressed in the Investor Protection Act – Dodd-Frank. This act primarily focuses on regulatory reforms and consumer protections rather than specific operational practices within financial institutions.

Choice D is incorrect. Fair and non-misleading communication with customers is a general principle that all businesses should adhere to, but it's not a specific provision of the Investor Protection Act – Dodd-Frank. The act does include provisions aimed at increasing transparency and accountability in financial markets, but these are broader than simply ensuring fair communication with customers.

Things to Remember

- The Investor Protection Act – Dodd-Frank was enacted in response to the 2008 financial crisis to increase transparency, accountability, and consumer protections in the financial industry.
- The Volcker Rule, established under the Dodd-Frank Act, aims to prevent banks from engaging in risky speculative activities that could endanger depositors and the financial system.
- Investor protection laws and regulations are crucial for maintaining market integrity and investor confidence in the financial markets.
- Regulatory reforms like the Dodd-Frank Act play a key role in safeguarding investors

against fraud, misconduct, and excessive risk-taking by financial institutions.

Reading 117: Supervisory Guidance on Model Risk Management

Q.4297 Which of the following is a primary way in which models can pose a significant risk to financial service firms?

- A. Models are costly
- B. Models can give inaccurate results
- C. Models are not time-sensitive
- D. Models take too long to be implemented

The correct answer is **B**.

Models that produce inaccurate results may lead to unexpected losses. The two primary ways in which models can pose a significant risk to financial services firms:

- Models can be manipulated, misunderstood, or misused; this leads to unexpected losses to the firm.
- Models can give inaccurate results, which leads to unexpected losses to the firm.

A is incorrect: Model's cost does not pose a significant risk to a financial institution.

C is incorrect: Model's time consumption does not pose any significant risk.

D is incorrect: Time of implementation does not pose a significant risk to a firm.

Things to Remember

- **Model Risk:** Model risk is the risk of financial loss resulting from using or relying on inaccurate, inappropriate, or misused models.
- **Model Validation:** Model validation is the process of ensuring that a model is performing as intended, accurately, and in compliance with regulatory requirements.
- **Model Governance:** Model governance refers to the framework, policies, and procedures put in place to oversee and manage the development, implementation, and usage of models within a financial institution.
- **Regulatory Requirements:** Financial regulators often require financial institutions to have robust model risk management frameworks in place to mitigate the risks associated with using models.

Q.4298 A risk management team at a bank is implementing a comprehensive validation framework for their newly developed credit risk models in preparation for regulatory review. According to best practices in model validation, which of the following validation processes would be MOST appropriate for determining whether the mathematical framework of the model is conceptually well-founded?

- A. Qualitative review of model documentation, derivation of formulae, and comparison with established theoretical frameworks
- B. Data quality checks focusing on the completeness, accuracy, and appropriateness of input data
- C. Systems implementation testing including user acceptance testing and code verification
- D. Management oversight involving senior leadership in reviewing model outputs and their application in business decisions

The correct answer is **A**.

Determining whether a credit risk model is conceptually sound requires a qualitative validation approach. Specifically, the mathematical framework needs to be reviewed for its theoretical rigor, accuracy, clarity in the derivation of formulas, and alignment with established theoretical and empirical research. This process involves ensuring that the model assumptions and methodologies have strong grounding in widely accepted financial theory and best industry practices.

Why other options are incorrect:

B (Data Cleaning) assesses data integrity but does not directly address the conceptual foundation of the mathematical framework.

C (Back-testing or numerical validation) assesses the model's accuracy against historical data, not its conceptual foundations.

D (Independent quantitative replication) verifies the model's calculations numerically rather than its theoretical basis.

Q.4299 Which of the following gives a reason why a firm should invest in model risk management?

- A. To give incentives to model developers to work faster
- B. To cater for losses brought about by a model
- C. To provide incentives to management

D. To ensure that the model is used as required

The correct answer is **D**.

Ensuring that the model is used as required is a fundamental reason for a firm to invest in model risk management. A strong model risk management framework relies on significant investment in supporting systems to guarantee data and reporting integrity and testing to ensure proper implementation of models, effective systems integration, and appropriate use. This investment is crucial to ensure that the models are used correctly and effectively, thereby reducing the risk of financial losses due to model errors or misuse.

Choice A is incorrect. While it may be beneficial for model developers to work faster, this is not a primary reason for investing in model risk management. The main purpose of model risk management is to ensure the integrity and effectiveness of financial models, not to speed up their development.

Choice B is incorrect. Although mitigating potential losses caused by a flawed or misused model can be an outcome of effective model risk management, it's not the primary reason for such investment. The main goal is to ensure that the models are used as intended and provide accurate results.

Choice C is incorrect. Providing incentives to management does not directly relate to the purpose of investing in model risk management. While good governance and oversight are important aspects of managing risks associated with financial models, they do not constitute a primary reason for such investments.

Things to Remember

- Model Risk Management (MRM) is the process of identifying, assessing, and managing the risks associated with the use of financial models within a firm.
- Model risk can arise from errors in the model design, data inputs, assumptions, implementation, or usage.
- Effective model risk management helps firms comply with regulatory requirements, improve decision-making processes, and enhance overall risk management practices.
- Key components of model risk management include model validation, ongoing monitoring, governance, and documentation.
- Model risk management is particularly important in industries such as banking, insurance, and asset management where financial models are extensively used for pricing, risk assessment, and strategic decision-making.

Q.4303 TCC bank has developed a set of models to analyze liquidity risk, market risk, as well as the credit risk of borrowers. Which of the following model risk management functions is *least likely* to be handled by the developers of these models?

- A. Coming up with a clear statement of purpose to ensure that model development is aligned with the intended use
- B. Rigorous assessment of data quality and relevance
- C. model testing
- D. Model validation

The correct answer is **D**.

The process of validation is crucial and it requires independence from model development and usage. It is generally advised that validation should be performed by individuals who are not involved in the development or usage of the model and do not have a vested interest in its validity. Independence is not a goal in itself, but it ensures that incentives are aligned with the objectives of model validation. Therefore, it is least likely to be handled by the developers of the models.

Choice A is incorrect. The individuals who develop the models are indeed responsible for coming up with a clear statement of purpose to ensure that model development is aligned with the intended use. This is because they have a deep understanding of the model's design and its intended application, making them best suited to define its purpose.

Choice B is incorrect. Rigorous assessment of data quality and relevance is also likely to be performed by those who developed the models. They need to ensure that the data used in their models are accurate, reliable, and relevant for their intended use.

Choice C is incorrect. Model testing involves checking if a model works as expected under different scenarios and conditions. This task would typically be performed by those who developed it since they understand how it should function in various situations.

Things to Remember

- Model Risk Management involves the identification, assessment, and mitigation of potential risks associated with the use of models in financial institutions.
- Model risk can arise from errors in model design, data quality issues, incorrect model assumptions, or inadequate model testing.
- Model validation is a critical component of model risk management, ensuring that models are fit for their intended purpose and comply with regulatory requirements.

- Model validation should be independent from model development and usage to maintain objectivity and ensure unbiased assessments.
 - Model developers are responsible for defining the model's purpose, assessing data quality, and conducting model testing, while validation is typically performed by a separate independent team.
-

Q.4304 Model development and implementation in risk management requires various best practices to ensure that the models are aligned to their intended use. Which of the following alternatives about these best practices is MOST ACCURATE?

- A. The merits and limitations of the model methodologies and processing components should be well explained.
- B. Developers should compare their models with alternative approaches and theories.
- C. The quality of data used to develop a model should be assessed and documented.
- D. All of the above

The correct answer is **D**.

All of the above-mentioned practices are indeed considered best practices in model development and implementation in risk management. Each of these practices contributes to the overall effectiveness and accuracy of the model. Choice A emphasizes the importance of understanding the strengths and weaknesses of the model methodologies and processing components. This understanding allows for better decision-making and risk assessment. Choice B highlights the importance of comparing the developed model with alternative approaches and theories. This comparison can provide insights into potential improvements or modifications that can enhance the model's performance. Choice C underscores the importance of assessing and documenting the quality of data used in the model development. High-quality data is crucial for the accuracy and reliability of the model. Therefore, all these practices are essential and contribute to the development of a robust and reliable risk management model.

Choice A is incorrect. While it is true that the merits and limitations of the model methodologies and processing components should be well explained, this alone does not encompass all the best practices in model development and implementation. Other aspects such as comparing models with alternative approaches and assessing the quality of data used are also crucial.

Choice B is incorrect. Comparing models with alternative approaches and theories is indeed a best practice in risk management modeling. However, this statement does not cover all other important practices such as explaining the merits and limitations of the model methodologies or assessing data quality.

Choice C is incorrect. Assessing and documenting the quality of data used to develop a model is an essential practice but it doesn't represent all best practices in risk management modeling. Other key aspects like explaining model methodologies' pros & cons or comparing models with alternatives are also necessary.

Things to Remember

- **Model Risk Management:** It is important to have a robust framework in place to identify, assess, and mitigate model risk throughout the model development and implementation process.
 - **Model Validation:** Independent validation of models is crucial to ensure their accuracy, reliability, and alignment with the intended use.
 - **Regulatory Compliance:** Models used in risk management must comply with regulatory requirements and guidelines to ensure transparency and accountability.
 - **Sensitivity Analysis:** Conducting sensitivity analysis helps in understanding the impact of changes in input variables on the model outputs, enhancing the model's robustness.
 - **Model Documentation:** Comprehensive documentation of models, including assumptions, methodologies, and data sources, is essential for transparency, auditability, and reproducibility.
-

Q.4305 Which of the following gives a common error in model use and management across all industries?

- A. Spending more than anticipated
- B. Sample bias in model data
- C. Users failing to keep documentation
- D. Model invalidation

The correct answer is **B**.

Sample bias in model data is a common error in model use and management across all industries. This error occurs when a nonrepresentative set of data is used during the development of a model. The use of such data can lead to incorrect model outcomes. This is because the model is trained on a biased sample, which does not accurately represent the

population. As a result, the model's predictions or classifications may also be biased and inaccurate. This can have significant implications, particularly in industries where models are used to make critical decisions or predictions. Therefore, it is crucial to ensure that the data used to train a model is representative of the population to avoid sample bias.

Choice A is incorrect. While overspending can be a concern in any business scenario, it does not specifically pertain to the use and management of models across various industries. Overspending is more related to budgeting and financial management rather than model use and management.

Choice C is incorrect. Although maintaining documentation is important for model use and management, users failing to keep documentation isn't a common mistake that's often encountered in this context. Documentation issues are more related to organizational practices rather than inherent issues with model use or management.

Choice D is incorrect. Model invalidation isn't a common mistake encountered in the context of model use and management across various industries. Invalidation of a model usually occurs when the assumptions or data upon which it was built change significantly, making the model no longer applicable or accurate.

Things to Remember

- **Model Risk Management:** It is important to have a robust model risk management framework in place to identify, assess, and mitigate risks associated with model use and management.
- **Data Quality:** Ensuring data quality is crucial for developing accurate and reliable models. Poor data quality can lead to biased models and incorrect outcomes.
- **Model Validation:** Regular validation of models is essential to ensure that they are still accurate and relevant. This process involves comparing model outputs with actual outcomes and assessing their performance.
- **Model Governance:** Establishing clear governance structures and processes for model development, implementation, and monitoring is key to effective model use and management.
- **Overfitting:** Overfitting occurs when a model is overly complex and fits the training data too closely, leading to poor generalization to new data. It is important to guard against overfitting when developing models.

Q.4306 Which of the given options identifies one challenge faced by model risk managers while designing and delivering effective model risk reporting?

- A. Lack of funds to fully implement the model
- B. Implementing the required infrastructure to deliver reporting
- C. Implementing the model
- D. Lack of personnel to test the model

The correct answer is **B**.

Implementing the required infrastructure to deliver reporting is indeed a significant challenge faced by model risk managers. Model risk reporting is a critical component of an organization's risk management framework. It provides insights into the model risk limit that the organization can tolerate. However, to deliver effective reporting, there is a need for appropriate infrastructure. This includes workflow tools and databases that enable the organization, management, and updating of data. Without the right infrastructure, it becomes difficult to measure the impact of models, define metrics linked to model risk appetite, determine an appropriate frequency of reporting, and aggregate reporting on individual models to provide a comprehensive and consistent view of model risk at a defined level of aggregation. Therefore, implementing the required infrastructure to deliver reporting is a significant challenge in model risk management.

Choice A is incorrect. While lack of funds can be a challenge in implementing the model, it does not directly relate to the challenges faced by model risk managers in designing and delivering effective model risk reporting. The question specifically asks about challenges related to reporting, not implementation.

Choice C is incorrect. Implementing the model itself is a part of the overall process but it doesn't represent a specific challenge that model risk managers might face while designing and delivering effective model risk reporting. The focus here is on issues related to reporting, not implementation.

Choice D is incorrect. Lack of personnel to test the model could be an issue in overall risk management but it doesn't specifically address the challenges faced by model risk managers in terms of designing and delivering effective reports on model risks.

Things to Remember

- **Model Risk Management:** Involves the identification, assessment, and mitigation of risks associated with the use of models in decision-making processes.
- **Model Risk Reporting:** Refers to the process of communicating information about model risk to stakeholders, including senior management, regulators, and the board of directors.

- **Model Risk Limit:** The maximum level of model risk that an organization is willing to accept or tolerate.
 - **Workflow Tools:** Software applications that help streamline and automate the flow of work within an organization, including data management and reporting processes.
 - **Data Infrastructure:** The underlying technology and systems that support the storage, management, and analysis of data required for model risk reporting.
-

Q.4443 The following are some of the supervisory guidance for model validation process, **EXCEPT:**

- A. Ensure that the model is used for decision making
- B. Ensure that documentation indicates where the internal model does not work effectively
- C. Model developers must also be involved in the model validation team
- D. None of the above.

The correct answer is **C**.

The statement that 'Model developers must also be involved in the model validation team' is incorrect. According to the supervisory guidance for model validation process, specifically SR 11-7, the model validation team should be independent. This means that the individuals who developed the model should not be involved in its validation. The reason for this is to ensure objectivity in the validation process. If the developers were involved in the validation, there could be a conflict of interest, as they may be biased towards their own work. Therefore, to maintain the integrity and reliability of the model, it is crucial that the validation team is independent of the development team.

Choice A is incorrect. The supervisory guidelines indeed emphasize the importance of using the model in decision making. This is to ensure that the model is not just a theoretical construct, but has practical applications and can guide business decisions effectively.

Choice B is incorrect. According to supervisory guidelines, it's crucial that documentation clearly indicates where the internal model does not work effectively. This helps in identifying potential areas of improvement and mitigating any risks associated with these limitations.

Choice D is incorrect. As explained above, both choices A and B accurately represent supervisory guidelines for model validation process, hence 'None of the above' cannot be correct.

Things to Remember

- **Model Risk Management:** Model validation is a key component of model risk management, which involves identifying, assessing, monitoring, and mitigating risks associated with the use of quantitative models in decision making.
 - **Model Governance:** Model validation is part of the broader framework of model governance, which includes policies, procedures, and controls to ensure the effective and responsible use of models within an organization.
 - **Regulatory Compliance:** Supervisory guidance, such as SR 11-7 from the Federal Reserve, provides a framework for financial institutions to comply with regulatory requirements related to model risk management and validation.
 - **Independence and Objectivity:** Independence of the model validation team is crucial to ensure objectivity in the validation process and mitigate potential conflicts of interest that may arise if model developers are involved in the validation.
-

Q.4444 Which of the following best describes a model?

- A. A spreadsheet that aggregates groups' trading positions for reporting
- B. A spreadsheet with what-if calculations for potential buyers
- C. A spreadsheet with coded probabilistic risk calculation that enables what-if scenarios to be run each day
- D. Both B and C

The correct answer is **C**.

A model is a tool that provides useful outputs to a firm given a set of inputs and can be reused day by day. A spreadsheet with coded probabilistic risk calculation that enables what-if scenarios to be run each day fits this definition perfectly. It is not just a simple spreadsheet, but a complex tool that can process inputs (data entered into the spreadsheet), apply a probabilistic risk calculation (the model's algorithm), and generate useful outputs (the results of the what-if scenarios). These outputs can then be used to make informed decisions about risk management. Furthermore, the fact that this tool can be used day after day, with different inputs, makes it a reusable model, which is a key characteristic of financial models.

Choice A is incorrect. While a spreadsheet that aggregates groups' trading positions for reporting can be a useful tool, it does not necessarily constitute a model in the context of

financial risk management. This is because it does not process inputs to generate outputs that can be used in various scenarios, which is an essential characteristic of a model.

Choice B is incorrect. A spreadsheet with what-if calculations for potential buyers may provide valuable insights, but it does not necessarily qualify as a model unless it also includes coded probabilistic risk calculations that enable what-if scenarios to be run each day.

Choice D is incorrect. As explained above, both options B and C do not fully encapsulate the definition of a model in financial risk management context. Only option C meets all the criteria required for something to be considered as a model.

Things to Remember

- Models in financial risk management are tools that provide useful outputs based on a set of inputs and can be reused day by day.
 - Models often involve complex algorithms, such as coded probabilistic risk calculations, to process inputs and generate outputs.
 - What-if scenarios are a common use of financial models, allowing users to test different scenarios and assess potential outcomes.
 - Models should be designed to be reusable, allowing for different inputs to be entered and analyzed on a regular basis.
 - Accuracy and reliability are crucial aspects of financial models, as they are used to make important risk management decisions.
-

Q.4446 Which of the following best describes the importance of an independent validation?

- A. It reduces the cost of the validation process
- B. It ensures that the bank doesn't spend much on incentives that model developers may require to validate the model
- C. It provides comfort to the CRO, and regulators
- D. All of the above

The correct answer is **C**.

Regulation required that banks should use independent validators. Besides being a requirement, it helps eradicate the risks as validators are experts; thus, the CRO is comfortable using an independent team of validators.

A is incorrect. Use of an independent validation does not reduce any cost that was meant for validation.

B is incorrect. Developers are not required to validate a model.

D is incorrect.

Things to Remember

- Independent validation is a crucial component of model risk management in financial institutions.
- It helps ensure that the models used by the institution are accurate, reliable, and in compliance with regulations.
- Independent validation provides an unbiased assessment of the model's performance and limitations.
- Regulators often require banks to have independent validation processes in place to mitigate model risk.
- The validation team should have the necessary expertise and independence to effectively evaluate the models.

Q.4447 The following are key components of the model development process, **EXCEPT**:

- A. Model lifecycle
- B. Data preparation
- C. Model audit
- D. Model assembly

The correct answer is **A**.

Model lifecycle is not a component of the model development process. Instead, the model development process is a part of the model lifecycle. The model lifecycle encompasses all stages of a model's existence, from its initial conception and development, through its deployment, use, and maintenance, to its eventual retirement. The model development process, on the other hand, is a specific stage within this lifecycle, focusing on the creation and testing of the model.

Choice B is incorrect. Data preparation is a crucial part of the model development process. It involves cleaning, transforming, and analyzing data to be used in the model. Without proper data preparation, the model may not accurately represent the situation it's intended to analyze.

Choice C is incorrect. Model audit is also an important component of the model development process. It ensures that the model has been developed correctly and functions as intended by checking for errors or inconsistencies in its design or implementation.

Choice D is incorrect. Model assembly refers to putting together all parts of a model including inputs, calculations and outputs which makes it an integral part of any modeling process.

Things to Remember

- Model validation: After the model development process, it is important to validate the model to ensure its accuracy and reliability.
- Model documentation: Proper documentation of the model development process is essential for transparency, reproducibility, and auditability.
- Model governance: Establishing clear governance policies and procedures around model development, validation, and implementation is crucial for risk management and compliance.
- Model monitoring: Continuous monitoring of the model's performance and outputs is necessary to detect any issues or drift over time.
- Model re-evaluation: Periodic re-evaluation of the model's assumptions, inputs, and outputs is important to ensure its continued relevance and effectiveness.

Q.4448 Which of the following should be found in the model documentation?

- A. Model validation team
- B. Data sources
- C. Individual model users
- D. None of the above

The correct answer is **B**.

Model documentation should include the data sources used in the model. This is crucial because it provides transparency about the origin of the data, which can affect the model's reliability and validity. The data sources section in the model documentation should detail the specific databases, files, or other sources from which the data was obtained. It should also include information about the data's quality and any issues that might affect its accuracy or reliability. Justification for using the chosen data should also be provided, explaining why this data is suitable for the model's purpose. This can include discussions about the data's relevance, timeliness, completeness, and other attributes that make it appropriate for the model. By including this information in the model documentation, users and reviewers of the model can better understand its workings and make informed decisions about its use.

Choice A is incorrect. While the model validation team plays a crucial role in verifying the accuracy and reliability of a financial model, they are not a component that needs to be included in the model documentation. The documentation should focus on the development, assumptions, and data inputs of the model.

Choice C is incorrect. Individual model users are not necessary to be included in the model documentation. The focus of such documentation should be on providing comprehensive information about how the model was developed, what assumptions were made during its creation, and what data sources were used.

Choice D is incorrect. As explained above, there are certain components like data sources that need to be included in a financial model's documentation.

Things to Remember

- Model validation team plays a crucial role in verifying the accuracy and reliability of a financial model.
- Model documentation should include data sources used in the model for transparency and reliability.
- Data quality and issues affecting accuracy should be detailed in the model documentation.

- Justification for using specific data should be provided in the documentation.
 - Model documentation focuses on development, assumptions, and data inputs of the model.
-

Q.4449 Which one of the following is a challenge faced by banks in the model validation process?

- A. Use of model users for validation
- B. Model developers requiring incentives to validate the model
- C. Use of vendor models
- D. All of the above

The correct answer is **C**.

The use of vendor models presents a significant challenge in the model validation process for banks. According to the SR 11-7 guidelines, all models, whether developed internally or purchased, should undergo the same rigorous validation process. However, vendors often lack transparency regarding their intellectual property, which can complicate the validation process. This lack of transparency may force banks to relax their validation standards and instead rely on methods such as benchmarking and outcome analysis. This approach, while necessary due to the circumstances, can potentially compromise the thoroughness and accuracy of the validation process.

Choice A is incorrect. The use of model users for validation does not necessarily present a challenge in the model validation process. Model users can provide valuable insights and feedback on the practicality and usability of the models, which can be beneficial for improving them.

Choice B is incorrect. While incentives might influence the behavior of model developers, it's not inherently a challenge in the model validation process. Incentives could potentially motivate developers to ensure their models are accurate and reliable, although they should ideally be motivated by professional integrity rather than financial gain.

Choice D is incorrect. As explained above, both options A and B do not inherently represent challenges that banks might encounter during the model validation process.

Things to Remember

- Model validation is a critical process for banks to ensure the accuracy and reliability of their risk models.
- Model validation involves assessing the conceptual soundness, data quality, model

methodology, and model performance.

- Regulatory guidelines, such as SR 11-7, provide a framework for banks to follow when validating their models.
 - Vendor models are often used by banks to supplement their internal models, but they present challenges due to lack of transparency and intellectual property issues.
 - Model users can provide valuable feedback on the practicality and usability of models during the validation process.
 - Incentives for model developers can influence their behavior, but professional integrity should be the primary motivator for accurate model validation.
-

Q.4450 Which of the following is an important element of the model risk management framework?

- A. Model lifecycle management
- B. Model risks
- C. Third-party models
- D. None of the above

The correct answer is **A**.

Model lifecycle management is indeed an essential element of the model risk management framework. It involves the comprehensive management of a model throughout its lifecycle, from its initial development to its eventual retirement. This process includes several stages such as model development, validation, deployment, monitoring, and decommissioning. Each stage requires careful management to ensure the model's effectiveness and reliability. For instance, during the development stage, the model's design and assumptions are thoroughly evaluated. Similarly, during the validation stage, the model's performance is rigorously tested against various scenarios to ensure its robustness. Therefore, model lifecycle management plays a crucial role in maintaining the quality and reliability of financial models, making it a vital component of the model risk management framework.

Choice B is incorrect. While model risks are indeed a significant concern in financial risk management, they are not a component of the model risk management framework itself. Instead, they are what the framework aims to identify, manage and mitigate.

Choice C is incorrect. Third-party models can be part of an organization's overall financial

modeling strategy, but they do not constitute a key element of the model risk management framework. The use of third-party models may introduce additional risks that need to be managed within this framework.

Choice D is incorrect. As explained above, both Model lifecycle management and Third-party models are related to the model risk management framework but neither constitutes its vital component.

Things to Remember

- Model risk management framework aims to identify, manage, and mitigate risks associated with financial models used in decision-making processes.
 - Model validation is a critical component of model risk management, ensuring that the model is accurate, reliable, and appropriate for its intended use.
 - Regulatory requirements often mandate the implementation of a robust model risk management framework to ensure the integrity of financial models.
 - Ongoing monitoring and review of models are essential to detect any potential model risks or issues that may arise during their lifecycle.
 - Model governance, including clear policies and procedures, is crucial for effective model risk management and compliance with regulatory standards.
-

Q.4451 The following are the essential components of a model, **EXCEPT**:

- A. A data input component
- B. A data processing component
- C. A data understanding component
- D. A reporting component

The correct answer is **C**.

A data understanding component is not considered an essential component of a model. While understanding the data is a crucial part of the model development process, it is not a component of the model itself. The model development process involves understanding the data, but once the model is developed, it does not require a data understanding component. Instead, the model requires data input, data processing, and reporting components to function effectively. The data input component is responsible for obtaining data from the user, the data processing component processes the data using statistical or numerical computations, and the reporting component provides the outcome or results after processing.

Choice A is incorrect. The data input component is an essential part of a model as it provides the raw data that will be processed and analyzed by the model. Without this component, the model would not have any information to work with.

Choice B is incorrect. The data processing component is also crucial in a model as it manipulates and processes the raw data into a form that can be easily understood and analyzed. This component helps in extracting meaningful insights from the raw data.

Choice D is incorrect. The reporting component of a model plays an important role in presenting the results derived from the processed data in an understandable manner to stakeholders or decision-makers. It aids in communicating findings effectively, hence it's considered essential.

Things to Remember

- Model Development Process
- Data Input Component
- Data Processing Component
- Reporting Component
- Model Output
- Model Validation

Q.4742 Which of the following parties is primarily responsible for ensuring a model has undergone appropriate validation and approval processes before being deployed?

- A. Internal audit function
- B. Model owner
- C. Senior management
- D. Independent validation team

The correct answer is **B**.

The model owner is accountable for the model's proper development, implementation, and use within the framework set by the bank's policies. This includes ensuring that the model undergoes appropriate validation and approval processes. The independent validation team conducts the validation but does not own the model. Senior management oversees the broader model risk management framework, while the internal audit function evaluates the effectiveness of the framework as a whole.

Q.4743 Which of the following is a primary element of a strong model validation framework?

- A. Good investment
- B. Ongoing monitoring
- C. Technology advancement
- D. Time efficiency

The correct answer is **B**.

Ongoing monitoring is a critical element of a reliable model validation framework. The main aim of this element is to confirm the appropriate implementation of the model, in addition to its usage and performance as intended.

Other key elements of comprehensive validation include:

- Evaluation of conceptual soundness: it entails the assessment of the quality of model design and its construction. There should always be documented evidence to provide support for all model choices.

- Outcomes analysis: This element highly relies on statistical tests and other quantitative measures. It involves a comparison of outcomes. The actual outcomes are compared with the model's outcomes.

A, C, D are incorrect: They are not elements of the validation process.

Things to Remember

- Model Risk: Model validation is crucial in managing model risk, which is the risk of financial loss resulting from reliance on inaccurate or misused models.
 - Regulatory Compliance: Model validation frameworks are often required by regulatory bodies to ensure the accuracy and reliability of models used in financial institutions.
 - Model Governance: Model validation is part of a broader model governance framework that includes model development, implementation, and monitoring.
 - Model Documentation: Proper documentation of models and validation processes is essential for transparency, auditability, and reproducibility.
-

Reading 118: Case Study: Model Risk and Model Validation

Q.5149 A newly hired risk manager analyzes the types of risk and also wishes to explain different ways that financial institutions can become exposed to model risk. Which of the following options **best** describes the ways that financial institutions can become exposed to model risk?

- A. By investing in low-risk assets.
- B. By avoiding complex financial instruments.
- C. By relying too heavily on a single model or failing to consider alternative models.
- D. By conducting regular stress tests and scenario analysis.

The correct answer is **C**.

Financial institutions can become exposed to model risk by relying too heavily on a single model or failing to consider alternative models. This can result in inaccurate or incomplete assessments of risk, which can have serious consequences for the institution.

A is incorrect. Investing in low-risk assets may help to minimize some types of risk, but it is not directly related to model risk.

B is incorrect. Avoiding complex financial instruments may help to reduce some types of risk, but it does not address the issue of model risk specifically.

D is incorrect. Conducting regular stress tests and scenario analysis is an important risk management practice, but it is not directly related to the ways that financial institutions can become exposed to model risk.

Things to Remember

- Model risk is the risk of financial loss resulting from errors in the development, implementation, or use of models and modeling techniques.
- Financial institutions can be exposed to model risk when they rely too heavily on a single model without considering alternative models or scenarios.
- Model risk can arise from various sources, including data quality issues, model limitations, incorrect assumptions, and inadequate validation processes.
- To mitigate model risk, financial institutions should have robust model governance frameworks, independent model validation processes, and regular model performance monitoring.

- Regulatory authorities often require financial institutions to have policies and procedures in place to manage and mitigate model risk.
-

Q.5150 A junior analyst at a bank wishes to understand more about the role of the model risk management function and best practices in model risk management. What is the role of model risk management (MRM) function in financial institutions, and how do they determine the frequency of model validation?

- A. MRM function validates models every year, regardless of their tier, to minimize risks.
- B. MRM function specifies the frequency of model validation, but the tier of the model is not taken into consideration.
- C. MRM function is responsible for reviewing and challenging models to minimize risks, and models are assigned to different tiers based on their risk level.
- D. MRM function monitors the performance of models through reports produced by model owners, but they do not conduct validations.

The correct answer is **C**.

The Model Risk Management (MRM) function in financial institutions is primarily responsible for reviewing and challenging models to minimize risks. This involves a thorough examination of the models to identify any potential issues or inaccuracies that could lead to significant financial losses or regulatory penalties. The MRM function also assigns models to different tiers based on their risk level. This tier-based system is crucial as it helps in prioritizing the models that require immediate attention and frequent validation. High-risk models, or those belonging to the top tier, are subjected to more frequent and detailed validation to ensure their accuracy and reliability. On the other hand, models with lower risk levels undergo less frequent validation. This approach allows the MRM function to effectively manage model risk by focusing their efforts on the models that pose the greatest risk to the institution.

Choice A is incorrect. While the MRM function does validate models, it does not do so every year regardless of their tier. The frequency of model validation is determined based on the risk level associated with each model, which is indicated by its assigned tier.

Choice B is incorrect. This statement incorrectly suggests that the MRM function does not consider the tier of a model when specifying the frequency of its validation. In reality, models are assigned to different tiers based on their risk level and this tier assignment plays a crucial role in determining how often they should be validated.

Choice D is incorrect. The MRM function's responsibilities extend beyond merely monitoring reports produced by model owners; they also actively review and challenge models to minimize risks and conduct validations as necessary based on each model's assigned risk level or tier.

Things to Remember

- Model Risk Management (MRM) function is crucial in financial institutions to identify and mitigate risks associated with models used for decision-making.
 - Models are assigned to different tiers based on their risk level, with high-risk models requiring more frequent and detailed validation.
 - Model validation frequency is determined based on the risk level of the model, not a fixed annual schedule.
 - MRM function reviews and challenges models to ensure accuracy and reliability, beyond just monitoring reports produced by model owners.
 - Regular communication and collaboration between MRM function and model owners is essential for effective model risk management.
-

Q.5151 What is the role of the first line of defense in the three lines of defense model in model risk management, and how do first-line QA/QC teams help mitigate model risk?

- A. The first line of defense is responsible for independently assessing the risk and risk management practices of the second line, while the first-line QA/QC teams monitor the performance of models.
- B. The first line of defense abdicates its own responsibilities to the second line, while the first-line QA/QC teams ensure models are validated at the appropriate frequency.
- C. The first line of defense generates the risk to which the organization is exposed and owns the risk. The first-line QA/QC teams play a pivotal role in mitigating model risk, especially execution risk.
- D. The first line of defense is responsible for validating models, while the first-line QA/QC teams conduct comprehensive backtesting.

The correct answer is **C**.

The first line of defense in the three lines of defense model in model risk management is responsible for generating the risk to which the organization is exposed. This line of defense is primarily composed of model developers and model owners. They are the ones who create and own the risk, and hence, they are in the best position to manage and mitigate it. The first-line QA/QC teams play a pivotal role in mitigating model risk, especially execution risk. They ensure that models are developed and implemented according to best practices. By doing so, they help in reducing the likelihood of errors and inaccuracies in the models, which can lead to significant

risks for the organization. Therefore, the first line of defense and the first-line QA/QC teams are integral to the effective management of model risk.

Choice A is incorrect. The first line of defense is not responsible for independently assessing the risk and risk management practices of the second line. Instead, it owns and manages the risks that are generated by its activities. The first-line QA/QC teams do monitor the performance of models, but their role extends beyond just monitoring to include mitigating model risk.

Choice B is incorrect. The first line of defense does not abdicate its responsibilities to the second line; rather, it actively manages and owns its risks. While ensuring models are validated at an appropriate frequency is important, this task typically falls under the purview of second or third lines of defense rather than being a responsibility of first-line QA/QC teams.

Choice D is incorrect. Model validation is generally not a responsibility assigned to the first line of defense; instead, this task usually falls under either second or third lines in order to maintain independence from those who develop and use these models daily. Similarly, while comprehensive backtesting can be part of model risk management process, it's typically performed by independent validation teams (second or third lines) rather than by first-line QA/QC teams.

Things to Remember

- **Three Lines of Defense Model:** This model is a risk management framework that divides responsibilities among three lines of defense -first line (business units), second line (risk management and compliance functions), and third line (internal audit).
- **Model Risk Management:** Refers to the process of identifying, assessing, mitigating, and monitoring the risks associated with the use of models in decision-making processes.
- **Model Validation:** The process of assessing a model's conceptual soundness, implementation, and outcomes to ensure it meets predefined requirements and is fit for its intended purpose.
- **Backtesting:** A method for validating the accuracy and reliability of a model by comparing its output to actual outcomes or historical data.
- **Execution Risk:** The risk that arises from errors or failures in the implementation or execution of a model, leading to adverse consequences.

Q.5152 The risk committee of a large bank has prepared a report on model risk management framework. In the report, it is stated that just like operational risk management (ORM), the

MRM applies the three lines of defense model. Which of the following is **correct** regarding the three lines of defense in the MRM framework?

- A. Model developers and model owners form the first line of defense
- B. The second line of defense works with the first line to assess all the activities of the first line of defense
- C. The second line of defense owns the risk
- D. The first line of defense oversees all the activities of the second line of defense

The correct answer is **A**.

In the context of model risk, model developers and model owners form the first line of defense. Hence, they generate the risk to which the organization is exposed.

B is incorrect. The first line owns the risk and should take all necessary steps to mitigate it, while the second line independently assesses the first line's risk and risk management practices.

C is incorrect. The first line owns the risk

D is incorrect. It should be the other way round, i.e., the second line of defense oversees the activities of the first line of defense.

Things to Remember

- Model Risk Management (MRM) framework is essential for identifying, assessing, and managing model risk within an organization.
- The Three Lines of Defense model is a widely accepted framework for effective risk management and control.
- In the Three Lines of Defense model, the first line of defense consists of business units that own and manage risks, the second line consists of risk management and compliance functions, and the third line consists of internal audit.
- Model developers and model owners in the MRM framework form the first line of defense as they are responsible for creating and managing the models that introduce risk to the organization.
- The second line of defense works independently to assess the risks identified by the first line and the effectiveness of risk management practices.

Q.5153 A bank's risk manager presents to the risk committee various case studies in which small errors and ignorance led to or nearly costed the firm huge losses. What lesson related to the collapse of the CDO market in 2008 did the bank's risk manager present to the risk committee?

- A. The importance of diversification in investments.
- B. The significance of credit ratings in selecting securities.
- C. The risk of relying solely on quantitative models in investment decisions.
- D. The necessity of hedging against market downturns.

The correct answer is **C**.

The 2008 financial crisis, particularly the collapse of the CDO market, was largely attributed to an over-reliance on quantitative models for evaluating and pricing securities. These models, while sophisticated and complex, failed to accurately capture the inherent risks associated with these securities. This led to the creation of overly complex and ultimately unsustainable financial products. The models underestimated the likelihood and potential impact of extreme market events, leading to a significant mispricing of risk. As a result, when the housing market collapsed, the models were unable to accurately predict the resulting losses, leading to a systemic failure of the financial markets. Therefore, the key lesson from the CDO market collapse is the risk associated with relying solely on quantitative models for investment decisions. It underscores the need for a more holistic approach to risk management, one that combines quantitative analysis with qualitative factors such as market conditions, regulatory environment, and human judgment.

Choice A is incorrect. While diversification is a key principle in investment, it was not the primary lesson from the CDO market collapse. The crisis occurred despite diversified investments because of the systemic risk that affected all types of securities.

Choice B is incorrect. The significance of credit ratings was indeed questioned during the crisis as many high-rated securities defaulted. However, this was not the main lesson emphasized by the risk manager. The issue lay more with over-reliance on these ratings and lack of independent assessment.

Choice D is incorrect. Hedging against market downturns is always important but it wasn't the key takeaway from this event. Even well-hedged portfolios suffered losses due to unprecedented scale and nature of this crisis.

Things to Remember

- Systemic risk in financial markets
- Quantitative models in risk management
- Complex financial products and their risks

- Holistic approach to risk management
 - Market conditions and their impact on investments
 - Regulatory environment and its influence on financial markets
 - Human judgment in investment decisions
-

Q.5154 A large bank has hired an expert to look into their newly developed model as good MRM practice. Which of the following options presents a case study in which banks and model users suffered huge losses due to their ignorance to assess the newly developed models before fully adopting them?

- A. Gaussian Copula and CDO pricing
- B. Barclays' acquisition of Lehman Brothers and the excel spreadsheet error
- C. NASA mars orbiter
- D. Lehman Brothers scandal

The correct answer is **A**.

This case study focuses on the collapse of Collateralized Debt Obligations (CDO) markets in 2008. In the early 2000s, David X. Li published a paper on pricing CDOs and how to price pools of assets without considering their correlations. Li's approach was based on the Gaussian copula and the use of Credit Default Swap (CDS) prices to infer the correlation of assets. Despite the limitations associated with Li's pricing model, it was widely adopted by both banks and model users. These parties failed to assess the limitations of the model before fully adopting it. When signs of weaknesses in the model began to emerge in 2008, the correlation implied by the CDSs and the CDO prices increased dramatically. This led to the collapse of the CDO market, resulting in substantial losses for banks and model users.

Choice B is incorrect. Barclays' acquisition of Lehman Brothers and the excel spreadsheet error is not a case of failure to adequately assess a newly developed model. Instead, it was an operational error in the use of an Excel spreadsheet that led to significant losses for Barclays.

Choice C is incorrect. NASA's Mars Orbiter loss was due to a unit conversion error, not due to inadequate assessment of a financial risk management model. This case study does not fit into the context of financial risk management.

Choice D is incorrect. The Lehman Brothers scandal involved fraudulent activities and misrepresentation of financial statements rather than issues with inadequately assessed risk models. Therefore, this choice does not exemplify the situation described in the question.

Things to Remember

- Gaussian Copula is a mathematical model used to estimate the correlation between different assets in a portfolio.
 - Collateralized Debt Obligations (CDOs) are financial products that repackage individual loans into a product that can be sold to investors.
 - Credit Default Swaps (CDS) are financial derivatives that allow investors to hedge against the risk of default on loans or bonds.
 - Operational errors, like the Excel spreadsheet error in the Barclays' acquisition of Lehman Brothers case, can also lead to significant losses for institutions.
 - Unit conversion errors, like the one in the NASA Mars Orbiter case, can have serious consequences but are not related to financial risk management models.
 - The Lehman Brothers scandal involved fraudulent activities and misrepresentation of financial statements, highlighting the importance of transparency and integrity in financial institutions.
-

Q.5155 In September 2008, Lehman Brothers collapsed, sparking the 2008 global financial crisis. In one incident not known to many, Barclays Capital almost bought 179 trading contracts from Lehman Brothers by accident. Which of the following lessons can be learned from this incident?

- A. MRM should challenge the assumptions and ensure users understand related limitations
- B. Even tools and models that seem so simple should be challenged and reviewed properly
- C. Even small errors, such as the use of wrong units, can lead to massive losses
- D. A good MRM should help minimize the misuse of models by helping users understand the limitations accompanying a model

The correct answer is **B**.

The incident involving Barclays Capital and Lehman Brothers underscores the importance of thorough review and challenge, even for tools and models that may appear simple on the

surface. In this case, a seemingly minor oversight - the failure to delete hidden rows - could have resulted in a significant financial loss for Barclays Capital. Although the loss did not materialize in this specific instance, it serves as a stark reminder that potential losses can arise from the simplest of errors. Therefore, it is crucial to challenge and review all tools and models, regardless of their perceived simplicity, to prevent such mistakes from occurring.

Choice A is incorrect. While it's true that Model Risk Management (MRM) should challenge assumptions and ensure users understand related limitations, this choice does not directly relate to the incident at hand. The Barclays-Lehman Brothers near-miss event was more about the importance of meticulousness in financial transactions rather than challenging assumptions or understanding model limitations.

Choice C is incorrect. This statement is generally true in risk management, but it doesn't encapsulate the lesson from the Barclays-Lehman Brothers incident. The issue wasn't about using wrong units but rather a lack of thorough review of tools and models.

Choice D is incorrect. Although a good MRM should indeed help minimize misuse of models by helping users understand their limitations, this choice doesn't directly address the key lesson from the incident - which emphasizes on thorough review and challenge even for seemingly simple tools and models.

Things to Remember

- Model Risk Management (MRM) is a crucial aspect of risk management that involves identifying, assessing, and mitigating risks associated with the use of models and tools in financial decision-making.
- Thorough review and challenge of models and tools, regardless of their complexity or simplicity, is essential to prevent errors and potential financial losses.
- Hidden risks, such as the oversight of not deleting hidden rows in a transaction, can have significant consequences and highlight the importance of attention to detail in financial transactions.
- Understanding the limitations of models and tools is important, but equally important is the need for meticulousness and scrutiny in financial dealings to avoid costly mistakes.

Q.5297 What is the primary role of model risk management in dealing with conceptual errors?

- A. To identify and correct all conceptual errors.
- B. To ensure transparency and clear communication of assumptions to users.
- C. To make models always valid in any context.

D. To prevent market conditions from deteriorating.

The correct answer is **B**.

The primary role of model risk management in dealing with conceptual errors is to ensure transparency and clear communication of assumptions to users. Conceptual errors in financial models can arise due to a variety of reasons, including incorrect assumptions, inappropriate use of models, or lack of understanding of the underlying financial concepts. These errors can lead to significant financial losses if not identified and addressed promptly. Model risk management plays a crucial role in mitigating these risks by ensuring that the assumptions used in the models are clearly communicated to the users. This helps in enhancing the understanding of the users about the model and its limitations, thereby enabling them to make informed decisions. Furthermore, transparency in communication also fosters trust among the users, which is essential for the effective use of financial models.

Choice A is incorrect. While model risk management does involve identifying and correcting conceptual errors, it is not its primary responsibility. The main focus of model risk management is to ensure that the risks associated with the use of financial models are properly managed and mitigated, which includes but is not limited to dealing with conceptual errors.

Choice C is incorrect. It's unrealistic and impractical to expect any financial model to be always valid in any context. Financial models are simplifications of reality and their validity depends on the assumptions made during their development. Therefore, ensuring that a model is always valid in any context goes beyond the scope of model risk management.

Choice D is incorrect. Model risk management cannot prevent market conditions from deteriorating as it has no control over external factors affecting market conditions such as economic trends or policy changes. Its role lies in managing risks associated with financial models rather than controlling market dynamics.

Things to Remember

- Model Risk Management (MRM) is a critical function in financial institutions that focuses on identifying, assessing, and managing the risks associated with the use of financial models.
- Conceptual errors in financial models can lead to incorrect decisions, financial losses, and regulatory issues.
- Transparency in model assumptions and limitations is essential for effective model risk management.
- Model risk management also involves validating models, assessing model performance, and ensuring compliance with regulatory requirements.

- Regular model reviews and independent validation are key components of a robust model risk management framework.
-

Reading 119: Stress Testing Banks

Q.2306 Jim Scott, a risk manager, has been tasked with creating a presentation on capital and liquidity for students at a high school. His introduction begins with a broad definition of the different types of capital and liquidity. In this regard, which of the following is not a type of capital/liquidity?

- A. The capital/liquidity you have.
- B. The capital/liquidity the regulators think that you have.
- C. The capital/liquidity you need.
- D. The capital/liquidity the regulators think that you need.

The correct answer is **B**.

There are three kinds of capital and liquidity: 1) the capital/liquidity you have; 2) the capital/liquidity you need (to support your business activities); and 3) the capital/liquidity the regulators *think* that you need.

Things to Remember

Capital and liquidity are fundamental concepts in finance and risk management. Capital refers to the financial resources that are available for use, while liquidity refers to the ease with which assets can be converted into cash. Understanding the different types of capital and liquidity is crucial for effective financial planning and risk management. The three primary types of capital and liquidity are: 1) the capital/liquidity you have, 2) the capital/liquidity you need, and 3) the capital/liquidity the regulators think that you need. These categories provide a comprehensive framework for assessing and managing financial resources, and are key to ensuring financial stability and success.

Q.2307 Stress testing is a simulation technique used in banking to determine the ability of a given financial instrument or financial institution to deal with an economic crisis. Various authorities conduct different stress test exercises. Among the following options, which one does not represent a valid stress test exercise conducted by an authoritative body?

- A. SCAP 2009 – Supervisory Capital Assessment Program
- B. CEBS 2010 – Committee of European Bank Supervisors
- C. TCAP 2009 – Treasury’s Capital Assistance Program
- D. CCAR 2011/2012 – Comprehensive Capital Analysis and Review

The correct answer is **C**.

Answer C is the only one which is not a stress test exercise, but a mechanism to supply capital to banks in need. It is also not connected to a particular year, and the correct full name is “U.S. Treasury’s CAP”.

Things to Remember

- Stress testing is a risk management technique used to evaluate the potential impact of financial or economic shocks on a firm's financial condition.
 - Stress tests are typically conducted by regulatory authorities to ensure that financial institutions have enough capital to withstand adverse scenarios.
 - SCAP 2009, CEBS 2010, and CCAR 2011/2012 are all examples of stress test exercises conducted by authoritative bodies to assess the resilience of banks.
 - Stress testing helps identify vulnerabilities in a financial system and allows institutions to take corrective actions to mitigate risks.
 - CCAR (Comprehensive Capital Analysis and Review) is an annual exercise by the Federal Reserve to assess the capital adequacy of large bank holding companies.
-

Q.2308 When modeling a bank's revenues, losses, and balance sheet, there are some vital measures to be taken into account. Which of the following is an important measure to the modelers?

- A. Asset values
- B. Accounting and economic profits and losses
- C. Cash inflows and cash outflows
- D. All of the above

The correct answer is **D**.

Modeling a bank's revenues, losses, and balance sheet makes use of asset values for modeling balance sheets, accounting and economic profits and losses for modeling losses and cash inflows and cash outflows for modeling revenues.

Things to Remember

- Asset Liability Management (ALM) is a key concept in banking that involves managing assets and liabilities to ensure the bank's financial stability.
 - Net Interest Income (NII) is an important measure for banks, representing the difference between interest income generated and the interest expenses incurred.
 - Provision for Credit Losses is a critical aspect in modeling bank losses, as it accounts for potential losses from loans and other credit exposures.
 - Liquidity Risk Management is essential for banks to ensure they have enough liquid assets to meet short-term obligations.
-

Q.2309 What is a Credit Value Adjustment (CVA) in the context of stress testing?

- A. A function of the expected default likelihood of the counterparty during normal operation.
- B. A function of the expected default likelihood of the counterparty under a stress scenario.
- C. A function of the expected default likelihood of the counterparty according to historical data.

D. A function of the expected default likelihood of the counterparty according to other variables.

The correct answer is **B**.

A Credit Value Adjustment (CVA) is indeed a function of the expected default likelihood of the counterparty under a stress scenario. Counterparty credit risk is a significant concern in derivative transactions. This risk arises when a bank, upon revaluing a derivative to the stress scenario, finds itself 'in the money' (i.e., it has a derivative receivable). However, the bank cannot be certain that the counterparty to the transaction will remain solvent to fulfill the payment. As a result, the value of the derivative is discounted. This discount is a function of the expected default likelihood of the counterparty under the stress scenario, which is presumably higher than the current scenario. This adjustment to the value of the derivative is referred to as a Credit Value Adjustment (CVA). Banks with substantial derivative activities manage CVA as a standard practice to mitigate counterparty credit risk.

Choice A is incorrect. While the expected default likelihood of the counterparty during normal operation is a factor in assessing counterparty credit risk, it does not represent an adjustment to the value of the derivative. The adjustment in question specifically pertains to a stress scenario, not normal operations.

Choice C is incorrect. The adjustment to the value of a derivative due to counterparty credit risk is not solely based on historical data. Although historical data can provide insights into past behavior and trends, it does not necessarily predict future outcomes or account for potential stress scenarios.

Choice D is incorrect. While other variables may influence the expected default likelihood of a counterparty, they do not constitute an adjustment to the value of a derivative in terms of counterparty credit risk. This adjustment specifically relates to potential default under stress scenarios.

Things to Remember

- Counterparty credit risk is the risk that the counterparty in a financial transaction will default before the final settlement of the transaction.
- Credit Value Adjustment (CVA) is just one component of the broader concept of Counterparty Risk, which also includes Debit Value Adjustment (DVA), Funding Value Adjustment (FVA), and Capital Value Adjustment (KVA).
- Stress testing is a risk management technique used to evaluate the potential impact of adverse events or scenarios on a financial institution's financial condition.
- Expected default likelihood is often estimated using credit risk models such as Credit Default Models (CDMs) or Credit Rating Models.

- Derivatives are financial instruments whose value is derived from an underlying asset, index, or rate, and they are commonly used for hedging or speculative purposes.
-

Q.2311 Regulatory and economic capital models are important instruments for measuring the amount of capital needed. One of the following statements is not true about regulatory and economic capital models. Which one?

Regulatory and economic capital models:

- A. Evolve very slowly.
- B. Evolve dynamically and quickly.
- C. Have difficulty adapting to financial innovation and rapidly changing macro conditions.
- D. Motivate some innovation in modeling due to their “one-size-fits-all” rules.

The correct answer is **B**.

Both regulatory and economic capital models (and especially the former) evolve very slowly and thus have difficulty adapting to financial innovation and rapidly changing macro conditions. Indeed, some of the innovation is motivated by those slowly evolving, one-size-fits-all regulatory capital rules.

Things to Remember

Regulatory and economic capital models are critical tools in financial risk management. They help financial institutions determine the amount of capital they need to hold to cover their risks. Here are some key points to remember about these models:

- They are designed to be robust and stable, and changes are made cautiously to avoid unintended consequences.
- Their slow evolution can be a challenge in a rapidly changing financial environment, as they may not be able to adapt quickly to new risks and conditions.
- Financial innovation often introduces new types of risks and opportunities, which may not be adequately captured by existing models.
- The 'one-size-fits-all' nature of these models can motivate financial institutions to develop their own models that are more tailored to their specific needs and conditions.
- These models also need to meet regulatory standards and requirements, which can be a challenge.

Reading 120: Risk Capital Attribution and Risk-Adjusted Performance Measurement

Q.2207 A Catalanian bank is looking to expand its business lines. The management decides that the primary condition for investment will be the highest RAROC (risk-adjusted return on capital). The possibilities being discussed are:

- I. Tarragona Construcciones, with an expected net profit of EUR 3,000,000 per year and economic capital of EUR 50,000,000; and
- II. Valencia Bonos, with an expected net profit of EUR 1,500,000 per year and economic capital of EUR 22,000,000.

Assuming the cost of equity is 0.062, based on RAROC, the bank would most likely invest in:

- A. Tarragona Construcciones
- B. Valencia Bonos
- C. Both projects
- D. None - neither of the two would be economically viable

The correct answer is **B**.

$$\text{RAROC} = \frac{\text{After-tax expected risk-adjusted net income}}{\text{Economic capital}}$$

	Expected Profit	Economic capital	Calculation
Tarragona Construcciones	3,000,000	50,000,000	$\frac{3,000,000}{50,000,000} = 0.06$
Valencia Bonos	1,500,000	22,000,000	$\frac{1,500,000}{22,000,000} = 0.068$

The RAROC for Valencia Bonos(0.068) is greater than the cost of equity (0.062), thus it is most likely the next investment opportunity for the bank.

Q.2210 A bank in Vermont is considering investing in one of four regional factories producing maple syrup. The bank intends to make a decision based on RAROC (risk-adjusted return on capital). The following information is available:

Factory A:	expected revenues of USD 150,000; expected losses of USD 8,000; economic capital of \$1, 400, 000.
Factory B:	expected revenues of USD 175,000; expected losses of USD 15,000; economic capital of \$1, 500, 000.
Factory C:	expected revenues of USD 200,000; expected losses of USD 15,000; economic capital of \$1, 800, 000.
Factory D:	expected revenues of USD 250,000; expected losses of USD 10,000; economic capital of \$2, 000, 000.

On the basis of the risk-adjusted return on capital for each factory, the bank will most likely pick:

- A. Factory A
- B. Factory B
- C. Factory C
- D. Factory D

The correct answer is **D**.

$$\text{RAROC} = \frac{(\text{expected revenues} - \text{costs} - \text{expected losses} - \text{taxes} + \text{return on risk capital} + / - \text{taxes})}{\text{economic capital}}$$

	Revenues	Expected losses	Economic capital	Calculation
Factory A	150,000	8, 000	1, 400,000	$\frac{(150,000 - 8,000)}{1,400,000} = 0.10143$
Factory B	175,000	15, 000	1, 500,000	$\frac{(175,000 - 15,000)}{1,500,000} = 0.10667$
Factory C	200,000	15, 000	1, 800,000	$\frac{(200,000 - 15,000)}{1,800,000} = 0.10278$
Factory D	250,000	10, 000	2, 000,000	$\frac{(250,000 - 10,000)}{2,000,000} = 0.12000$

Q.2211 The difference between risk capital and regulatory capital is that:

- A. Regulatory capital only applies to a few closely monitored industries like banking and insurance.
- B. While risk capital depends on an institution's individual characteristics and investment choices, regulatory capital is calculated based on industry-wide rules.

C. Aggregate risk capital and regulatory capital may be equal at firm level, but different at business lines level.

D. All of the above.

The correct answer is **D**.

Risk capital and regulatory capital are indeed different in several ways. Firstly, regulatory capital is a concept that is primarily applicable to certain industries where the protection of investors or depositors is of utmost importance. These industries, such as banking and insurance, are closely monitored and regulated, and hence, the concept of regulatory capital is more relevant to them. On the other hand, risk capital is a broader concept that is applicable across all risk-taking businesses, irrespective of the industry they operate in. Secondly, the calculation of risk capital and regulatory capital is based on different parameters. Risk capital is determined by the specific characteristics of a firm and its investment choices. It is a more individualized measure that takes into account the unique risk profile of a firm. In contrast, regulatory capital is calculated based on standardized, industry-wide rules. It is a more uniform measure that is designed to ensure a minimum level of capital adequacy across all firms in a regulated industry. Lastly, while the aggregate figures for risk capital and regulatory capital may be similar at the firm level, there can be significant differences at the business line or department level. Some departments or business lines may be subject to more regulatory measures than others, and hence, their regulatory capital requirements may be higher. Therefore, all the statements in the options are correct, making Choice D the correct answer.

Things to Remember

- Risk capital is the amount of capital that a firm needs to put aside to cover potential losses from its risk-taking activities.
- Regulatory capital is the minimum amount of capital that a financial institution is required to hold by regulatory authorities to ensure its solvency and stability.
- Regulatory capital requirements are set by regulatory bodies such as the Basel Committee on Banking Supervision.
- Risk capital is often used by firms to determine their risk appetite and make decisions on investments and risk management strategies.
- Regulatory capital ratios, such as the Basel III capital adequacy ratios, are used to assess a financial institution's capital adequacy and financial stability.

Q.2212 Kimberley Excavations, a diamond-mining company from South Africa, has implemented

a RAROC (risk-adjusted return on capital) system for future strategic investments. Kimberley Excavations owns several diamond mines which have been showing signs of a decrease in yield, with sharp rises and drops. Management of the mines is deeply dissatisfied with the new system, complaining that RAROC is lacking fairness in attributing economic capital (EC) to their businesses – namely that the EC is too high. What is the correct course of action for senior management in this case?

- A. Neglect dissatisfaction in the local management and enforce the RAROC system at all costs.
- B. Display commitment to RAROC, expand in-house communication and education regarding the system.
- C. Disregard RAROC and return to the old, tried and proven risk management system.
- D. Allocate resources to each of the businesses for purpose of stricter employee monitoring.

The correct answer is **B**.

The senior management should display commitment to the RAROC system and expand in-house communication and education regarding the system. This is because RAROC is not just a common language of risk, but a quantitative technique that can be used to maximize shareholder value. It can be thought of as an internal capital market where businesses compete with each other for scarce balance sheet resources. This makes RAROC a useful tool for capital allocation, both for banks and nonbank corporations. Therefore, instead of disregarding the concerns of the local management, the senior management should take steps to educate them about the benefits and workings of the RAROC system. This would help in ensuring a smooth transition to the new system and in addressing any misconceptions or misunderstandings about it.

Choice A is incorrect. Neglecting the dissatisfaction of local management and enforcing the RAROC system at all costs may lead to further discontent and resistance, which could negatively impact the overall operations of the mines. It's important for senior management to address concerns and provide clarity on any new systems implemented.

Choice C is incorrect. Disregarding RAROC and returning to an old risk management system may not be beneficial in the long run as it might not adequately account for risk-adjusted returns on capital. The adoption of RAROC was likely due to its ability to better manage risks associated with strategic investments, hence reverting back might expose Kimberley Excavations to higher risks.

Choice D is incorrect. Allocating resources for stricter employee monitoring does not directly address the issue at hand - dissatisfaction with RAROC's high economic capital attribution. Instead, it could potentially create a hostile work environment by implying mistrust towards employees, which can further exacerbate existing issues.

Things to Remember

- Risk-Adjusted Return on Capital (RAROC) is a financial performance measure used to

evaluate the efficiency of an investment or to compare the efficiency of a number of different investments. It takes into account the risk of an investment and compares it to the expected return.

- Economic Capital (EC) is the amount of capital that a firm needs to ensure its solvency and to cover unexpected losses. It is a measure of a company's ability to absorb losses without jeopardizing its ongoing operations.
 - Internal Capital Market is a system within a firm where business units compete for financial resources. RAROC can be used as a tool for capital allocation within this internal capital market.
 - Effective communication and education are key factors in implementing new systems or strategies within an organization. Ensuring that all stakeholders understand the benefits and workings of a new system can help in its successful adoption.
 - Strategic investments require careful consideration of risk and return. RAROC helps in evaluating the risk-adjusted returns on such investments, making it a valuable tool for decision-making.
-

Q.2213 An Indian bank is in the process of calculating its risk capital. The main purpose of risk capital calculation is:

- A. To show the level of expected losses that the bank could absorb.
- B. To find differences between regulatory and risk capital.
- C. To show the level of unexpected losses the bank could absorb.
- D. All of the above.

The correct answer is **C**.

The primary purpose of calculating risk capital is to demonstrate the level of unexpected losses that the bank could absorb. Unexpected losses refer to the losses that exceed what is expected or predicted. These losses are not anticipated and hence, are not included in the regular loss provisions. The risk capital is essentially a buffer to protect the bank against these unexpected losses. It is calculated based on a level of confidence that aligns with the requirements of the

bank's various stakeholders. This level of confidence is typically high to ensure that the bank can withstand significant unexpected losses. Therefore, the risk capital serves as a safety net for the bank, providing financial stability and resilience in the face of unexpected adverse events.

Choice A is incorrect. The calculation of risk capital does not primarily aim to show the level of expected losses that a bank could absorb. Expected losses are typically covered by provisions and reserves, not risk capital.

Choice B is incorrect. While finding differences between regulatory and risk capital can be part of the overall financial management process, it is not the primary objective of calculating risk capital. Risk capital is calculated to measure unexpected losses that may occur due to extreme events.

Choice D is incorrect. As explained above, choices A and B do not accurately represent the primary objective of calculating risk capital, therefore an option stating all choices are correct would also be incorrect.

Things to Remember

- Expected losses are losses that are anticipated and can be covered by provisions and reserves.
 - Unexpected losses are losses that exceed what is expected or predicted and are not included in regular loss provisions.
 - Risk capital is a buffer to protect the bank against unexpected losses and is calculated based on a high level of confidence.
 - Risk capital provides financial stability and resilience to the bank in the face of unexpected adverse events.
 - Regulatory capital requirements are set by regulatory bodies to ensure the financial stability and solvency of banks.
-

Q.2214 A certain bank is calculating RAROC for some of its business lines. The available data gives information on: expected revenues, costs, taxes, return on risk capital, transfers, and economic capital. What type of data is missing?

- A. Sharpe ratio
- B. Expected losses
- C. Net present value
- D. VaR (Value-at-risk)

The correct answer is **B**.

Risk-Adjusted Return on Capital (RAROC) is a risk-based profitability measurement framework for analysing risk-adjusted financial performance and providing a consistent view of profitability across businesses. The basic formula for calculating RAROC is:

$$\text{RAROC} = \frac{(\text{expected revenues} - \text{costs} - \text{expected losses} - \text{taxes} + \text{return on risk capital} + / - \text{transfers})}{\text{economic capital}}$$

In the given scenario, the bank has information on expected revenues, costs, taxes, return on risk capital, transfers, and economic capital. However, the data on expected losses is missing. Expected losses are an integral part of the RAROC calculation as they represent the losses that the bank expects to incur as a result of its business activities. These losses could be due to credit risk, market risk, operational risk, or other types of risk that the bank is exposed to. Therefore, without this data, the bank cannot accurately calculate its RAROC.

Choice A is incorrect. The Sharpe ratio is not required for the calculation of RAROC. The Sharpe ratio measures the performance of an investment compared to a risk-free asset, after adjusting for its risk. It does not provide any information about expected losses or gains which are necessary for calculating RAROC.

Choice C is incorrect. Net present value (NPV) is also not needed in the calculation of RAROC. NPV is a method used in capital budgeting to analyze the profitability of an investment or project, but it does not contribute to determining expected losses or gains which are integral components in calculating RAROC.

Choice D is incorrect. VaR (Value-at-risk) measures the potential loss that could occur in an investment portfolio over a specific period with a given confidence level, but it's not directly involved in computing RAROC as it doesn't provide information on expected losses or gains.

Q.2734 Determine the RAROC using the following information about a loan.

Loan Value	\$ 2 million
Gross Revenue	\$ 250,000
Expected Loss	300 bps
Interest Expense	\$ 100,000
Operating Costs	\$ 60,000
Return on invested economic capital	\$ 10,000
Economic capital required	\$ 400,000

- A. 10.00%
- B. 10.50%
- C. 11.00%
- D. 12.50%

The correct answer is **A**.

$$\text{RAROC} = \frac{\text{Revenues} - \text{Expected loss} - \text{Expenses} + \text{Return on capital} + / - \text{Transfer price}}{\text{Economic capital}}$$

$$\text{Expected loss} = 0.0300 \times 2,000,000 = 60,000$$

$$\text{RAROC} = \frac{250,000 - 60,000 - 60,000 + 10,000 - 100,000}{400,000} = 10\%$$

Q.2735 Given that the RAROC on a project is 12%, the risk-free rate is 4%, the return on the market portfolio is 10%, and the firm's equity beta is 1.25, calculate the adjusted RAROC for the project and determine whether it should be accepted or rejected.

- A. 6.4%; rejected
- B. 4.5%; accepted
- C. 6.0%; accepted
- D. 6.0%; rejected

The correct answer is **B**.

RAROC is a profitability measure for analyzing risk-adjusted financial performance. For acceptance, a project must earn a return that's higher than the firm's hurdle rate - a benchmark rate of return set taking into account the firm's cost of both common and preferred equity. However, exclusively accepting only the projects whose RAROC > hurdle rate can result in a portfolio of high-risk projects that could ultimately result in losses and reduce the value of the firm. What's more lower return projects that have a RAROC < hurdle rate (rejected projects) also come with low risk that could provide steady returns and increase the value of the firm.

For these reasons, we adjust RAROC for systematic risk, giving rise to ARAROC, where:

$$\text{Adjusted RAROC} = \text{RAROC} - \beta_e(R_m - R_f)$$

Where:

R_m = expected return on the market

R_f = risk-free rate

β_e = firm's equity beta

$$= 0.12 - 1.25(0.10 - 0.04) = 0.045$$

The project can be accepted if ARAROC > risk-free rate.

Since 4.5% > 4%, this particular projected can be accepted.

Q.2964 Supposing we are given the following information for a loan:

- \$89.5 million is the expected revenue
- \$8.89 million is the operating cost
- \$50.98 million is the tax expense
- \$10 million is the expected loss
- \$24.01 million is the return on risk capital
- \$69.5 million is the economic capital

What is the RAROC for the loan?

- A. 0.7867
- B. 0.4537
- C. 0.6279
- D. 0.8794

The correct answer is C.

$$\text{RAROC} = \frac{\text{Revenues} - \text{Expected loss} - \text{Expenses} + \text{Return on capital} + / - \text{Transfer price}}{\text{Economic capital}}$$

Therefore:

$$\text{RAROC} = \frac{89.5 - 8.89 - 50.98 - 10 + 24.01}{69.5} = 0.6279$$

Q.3136 Sigma Inc. has an equity beta of 1.18. In addition, the risk-free rate is 2%, the expected market return is 7.932% and the RAROC on the proposed project is 10%. If the beta of the proposed project is the same as that of Sigma Inc, then, in order to increase the shareholders' wealth, ARAROC should increase by more than?

- A. 1.446%
- B. 1.592%
- C. 0.0000%
- D. 2.000%

The correct answer is **C**.

Shareholders' wealth increases when ARAROC is greater than the risk-free rate
ARAROC can be computed using the following formula:

$$\begin{aligned}\text{ARAROC} &= \text{RAROC} - \beta(R_M - r_f) \\ &= 10\% - 1.18(7.932\% - 2\%) \\ &= 3\%\end{aligned}$$

In order for shareholders to increase the value of their wealth, ARAROC should be greater than the risk-free rate. Clearly, this condition has been met and, therefore there is no need to increase it.

Q.3206 Samar Vance is an equity strategist at Jumbo Capital. She has been given with the following information about an investee banking company whose:

- Gross revenue: \$12 million
- Economic capital: 80 million
- Return on invested economic capital: 700,000
- Operating costs associated with making the loan: \$2.3 million
- Expected loss on the loan: 1,600,000

Based on the above information, the RAROC is closest to?

- A. 30%.
- B. 23%.
- C. 12%.
- D. 11%.

The correct answer is **D**.

$$\begin{aligned}\text{RAROC} &= \frac{\text{Expected revenues} - \text{Costs} - \text{Expected losses} - \text{Taxes} + \text{Return on risk capital} \pm \text{Tr}}{\text{Economic Capital}} \\ &= \frac{(12 - 2.3 - 1.6 - 0 + 0.7 \pm 0)}{80} \\ &= 0.11 \text{ or } 11\%\end{aligned}$$

Q.3207 Larry Sing is considering to invest in an Oil Marketing Company stock named Hudson Petroleum. If its RAROC is 17%, the company's beta is 1.2, the return on the market is 12%, and the risk-free rate is 8% what will be the adjusted RAROC for a Hudson?

- A. 13.2%.
- B. 16%.
- C. 12.2%.
- D. 5.9%.

The correct answer is **C**.

$$ARAROC = RAROC - \text{Beta}(R_m - R_f) = 17 - 1.2(12 - 8) = 12.2\%$$

Q.3209 Henry Campbell is equity analyst at Four Brothers Financials. He is currently analyzing a new project for expanding in new markets. His calculated RAROC is 13%, the risk-free rate is 6%, the market return is 14%, the firm's required return on equity is 12%, and the firm's beta is 1.5. What is the ARAROC and should the project be accepted?

- A. 11%; accept.
- B. 5.5%; reject.
- C. 6.2%; accept.
- D. 1.0%; reject.

The correct answer is **D**.

Adj RAROC = RAROC - Beta (Rm-Rf)

Decision rule: Accept (reject) projects whose adjusted RAROC is greater (smaller) than Rf.

Adjusted RAROC = 13% - 1.5(14% - 6%) = 1%

Since 1% < 6%, the project should be rejected

A note on the formula used

Old mock exams and study material used a slightly different formula, i.e.

$ARAROC = (RAROC - R_f) / \text{Beta}$

In fact, both formulas will lead to the same decision, but there are conditions.

First case: $ARAROC = (RAROC - R_f) / \text{Beta} \Rightarrow$ to be compared with $R_m - R_f$

Accept (reject) projects whose adjusted RAROC is greater (smaller) than $(R_m - R_f)$.

Second case: $ARAROC = RAROC - \text{Beta} (R_m - R_f) \Rightarrow$ to be compared with R_f
Accept (reject) projects whose adjusted RAROC is greater (smaller) than R_f .

Applying the second (newer) approach is recommended.

Things to Remember

- Risk-Adjusted Return on Capital (RAROC) is a financial performance measure used to evaluate the efficiency of a capital investment.
 - The risk-free rate is the theoretical rate of return of an investment with zero risk of financial loss.
 - Market return is the return on a portfolio that reflects the risk and return of the market as a whole.
 - Required return on equity is the minimum return that an investor expects to achieve by investing in a company's stock.
 - Beta is a measure of a stock's volatility in relation to the overall market.
 - Adjusted RAROC (ARAROC) is calculated by adjusting the RAROC for the systematic risk of the investment.
 - When evaluating projects, it is important to compare the ARAROC to the risk-free rate to make decisions on whether to accept or reject the project.
-

Q.5388 What correct observations could a quantitative model validator make about ABC Bank's rating migration matrix model, considering that it is based on data from 2009 to 2024, a period characterized by economic growth and mild recessions in the country where the bank operates, but the country is anticipated to face a severe recession in 2025, unlike anything seen since the 1990s?

- A. The rating migration matrix adopts a point-in-time approach to data management, leading to an underestimation of the default probabilities for the year 2025.
- B. By utilizing the point-in-time approach to data management, the rating migration matrix will overstate the default probabilities for the year 2025.
- C. The utilization of the through-the-cycle methodology in the rating migration matrix results in an underestimation of the default probabilities for the year 2025.

D. The rating migration matrix, employing the through-the-cycle approach to data management, will overstate the default probabilities for the year 2025.

The correct answer is C.

Due to the rating migration matrix employing the “through-the-cycle” data approach, the observed migration during a recession will be more pronounced compared to an average period of stability. As a result, the default probabilities for the year 2023 will be underestimated.

A is incorrect. The management of data in the rating migration employs a through-the-cycle approach rather than a point-in-time approach.

B is incorrect. The management of data in the rating migration employs a through-the-cycle approach rather than a point-in-time approach.

D is incorrect. In a recession, the rating migration matrix will underestimate, rather than overestimate, the defaults, considering that the observed migration tends to be more severe.

Things to Remember

- Rating migration matrix: A tool used in credit risk management to track changes in credit ratings of borrowers over time.
 - Point-in-time approach: A method that captures data at a specific point in time, providing a snapshot of credit risk at that moment.
 - Through-the-cycle approach: A method that considers the entire economic cycle, smoothing out fluctuations to provide a more stable view of credit risk.
 - Default probabilities: The likelihood that a borrower will fail to meet their debt obligations.
 - Recession: A period of economic decline characterized by a decrease in GDP, employment, and other economic indicators.
 - Severe recession: A particularly intense and prolonged economic downturn with significant negative impacts on the economy.
-

Reading 121: Range of Practices and Issues in Economic Capital Frameworks

Q.2216 The main challenge faced by financial institutions while choosing the risk measure to use for economic capital purposes is that:

- A. There are no generally accepted properties of a good risk measure.
- B. Most risk measures are too complex, which means implementation and eventual communication to stakeholders can be quite difficult.
- C. There is a general lack of relevant and reliable data that can be used to assess risks.
- D. There is no singularly preferred risk measure for economic capital purposes.

The correct answer is **D**.

The primary challenge that financial institutions face when selecting a risk measure for economic capital purposes is that there is no singularly preferred risk measure. While there are several risk measures available, none of them can be considered as the ideal choice for every institution. Each risk measure has its own strengths and weaknesses, and the choice of a risk measure often depends on the specific circumstances and requirements of the institution. Therefore, it is not possible to single out one risk measure as the preferred choice for all institutions. This lack of a universally preferred risk measure makes the selection process challenging for financial institutions.

Choice A is incorrect. There are indeed generally accepted properties of a good risk measure, such as sub-additivity, monotonicity, and translation invariance. These properties are widely recognized and used in the field of risk management.

Choice B is incorrect. While it's true that some risk measures can be complex to implement and communicate to stakeholders, this does not represent the primary challenge faced by financial institutions when selecting an appropriate risk measure for economic capital purposes. The complexity of a risk measure does not necessarily make it inappropriate or unsuitable for use.

Choice C is incorrect. Although there may be challenges associated with obtaining relevant and reliable data for assessing risks, this issue doesn't primarily hinder the selection of an appropriate risk measure for economic capital purposes. In fact, most financial institutions have systems in place to collect and analyze relevant data.

Things to Remember

- Sub-additivity, monotonicity, and translation invariance are generally accepted properties of a good risk measure.
- Financial institutions often use a combination of risk measures to address different

aspects of risk.

- Regulatory requirements and industry standards may influence the choice of risk measure for economic capital purposes.
 - Scenario analysis and stress testing are important tools used in conjunction with risk measures for assessing potential risks.
-

Q.2217 While developing an economic capital framework, it is important to come up with the aggregate risk facing the institution as a whole. However, aggregate risk can be erroneous and inaccurate in light of certain circumstances. These include:

- A. Presence of too many autonomous business units.
- B. Use of different risk assessment models by different business units.
- C. Recognition of benefits of diversification across the organization.
- D. Failure to recognize correlations between different risks.

The correct answer is **D**.

The failure to recognize correlations between different risks can lead to a significant underestimation of the total risk facing an organization. In reality, risks do not exist in isolation and there are often interactions between different types of risks. For example, there can be a correlation between market risk and credit risk. If these correlations are not taken into account during the aggregation of risks, it can result in a gross underestimation of the total risk. Therefore, recognizing and accounting for these correlations is crucial for accurate risk aggregation.

Choice A is incorrect. The presence of too many autonomous business units does not necessarily lead to inaccuracies in the calculation of aggregate risk. While it may make the process more complex, as long as each unit accurately assesses and reports its risks, the overall risk can be correctly aggregated.

Choice B is incorrect. The use of different risk assessment models by different business units can potentially lead to inconsistencies in risk measurement across the organization. However, this does not inherently result in inaccuracies in calculating aggregate risk if these models are properly calibrated and validated.

Choice C is incorrect. Recognizing benefits of diversification across an organization actually improves accuracy in calculating aggregate risk because it takes into account that some risks may offset each other due to their negative correlation.

Things to Remember

- Correlation between risks: Risks are often interrelated and can have correlations with each other, impacting the overall risk profile of an organization.
 - Diversification benefits: Diversification across different business units or asset classes can help in reducing overall risk by offsetting some risks with others.
 - Risk aggregation: The process of combining individual risks to assess the total risk exposure of an organization is crucial for effective risk management.
 - Risk assessment models: Consistent use of validated risk assessment models across business units is important to ensure accurate measurement and aggregation of risks.
-

Q.2219 When examining a firm's capital adequacy, it's always important to establish the dependency (correlation) between obligors. However, correlation estimates provided by current models are usually inaccurate and unstable – mainly because of:

- A. A lack of well-developed computer algorithms.
- B. Scarcity of skilled personnel to do the calculations.
- C. Overdependence on model assumptions.
- D. The use of irrelevant input data.

The correct answer is **C**.

Overdependence on model assumptions is the primary reason for the inaccuracy and instability in correlation estimates provided by current models. Models used to estimate the correlation between obligors in the context of a firm's capital adequacy are heavily reliant on both explicit and implicit assumptions. These assumptions may include factors such as the obligors' creditworthiness, the economic environment, and the firm's financial health. However, these assumptions may not always hold true in real-world scenarios, leading to inaccurate and unstable correlation estimates. Furthermore, the overreliance on these assumptions limits the model's adaptability to changing market conditions and obligor behaviors, further exacerbating the inaccuracy and instability of the correlation estimates.

Choice A is incorrect. While well-developed computer algorithms are important for accurate calculations, their absence does not primarily contribute to the instability and inaccuracy of correlation estimates. The issue lies more with the assumptions made within these models rather than the computational tools used.

Choice B is incorrect. Although skilled personnel are necessary for accurate calculations, their scarcity is not the primary reason for inaccurate and unstable correlation estimates. Even with

highly skilled personnel, if a model's assumptions are flawed or overly simplistic, it will produce inaccurate results.

Choice D is incorrect. The use of irrelevant input data can indeed lead to inaccurate results; however, this isn't typically the primary cause of instability in correlation estimates. The main issue often lies in overdependence on model assumptions which may not hold true under all circumstances.

Things to Remember

- Correlation estimates play a crucial role in assessing a firm's capital adequacy by helping to understand the interdependencies between obligors.
 - Model assumptions, such as obligors' creditworthiness, economic environment, and firm's financial health, heavily influence correlation estimates.
 - Market conditions and obligor behaviors can impact the accuracy and stability of correlation estimates.
 - Well-developed computer algorithms are important for accurate calculations, but the primary issue lies in the assumptions made within the models.
 - Skilled personnel are necessary for accurate calculations, but the scarcity of skilled personnel is not the primary reason for inaccurate correlation estimates.
 - The use of irrelevant input data can lead to inaccurate results, but overdependence on model assumptions is typically the main cause of instability in correlation estimates.
-

Q.2220 A bank-wide view of counterparty credit risk for economic capital purposes can be a challenge mainly because:

- A. It involves large-scale gathering of data and transactions monitoring, which can easily strain human resources.
- B. It requires the use of expensive software to track transactions.
- C. It relies heavily on independent opinions of credit rating agencies, some of which can be compromised.
- D. It requires cooperation among all business divisions, some of which could be autonomous.

The correct answer is **A**.

The process of assessing counterparty credit risk on a bank-wide scale is indeed a complex and resource-intensive task. It involves the large-scale gathering of data from multiple systems and the continuous monitoring of numerous risk exposures, which can sometimes number in the millions. Furthermore, the duration of these transactions can vary greatly, with some concluding overnight while others may run for several decades. This extensive and continuous monitoring and data gathering can easily strain human resources, making it a significant challenge for banks. Therefore, the assertion that a bank-wide view of counterparty credit risk for economic capital purposes can be a challenge mainly because it involves large-scale gathering of data and transactions monitoring, which can easily strain human resources, is accurate.

Choice B is incorrect. While the use of software can indeed be a part of the process, it is not the primary challenge in assessing counterparty credit risk for economic capital. The main challenge lies in the large-scale gathering of data and monitoring transactions, which can strain human resources.

Choice C is incorrect. Although credit rating agencies' opinions can play a role in assessing counterparty credit risk, they are not heavily relied upon for this purpose. The assessment primarily involves internal processes such as data collection and transaction monitoring.

Choice D is incorrect. Cooperation among business divisions could be beneficial but it's not the primary challenge when conducting a comprehensive assessment of counterparty credit risk for economic capital purposes. The main issue arises from the extensive data gathering and transaction monitoring required.

Things to Remember

- Counterparty credit risk is the risk that a counterparty in a financial transaction will default before the final settlement of the transaction's cash flows.
- Economic capital is the amount of capital that a bank or financial institution needs to set aside to cover the risks it faces.
- Data gathering for counterparty credit risk assessment involves collecting information on counterparties, exposure amounts, collateral, and other relevant factors.
- Transaction monitoring is the process of continuously tracking and evaluating the risks associated with ongoing transactions.
- Internal processes such as stress testing and scenario analysis are crucial for assessing counterparty credit risk.

Q.2221 One of the main challenges in the calculation of economic capital for interest rate risk in the banking book relates to:

- A. The long holding period of balance sheet assets and liabilities.
- B. Varying market forces of supply and demand.
- C. The unpredictable nature of regulatory action by central banks.
- D. The presence of a large bouquet of products, all priced differently.

The correct answer is **A**.

The long holding period of balance sheet assets and liabilities indeed poses a significant challenge in the calculation of economic capital for interest rate risk in the banking book. This is because most assets and liabilities in a bank's balance sheet have long holding periods. Predicting interest rates over such extended periods (10-20 years or more) is speculative at best. Therefore, determining the level of economic capital required to mitigate interest rate risk becomes a complex task. The economic capital serves as a buffer against potential losses that could arise from adverse movements in interest rates. However, the uncertainty surrounding long-term interest rate movements makes it difficult to accurately estimate the amount of economic capital needed. This uncertainty is further compounded by the fact that the value of assets and liabilities can significantly change over their long holding periods due to various factors, including changes in market conditions and the bank's own creditworthiness.

Choice B is incorrect. While varying market forces of supply and demand can influence the interest rates, they do not directly impact the computation of economic capital for managing interest rate risk in banking. The calculation process primarily depends on the risk profile of assets and liabilities, not external market forces.

Choice C is incorrect. The unpredictable nature of regulatory action by central banks can indeed pose a challenge to banks, but it does not directly affect the computation of economic capital for managing interest rate risk. Regulatory actions are more related to policy changes which may indirectly influence interest rates but are not a direct factor in calculating economic capital.

Choice D is incorrect. Although a large bouquet of products priced differently can add complexity to asset-liability management, it does not present a direct challenge in computing economic capital for managing interest rate risk. The diversity in product pricing might affect profitability or liquidity management but doesn't significantly impact the calculation process for economic capital related to interest rate risk.

Things to Remember

- **Interest Rate Risk:** The risk that changes in interest rates will adversely affect a bank's financial condition. It can impact a bank's net interest income, market value of equity, and economic capital.

- **Economic Capital:** The amount of capital that a bank needs to hold to cover potential losses from various risks, including interest rate risk. It acts as a buffer against unexpected losses.
 - **Asset-Liability Management (ALM):** The process of managing risks that arise due to differences in the maturity and interest rate sensitivities of a bank's assets and liabilities.
 - **Duration Gaps:** The difference between the durations of a bank's assets and liabilities. Managing duration gaps is crucial in mitigating interest rate risk.
 - **Stress Testing:** A risk management technique used to evaluate a bank's resilience to adverse economic scenarios, including changes in interest rates.
-

Q.2222 In the context of banking, certain financial products can significantly complicate the process of determining a bank's economic capital. This complexity arises due to the inherent risks and uncertainties associated with these products. Among the following options, which financial product is considered to be the most challenging when it comes to calculating a bank's economic capital?

- A. Ordinary stocks
- B. Preference shares
- C. Bonds with embedded options
- D. Fixed-rate interest rate loans

The correct answer is **C**.

Embedded optionality in banking brings about indeterminate cash flows on both the asset and liability sides. It's normally not easy to predict whether or not outstanding options will be exercised. Such products pose risks that are significantly greater than most measures suggest.

Things to Remember

- Embedded options in financial products can introduce uncertainty in cash flows
 - Calculating economic capital for banks involves considering various risks associated with different financial products
 - Preference shares may have characteristics that make them less risky compared to other financial products
 - Fixed-rate interest rate loans have more predictable cash flows compared to products with embedded options
-

Q.2223 Economic capital is best defined as:

- A. The amount of money invested in various risk-taking activities.
- B. The amount of reserve cash held by a bank, which is used to absorb losses resulting from credit risk.
- C. Practices that allow institutions to assess risk and attribute capital to the economic effects of risk-taking activities.
- D. Practices that allow institutions to set aside sufficient funds to mitigate risks emanating from future uncertainties.

The correct answer is **C**.

Economic capital refers to the amount of capital that an institution needs to hold in order to cover potential losses resulting from various risk-taking activities. This capital is not just the regulatory capital required by authorities but also includes additional funds based on the institution's internal risk assessment. Economic capital helps institutions to understand and manage their exposure to risks effectively. Option C best defines economic capital as it emphasizes the practices of assessing risk and allocating capital accordingly.

A is incorrect. Economic capital is a measure of the institution's capacity to absorb losses, not the actual amount invested in risky ventures.

B is incorrect. While it is true that economic capital involves setting aside funds to absorb losses from risk, it is not limited to losses resulting from credit risk alone.

D is incorrect. Economic capital takes into account both known and potential risks and is not limited to uncertainties in the future.

Things to Remember

- Regulatory capital vs. economic capital
- Factors influencing economic capital calculation
- Importance of stress testing in determining economic capital
- Relationship between economic capital and risk appetite
- Integration of economic capital into overall risk management framework

Q.2965 Fidelity Bank uses models based on the asymptotic single risk factor (ASRF) model for credit risk. In particular, the model is based on Basel II risk weights. What is the effect to the

capital charge for an exposure based on this ASRF model?

- A. The capital charge depends on the composition of the portfolio to which the exposure is added.
- B. The capital charge for an exposure depends on risk characteristics of the exposure only.
- C. The capital charge captures general types of tendencies as opposed to the Gaussian copula models.
- D. All the above answers are correct.

The correct answer is **B**.

The ASRF model, which is based on Basel II risk weights, is used by Fidelity Bank for credit risk management. The capital charge for an exposure in this model is determined solely by the risk characteristics of the exposure. This is because the ASRF model is derived from 'ordinary' credit portfolio models through the law of large numbers. When a portfolio comprises a large number of relatively small exposures, the idiosyncratic risks associated with individual exposures tend to cancel each other out. As a result, only systematic risks that affect many exposures have a significant impact on portfolio losses. In the ASRF model, all systematic (or system-wide) risks that affect all borrowers to a certain degree, such as industry or regional risks, are modeled with only one (the 'single') systematic risk factor. This modeling approach allows for the use of banks' correlation estimates or multiple systematic risk factors for correlations to be addressed. Therefore, the capital charge for an exposure depends solely on the risk characteristics of the exposure, not on the composition of the portfolio to which the exposure is added.

Choice A is incorrect. The ASRF model, as per Basel II risk weights, does not consider the composition of the portfolio to which the exposure is added. Instead, it focuses on individual risk characteristics of each exposure.

Choice C is incorrect. The capital charge in an ASRF model does not capture general types of tendencies; rather it depends on specific risk characteristics of the exposure. Gaussian copula models are a different type of credit risk models that capture dependencies between different exposures but are not directly related to how capital charges are calculated in an ASRF model.

Choice D is incorrect. As explained above, both choices A and C do not accurately describe how the capital charge for a specific exposure gets influenced in an ASRF model.

Things to Remember

- **Basel II framework:** Basel II is an international regulatory framework for banks that aims to ensure financial stability by setting minimum capital requirements, supervisory review processes, and market discipline standards.
- **Systematic risk:** Systematic risk refers to the risk inherent to the entire market or an

entire market segment. It cannot be diversified away through the holding of a diversified portfolio of assets.

- **Idiosyncratic risk:** Idiosyncratic risk, also known as unsystematic risk, is the risk that is specific to a particular asset or company. It can be mitigated through diversification.
 - **Gaussian copula models:** Gaussian copula models are used in credit risk modeling to assess the dependency structure between different credit risks. They are based on the assumption of a Gaussian distribution of risks.
 - **Law of large numbers:** The law of large numbers states that as the number of exposures in a portfolio increases, the average outcome will tend to converge to the expected value. This is a key concept in credit risk modeling.
-

Q.2967 Copulas combine the marginal probability distributions into a joint distribution. Which of the following is an advantage of copulas as a form of risk aggregation methodology?

- A. The effect of fixed diversification is sensitive to underlying interactions between the different components.
- B. The method is easy to use as it easily estimates inter-risk correlations and does not capture nonlinearities.
- C. Simulation of common drivers provides for calculating the distribution of outcomes and economic capital risk measure.
- D. Is more flexible than a covariance matrix and allows for nonlinearities and higher-order dependencies.

The correct answer is **D**.

Copulas are more flexible than a covariance matrix and allow for nonlinearities and higher-order dependencies. This flexibility is a significant advantage of copulas. Unlike covariance matrices, which only capture linear dependencies, copulas can capture both linear and nonlinear dependencies. This means they can model complex relationships between different risk factors more accurately. Additionally, copulas can capture higher-order dependencies, which are relationships involving more than two variables. This is particularly useful in risk aggregation, where multiple risk factors need to be considered simultaneously. Therefore, the flexibility of copulas in handling nonlinearities and higher-order dependencies makes them a powerful tool in risk aggregation methodology.

Choice A is incorrect. The statement is not accurate as the effect of fixed diversification is not sensitive to underlying interactions between different components when using copulas. In fact, one of the advantages of copulas is that they can capture the dependence structure independently from the marginal distributions.

Choice B is incorrect. This statement contradicts one of the main advantages of using copulas in risk aggregation methodology. While it's true that estimating inter-risk correlations can be challenging, copulas are actually capable of capturing nonlinearities, which makes them more sophisticated and flexible than traditional correlation matrices.

Choice C is incorrect. Although simulation of common drivers does provide for calculating distribution outcomes and economic capital risk measures, this isn't a unique advantage to copulas nor does it accurately represent their primary function in risk aggregation methodology.

Things to Remember

- Copulas are used to model the dependence structure between random variables, allowing for more accurate risk aggregation.
 - They are particularly useful in capturing complex relationships and dependencies that may not be linear.
 - Copulas can handle higher-order dependencies, which involve relationships between more than two variables.
 - Using copulas can improve the accuracy of risk measures and help in better understanding the joint distribution of risks.
 - It is important to select the appropriate copula function based on the characteristics of the data and the relationships between variables.
-

Q.2968 Broadways Bank uses the unit of account as a component of risk aggregation methodology. Which of the following is NOT a characteristic of the unit of risk accounting?

- A. Risk metric
- B. Confidence level
- C. Complex simulation
- D. Time horizon

The correct answer is **C**.

Complex simulation is not a characteristic of the unit of risk accounting. The unit of risk accounting is a component of risk aggregation methodology, which is used to quantify and manage risk. It includes several defining characteristics such as risk metric, confidence level, and time horizon. However, complex simulation is not one of these characteristics. Complex simulations are often used in risk management to model potential outcomes and scenarios, but they are not a defining characteristic of the unit of risk accounting itself. The unit of risk accounting is more about the metrics and parameters used to quantify and aggregate risk, rather than the methods or simulations used to model or predict risk.

Choice A is incorrect. The risk metric is indeed a characteristic of the unit of account in risk aggregation methodology. It refers to the specific measure used to quantify risk, such as Value at Risk (VaR), Expected Shortfall (ES), etc.

Choice B is incorrect. Confidence level is also a defining characteristic of the unit of account in risk aggregation methodology. It represents the statistical confidence with which we can state that losses will not exceed a certain amount.

Choice D is incorrect. Time horizon forms an integral part of the unit of account in risk aggregation methodology as it defines over what period we are assessing our risks.

Things to Remember

- Risk aggregation methodology is a process used by financial institutions to consolidate and quantify different types of risks across the organization.
- Unit of risk accounting is a key component of risk aggregation methodology that helps in standardizing risk measurement and management practices.
- Risk metric is a crucial aspect of risk measurement and refers to the specific measure used to quantify risk, such as Value at Risk (VaR) or Expected Shortfall (ES).
- Confidence level is the statistical measure of certainty associated with a risk estimate, indicating the level of confidence in the accuracy of the risk assessment.
- Time horizon is an important consideration in risk management as it defines the period over which risks are assessed and managed.
- Complex simulations are advanced modeling techniques used in risk management to simulate various scenarios and outcomes, but they are not a defining characteristic of the unit of risk accounting.

Q.2969 Which of the following risk measures is the least commonly used measure in the practice

of risk management?

- A. Standard deviation
- B. Spectral risk measures
- C. Value at risk
- D. Expected shortfall

The correct answer is **B**.

Spectral risk measures are a relatively new class of risk measures that allow for different weights to be assigned to the quantiles of a loss distribution, rather than assuming equal weights for all observations, as is the case for Expected shortfall. This flexibility allows for a more nuanced understanding of risk, as it can account for the fact that different losses may have different impacts on an organization's overall risk profile. However, despite their theoretical appeal, spectral risk measures are not widely used in practice. This is largely due to their complexity and the computational challenges associated with their implementation. As a result, they are currently largely of academic interest, with their practical application being relatively limited.

Choice A is incorrect. Standard deviation is a widely used measure in risk management scenarios. It quantifies the amount of variation or dispersion of a set of values, which helps in understanding the volatility and thus, the risk associated with an investment.

Choice C is incorrect. Value at Risk (VaR) is also commonly used in practical risk management scenarios. VaR measures the potential loss that could happen in an investment portfolio over a specific period for a given confidence interval.

Choice D is incorrect. Expected shortfall (ES), also known as Conditional Value at Risk (CVaR), although being more conservative than VaR, it's still frequently utilized due to its ability to provide information about tail risks and extreme market conditions.

Things to Remember

- Standard deviation is a measure of the dispersion of a set of values and is commonly used in risk management to understand volatility.
- Value at Risk (VaR) is a widely used measure that estimates the potential loss in an investment portfolio over a specific period at a given confidence level.
- Expected shortfall (ES) or Conditional Value at Risk (CVaR) is a risk measure that provides information about tail risks and extreme market conditions.
- Spectral risk measures are a newer class of risk measures that allow for different

weights to be assigned to quantiles of a loss distribution, but they are not widely used in practice due to complexity and computational challenges.

Q.3211 Which of the following categories of BIS recommendations specifically refers to the need to consider using additional methods, such as stress testing, to help cover all exposures?

- A. Risk aggregation.
- B. Interest rate risk in the banking book.
- C. Netting
- D. Counterparty credit risk.

The correct answer is **D**.

Counterparty credit risk refers to the risk that a counterparty in a financial contract will not live up to their contractual obligations. In the context of the Basel Committee on Banking Supervision (BCBS) recommendations, counterparty credit risk is the category that specifically refers to the need to consider using additional methods, such as stress testing, to help cover all exposures. Stress testing is a risk management technique used to evaluate the potential impact of an adverse event or market condition. It is a powerful tool that allows financial institutions to assess their risk exposure under extreme scenarios. By incorporating stress testing into their risk management practices, banks can ensure that they are adequately prepared for even the most severe economic shocks. This is why the BCBS recommends the use of stress testing as a supplementary method for measuring counterparty credit risk.

Choice A is incorrect. Risk aggregation is a process of consolidating all risks across an organization to get a holistic view. While it's an important part of risk management, it's not a specific category of BCBS recommendations.

Choice B is incorrect. Interest rate risk in the banking book refers to the potential change in net interest income due to changes in interest rates. Although BCBS has guidelines on this, it's not what the question is referring to.

Choice C is incorrect. Netting refers to offsetting positions or obligations by aggregating multiple positions or obligations with the aim of reducing exposures. This concept, while important in risk management and covered under BCBS guidelines, does not match with what was described in the question.

Things to Remember

- Counterparty credit risk is the risk that a counterparty will not fulfill their obligations in a financial contract.

- Stress testing is a risk management technique used to evaluate the impact of adverse events or market conditions.
 - Risk aggregation involves consolidating all risks across an organization for a holistic view.
 - Interest rate risk in the banking book refers to potential changes in net interest income due to interest rate fluctuations.
 - Netting is the practice of offsetting positions or obligations to reduce exposures.
-

Reading 122: Capital Planning at Large Bank Holding Companies: Supervisory Expectations and Range of Current Practice

Q.2224 Oak Creek bank, part of a Bank Holding Company (BHC), is preparing for its annual CCAR (Comprehensive Capital Analysis and Review). After careful consideration, analysts have identified a wrongly implemented principle of capital adequacy process in the bank. Which of the following principles is not part of the CCAR?

- A. Effective loss-estimation methodologies
- B. Sufficient capital adequacy impact assessment
- C. Adequate IT resources
- D. Robust internal controls

The correct answer is **C**.

Adequate IT resources is not a principle of an effective capital adequacy process. The Comprehensive Capital Analysis and Review (CCAR) is a regulatory framework introduced by the Federal Reserve in the United States to supervise, assess, and regulate the capital adequacy processes of large, complex Bank Holding Companies (BHCs). The CCAR aims to ensure that these institutions have robust, forward-looking capital planning processes that account for their unique risks and sufficient capital to continue operations throughout times of economic and financial stress. The seven principles of an effective capital adequacy process under the CCAR are: sound foundational risk management, effective loss-estimation methodologies, solid resource-estimation methodologies, sufficient capital adequacy impact assessment, comprehensive capital policy and capital planning, robust internal controls, and effective governance. While IT resources are crucial for the operational efficiency of a bank, they do not form a principle of the capital adequacy process under the CCAR.

Choice A is incorrect. Effective loss-estimation methodologies are indeed a component of the CCAR. These methodologies help banks to estimate potential losses under various stress scenarios, which is crucial for assessing capital adequacy.

Choice B is incorrect. Sufficient capital adequacy impact assessment is also a part of the CCAR process. This involves evaluating the impact of different stress scenarios on a bank's capital position and ensuring that it maintains adequate capital even in adverse conditions.

Choice D is incorrect. Robust internal controls are an essential part of the CCAR process as well. These controls ensure that all processes related to risk management and capital planning are functioning effectively and accurately, thereby reducing the likelihood of errors or misjudgments in these critical areas.

Things to Remember

- Comprehensive Capital Analysis and Review (CCAR) is a regulatory framework by the Federal Reserve to supervise, assess, and regulate the capital adequacy processes of

large Bank Holding Companies (BHCs).

- The seven principles of an effective capital adequacy process under CCAR include sound foundational risk management, effective loss-estimation methodologies, solid resource-estimation methodologies, sufficient capital adequacy impact assessment, comprehensive capital policy and planning, robust internal controls, and effective governance.
 - IT resources are crucial for operational efficiency in banks but are not a principle of the capital adequacy process under CCAR.
 - Effective loss-estimation methodologies help banks estimate potential losses under various stress scenarios, aiding in assessing capital adequacy.
 - Sufficient capital adequacy impact assessment involves evaluating the impact of stress scenarios on a bank's capital position to ensure it maintains adequate capital in adverse conditions.
 - Robust internal controls are essential in the CCAR process to ensure effective and accurate risk management and capital planning processes.
-

Q.2225 The Great Falls Bank of Montana, USA, part of a Bank Holding Company (BHC), is performing an annual CCAR (Comprehensive Capital Analysis and Review). During the process, it is revealed that one of the existing models has not been appropriately validated nor independently reviewed. Which principle of effective capital adequacy has been violated?

- A. Robust internal controls
- B. Sufficient capital adequacy impact assessment
- C. Effective loss-estimation methodologies
- D. Effective governance

The correct answer is **A**.

The failure to appropriately validate or independently review the model indicates a breach in internal controls. Robust internal controls are essential to ensure that models used in the capital

adequacy assessment are reliable, accurate, and free from errors. Proper validation and independent review are critical components of internal controls that help in identifying and mitigating risks associated with model inaccuracies.

Choice B is incorrect. This principle focuses on the assessment of capital adequacy impact, ensuring that a bank holds enough capital to cover its risks under various scenarios, including stress conditions.

Choice C is incorrect. Effective loss-estimation methodologies pertain to the processes and techniques used to estimate potential losses under different scenarios.

Choice D is incorrect. Principle 7 (Effective governance) focuses on effective governance and underscores the necessity of board and senior management oversight. This principle ensures that the capital adequacy process is not only robust but also aligned with the BHC's overall risk management framework.

Things to Remember

- Robust internal controls are critical for ensuring the reliability of models used in capital adequacy assessments. This includes model validation, which verifies the accuracy of the model's outputs, and independent review, which provides an unbiased assessment of the model's effectiveness.
- Model risk, which arises from the use of inaccurate or misapplied models, is a significant concern in capital planning. Effective internal controls help mitigate this risk by enforcing validation and review processes.
- The Federal Reserve's CCAR process emphasizes strong internal controls as a foundation for effective capital planning, ensuring that all models are rigorously tested and reviewed.

Q.2226 Minnetonka Bank, part of a Bank Holding Company (BHC), is involved in comprehensive capital analysis and review. During the process, it is confirmed that one of their processes for translating risk measures into estimates of potential losses does not encompass a satisfactory range of stressful scenarios and environments. Which principle of an effective capital adequacy process has been violated?

- A. Sound foundational risk management
- B. Sufficient capital adequacy impact assessment
- C. Effective governance

D. Effective loss-estimation methodologies

The correct answer is **D**.

The principle of effective loss-estimation methodologies is a crucial component of an effective capital adequacy process. This principle emphasizes that a Bank Holding Company (BHC) should have robust processes in place for converting risk measures into estimates of potential losses. These estimates should cover a wide range of stressful scenarios and environments. Furthermore, the BHC should be capable of aggregating these estimated losses across the entire organization. In the case of Minnetonka Bank, their failure to encompass a satisfactory range of stressful scenarios and environments in their loss-estimation methodology indicates a violation of this principle. Therefore, choice D is the correct answer.

Choice A is incorrect. Sound foundational risk management refers to the basic principles and practices that a bank should follow to manage its risks effectively. It includes identifying, measuring, monitoring, and controlling risk. In this case, the bank has not failed in its foundational risk management but rather in its loss-estimation methodologies.

Choice B is incorrect. Sufficient capital adequacy impact assessment refers to the process of evaluating how different scenarios might affect a bank's capital adequacy ratio (CAR). While it's true that an inadequate loss-estimation methodology could potentially impact this assessment, the question specifically points out a failure in converting risk measures into potential loss estimates which falls under effective loss-estimation methodologies.

Choice C is incorrect. Effective governance refers to the systems and processes used by a bank to make decisions and oversee its operations. This includes setting strategy, managing risks, ensuring compliance with laws and regulations etc., The problem identified here does not relate directly to governance but rather specific technical aspects of their capital analysis process i.e., their loss estimation methodology.

Things to Remember

- **Capital Adequacy:** Refers to the amount of capital a bank or financial institution must hold to cover its risks and potential losses.
- **Stress Testing:** Involves subjecting a bank's financials to hypothetical scenarios to assess its resilience under adverse conditions.
- **Risk Measures:** Quantitative measures used to assess the potential risks faced by a bank, such as Value at Risk (VaR) or Expected Shortfall (ES).
- **Aggregation of Risk:** The process of combining individual risk measures across different business units to get a comprehensive view of the overall risk exposure.
- **Comprehensive Capital Analysis and Review (CCAR):** A regulatory framework that

assesses the capital adequacy and risk management practices of large U.S. banks.

Q.2227 A certain bank based in New York is assessing risks as part of its preparation for the annual CCAR (Comprehensive Capital Analysis and Review). During the process of stress-testing, several risk categories are defined, particularly those that are difficult to quantify or not directly attributable to any of the specific integrated firm-wide risk categories. Which of the following risks would not fall under such a category?

- A. Compliance risk
- B. Credit risk
- C. Reputational risk
- D. Strategic risk

The correct answer is **B**.

Credit risk does not fall under the category of risks that are difficult to quantify or not directly attributable to any specific integrated firm-wide risk categories. Credit risk is a well-defined and quantifiable risk that arises from the potential that a borrower or counterparty will fail to meet its obligations in accordance with agreed terms. It is a fundamental risk category that banks face and is directly attributable to their lending and investment activities. Banks have well-established methodologies for quantifying credit risk, including credit scoring systems, credit risk models, and provisions for loan losses. Therefore, credit risk does not fall into the category of 'other risks' that are difficult to quantify or not directly attributable to specific firm-wide risk categories.

Choice A is incorrect. Compliance risk refers to the potential for financial loss, legal penalties, or regulatory sanctions due to non-compliance with laws or regulations. It is a type of risk that is challenging to quantify as it depends on various factors such as changes in regulatory environment, effectiveness of internal controls and so on.

Choice C is incorrect. Reputational risk refers to the potential for negative publicity, public perception, or uncontrollable events that could damage a company's reputation and affect its revenue. This type of risk is also difficult to quantify because it relies heavily on subjective factors like public opinion and media coverage.

Choice D is incorrect. Strategic risk arises from poor business decisions, improper implementation of decisions, or lack of responsiveness to industry changes. This kind of risk can be hard-to-quantify as it involves assessing the impact of indirect factors such as market competition and strategic positioning which are not directly attributable to any specific integrated firm-wide risk categories.

Things to Remember

- Credit risk is the risk of loss resulting from the failure of a borrower or counterparty to meet their financial obligations.
 - Compliance risk is the risk of legal or regulatory sanctions, financial loss, or damage to reputation resulting from failure to comply with laws, regulations, or internal policies.
 - Reputational risk is the risk of negative public perception or damage to a company's reputation, which can impact its ability to attract customers or investors.
 - Strategic risk is the risk arising from poor business decisions, improper implementation of decisions, or lack of responsiveness to industry changes.
 - Stress-testing is a risk management technique used to evaluate the potential impact of adverse events or scenarios on a financial institution's financial condition.
 - CCAR (Comprehensive Capital Analysis and Review) is a regulatory framework introduced by the Federal Reserve to assess the capital adequacy and capital planning processes of large U.S. bank holding companies.
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Q.2229 Fairgrounds Bank forms part of a Bank Holding Company (BHC). The bank has been very successful in a business line that was established about 6 months ago. The bank intends to stress test models for the business line for a longer period. As part of best practice during stress testing, the bank should:

- A. Ensure minimal variation from established internal data patterns.
- B. Test a wide range of adverse effects reaching outside the established data patterns.
- C. Only use the data which reflects the most positive outcomes.
- D. Only use the data which reflects the most negative outcomes.

The correct answer is **B**.

The bank should test a wide range of adverse effects reaching outside the established data patterns. Stress testing is a simulation technique used in banking to determine the ability of a financial institution to deal with an economic crisis. Instead of looking at the past performance and expecting the same pattern to continue, the bank should consider a wide range of possible outcomes, including those that are outside of the established data patterns. This is because the future is uncertain and the past performance of the bank may not necessarily indicate its future

performance, especially in the event of a crisis. Therefore, it is important for the bank to test a wide range of adverse effects to ensure its resilience to stressful conditions. This approach is in line with the Federal Reserve's expectation for BHCs to apply generally conservative assumptions throughout the stress testing process.

Choice A is incorrect. While it's important to consider established internal data patterns, stress testing should not be limited to these. Stress tests are designed to evaluate the bank's resilience under extreme conditions, which often means going beyond established patterns and considering a wide range of adverse effects.

Choice C is incorrect. Using only the data which reflects the most positive outcomes would not provide a comprehensive view of potential risks and vulnerabilities. Stress testing should include scenarios that reflect a variety of outcomes, including adverse ones.

Choice D is incorrect. Similarly, using only the data which reflects the most negative outcomes could lead to an overly pessimistic view and may not accurately represent all possible scenarios. A balanced approach that includes both positive and negative outcomes would be more appropriate for stress testing.

Things to Remember

- Stress testing is a key tool used by banks to assess their resilience to adverse economic conditions and shocks.
 - Stress testing involves simulating a wide range of scenarios, including extreme and unlikely events, to understand the potential impact on the bank's financial health.
 - Regulatory bodies like the Federal Reserve expect banks, especially those within Bank Holding Companies (BHCs), to conduct stress tests regularly to ensure their stability and ability to withstand crises.
 - Stress testing should not be limited to historical data patterns but should consider a variety of outcomes, both positive and negative, to provide a comprehensive assessment of risks.
 - It is important for banks to use stress testing results to inform their risk management practices and capital planning processes.
-

Q.2230 Clayton bank forms part of a Bank Holding Company (BHC) and has been requested, by the Federal Reserve, to compile documentation regarding its estimation practices. What are the main guidelines that should be followed by the bank while documenting its estimates?

- A. Extremely detailed explanations of key methodologies with every bit of data available being presented.
- B. Limited documentation emphasizing the importance of management's judgments in estimates.
- C. Limited documentation encompassing theoretical assumptions and quantitative estimates.
- D. Concisely explained key methodologies and assumptions presented in a well-organized manner.

The correct answer is **D**.

The Federal Reserve expects BHCs to clearly document their key methodologies and assumptions used to estimate losses, revenues, and expenses. BHCs with stronger practices provided documentation that concisely explained methodologies, with relevant macroeconomic or other risk drivers, and demonstrated relationships between these drivers and estimates. Documentation should clearly delineate among model outputs, qualitative overlays to model outputs, and purely qualitative estimates. BHCs with weaker practices often had limited documentation that was poorly organized and that relied heavily on subjective management judgment for key model inputs with limited empirical support for and documentation of these adjustments.

Things to Remember

- Bank Holding Company (BHC) regulations by the Federal Reserve
- Estimation practices in financial institutions
- Importance of documenting key methodologies and assumptions
- Role of management judgments in estimates
- Clear delineation among model outputs, qualitative overlays, and purely qualitative estimates

Q.2231 Highlands Bank forms part of a Bank Holding Company (BHC). The bank is computing loss estimates on a number of its business lines. What are the components that the bank should take into account when estimating losses?

- A. Probability of default (PD), time value of money (TM), and loss given default (LGD).
- B. Probability of default (PD), loss given default (LGD), and exposure at default (EAD).
- C. Probability of default (PD), time value of money (TM), and exposure at default (EAD).
- D. Loss given default (LGD), exposure at default (EAD), and credit rating (CR).

The correct answer is **B**.

The expected loss approach, which is the most appropriate method for estimating losses in the banking sector, involves three key components: Probability of Default (PD), Loss Given Default (LGD), and Exposure at Default (EAD). PD refers to the likelihood that a borrower will default on their loan obligations. LGD is the amount that a bank stands to lose if a default occurs, taking into account any potential recoveries from the sale of collateral or other recovery processes. EAD, on the other hand, is the total value that a bank is exposed to at the time of a default. These three components can be estimated at either a segment level or an individual loan level, using various models or assumptions. Therefore, when Highlands Bank is estimating losses on its business lines, it should take into account PD, LGD, and EAD.

Choice A is incorrect. While Probability of Default (PD) and Loss Given Default (LGD) are key components in loss estimation, Time Value of Money (TM) is not typically used in this context. The time value of money refers to the concept that money available today is worth more than the same amount in the future due to its potential earning capacity. This principle suggests that the value of a unit of currency will decrease over time because it could have earned interest or investment income if it was invested.

Choice C is incorrect. Similar to Choice A, while Probability of Default (PD) and Exposure at Default (EAD) are important factors for loss estimation, Time Value of Money (TM) does not play a direct role in estimating potential losses for a bank's business lines.

Choice D is incorrect. Although Loss Given Default (LGD), and Exposure at Default (EAD), are crucial components for calculating potential losses, Credit Rating (CR), though important for assessing credit risk, does not directly factor into the calculation process for estimating losses.

Things to Remember

- Expected Loss Approach is a key method for estimating losses in the banking sector.
- Probability of Default (PD) is the likelihood that a borrower will default on their loan obligations.
- Loss Given Default (LGD) is the amount that a bank stands to lose if a default occurs.
- Exposure at Default (EAD) is the total value that a bank is exposed to at the time of a default.

- Estimating PD, LGD, and EAD can be done at either a segment level or an individual loan level.
 - Models or assumptions are used to estimate these components for loss estimation.
-

Q.2232 Fetterman Bank is currently engaged in the task of projecting its revenue and expenses for a future period. According to the regulations set forth by the Federal Reserve's Capital Plan Rule, what is the specified duration for which should these financial estimates be made?

- A. Nine quarters
- B. Eight quarters
- C. Twelve quarters
- D. There is no explicit length of time defined by the Federal Reserve

The correct answer is **A**.

The Capital Plan Rule requires BHCs to estimate revenue and expenses over the nine-quarter planning horizon.

Things to Remember

- Financial projections are crucial for banks to plan for their capital requirements and ensure they have enough liquidity to meet their obligations.
 - The Federal Reserve's Capital Plan Rule is designed to ensure that large bank holding companies (BHCs) have adequate capital to withstand economic downturns and financial stress.
 - Estimating revenue and expenses over a specific time horizon helps banks assess their financial health and make strategic decisions.
 - The nine-quarter planning horizon specified by the Federal Reserve allows BHCs to make informed capital planning decisions and demonstrate their ability to maintain capital adequacy.
-

Q.2234 Bank of Elmwood, part of a Bank Holding Company (BHC), is preparing for independent model review and validation. What's included in such a process?

- A. An evaluation of conceptual soundness
- B. Ongoing monitoring that includes verification of processes and benchmarking
- C. An "outcome analysis"
- D. All of the above

The correct answer is **D**.

The model review and validation process should include:

- an evaluation of conceptual soundness;
- ongoing monitoring that includes verification of processes and benchmarking; and
- an "outcomes analysis"

Things to Remember

Model review and validation is a critical process in financial institutions, particularly for Bank Holding Companies (BHCs). It is a regulatory requirement and a best practice to ensure that the models used for decision-making are reliable, accurate, and robust. The process includes several key components:

- **Evaluation of Conceptual Soundness.** This involves a thorough review of the model's design and underlying mathematics to ensure that they are theoretically sound and appropriate for the intended use.
- **Ongoing Monitoring.** This includes verification of processes and benchmarking to ensure that the model continues to perform as expected and that any changes or updates to the model are properly implemented and tested.
- **Outcome Analysis.** This is a comparison of the model's predictions with actual outcomes to assess the model's performance.

Each of these components is crucial for ensuring that the model is reliable and effective. Neglecting any one of these components could lead to inaccurate predictions and poor decision-making.

Q.2235 A BHC is having a supervisory review performed on its modeling practices for capital planning. Following the review, the company receives very positive feedback regarding its model documentation as part of risk management. What could be the reason for the positive feedback?

- A. Large-scale collection of relevant historical data for use as input data.
- B. Presence of an updated inventory of all models used in the process.
- C. Presence of qualified modeling staff.
- D. Adoption of large-scale sensitivity testing and stress testing.

The correct answer is **B**.

The presence of an updated inventory of all models used in the process is a key factor in the positive feedback received by the BHC. This is because maintaining an updated inventory of all models used in the modeling process is a best practice in risk management. It ensures that all models are accounted for and can be easily accessed when needed. This practice also helps in tracking the performance of each model and identifying any issues that may arise. Furthermore, it aids in the decision-making process as it provides a comprehensive overview of all the models in use. Therefore, the presence of an updated inventory of all models used in the process is a critical component of effective model documentation and risk management.

Choice A is incorrect. While the collection of relevant historical data is important for model development and validation, it does not directly relate to model documentation. Model documentation involves detailing the development process, assumptions, limitations, and performance of a model which is independent of the volume of data collected.

Choice C is incorrect. Having qualified modeling staff is crucial for developing and maintaining robust models but it doesn't necessarily mean that their model documentation would be commendable. The quality of model documentation depends on how well the modeling process, assumptions, limitations and performance are recorded and communicated rather than just having qualified staff.

Choice D is incorrect. Large-scale sensitivity testing and stress testing are part of risk management strategy but they do not directly contribute to the quality of model documentation. These tests help in understanding how a financial system or a specific portfolio might behave under different scenarios but do not necessarily improve the clarity or comprehensiveness of model documentation.

Things to Remember

- **Model Inventory Management:** Keeping track of all models used in the process is crucial for effective risk management and regulatory compliance.
- **Model Documentation:** Detailed documentation of model development, assumptions, limitations, and performance is essential for transparency and auditability.

- **Model Validation:** Independent validation of models by qualified personnel is necessary to ensure accuracy and reliability.
 - **Regulatory Compliance:** Meeting regulatory requirements for model risk management is a key aspect of capital planning for financial institutions.
 - **Model Governance:** Establishing clear policies and procedures for model development, implementation, and monitoring is essential for effective risk management.
-

Q.2236 Campbell bank, part of a Bank Holding Company (BHC), has not had its risk infrastructure, nor its loss-estimation methodologies reviewed for more than a year. Which principle of an effective capital adequacy process does this violate?

- A. Robust internal controls
- B. Sound foundational risk management
- C. Effective governance
- D. Sufficient capital adequacy impact assessment

The correct answer is **C**.

Effective governance is a crucial principle of an effective capital adequacy process. It necessitates the board and senior management's effective oversight of the capital adequacy process. This includes periodic reviews of the BHC's risk infrastructure and loss- and resource-estimation methodologies. It also involves the evaluation of capital goals, the assessment of the appropriateness of stressful scenarios considered, regular reviews of any limitations and uncertainties in all aspects of the Capital Adequacy Process (CAP), and the approval of capital decisions. In the case of Campbell bank, the lack of review of its risk infrastructure and loss-estimation methodologies for over a year indicates a violation of the principle of effective governance.

Choice A is incorrect. Robust internal controls refer to the systems and procedures put in place to ensure the integrity of financial and accounting information, meet operational and profitability targets, and achieve compliance with laws, regulations, and policies. While regular reviews of risk infrastructure could be part of a robust internal control system, it is not directly related to the scenario described.

Choice B is incorrect. Sound foundational risk management refers to having a solid base for identifying, assessing, managing, monitoring and reporting risks. Although reviewing risk infrastructure can be part of this process, it does not necessarily mean that if Campbell bank has not undergone a review for over a year its foundational risk management is unsound.

Choice D is incorrect. Sufficient capital adequacy impact assessment refers to evaluating how different scenarios or decisions will affect an institution's capital adequacy ratios. The scenario provided does not provide enough information about Campbell bank's capital adequacy impact assessments.

Things to Remember

- Effective governance is essential for overseeing the capital adequacy process in a financial institution.
 - Regular reviews of risk infrastructure and loss-estimation methodologies are crucial components of effective governance.
 - Board and senior management play a key role in ensuring effective governance in the capital adequacy process.
 - Robust internal controls are important for maintaining the integrity of financial information and ensuring compliance with regulations.
 - Sound foundational risk management involves identifying, assessing, and managing risks effectively.
 - Sufficient capital adequacy impact assessment is necessary to evaluate the impact of different scenarios on capital ratios.
-

Q.2237 What do you understand by “feeder models” as used in modeling by BHCs?

- A. Models outsourced from external sources for the purpose of performance comparison with internal models.
- B. Models used to produce projections or estimates that can then be used in another model to generate final figures for expected losses, expenses and revenue.
- C. Models whose outcome has been disputed by experts and analysts at firm level.
- D. Models used to generate the final projected figures for losses, expenses, and revenues.

The correct answer is **B**.

BHCs should maintain an inventory of all models used in the capital planning process, including all input or “feeder” models that produce projections or estimates used by the models that generate the final loss, revenue or expense projections.

Things to Remember

1. Feeder models are a crucial part of the capital planning process in Bank Holding Companies (BHCs). They are used to generate projections or estimates that are then used in another model to generate the final figures for expected losses, expenses, and revenue.
 2. The use of feeder models allows for a more detailed and comprehensive analysis as it breaks down the overall modeling process into smaller, more manageable parts. This increases the accuracy and reliability of the overall projections.
 3. Feeder models are not used for performance comparison with external models, nor are they defined by the level of agreement or disagreement regarding their outcomes. They are also not used to generate the final projected figures for losses, expenses, and revenues directly.
-

Q.2238 A BHC in Mississippi, USA, was recently subjected to a supervisory review of its model risk management. Following the exercise, the company received negative feedback. Which of the following could have led to such an outcome?

- A. Using models without validation or models that had identified weaknesses.
- B. Using benchmark or challenger models to help assess the reasonableness of the primary model output.
- C. Employing independent validation staff to critically evaluate the models.
- D. Being too transparent about the validation status of all models used for capital planning.

The correct answer is **A**.

The use of models without validation or models that have identified weaknesses could have led to the negative feedback received by the BHC. In the context of model risk management, validation is a critical process that ensures the accuracy and reliability of a model's output. It involves a thorough examination of the model's conceptual soundness, ongoing monitoring, and outcome analysis. If a model is used without undergoing this rigorous process, it may produce inaccurate or misleading results, leading to poor decision-making and potential financial losses. Furthermore, if a model has identified weaknesses, it indicates that there are known issues that could affect the model's performance and reliability. Using such a model without addressing these weaknesses could lead to similar adverse outcomes. Therefore, the use of unvalidated models or models with known weaknesses is a significant lapse in model risk management, which could have resulted in the negative feedback received by the BHC.

Choice B is incorrect. Using benchmark or challenger models to assess the reasonableness of the primary model output is actually a good practice in model risk management. It helps in identifying any potential errors or biases in the primary model, thereby enhancing its reliability and accuracy.

Choice C is incorrect. Employing independent validation staff to critically evaluate the models is also a recommended practice for managing model risk. Independent validation ensures that there are no conflicts of interest and that the evaluation of the models is unbiased and objective.

Choice D is incorrect. Being transparent about the validation status of all models used for capital planning cannot be a reason for negative feedback from a supervisory review. Transparency in this context would mean that all stakeholders are aware of whether each model has been validated, which can help prevent misuse or over-reliance on unvalidated models.

Things to Remember

- Model Risk Management is the process of identifying, assessing, and managing the risks associated with the use of models in decision-making.
- Model validation is a critical component of model risk management and involves a thorough review and testing of the model's accuracy and reliability.
- Benchmark or challenger models are used to compare and validate the outputs of the primary model, enhancing the overall robustness of the modeling process.
- Independent validation staff help ensure objectivity and unbiased evaluation of models, reducing the risk of conflicts of interest.
- Transparency in model validation status is important for stakeholders to understand the reliability of the models used in decision-making processes.

Q.2971 Internal controls in bank holding companies (BHCs) should ensure that there is integrity of reported results and the documentation, review, and approval of all material changes to the capital planning process and its components. Such controls as ensured by BHCs should exist at all levels of the capital planning process, with specific control measures to perform all the following roles apart from:

- A. Making sure that there is sufficient robustness in MIS for capital analysis and decision making to be supported, with sufficient flexibility to run ad-hoc analysis whenever necessary.
- B. Provide for reconciliation and data integrity process for all key reports.
- C. Enable the addressing of presentation of aggregate, enterprise-wide capital planning results that gives the description of manual adjustments created in the aggregation process and identified weaknesses are compensated by these adjustments.
- D. Ensure that the documentation provides evidence that results and recommendations can be challenged by the Board.

The correct answer is **D**.

The role of ensuring that the documentation provides evidence that results and recommendations can be challenged by the Board is not typically a function of the specific control measures in place at Bank Holding Companies (BHCs). While it is important for the Board to have the ability to challenge results and recommendations, this is not usually achieved through the specific control measures associated with the capital planning process. Instead, this is more likely to be a function of the broader governance and oversight mechanisms in place at the BHC. These might include, for example, the establishment of a robust internal audit function, the implementation of a strong risk management framework, and the cultivation of a culture of transparency and accountability.

Choice A is incorrect. The control measures in BHCs are indeed designed to ensure robustness in the Management Information System (MIS) for capital analysis and decision making. This includes the ability to run ad-hoc analysis whenever necessary, which is crucial for effective capital planning.

Choice B is incorrect. Reconciliation and data integrity processes for all key reports are also a part of the control measures implemented by BHCs. These processes ensure that all data used in capital planning is accurate and reliable, thereby enhancing the credibility of the results.

Choice C is incorrect. The control measures also enable addressing presentation of aggregate, enterprise-wide capital planning results including description of manual adjustments created in aggregation process and identified weaknesses compensated by these adjustments. This ensures transparency and accountability in the capital planning process.

The role of internal controls at a Bank Holding Company (BHC) does not typically include ensuring that documentation provides evidence that results and recommendations can be challenged by the Board. While it's important for a board to have oversight over an organization's operations, this does not typically fall under internal controls' responsibilities

within a BHC's capital planning process.

Things to Remember

- Internal controls are processes implemented by organizations to ensure the integrity of financial and operational information, compliance with regulations, and the achievement of organizational objectives.
 - Internal controls help in preventing fraud, errors, and inefficiencies, and also provide assurance to stakeholders regarding the reliability of financial reporting.
 - Key components of internal controls include control activities, risk assessment, control environment, information and communication, and monitoring activities.
 - Internal controls should be designed to address specific risks and objectives of an organization, such as capital planning in the case of Bank Holding Companies (BHCs).
 - Effective internal controls require a strong tone at the top, clear policies and procedures, segregation of duties, regular monitoring and evaluation, and a culture of compliance and ethical behavior.
-

Reading 123: Capital Regulation Before the Global Financial Crisis

Q.2334 In 1992, Germany was under Basel I regulations. Eintracht Bank from Frankfurt has had the following portfolio structure (in Deutsche Marks (DM)):

Loans to corporations – 1 billion (600 million in uninsured residential real estate)

OECD countries government's exposures – 500 million

Cash, balance with a central bank – 200 million

Risk weights were as follows:

Risk Weight (%)	Asset category
0	Cash, gold bullion, claims on OECD governments such as Treasury bonds or insured residential mortgages
20	Claims on OECD banks and OECD public sector entities such as securities issued by U.S. government agencies or claims on municipalities
50	Uninsured residential mortgage loans
100	All other claims such as corporate bonds and less-developed country debt, claims on non-OECD banks

The risk-weighted assets of Eintracht bank were closest to which of the following?

A. 700 million

B. 1 billion

C. 500 million

D. 1.2 billion

The correct answer is **A**.

Risk weighted assets should be calculated as follows:

Loans to corporate with uninsured residential real-estate as collateral = $600\text{m} * 50\%$

Loans to corporate without collateral = $400\text{m} * 100\%$

OECD countries government's exposures = $500\text{m} * 0\%$

Cash, balance with a central bank = $200\text{m} * 0\%$

Total risk weighted assets = 700m

Q.2335 In 1992, Italy was under Basel I regulations. Scala Bank from Milan had the following portfolio structure (in USD):

Loans to corporations: \$1.5 billion (\$600 million in commercial real estate)

OECD countries government's exposures: \$300 million

Cash, balance with a central bank: \$500 million

Risk weights were as follows:

Risk Weight (%)	Asset category
0	Cash, gold bullion, claims on OECD governments such as Treasury bonds or insured residential mortgages
20	Claims on OECD banks and OECD public sector entities such as securities issued by U.S. government agencies or claims on municipalities
50	Uninsured residential mortgage loans
100	All other claims such as corporate bonds and less-developed country debt, claims on non-OECD banks

The risk-weighted assets of Scala Bank were closest to which of the following?

- A. \$1.2 billion
- B. \$1.5 billion
- C. \$2.3 billion
- D. \$1.65 billion

The correct answer is **B**.

Risk-weighted assets should be calculated as follows:

Loans to corporate = $1.5b \times 100\%$

OECD countries government's exposures = $300m \times 0\%$

Cash, balance with a central bank = $500m \times 0\%$

Total risk weighted assets = 1.5b

Q.2336 Arrenberg bank from Rotterdam, Netherlands, has to calculate its RWA under Basel I for its exposure in an over-the-counter FX swap agreement. The data on the swap exposure is as follows:

Add-on factor – 1%

Notional amount – EUR 500 million

Current value – EUR 1 million

Risk-weighted factor for counterparty – 100%

The RWA is equal to:

- A. EUR 501 million
- B. EUR 6 million
- C. EUR 5.01 million
- D. EUR 1 million

The correct answer is **B**.

The RWA should be calculated as follows:

$$\begin{aligned}\text{RWA} &= (\text{notional amount} \times \text{add-on factor} + \text{current value}) \times \text{risk weighted factor} \\ &= (500\text{m} \times 1\% + 1) \times 100\% = \text{EUR 6 million}\end{aligned}$$

Q.2337 Calc Bank from Frankfurt, Germany, had to calculate its risk-weighted assets (RWA) under Basel I for its exposure in over-the-counter interest rate swap agreement. The data on the swap exposure is as follows:

Add-on factor: 1.5%

Notional amount: EUR 1 billion

Current value: EUR -2 million

Risk-weighted factor for counterparty: 100%

The RWA is equal to:

A. EUR 13 million

B. EUR 0

C. EUR 15 million

D. EUR 1 billion

The correct answer is **C**.

RWA should be calculated as follows:

$$\begin{aligned} \text{RWA} &= (\text{notional amount} \times \text{add-on factor} + \max(\text{current value}; 0)) \times \text{risk weighted factor} \\ &= (1\text{b} \times 1.5\% + 0) \times 100\% = \text{EUR 15 million} \end{aligned}$$

Q.2338 Kediray Bank from Izmir, Turkey is calculating its regulatory capital under Basel I regulations. It has the following capital instruments: equity, noncumulative perpetual preferred stocks, and subordinated debt with a maturity of over 5 years. What is the structure of its regulatory capital?

A. Tier 1 capital includes equity, noncumulative perpetual preferred stocks, and subordinated debt.

B. Tier 1 capital includes equity, and Tier 2 capital includes noncumulative perpetual preferred stocks and subordinated debt.

C. Tier 1 capital includes equity, and Tier 2 includes noncumulative perpetual preferred stocks; subordinated debt is not included in regulatory capital.

D. Tier 1 capital includes equity, noncumulative perpetual preferred stocks, and Tier 2 includes subordinated debt.

The correct answer is **D**.

Under Basel I regulations, the regulatory capital of a bank is divided into two tiers: Tier 1 and Tier 2. Tier 1 capital, also known as core capital, includes the most liquid forms of capital and is a key indicator of a bank's financial strength from a regulator's point of view. It primarily consists of equity and noncumulative perpetual preferred stocks. Equity represents the value of an ownership interest in the bank, while noncumulative perpetual preferred stocks are a type of preferred stock where the issuer has the right to withhold dividend payments. When dividends are withheld, they are not accumulated for future payment. These forms of capital are considered high-quality because they are readily available to cover losses. On the other hand, Tier 2 capital, also known as supplementary capital, includes less liquid forms of capital like subordinated debt with a maturity of over 5 years. Subordinated debt refers to loans or securities that rank below other loans or securities with regard to claims on assets or earnings. In the event of a liquidation, subordinated debt is only repaid after other debts have been paid. Therefore, in the context of Kediray Bank, its equity and noncumulative perpetual preferred stocks constitute its Tier 1 capital, while its subordinated debt forms part of its Tier 2 capital.

Choice A is incorrect. According to Basel I regulations, subordinated debt with a maturity period exceeding 5 years is not included in Tier 1 capital but rather in Tier 2 capital. Therefore, it's incorrect to state that Tier 1 capital includes equity, noncumulative perpetual preferred stocks, and subordinated debt.

Choice B is incorrect. This choice incorrectly categorizes noncumulative perpetual preferred stocks as part of Tier 2 capital. Under Basel I regulations, noncumulative perpetual preferred stocks are considered part of Tier 1 capital along with equity.

Choice C is incorrect. While this choice correctly identifies that equity forms part of the bank's Tier 1 regulatory capital and that subordinated debt does not form part of the regulatory capital under Basel I regulations, it incorrectly states that noncumulative perpetual preferred stocks are included in Tier 2 instead of being a component of the bank's Tier 1 regulatory capital.

Things to Remember

- Basel I regulations divide regulatory capital into Tier 1 and Tier 2 capital.
- Tier 1 capital includes the most liquid forms of capital such as equity and noncumulative perpetual preferred stocks.
- Equity represents ownership in the bank, while noncumulative perpetual preferred stocks have dividend payment characteristics.
- Tier 2 capital includes less liquid forms of capital like subordinated debt with a maturity of over 5 years.
- Subordinated debt ranks below other debts in terms of claims on assets or earnings and is only repaid after other debts in the event of liquidation.

Q.2339 Banat Bank from Timisoara, Romania, is calculating its regulatory capital under Basel I regulations. It has the following structure of capital instruments (in EUR):

Equity: 150m

Subordinated debt (over 5 years maturity): 50m

Cumulative preferred stocks: 20m

What is the structure of its Tier 1 and Tier 2 capital?

A. Tier 1: 170m; Tier 2: 50m

B. Tier 1: 150m, Tier 2: 70m

C. Tier 1: 150m, Tier 2: 20m

D. Tier 1: 170m

The correct answer is **B**.

The capital has two components:

1. Tier 1: This consists of items such as equity and noncumulative perpetual preferred stock. (Goodwill is subtracted from equity.)
2. Tier 2: This is sometimes referred to as Supplementary Capital. It includes instruments such as cumulative perpetual preferred stock, certain types of 99-year debenture issues, and subordinated debt (i.e. debt subordinated to depositors) with an original life of more than five years.

In this example, Tier 1 = Equity (150m); and Tier 2 = Subordinated debt (50m) + Cumulative preferred stocks (20m) = 70m

Things to Remember

- Basel I regulations were the first international set of regulations on bank capital adequacy and were introduced by the Basel Committee on Banking Supervision in 1988.
- Tier 1 capital is the core measure of a bank's financial strength from a regulator's point of view. It consists of shareholders' equity, retained earnings, and some types of preferred shares.
- Tier 2 capital is supplementary capital that includes items such as subordinated debt, hybrid instruments, and other forms of less permanent capital.

- Regulatory capital requirements are set to ensure that banks have enough capital to absorb unexpected losses and continue operations without endangering the stability of the financial system.
-

Q.2340 Osijek Commercial Bank from Croatia has to calculate its Tier 1 and Tier 2 capital under Basel I regulations. It has the following capital structure (in EUR):

Equity: 50m

Subordinated debt (over 5 years maturity): 30m

Cumulative preferred stocks: 5m

Noncumulative preferred stocks: 10m

What is the structure of the bank's Tier 1 and Tier 2 capital?

- A. Tier 1 capital: 60m; Tier 2: 35m
- B. Tier 1 capital: 55m; Tier 2: 40m
- C. Tier 1 capital: 55m; Tier 2: 10m
- D. Tier 1 capital: 60m; Tier 2: 5m

The correct answer is **A**.

Tier 1 = Equity (50m) + Noncumulative preferred stocks (10m) = 60m

Tier 2 = Subordinated debt (30m) + Cumulative preferred stocks (5m) = 35m

Things to Remember

- Tier 1 capital is the core capital of a bank, consisting of equity and disclosed reserves.
- Tier 2 capital includes subordinated debt, undisclosed reserves, and hybrid instruments.
- Basel I regulations were the first international set of banking regulations on capital adequacy and were introduced in 1988.
- Equity is the most loss-absorbing component of a bank's capital structure.
- Preferred stocks are a type of equity security with characteristics of both equity and debt.

Q.2341 Basel II introduced a capital requirement for one “new” risk in Pillar 1. Which one?

- A. Interest rate risk in banking book
- B. Market risk in trading book
- C. Operational risk
- D. Credit risk

The correct answer is **C**.

Things to Remember

1. The Basel II framework was developed by the Basel Committee on Banking Supervision to enhance the banking regulatory framework. It consists of three pillars: Pillar 1 - Minimum Capital Requirements, Pillar 2 - Supervisory Review Process, and Pillar 3 - Market Discipline.
2. Under Pillar 1, Basel II introduced a capital requirement for operational risk, recognizing the importance of managing this type of risk in banking institutions.
3. The capital requirement for operational risk is calculated based on three methods: the Basic Indicator Approach (BIA), the Standardized Approach (TSA), and the Advanced Measurement Approach (AMA).
4. The capital requirements for credit risk and market risk were not introduced in Basel II. They were introduced in the Basel I Capital Accord and the Basel I Amendment from 1996, respectively.
5. Interest rate risk in the banking book is typically addressed under Pillar 2 of the Basel framework, which deals with the supervisory review process.

Q.2342 NYC Bank from New York, USA, is one of the largest banks in the USA. At the moment of the introduction of Basel II standards in the USA, it was free to choose the approach to use so as to meet credit risk capital requirements. What options did the bank have in this regard?

- A. Standardized approach, Internal rating based approach, and Advanced IRB approach.
- B. None, because the USA chose not to apply Basel II.
- C. NYC Bank could only use standardized approach.
- D. NYC Bank could only use the Foundation IRB approach.

The correct answer is **A**.

When Basel II standards were introduced in the USA, banks had the option to choose from

various approaches to meet credit risk capital requirements. The options available to NYC Bank, as one of the largest banks in the USA, would typically include:

1. Standardized approach: Under this approach, banks use standardized risk weights assigned by regulators to various asset classes to calculate their capital requirements.
2. Internal rating based (IRB) approach: This approach allows banks to use their internal rating systems to assess credit risk and calculate capital requirements accordingly. There are two variations of the IRB approach:
 - a. Foundation IRB approach: This allows banks to use their own internal estimates of probability of default (PD) for eligible exposures, while other parameters such as loss given default (LGD) and exposure at default (EAD) are determined by regulators.
 - b. Advanced IRB approach: This approach permits banks to use their own internal estimates for PD, LGD, and EAD for eligible exposures, subject to approval and oversight by regulators.

Things to Remember

- Basel II: Basel II is an international regulatory framework that provides guidelines on capital adequacy requirements, risk management, and supervision for banks.
- Credit Risk: Credit risk is the risk of loss resulting from the failure of a borrower to repay a loan or meet contractual obligations.
- Capital Requirements: Capital requirements refer to the amount of capital that banks are required to hold to cover potential losses from their risk exposures.
- Standardized Approach: In the standardized approach, banks use predefined risk weights assigned by regulators to calculate their capital requirements.
- Internal Rating Based (IRB) Approach: The IRB approach allows banks to use their internal models and ratings to assess credit risk and determine capital requirements.
- Foundation IRB Approach: Under the Foundation IRB approach, banks use their internal estimates of probability of default (PD) for eligible exposures.
- Advanced IRB Approach: The Advanced IRB approach permits banks to use their internal estimates for PD, LGD, and EAD for eligible exposures, subject to regulatory approval.

Q.2343 PSV Bank, a small regional bank from Eindhoven, in North Brabant, in the Netherlands,

is in process of calculating its capital requirements. Which of the following statements is true?

- A. The bank must use the standardized approach for calculating regulatory capital, because of its size.
- B. The bank can choose between standardized, IRB, and advanced IRB approaches under EU regulation.
- C. The bank is under Basel I regulations which set out specific capital adequacy requirements for different types of assets.
- D. The bank is under both Basel I and Basel II regulations which require banks to hold capital based on specific risk-weighted assets.

The correct answer is **B**.

Under the regulations of the European Union, banks, regardless of their size or international activity, have the option to choose between the standardized approach, the Internal Ratings-Based (IRB) approach, and the Advanced IRB (AIRB) approach for calculating their regulatory capital requirements. This choice is not arbitrary but is based on a variety of factors such as the bank's risk profile, its level of sophistication, and other factors that are subject to supervisory approval. The standardized approach is the simplest and least risk-sensitive of the three, while the IRB and AIRB approaches allow for more sophisticated risk management and capital calculation techniques. The IRB approach allows banks to use their own internal models to calculate credit risk, while the AIRB approach allows banks to use their own internal models to calculate both credit risk and operational risk. This flexibility allows banks to align their capital requirements more closely with their actual risk profile, thereby promoting more efficient use of capital.

Choice A is incorrect. The size of the bank does not necessarily determine the approach it must use for calculating regulatory capital. Under EU regulation, banks have the option to choose between standardized, IRB (Internal Ratings-Based), and advanced IRB approaches.

Choice C is incorrect. While Basel I regulations do set out specific capital adequacy requirements for different types of assets, PSV Bank is based in the Netherlands which falls under EU jurisdiction. Therefore, it follows Basel III regulations which allow banks to choose between standardized, IRB and advanced IRB approaches for calculating regulatory capital requirements.

Choice D is incorrect. Although Basel I and II regulations do require banks to hold capital based on specific risk-weighted assets, this statement does not accurately reflect the current regulatory environment for PSV Bank as it operates under EU jurisdiction where Basel III regulations are applicable.

Things to Remember

- Basel III regulations provide a comprehensive set of reform measures to strengthen the regulation, supervision, and risk management within the banking sector.

- The standardized approach assigns fixed risk weights to different categories of assets, while the IRB approach allows banks to use their own internal models to calculate credit risk.
 - The Advanced IRB (AIRB) approach extends the IRB approach by allowing banks to use their own internal models to calculate both credit risk and operational risk.
 - Regulatory capital requirements are designed to ensure that banks have enough capital to absorb unexpected losses and continue operating in a safe and sound manner.
 - Supervisory approval is required for banks to use the IRB and AIRB approaches, as these methods involve more complex risk management techniques.
-

Q.2345 Bethlenbank from Kecskemet, Hungary, has to calculate its capital requirement for credit risk. The bank has decided to use the standardized approach and has managed to gather data on: exposure, collateral, probability of default, and credit rating of the debtor. Which piece of data is missing so as to proceed with the required calculations smoothly?

- A. Loss given default (LGD)
- B. Exposure at default (EAD)
- C. Not a number
- D. Maturity

The correct answer is **C**.

"Not a number" is a placeholder value that is used in computations when a numerical result cannot be produced. In the context of this question, 'nan' is the correct answer because all the other options - Loss given default (LGD), Exposure at default (EAD), and Maturity - are not missing from the data gathered by Bethlenbank. Under the standardized approach (Basel II), risk-weighted assets are calculated as the product of exposure and a weighted factor which depends on the credit rating of the debtor. LGD and maturity are used for the calculation of capital requirement for credit risk under the IRB approach, not the standardized approach that Bethlenbank is using. Therefore, these are not the missing data that the bank needs to collect. The 'nan' option indicates that no additional data is required for the bank to proceed with its calculations.

Choice A is incorrect. Loss given default (LGD) is not a required input for the standardized approach to credit risk. LGD is used in the Internal Ratings-Based (IRB) approach, which allows banks to estimate their own parameters for risk weight functions.

Choice B is incorrect. Exposure at default (EAD) has already been collected by Bethlenbank as part of its data gathering process, so it does not represent the missing data that needs to be collected.

Choice D is incorrect. Maturity of the loan or exposure is also not a required input under the standardized approach for credit risk calculation. Like LGD, maturity factor plays a role in IRB approach but not in standardized one.

Things to Remember

- Loss Given Default (LGD) is the proportion of an exposure that is not recovered in the event of a default.
 - Exposure at Default (EAD) is the amount of exposure that a bank has at the time of default by a counterparty.
 - Maturity refers to the remaining time until the final repayment of a loan or financial instrument.
 - The Probability of Default (PD) is a key input in credit risk modeling and represents the likelihood that a borrower will default on a loan or debt obligation.
 - Credit Rating of the debtor is used to determine the creditworthiness of the borrower and plays a crucial role in calculating the capital requirement for credit risk.
-

Q.2346 Astoria Bank from Marseille, France, has chosen the IRB approach to calculate its capital requirement for credit risk. In line with standard practice, the bank should calculate its:

- A. Value at risk with a time horizon of 1 year and a confidence interval of 99.9%.
- B. Value at risk with a time horizon of 1 year and a confidence interval of 99%.
- C. Value at risk with a time horizon of 1 month and a confidence interval of 99.9%.
- D. Value at risk with a time horizon of 10 days and a confidence interval of 99%.

The correct answer is **A**.

The capital requirement for credit risk under the IRB approach is based on the Value at Risk (VaR) calculated using a one-year time horizon and a 99.9% confidence level. This is a standard practice recognized by regulators. The VaR is a statistical technique used to measure and quantify the level of financial risk within a firm or investment portfolio over a specific time

frame. In this case, the time frame is one year and the confidence level is 99.9%, meaning the bank is 99.9% confident that its losses will not exceed the VaR over a one-year period. The capital required is therefore the VaR minus the expected loss. Expected losses are usually covered by the way a financial institution prices its products. For example, the interest charged by a bank on a loan is designed to recover expected loan losses. Therefore, the capital requirement is essentially a buffer against unexpected losses, which are the losses that exceed the expected losses.

Choice B is incorrect. While the Value at Risk (VaR) model is indeed used in risk management, the Basel II regulations specify a confidence interval of 99.9% for the IRB approach, not 99%. Therefore, this choice does not align with the standard procedures outlined by Basel II for banks using the IRB approach.

Choice C is incorrect. This option suggests a time horizon of 1 month which is not in accordance with Basel II regulations for banks using the IRB approach. The correct time horizon as per these regulations should be 1 year and not shorter periods like one month.

Choice D is incorrect. Similar to Choice C, this option also proposes an inappropriate time horizon - 10 days in this case - which does not comply with Basel II standards for banks that have opted for the IRB method. The correct time frame should be one year as per these guidelines.

Things to Remember

- Internal Ratings-Based (IRB) approach: A method used by banks to calculate their capital requirement for credit risk based on their own internal risk assessment models.
 - Value at Risk (VaR): A statistical technique used to measure and quantify the level of financial risk within a firm or investment portfolio over a specific time frame.
 - Expected loss: The anticipated losses that a financial institution expects to incur based on historical data and risk assessments.
 - Unexpected loss: The losses that exceed the expected losses and for which the capital requirement serves as a buffer.
 - Basel II regulations: International banking regulations that provide guidelines for capital adequacy requirements and risk management practices for banks.
-

Q.2737 A bank's annual financial statements showed the following assets:

	(in \$ million)
Cash	50
Treasury bills	100
Loans to corporations	750
Uninsured Residential mortgages	100

Calculate the bank's risk-weighted assets based on the Basel I guidelines.

- A. \$850 million
- B. \$700 million
- C. \$750 million
- D. \$800 million

The correct answer is **D**.

According to Basel I the risk weights for different assets are:

Cash and Treasury bills	0%
Uninsured residential mortgages	50%
Loans to corporations	100%

The risk-weighted assets for the bank can be calculated as:

$$50 \times 0\% + 100 \times 0\% + 750 \times 100\% + 100 \times 50\% = 800 \text{ million}$$

Q.2738 Which of the following approaches is NOT appropriate for calculating credit risk capital under Basel II?

- A. Standardized Approach
- B. Foundation IRB Approach
- C. Advanced IRB Approach
- D. Advanced Measurement Approach

The correct answer is **D**.

The Advanced Measurement Approach (AMA) is not an appropriate method for calculating credit risk capital under Basel II. The AMA is actually used for calculating the operational risk of a bank, not credit risk. Operational risk refers to the risk of loss resulting from inadequate or failed internal processes, people, and systems, or from external events. This includes legal risk, but excludes strategic and reputational risk. The AMA for operational risk allows banks to develop their own empirical model to quantify required capital for operational risk, subject to certain minimum conditions and oversight from their national regulator. Therefore, while the AMA is a recognized approach under Basel II, it is not used for calculating credit risk capital.

Choice A is incorrect. The Standardized Approach is indeed a method for calculating credit risk capital as per the Basel II guidelines. Under this approach, banks use ratings from external credit assessment institutions to quantify required capital for credit risk.

Choice B is incorrect. The Foundation IRB (Internal Ratings-Based) Approach is also a suitable method under Basel II guidelines. In this approach, banks are allowed to develop their own empirical model to estimate PD (Probability of Default), but they must use prescribed values for other parameters such as Loss Given Default (LGD) and Exposure at Default (EAD).

Choice C is incorrect. The Advanced IRB Approach allows banks more flexibility in estimating the necessary risk parameters including PD, LGD and EAD based on their internal models, which makes it another suitable method under Basel II guidelines.

Things to Remember

- Basel II framework provides guidelines for calculating credit risk capital for banks.
- Operational risk is a separate category of risk that is calculated using the Advanced Measurement Approach (AMA).
- The Standardized Approach, Foundation IRB Approach, and Advanced IRB Approach are all appropriate methods for calculating credit risk capital under Basel II.
- Each approach has different levels of flexibility in estimating risk parameters such as Probability of Default (PD), Loss Given Default (LGD), and Exposure at Default (EAD).
- Basel II aims to ensure that banks maintain adequate capital reserves to cover potential losses from credit and operational risks.

Q.2740 All of these are pillars of sound bank management under the Basel II framework, except:

- A. Minimum capital requirements
- B. Sound corporate governance
- C. Supervisory review

D. Market discipline

The correct answer is **B**.

Sound corporate governance is not one of the three pillars of the Basel II framework. While sound corporate governance is indeed a crucial aspect of any banking institution's operations, it is not explicitly listed as one of the three pillars under the Basel II framework. The Basel II framework is specifically focused on three areas: Minimum Capital Requirements, Supervisory Review, and Market Discipline. These three pillars are designed to ensure that banks have adequate capital on hand to absorb losses, that they are subject to effective supervision, and that they adhere to market discipline through transparency and disclosure. However, it should be noted that sound corporate governance can indirectly contribute to these three pillars by promoting responsible and ethical decision-making, effective risk management, and transparency.

Choice A is incorrect. Minimum capital requirements is indeed one of the three pillars of the Basel II framework. It sets out the minimum capital requirements that banks must hold to cover their risk-weighted assets.

Choice C is incorrect. Supervisory review, which encourages banks to develop and use better risk management techniques in monitoring and managing their risks, is also one of the three pillars of Basel II.

Choice D is incorrect. Market discipline, which aims to achieve a more stable banking system by making banks' financial conditions more transparent and thus subjecting them to market discipline, constitutes the third pillar of Basel II.

Things to Remember

- **Basel II framework:** Basel II is an international regulatory framework that sets out the minimum capital requirements for financial institutions, with the goal of promoting a more resilient banking system.
- **Risk-weighted assets:** Basel II uses a risk-based approach to determine the amount of capital that banks need to hold based on the riskiness of their assets.
- **Capital adequacy:** Ensuring that banks have enough capital to cover potential losses is a key focus of the Basel II framework.
- **Supervisory review process:** This pillar of Basel II involves regulators assessing banks' risk management processes and ensuring they have adequate capital to support their operations.
- **Market discipline:** Market discipline encourages transparency and disclosure by banks,

allowing market forces to influence their behavior and risk-taking activities.

Q.2994 Suppose that G&R Bank's assets are made up of \$267 million of corporate loans, \$17 million of OECD government bonds, and \$79 million of residential mortgages. We are also given that corporate loans have a risk weight of 100%, loans to government agencies and banks in OECD countries carry a risk weight of 20%, and mortgages have a risk weight of 50%. Compute the total risk-weighted assets under Basel I regulation.

- A. \$520.7 million
- B. \$306.5 million
- C. \$267.4 million
- D. \$487.6 million

The correct answer is **B**.

Recall that the total risk-weighted assets for N on the balance-sheet items is given by the following expression:

$$\sum_{i=1}^N L_i W_i$$

Therefore:

$$\text{The total risk weighted assets} = 267 \times 1 + 79 \times 0.5 = \$306.5 \text{ Million}$$

Note: Cash and securities issued by governments of OECD countries (members of the Organisation of Economic Co-operation and Development) are considered to have virtually zero risk and have a risk weight of zero. It is loans to banks and government agencies in OECD countries that have a risk weight of 20%.

Q.2995 The following table shows a portfolio of three derivatives (in EUR million) possessed by a bank with a particular counterparty:

Transaction	Principal L_i	Current Value V_i
2-year interest rate swap	1000	95
5-year foreign exchange forward	1000	-35
8-month option on a stock	700	80

Calculate the net replacement ratio.

- A. 0.74
- B. 0.63
- C. 0.8
- D. 1.31

The correct answer is **C**.

Recall that:

$$\text{NRR} = \frac{\max(\sum_{i=1}^N V_i, 0)}{\sum_{i=1}^N \max(V_i, 0)}$$

The current exposure with netting (the numerator) is computed as:

$$95 + 80 - 35 = 140$$

The current exposure without netting (the denominator) is computed as:

$$95 + 0 + 80 = 175$$

Therefore:

$$\text{NRR} = \frac{140}{175} = 0.8$$

Q.3234 Jinshi&Houshi Corporation is a large commercial bank operating in mainland China. It has adopted the Basel I framework and had made the following transactions during the year:

(a) A seven-year interest rate swap with a notional principal of \$400 million and a current market value of -\$3 million.

(b) A three-year interest rate swap with a notional principal of \$170 million and a current value of \$7 million.

(c) A four-month derivative on a commodity with a principal of \$80 million that is currently worth \$4 million.

Given the above information, what is the net replacement ratio (NRR) under Basel I assuming that the 1995 netting amendment applies?

A. 1.375

B. 1.0

C. 0.727

D. 0.636

The correct answer is **C**.

The current exposure with netting is $-3 + 7 + 4 = 8$

The current exposure without netting is $0 + 7 + 4 = 11$

The net replacement ratio is given by:

$$\begin{aligned} \text{NRR} &= \frac{\max(\sum_{i=1}^N (V_i, 0))}{\sum_{i=1}^N (V_i, 0)} \\ &= \frac{8}{11} = 0.727 \end{aligned}$$

Q.3238 Paul Hales is a risk consultant at Kimpala Leasing Bank. The assets of the bank consist of \$690 million retail loans (not mortgages), mostly fleets of multinational companies financed by Kimpala. The bank's actuary has projected that the probability of default (PD) is 1% and the loss given default (LGD) is 40%.

Based on this information, what is the worst-case default rate at 99.9% certainty and the expected loss under the Basel II IRB approach? (Note: In this case, correlation $\rho = 0.1216$.)

A. 0.0907 and \$2.76 million

- B. 0.1216 and \$44.15 million
- C. 0.8784 and \$4.83 million
- D. 0.0907 and \$25.03 million

The correct answer is **A**.

WCDR (T,X) Or WCDR(X, T) indicates the Xth percentile of the default rate distribution during a period of length T. Its components are as follows:

$$\text{WCDR} = N \left[\frac{N^{-1}(\text{PD}) + \sqrt{\rho} N^{-1}(X)}{\sqrt{1 - \rho}} \right]$$

Given:

- Probability of default (PD): 1% or 0.01
- Correlation parameter (ρ): 0.1216

Using a standard normal distribution table or a calculator, we find:

- $N^{-1}(0.01) \approx -2.33$
- $N^{-1}(0.999) \approx 3.09$

Plugging these values into the formula, we get:

$$\begin{aligned} \text{WCDR} &= N \left(\frac{-2.33 + \sqrt{0.1216} \cdot 3.09}{\sqrt{1 - 0.1216}} \right) \\ &= N(-1.3364) \end{aligned}$$

Using the standard normal cumulative distribution function, we find:

$$\text{WCDR} \approx N(-1.3364) \approx 1 - 0.9093 = 0.0907$$

$$\begin{aligned} \text{EL} &= \sum \text{EAD}_i \times \text{LGD}_i \times \text{PD}_i \\ &= 690 \times 0.4 \times 0.01 = 2.76 \end{aligned}$$

Q.3239 Paul Hales is a risk consultant at Kimpala Leasing Bank. The assets of the bank consist of \$690 million retail loans (not mortgages), mostly fleets of multinational companies financed by

Kimpala. The bank's actuary has projected that the probability of default (PD) is 1% and the loss given default (LGD) is 40%. The correlation parameter is 0.1216. Based on the Basel II accord, what is the default rate at the 99.9th percentile for the bank?

- A. 9.55%
- B. 4.53%
- C. 9.07%
- D. 11.90%

The correct answer is **C**.

The default rate at the 99.9th percentile ($DR_{99.9}$) for Kimpala Leasing Bank is calculated using the formula:

$$DR_{99.9} = N \left(\frac{N^{-1}(PD) + \sqrt{\rho} \cdot N^{-1}(0.999)}{\sqrt{1 - \rho}} \right)$$

Given:

- Probability of default (PD): 1% or 0.01
- Correlation parameter (ρ): 0.1216

Using a standard normal distribution table or a calculator, we find:

- $N^{-1}(0.01) \approx -2.33$
- $N^{-1}(0.999) \approx 3.09$

Plugging these values into the formula, we get:

$$\begin{aligned} DR_{99.9} &= N \left(\frac{-2.33 + \sqrt{0.1216} \cdot 3.09}{\sqrt{1 - 0.1216}} \right) \\ &= N(-1.3364) \end{aligned}$$

Using the standard normal cumulative distribution function, we find:

$$DR_{99.9} \approx N(-1.3364) \approx 1 - 0.9093 = 0.0907$$

Therefore, the default rate at the 99.9th percentile for Kimpala Leasing Bank is approximately **9.07%**.

Q.3240 Python Commercial Bank uses the standardized approach to arrive at an estimate of total risk-weighted credit risk exposure. An external credit rating agency assigned the following weights to the bank's risk exposures.

Risk Exposure	Weight
\$24 million	80%
\$12 million	120%
\$18million	70%
\$17 million	30%
\$3 million	10%

According to the Basel II Accord, as a rough approximation, the bank is mandated to maintain a minimum capital of:

- A. \$51.6 million.
- B. \$1.792 million.
- C. \$4.128 million.
- D. \$5.920 million.

The correct answer is **C**.

$$\begin{aligned}\text{Minimum capital required} &= 0.08 \times (0.80 \times \$24 \text{ million} + 1.20 \times \$12 \text{ million} + 0.70 \times \$18 \text{ million} \\ &\quad + 0.30 \times \$17 \text{ million} + 0.10 \times \$3 \text{ million}) \\ &= 0.08 \times \$51.6 \text{ million} \\ &= \$4.128 \text{ million.}\end{aligned}$$

According to the Basel II Accord, the bank is mandated to maintain a capital of at least 8% of total risk-weighted assets.

Q.4216 Which of the following statements gives one of the reasons for the introduction of Basel I accord?

- A. The continuity of international financial transactions even after the Herstatt Bank failure
- B. The growing competition between the banks in different countries due to the varied level of capital requirements
- C. All of the above

D. None of the above

The correct answer is **C**.

The Basel I accord was introduced due to a combination of factors. One of the primary reasons was the continuity of international financial transactions even after the failure of Herstatt Bank. The bank's failure highlighted the need for a more robust regulatory framework to ensure the stability of the international banking system. Additionally, there was growing competition between banks in different countries due to the varied level of capital requirements. This competition was seen as potentially destabilizing, as banks with lower capital requirements could take on more risk, potentially leading to financial instability. Therefore, the Basel I accord was introduced to create a level playing field and ensure that all banks had sufficient capital to cover potential losses.

Choice A is incorrect. While the failure of Herstatt Bank did highlight the need for better risk management in international banking, it was not a direct reason for the establishment of Basel I. The Basel I accord was primarily introduced to address inconsistencies in capital requirements across different countries, which is not directly related to ensuring continuity of international financial transactions after a bank failure.

Choice B is incorrect. Although growing competition between banks due to varied levels of capital requirements was indeed an issue, it was not the sole reason that led to the establishment of Basel I. The accord aimed at addressing several issues including credit risk and market risk along with creating a level playing field by standardizing capital adequacy ratios.

Choice D is incorrect. As explained above, both options A and B represent some aspects that led to the establishment of Basel I but they were not the only reasons. Therefore, saying none of them were reasons would be inaccurate.

Things to Remember

- Basel I accord was the first global framework for regulating bank capital adequacy and risk management.
- The accord focused on credit risk and market risk, setting minimum capital requirements for banks.
- Basel I aimed to create a level playing field by standardizing capital adequacy ratios across different countries.
- The accord helped improve the stability and resilience of the international banking system.
- Basel I laid the foundation for subsequent Basel accords, such as Basel II and Basel III, which further refined and expanded regulatory requirements for banks.

Q.4217 What was the main goal of Basel I accord?

- A. Develop a common currency for all the banks
- B. Maintenance of sufficient capital for the banks to remain solvent in time of distress
- C. Raising the solvency level of small banks to match that of a big bank
- D. None of the above

The correct answer is **B**.

The main goal of the Basel I accord was to ensure that banks maintain sufficient capital to remain solvent in times of distress. This was done to promote the stability of the financial system and prevent bank failures that could trigger a financial crisis. The accord introduced the concept of risk-weighted assets, which required banks to hold capital in proportion to the different risk levels of their assets. This was a significant step towards risk-based regulation, which is now a fundamental principle in banking supervision. The Basel I accord was the first of three Basel accords issued by the Basel Committee on Banking Supervision, which provides a forum for regular cooperation on banking supervisory matters. Its objective is to enhance understanding of key supervisory issues and improve the quality of banking supervision worldwide.

Choice A is incorrect. The Basel I accord was not designed to develop a common currency for all the banks. It was primarily focused on ensuring that banks maintain sufficient capital to remain solvent during times of financial distress.

Choice C is incorrect. While the Basel I accord did aim to improve the solvency of banks, it did not specifically target raising the solvency level of small banks to match that of big banks. Its main objective was to ensure overall stability in the banking sector by requiring all banks, regardless of their size, to maintain adequate capital.

Choice D is incorrect. As explained above, the primary objective of Basel I accord was indeed related to banking regulation and it wasn't 'None of the above'. It aimed at maintaining sufficient capital for all types of bank so as they can remain solvent during times of financial distress.

Things to Remember

- Basel I accord was introduced in 1988 by the Basel Committee on Banking Supervision.
- It established the minimum capital requirements that banks needed to hold to cover credit risk.
- The accord classified assets into five risk categories, ranging from 0% to 100% risk weight, based on perceived credit risk.

- Basel I set the minimum capital adequacy ratio at 8%, with 4% Tier 1 capital and 4% Tier 2 capital.
 - It was a significant step towards standardizing banking regulations internationally.
-

Q.4218 Which of the following ratios did the Basel I used to establish the capital sufficiency of the banks?

- A. Leverage ratio
- B. Risk-based capital ratio
- C. All of the above
- D. None of the above

The correct answer is **B**.

The Basel I accord primarily used a risk-based capital ratio to determine the capital adequacy of banks. This ratio is the ratio of a bank's capital to its risk-weighted assets (RWA). The risk-weighted assets include both on-balance sheet assets (based on accounting conventions) and off-balance sheet exposures such as loan commitments and derivatives exposures. The risk-based capital ratio is designed to ensure that a bank has sufficient capital to cover the risks associated with its assets. The higher the risk associated with an asset, the higher the capital requirement. This approach encourages banks to manage their risk exposures and maintain a strong capital base, thereby promoting the stability and efficiency of financial systems.

Choice A is incorrect. While the leverage ratio is a financial metric used to evaluate a bank's financial health by measuring its capital against its consolidated assets, it was not the primary ratio used by Basel I for determining capital adequacy. The leverage ratio does not take into account the riskiness of a bank's assets, which was a key focus of Basel I.

Choice C is incorrect. This choice suggests that both the leverage ratio and risk-based capital ratio were primarily used by Basel I to determine capital adequacy. However, as explained above, while both ratios are important in assessing a bank's financial health and stability, only the risk-based capital ratio was primarily used under Basel I.

Choice D is incorrect. This choice implies that neither the leverage nor risk-based capital ratios were utilized by Basel I for determining a bank's capital adequacy. This contradicts with the fact that Basel I primarily relied on the risk-based capital ratio for this purpose.

Things to Remember

- Basel I was the first international banking accord introduced by the Basel Committee on Banking Supervision in 1988.

- The risk-based capital ratio is calculated as the ratio of a bank's capital to its risk-weighted assets (RWA).
 - Risk-weighted assets include both on-balance sheet assets and off-balance sheet exposures like loan commitments and derivatives.
 - The leverage ratio measures a bank's capital against its consolidated assets without considering the riskiness of the assets.
 - Basel I aimed to ensure that banks had enough capital to cover the risks associated with their assets, promoting financial stability.
-

Q.4219 Under the Basel I framework, what is the required value of the ratio of Tier 1 capital to risk-weighted assets (RWA)?

- A. Greater than 4%
- B. Less than 4%
- C. Greater than 8%
- D. Less than 8%

The correct answer is **A**.

The Basel I framework, which was introduced by the Basel Committee on Banking Supervision in 1988, required banks to maintain a minimum ratio of Tier 1 capital to risk-weighted assets (RWA) of 4%. Tier 1 capital, also known as core capital, includes the highest quality capital a bank possesses, such as common stock and disclosed reserves. Risk-weighted assets are the total of all assets held by a bank, adjusted for their associated risks. The higher the risk associated with an asset, the higher its weight. The purpose of this ratio is to ensure that banks have enough high-quality capital to absorb losses, thereby reducing the risk of bank failure and protecting depositors. The ratio is calculated as follows:

$$\frac{\text{Tier 1 Capital}}{\text{RWA}} > 4\%$$

Choice B is incorrect. The Basel I framework does not require the ratio of Tier 1 capital to RWA to be less than 4%. This would imply a lower level of financial strength, which contradicts the purpose of the Basel regulations that aim to ensure sufficient capital adequacy in banks.

Choice C is incorrect. While it's true that a higher ratio indicates greater financial strength, the Basel I framework does not set forth a requirement for this ratio to be greater than 8%. The standard set by Basel I was actually lower.

Choice D is incorrect. Similar to Choice C, this option misrepresents the standards set by Basel I. A requirement for this ratio to be less than 8% would suggest a relatively low threshold for financial strength, which isn't consistent with the objectives of these regulatory standards.

Q.4220 Based on the Basel I framework, which of the following expressions is incorrect?

- A. Tier 1 capital > 4%(RWA)
- B. Total capital > 8%(RWA)
- C. Tier 1 capital + Tier 2 capital = Total capital
- D. Tier 1 capital + Tier 2 capital > Total capital

The correct answer is **D**.

The Basel I framework does not state that the sum of Tier 1 and Tier 2 capital should be greater than the total capital. In fact, the total capital under the Basel I framework is defined as the sum of Tier 1 and Tier 2 capital. Therefore, it is not possible for the sum of Tier 1 and Tier 2 capital to be greater than the total capital. This statement is a contradiction and is therefore incorrect.

Choice A is incorrect. The Basel I framework indeed stipulates that Tier 1 capital should be greater than 4% of risk-weighted assets (RWA). This requirement ensures that banks maintain a certain level of high-quality, liquid assets to absorb losses.

Choice B is incorrect. According to the Basel I framework, the total capital (Tier 1 + Tier 2) should be greater than or equal to 8% of RWA. This rule was established to ensure that banks have sufficient overall capital reserves relative to their risk exposure.

Choice C is incorrect. Under the Basel I framework, Total Capital does indeed consist of Tier 1 and Tier 2 capital combined. Therefore, it's accurate to say that Tier 1 Capital + Tier 2 Capital equals Total Capital.

Things to Remember

- The Basel I framework was the first international capital adequacy standard for banks, introduced in 1988 by the Basel Committee on Banking Supervision.
- Tier 1 capital is the core capital of a bank, including common equity and disclosed reserves, while Tier 2 capital includes supplementary capital like subordinated debt and hybrid instruments.
- Risk-weighted assets (RWA) are a measure of a bank's total assets adjusted for risk, with higher-risk assets assigned higher weights.
- The Basel I framework aimed to ensure that banks maintain adequate capital to cover their risks and absorb potential losses, promoting financial stability.
- Regulatory capital requirements help protect depositors and maintain confidence in the banking system by ensuring banks have enough capital to withstand adverse events.

Q.4221 According to Basel I classification of capital, which of the following is NOT a constituent of Tier 2 capital?

A. Undisclosed reserves

- B. Common equity
- C. Hybrid instruments
- D. Loan loss reserves not allocated to non-performing assets

The correct answer is **B**.

Under the Basel I framework, Tier 1 Capital, also known as core capital, includes common equity and disclosed reserves minus goodwill. Common equity is the most fundamental form of capital and represents the ownership interest in a bank. It includes ordinary shares and retained earnings. This type of capital is considered the highest quality because it is fully available to cover losses. Therefore, common equity is not a part of Tier 2 capital, which is supplementary capital and includes other forms of capital that are less secure and reliable than Tier 1 capital.

Choice A is incorrect. Undisclosed reserves are indeed a part of Tier 2 capital under the Basel I framework. These reserves are created or increased through appropriations of retained earnings or other surplus, such as share premiums, general reserves, and legal reserves.

Choice C is incorrect. Hybrid instruments are also included in Tier 2 capital under Basel I framework. These instruments combine characteristics of both debt and equity and provide a supplementary source of capital for banks.

Choice D is incorrect. Loan loss reserves not allocated to non-performing assets form part of Tier 2 Capital as per Basel I norms. They serve as buffers against potential losses from loans that may default in future.

Things to Remember

- Basel I framework was the first international framework for banking regulations developed by the Basel Committee on Banking Supervision.
- Tier 1 Capital is considered the highest quality capital as it is fully available to cover losses and includes common equity and disclosed reserves minus goodwill.
- Tier 2 Capital is supplementary capital that includes undisclosed reserves, hybrid instruments, and certain types of loan loss reserves.
- Common equity represents the ownership interest in a bank and is not a part of Tier 2 capital.
- Undisclosed reserves are created or increased through appropriations of retained earnings or other surplus and are part of Tier 2 capital.
- Hybrid instruments combine characteristics of debt and equity and provide additional

capital for banks.

- Loan loss reserves not allocated to non-performing assets serve as buffers against potential losses and are part of Tier 2 capital.
-

Q.4222 According to Basel I, to create a risk-sensitive ratio, the risk-weighted assets are used as the denominator. Which of the following is assigned a risk weight of 0%?

- A. Uninsured residential mortgages
- B. Commercial and consumer loans
- C. Claims on Organization for Economic Cooperation and Development (OECD) government bonds
- D. Claims on OECD banks and public sector entities

The correct answer is **C**.

Under the Basel I framework, claims on Organization for Economic Cooperation and Development (OECD) government bonds are assigned a risk weight of 0%. This is based on the assumption that OECD governments are highly unlikely to default on their obligations. The 0% risk weight reflects the perceived safety and stability of these bonds, which are backed by the full faith and credit of OECD governments. These governments are typically characterized by strong and stable economies, robust institutional frameworks, and high levels of public sector transparency and accountability. Therefore, the risk of default is considered to be extremely low, justifying the 0% risk weight.

Choice A is incorrect. Uninsured residential mortgages are not assigned a risk weight of 0% under the Basel I framework. They carry credit risk and hence, are assigned a higher risk weight.

Choice B is incorrect. Commercial and consumer loans also carry credit risk and therefore, they are not assigned a 0% risk weight under the Basel I framework.

Choice D is incorrect. Claims on OECD banks and public sector entities do not have a 0% risk weight under the Basel I framework as they also carry some level of credit risk.

Things to Remember

- Risk-weighted assets are used in Basel I to calculate capital adequacy ratios, with different assets assigned different risk weights based on their credit risk.
- Basel I was the first international banking accord that aimed to establish minimum

capital requirements for banks to ensure financial stability.

- The risk weights assigned to different assets under Basel I are based on the perceived credit risk associated with those assets.
 - Claims on OECD government bonds are considered to have a 0% risk weight due to the assumption of low default risk associated with OECD governments.
 - Uninsured residential mortgages and commercial/consumer loans typically carry credit risk and are assigned higher risk weights under Basel I.
-

Q.4223 The constituents of an American bank are \$200 million of American government bonds, \$500 million of loans to corporations, \$300 million of uninsured residential mortgages, and \$250 million of residential mortgages issued by the public sector. What is the value of risk-weighted assets (RWA) based on Basel I accord?

- A. \$1250 million
- B. \$600 million
- C. \$700 million
- D. \$850 million

The correct answer is **C**.

Using the weight ratios under the Basel I accord, the RWA is given by:

$$\text{RWA} = 0\% \times 200 + 100\% \times 500 + 50\% \times 300 + 20\% \times 250 = \$700 \text{ million}$$

Q.4227 Under the Basel I framework, which of the following is one of the methods of measuring market risk?

- A. Current exposure method
- B. Original exposure method
- C. Standardized approach methods
- D. The foundations of internal ratings-based

The correct answer is **C**.

The Basel I framework, which was introduced in 1988, was primarily focused on credit risk. However, in 1996, the Basel Committee on Banking Supervision amended the Basel I framework to include market risk. The amendment provided two methodologies for measuring market risk: a standardized approach and an internal model-based approach. The standardized approach is a method that uses pre-determined risk weights to calculate the capital requirement for each type of market risk. This approach is simpler and less resource-intensive than the internal model-based approach, making it suitable for smaller banks or banks with less complex trading activities. The standardized approach methods are therefore correctly identified as a method of measuring market risk under the Basel I framework.

Choice A is incorrect. The Current Exposure Method (CEM) is a measure of counterparty credit risk in over-the-counter derivatives, not a method for measuring market risk under the Basel I framework.

Choice B is incorrect. The Original Exposure Method refers to the original amount of exposure at the time of transaction and it does not pertain to market risk measurement under Basel I.

Choice D is incorrect. The foundations of internal ratings-based approach are part of Basel II and III frameworks, which focus on credit risk, not market risk measurement under Basel I.

Things to Remember

- Basel I framework primarily focused on credit risk before the amendment in 1996 to include market risk.
- Market risk refers to the risk of losses in on and off-balance sheet positions arising from movements in market prices.
- Standardized approach in Basel I uses pre-determined risk weights to calculate capital requirements for market risk.
- Internal model-based approach in Basel I allows banks to use their own risk models to calculate capital requirements for market risk.
- Basel II and Basel III frameworks also include more advanced methods for measuring market risk.

Q.4401 The Solvency II uses both standardized and internal model-based approaches to compute SCR. However, if an insurance company decides to use internal models, the models must satisfy certain conditions. Which of the following is one of the conditions?

- A. The size of the data used should be small
- B. The model used to must be applicable to real business decision making
- C. All of the above
- D. None of the above

The correct answer is **B**.

The internal models used by insurance companies under Solvency II must be applicable to real business decision making. This means that the models should not only be theoretically sound, but also practical and useful in the context of the company's operations. They should be able to accurately represent the company's risk profile, and should be integrated into the company's decision-making process. This ensures that the models are not just used for regulatory compliance, but also contribute to the company's risk management and strategic planning. The models should be regularly updated and tested to ensure their continued relevance and accuracy.

Choice A is incorrect. The size of the data used in internal models should not necessarily be small. In fact, larger datasets can often provide more accurate and reliable results as they allow for a more comprehensive analysis of risk factors.

Choice C is incorrect. As explained above, not all the options listed are prerequisites for an insurance company that opts to use internal models under Solvency II regulations.

Choice D is incorrect. There are indeed certain prerequisites that these models must meet under Solvency II regulations, so it's not correct to say that none of the options listed are prerequisites.

Things to Remember

- Internal models under Solvency II must be approved by the relevant regulatory authorities before they can be used to calculate the Solvency Capital Requirement (SCR).
- The models should be transparent, well-documented, and subject to regular validation and review processes to ensure their accuracy and reliability.
- Insurance companies using internal models must have robust governance frameworks in place to oversee the development, implementation, and use of these models.
- The models should cover all material risks faced by the insurance company and should be able to capture the interactions between different risk factors.
- Regular stress testing and scenario analysis should be conducted using the internal

models to assess the company's resilience to adverse market conditions and other risk events.

Q.4403 A bank majors in four business lines whose corresponding multipliers and gross income (in millions) for three years are given in the table below:

Business Line	Multiplier	Annual	Gross	Income
		Year 1	Year 2	Year 3
Retail Banking	13%	6	18	8
Asset Management	14%	8	10	18
Trading and Sales	19%	9	18	28
Corporate Finance	18%	42	25	20

Based on the Basel II accord, what is the value of the required capital for operational risk under the Basic Indicator approach?

- A. 7.2
- B. 4.0
- C. 10.2
- D. 10.5

The correct answer is **D**.

This method computes the capital for the operational risk as the 15% of the bank's average annual gross income over the past three years while ignoring years that resulted in negative gross income.

So,

Business Line	Annual	Gross	Income
	Year 1	Year 2	Year 3
Retail Banking	6	18	8
Asset Management	8	10	18
Trading and Sales	9	18	28
Corporate Finance	42	25	20
Sum	65	71	74

Note that the multiplier column has been excluded since we do not need it here. Therefore, the required capital for the operational risk is given by:

$$0.15 \left[\frac{65 + 71 + 74}{3} \right] = 10.5 \text{ million}$$

Q.4405 A Canadian bank has assets consisting of CAD 300 million BB-rated drawn loans. The probability of default is estimated (PD) to be 0.01, the LGD is 30%, and DR is estimated to be 0.10. Assuming a maturity adjustment factor of 2.5, what is the RWA for the bank with regard to the Basel II accord?

- A. CAD 100.34 million
- B. CAD 125.53 million
- C. CAD 125.43 million
- D. CAD 253.13 million

The correct answer is **D**.

Recall that retail exposures were calculated similarly to that of advanced IRB only that there is no maturity adjustment. So,

$$\begin{aligned} \text{RWA} &= 12.5 \times \text{EAD} \times \text{LGD} \times (\text{DR} - \text{PD}) \times \text{MA} \\ &= 12.5 \times 300 \times 0.30 \times (0.10 - 0.01) \times 2.5 \\ &= \text{CAD } 253.125 \text{ million} \end{aligned}$$

Note.

- Under Basel II, banks are required to maintain a total capital ratio (Tier 1 + 2 + 3) of minimum 8%. 12.5 is the inverse of 8%. The multiplier has the effect of turning a capital requirement into a RWA measure.
 - EAD = Exposure at Default
 - LGD = Loss Given Default
 - DR = the default rate at the 99.9th percentile for a large portfolio of assets of type i.
 - PD = The bank's own probability of default
-

Q.4406 The bank's probability of default (PD) is estimated to be 0.01. What is the approximated value of the asset correlation in the context of the Basel II framework?

- A. 0.1562
- B. 0.1453
- C. 0.1928
- D. 0.2341

The correct answer is **C**.

Basel II model assumes the Lopez's model given by:

$$\rho = 0.12 \left[\frac{1 - e^{-50PD}}{1 - e^{-50}} \right] + 0.24 \left[1 - \frac{1 - e^{-50PD}}{1 - e^{-50}} \right]$$

Since we are given PD=0.01, then the asset correlation is given by:

$$\rho = 0.12 \left[\frac{1 - e^{-50 \times 0.01}}{1 - e^{-50}} \right] + 0.24 \left[1 - \frac{1 - e^{-50 \times 0.01}}{1 - e^{-50}} \right] = 0.1928$$

Reading 124: Solvency, Liquidity and Other Regulation After the Global Financial Crisis

Q.2347 BastaBank from Bari, Italy, has just adopted Basel II.5 regulations after years of Basel II compliance. The bank's risk management team wants to bring the directors up to speed, particularly with regard to the new requirements under Basel II.5. The team has prepared a report highlighting the main changes. These most likely have a lot to do with:

- A. Calculation of capital requirement for liquidity risk, calculation of stressed VaR, and a new methodology of capital calculation.
- B. Calculation of stressed VaR, a new incremental risk charge, and a comprehensive risk measure for instruments dependent on credit correlation.
- C. A new incremental risk charge, a comprehensive risk measure for instruments dependent on credit correlation, and a new methodology of capital calculation.
- D. A new incremental risk charge, new requirements for IRB parameters calculation, and new requirements for liquidity measurement.

The correct answer is **B**.

The transition from Basel II to Basel II.5 brought about three significant changes. These include the calculation of a stressed Value at Risk (VaR), the introduction of a new incremental risk charge, and a comprehensive risk measure for instruments that are dependent on credit correlation. The stressed VaR is a risk measure that estimates the potential losses a bank could incur under extreme market conditions. The incremental risk charge is a capital charge that covers potential losses arising from changes in credit quality. Lastly, the comprehensive risk measure is a capital requirement that covers potential losses from correlation trading activities. These changes were introduced to enhance the risk sensitivity of the capital framework and to address the shortcomings of the Basel II framework that were exposed during the financial crisis.

Choice A is incorrect. While Basel II.5 does introduce the calculation of stressed VaR, it does not include a new methodology for capital calculation or a specific capital requirement for liquidity risk. These aspects are more associated with Basel III regulations.

Choice C is incorrect. Although Basel II.5 introduces an incremental risk charge and a comprehensive risk measure for instruments dependent on credit correlation, it does not introduce a new methodology of capital calculation.

Choice D is incorrect. The introduction of an incremental risk charge is indeed part of Basel II.5, but the new requirements for IRB parameters calculation and liquidity measurement are not included in this regulatory update.

Things to Remember

- Basel II.5 is an update to the Basel II framework, aimed at addressing the shortcomings

exposed during the financial crisis.

- Stressed Value at Risk (VaR) is a risk measure that estimates potential losses under extreme market conditions.
 - Incremental risk charge covers potential losses from changes in credit quality.
 - Comprehensive risk measure covers potential losses from correlation trading activities.
 - Basel II.5 enhances risk sensitivity of the capital framework.
 - Basel III regulations focus more on liquidity risk and capital calculation methodologies.
-

Q.2348 Cosomora Bank from Eindhoven, in the Netherlands, is one of the largest European banks with a large trading book. The bank has been under Basel II and is currently in the later stages of Basel II.5 implementation. What will be the main effect of shifting from Basel II to Basel II.5?

- A. Capital charges for credit risk will increase.
- B. Capital charges for credit risk will be reduced.
- C. Capital charges for market risk will increase.
- D. Sweeping changes in liquidity measurement techniques.

The correct answer is **C**.

The transition from Basel II to Basel II.5 primarily results in an increase in the capital charges for market risk. This is because Basel II.5 introduces enhancements to the market risk framework, which include the incorporation of stressed Value-at-Risk (VaR) and the incremental risk charge (IRC). These enhancements aim to capture the risk of mark-to-market losses on the trading book due to changes in market conditions. As a result, banks are required to hold more capital against market risk to absorb potential losses, leading to an increase in capital charges for market risk. This is particularly relevant for large banks like Cosomora Bank, which have a significant trading book.

Choice A is incorrect. The transition from Basel II to Basel II.5 primarily impacts the capital charges for market risk, not credit risk. While Basel II.5 does introduce some changes in the calculation of credit risk, it is not the primary focus of this regulatory update.

Choice B is incorrect. Similar to Choice A, this option incorrectly focuses on credit risk instead of market risk. Furthermore, under Basel II.5 framework, there isn't a general reduction in capital charges for credit risk; rather it introduces more sophisticated and sensitive measures for

calculating these risks.

Choice D is incorrect. Although Basel III (the next iteration after Basel II and II.5) places significant emphasis on liquidity measurement techniques, this is not a primary impact of transitioning from Basel II to Basel II.5 which mainly addresses market risks.

Things to Remember

- Basel II.5 is an update to the Basel II framework that primarily focuses on enhancing the market risk framework for banks with significant trading activities.
 - Basel II.5 introduces stressed Value-at-Risk (VaR) and the incremental risk charge (IRC) to capture the risk of mark-to-market losses on the trading book due to changes in market conditions.
 - The transition from Basel II to Basel II.5 results in an increase in capital charges for market risk, as banks are required to hold more capital against market risk to absorb potential losses.
 - Basel II.5 does introduce some changes in the calculation of credit risk, but the primary focus is on enhancing market risk measurement techniques.
 - Basel III, the next iteration after Basel II and II.5, places significant emphasis on liquidity measurement techniques, which is not the primary impact of transitioning from Basel II to Basel II.5.
-

Q.2349 With the introduction of Basel II.5, the Basel Committee requires banks to calculate the so-called stressed VaR. Stressed VaR was introduced mainly because of:

- A. Very high capital requirements because of high volatility of market variables.
- B. Too low VaR as a result of low volatility of market variables.
- C. Increased capital charges for credit risk.
- D. None of the above.

The correct answer is **B**.

The period from 2003 to 2006 was characterized by low volatilities of most market variables.

Consequently, the market risk Value at Risk (VaR) calculated during this period for regulatory capital purposes were also low. Even after the onset of the financial crisis, the VaRs remained low for a while because much of the data used to calculate them still came from the low-volatility period. This led to an underestimation of the actual risk, which could potentially lead to insufficient capital allocation. Therefore, the Basel Committee introduced the concept of 'stressed VaR' under Basel II.5 to address this issue. The 'stressed VaR' is designed to provide a more realistic estimate of potential losses in periods of stress, thereby ensuring that banks hold adequate capital to cover these potential losses.

Choice A is incorrect. The Basel II.5 was not introduced due to very high capital requirements because of high volatility of market variables. In fact, the stressed VaR was introduced to ensure that banks have sufficient capital during periods of stress when market variables are highly volatile.

Choice C is incorrect. The introduction of Basel II.5 and the requirement for 'stressed VaR' were not primarily related to increased capital charges for credit risk. While Basel II did introduce more sophisticated approaches for credit risk, the stressed VaR specifically addresses market risk under stressful conditions.

Choice D is incorrect. There were specific reasons behind the introduction of Basel II.5 and the requirement for 'stressed VaR', hence 'None of the above' does not provide a correct explanation.

Things to Remember

- Basel II.5 was introduced as an amendment to the Basel II framework to address the shortcomings and lessons learned from the global financial crisis of 2007-2008.
- Stressed VaR is a risk measure that aims to capture the potential losses a bank could face during periods of market stress or extreme volatility.
- Market risk VaR is a key component of regulatory capital requirements for banks, and accurate estimation is crucial for maintaining financial stability.
- The Basel Committee on Banking Supervision plays a critical role in setting global standards for bank capital adequacy and risk management.
- Periods of low volatility can lead to underestimation of risk, highlighting the importance of stress testing and scenario analysis in risk management.

Q.2353 Katerini Bank from Greece is in the process of implementing Basel III regulations. One of the first assignments of its risk management team is to calculate the required regulatory capital. In line with Basel III, the bank should have the following categories of capital, **except**:

A. Tier 1 capital

- B. Tier 2 capital
- C. Tier 3 capital
- D. Additional Tier I capital

The correct answer is **C**.

Under the Basel III regulations, there is no category known as Tier 3 capital. The Basel III framework, which was developed by the Basel Committee on Banking Supervision (BCBS) to strengthen the regulation, supervision, and risk management within the banking sector, outlines three main categories of capital that banks are required to maintain. These include Tier 1 equity capital, Additional Tier 1 capital, and Tier 2 capital. Tier 1 capital, which is the highest quality of capital a bank possesses, is further divided into Common Equity Tier 1 (CET1) and Additional Tier 1 (AT1) capital. CET1 capital includes ordinary shares and retained earnings, while AT1 capital consists of instruments that are not CET1 but satisfy other criteria. Tier 2 capital, on the other hand, is a lower form of capital that includes items such as undisclosed reserves, revaluation reserves, general loan-loss reserves, hybrid (debt/equity) capital instruments, and subordinated debt. The concept of Tier 3 capital, which was present in the earlier Basel I and Basel II frameworks, was removed in Basel III. Therefore, the statement that the bank should have Tier 3 capital is incorrect.

Choice A is incorrect. Tier 1 capital, also known as core capital, is a key component of Basel III regulations. It includes the highest quality capital that a bank possesses, such as common equity and disclosed reserves.

Choice B is incorrect. Tier 2 capital, or supplementary capital, is also part of Basel III requirements. This category includes items like undisclosed reserves, revaluation reserves, general loan-loss reserves and hybrid (debt/equity) capital instruments.

Choice D is incorrect. Additional Tier I (AT1) Capital forms part of the regulatory framework under Basel III and comprises securities that are subordinated to most other debts, have no fixed maturity date and no encumbrances.

Things to Remember

- Basel III regulations were introduced to strengthen the regulation, supervision, and risk management within the banking sector.
- Basel III framework outlines three main categories of capital that banks are required to maintain: Tier 1 capital, Tier 2 capital, and Additional Tier 1 capital.
- Tier 1 capital is further divided into Common Equity Tier 1 (CET1) and Additional Tier 1 (AT1) capital.
- Tier 2 capital includes items such as undisclosed reserves, revaluation reserves,

general loan-loss reserves, hybrid (debt/equity) capital instruments, and subordinated debt.

- Tier 3 capital, present in earlier Basel frameworks, was removed in Basel III and is not a category of capital under these regulations.
-

Q.2354 Which of the following presents a component of Tier 1 capital?

- A. Subordinated debt.
- B. Share capital
- C. Goodwill
- D. General loan loss reserves

The correct answer is **B**.

Share capital is a primary component of Tier 1 capital. It refers to the funds that a company raises in exchange for shares. It is a key measure of a company's liquidity and overall financial health. In the context of banking, share capital can be used to absorb losses, thereby protecting depositors and other creditors. It is considered a high-quality capital because it is fully paid-up and available to the bank without any obligations. Therefore, it provides a strong buffer against losses. Share capital is also permanent in nature, meaning it does not have a maturity date and is available to the bank as long as it is in operation. This permanence further enhances its ability to absorb losses. In addition, share capital is freely available to cover losses, as it is not encumbered by contractual or regulatory obligations that could limit its availability in times of stress.

Choice A is incorrect. Subordinated debt is a component of Tier 2 capital.

Choice C is incorrect. Goodwill, while an asset on a company's balance sheet, is not included in Tier 1 capital calculations. This is because goodwill can be highly subjective and difficult to accurately value, making it less reliable as an indicator of financial strength.

Choice D is incorrect. General loan loss reserves is a component of Tier 2 capital.

Things to Remember

- Retained earnings are profits that a company has earned but not distributed to its shareholders. They can be an important component of a bank's capital structure.
- Regulatory capital requirements are set by regulatory bodies to ensure that banks

maintain adequate capital to absorb potential losses and protect depositors.

- Tier 1 capital is considered the core measure of a bank's financial strength and consists of high-quality, permanent capital.
 - Common equity Tier 1 (CET1) capital is a subset of Tier 1 capital and includes common shares and retained earnings.
 - Additional Tier 1 (AT1) capital includes instruments such as perpetual bonds that can absorb losses in times of financial stress.
-

Q.2356 In Basel III, the Basel Committee introduced, among others, a new requirement named *leverage ratio*. The main reason for its introduction was that:

- A. Capital adequacy ratio was too high for many banks.
- B. Banks had too much discretion in the way risk-weighted assets were calculated.
- C. Banks were too undercapitalized.
- D. Banks would have unlimited discretion while calculating their regulatory capital.

The correct answer is **B**.

The Basel Committee introduced the leverage ratio because regulators thought that banks had too much discretion in the way risk-weighted assets were calculated. They have far less discretion in the way “total exposure” is calculated.

Things to Remember

1. Basel III is a global regulatory standard on bank capital adequacy, stress testing, and market liquidity risk. It was introduced by the Basel Committee on Banking Supervision following the financial crisis of 2007-2008 to strengthen the regulation, supervision, and risk management of banks.
 2. The leverage ratio is a non-risk-based capital adequacy measure introduced under Basel III. It is calculated by dividing Tier 1 capital by the bank's total exposures, which include on-balance sheet items, derivative exposures, and off-balance sheet items.
 3. The leverage ratio serves as a backstop to the risk-based capital ratios. It helps ensure that banks have an adequate capital buffer against losses and restricts the level of leverage that banks can take on.
 4. The main reason for the introduction of the leverage ratio was that banks had too much discretion in the way risk-weighted assets were calculated. This could lead to an underestimation of risk and lower capital requirements.
 5. The calculation of the leverage ratio is more straightforward and leaves less room for manipulation compared to the calculation of risk-weighted assets.
-

Q.2358 Berthold Bruhne, a risk manager for the bank of Salzburg, was attending a board meeting where he presented the results of the liquidity coverage ratio (LCR) calculation. According to him, the bank's LCR stood at 152% as of December 31st, 2016, safely above the required minimum. His conclusion was that the bank could survive liquidity disruptions in the next:

- A. 1 year
- B. 60 days
- C. 30 days

D. 15 days

The correct answer is C.

The Liquidity Coverage Ratio (LCR) is a regulatory requirement under the Basel III framework, designed to ensure that banks hold enough high-quality liquid assets to survive a severe liquidity disruption over a 30-day period. The LCR is calculated as the ratio of a bank's High Quality Liquid Assets (HQLA) to its total net cash outflows over the next 30 days. A ratio of 100% or more indicates that the bank has enough liquid assets to meet its short-term obligations, even in a severe liquidity stress scenario. In this case, the Bank of Salzburg's LCR of 152% suggests that it has more than enough liquid assets to survive a 30-day liquidity disruption.

Choice A is incorrect. The liquidity coverage ratio (LCR) is a short-term liquidity measure designed to ensure that banks can withstand a 30-day stress scenario, not a full year. Therefore, an LCR of 152% does not imply that the bank could withstand liquidity disruptions for one year.

Choice B is incorrect. While the LCR is indeed a measure of short-term liquidity, it specifically pertains to a 30-day stress scenario as per Basel III regulations. An LCR of 152% does not indicate that the bank could survive liquidity disruptions for 60 days without additional measures.

Choice D is incorrect. The LCR's purpose is to ensure that banks have enough high-quality liquid assets to survive significant cash outflows over a period of 30 days under stressed conditions, not just for 15 days.

Things to Remember

- The Liquidity Coverage Ratio (LCR) is a key liquidity risk management tool introduced under the Basel III framework to ensure banks have enough high-quality liquid assets to survive a 30-day liquidity disruption.
- An LCR ratio of 100% or more indicates that a bank has sufficient liquid assets to meet its short-term obligations even under severe stress scenarios.
- High Quality Liquid Assets (HQLA) are assets that can be easily and quickly converted into cash with little to no loss in value, such as cash, central bank reserves, and certain government securities.
- Net cash outflows are calculated by considering expected cash outflows minus expected cash inflows over the next 30 days, taking into account various scenarios and assumptions.
- Having a higher LCR ratio than the regulatory minimum indicates a bank's ability to

withstand liquidity disruptions and meet its obligations without relying on additional funding sources.

Q.2359 In line with Basel III, the LCR is calculated as the:

- A. Ratio between stable funding and high-quality liquid assets.
- B. Ratio between high-quality liquid assets and total assets.
- C. Ratio between high-quality liquid assets and net cash outflows in a 30-day period.
- D. Ratio between stable funding and net cash outflows in a 30-day period.

The correct answer is **C**.

The Liquidity Coverage Ratio (LCR) is indeed calculated as the ratio between high-quality liquid assets and net cash outflows over a 30-day period. This ratio is a key component of Basel III, a set of international banking regulations developed by the Basel Committee on Banking Supervision. The LCR is designed to ensure that banks have enough high-quality liquid assets on hand to survive a severe liquidity stress scenario lasting 30 days. High-quality liquid assets are those that can be easily and immediately converted into cash with little or no loss of value. Net cash outflows are the total expected cash outflows minus total expected cash inflows occurring in the next 30 days. The LCR must be greater than or equal to 100%, indicating that a bank has enough liquid assets to cover its total net cash outflows for 30 days.

Choice A is incorrect. The ratio between stable funding and high-quality liquid assets is not the Liquidity Coverage Ratio (LCR) according to Basel III. This ratio refers more to the Net Stable Funding Ratio (NSFR), another standard introduced by Basel III, which aims to promote resilience over a longer-term structural horizon by requiring banks to fund their activities with sufficiently stable sources of funding.

Choice B is incorrect. The LCR does not involve a ratio between high-quality liquid assets and total assets. Total assets include both liquid and illiquid assets, whereas the LCR specifically focuses on high-quality liquid assets that can be readily converted into cash in times of stress.

Choice D is incorrect. The ratio between stable funding and net cash outflows in a 30-day period does not define the LCR as per Basel III regulations. This choice seems to mix elements from both the LCR and NSFR standards but does not accurately represent either.

Things to Remember

- Basel III: Basel III is a global regulatory framework on bank capital adequacy, stress testing, and market liquidity risk. It aims to strengthen the regulation, supervision, and risk management within the banking sector.

- High-Quality Liquid Assets (HQLA): These are assets that can be easily and quickly converted into cash without significant loss of value. Examples include cash, central bank reserves, and certain government securities.
 - Net Cash Outflows: This refers to the total expected cash outflows minus total expected cash inflows over a specified period, such as 30 days. It helps measure a bank's liquidity risk exposure.
 - Net Stable Funding Ratio (NSFR): The NSFR is another liquidity standard under Basel III that focuses on the stability of a bank's funding sources over a longer-term horizon (one year). It aims to ensure that banks maintain a stable funding profile in relation to the composition of their assets and off-balance sheet activities.
-

Q.2360 CIB Bank from Oklahoma City, USA, is a G-SIB, as classified by the Financial Stability Board. This implies that:

- A. The bank is recognized globally as a “Solid Investment Bank”.
- B. The bank’s failure could lead to a global economic crisis.
- C. The bank has a global reach and has been successful for an extended period.
- D. The bank’s “Sustained International Business” ratio is high in all operational countries.

The correct answer is **B**.

The term G-SIB stands for global systemically important bank. Their failure could be nearly catastrophic, triggering a market-wide disruption that could lead to a financial crisis. The systemic importance of a bank or other financial institution depends on the effect that its failure could have on the global financial system. This, in turn, depends on the nature of its activities and the contracts it has entered into with other financial institutions globally.

Things to Remember

- Global Systemically Important Banks (G-SIBs) are institutions whose failure could have a significant negative impact on the global financial system.
- G-SIBs are subject to higher capital requirements and stricter regulatory oversight due to their systemic importance.
- The Financial Stability Board (FSB) identifies and classifies G-SIBs based on criteria such as size, interconnectedness, global activity, complexity, and substitutability.
- G-SIBs are expected to have robust risk management practices and contingency plans in place to mitigate the risks they pose to the financial system.

Q.2741 All of these are changes that were implemented through Basel 2.5, except:

- A. Calculation of a stressed VaR.
- B. Implementation of a new incremental risk change (IRC).
- C. A comprehensive risk measure (CRM) for instruments sensitive to correlations between default risks of various instruments.

D. Calculation of the net stable funding ratio (NSFR) and the liquidity coverage ratio (LCR).

The correct answer is **D**.

The calculation of the net stable funding ratio (NSFR) and the liquidity coverage ratio (LCR) were not changes implemented through Basel 2.5. These are liquidity measures that were introduced later, under Basel III. Basel III was developed in response to the deficiencies in financial regulation revealed by the financial crisis of 2007–2008. It is intended to strengthen bank capital requirements by increasing bank liquidity and decreasing bank leverage. The NSFR is designed to ensure that long-term assets are funded with at least a minimum amount of stable liabilities in order to mitigate the risk of future funding stress. The LCR is a requirement that banks hold an amount of high-quality liquid assets that's enough to cover its total net cash outflows over 30 days. Both of these measures aim to improve the banking sector's ability to absorb shocks arising from financial and economic stress, thus reducing the risk of spill over from the financial sector to the real economy.

Choice A is incorrect. Basel 2.5 did introduce the calculation of a stressed VaR (Value at Risk). This was done to ensure that banks have enough capital to cover extreme losses during periods of significant financial stress.

Choice B is incorrect. The implementation of a new incremental risk charge (IRC) was indeed part of the changes brought about by Basel 2.5. The IRC was designed to capture the default and migration risks for unsecuritized credit products.

Choice C is incorrect. Basel 2.5 also introduced a comprehensive risk measure (CRM) for instruments sensitive to correlations between default risks of various instruments, in order to better manage and mitigate systemic risk.

Things to Remember

- Basel II.5 was an update to the Basel II framework, aimed at addressing some of the shortcomings and weaknesses identified in the original Basel II regulations.
- Stressed VaR is a risk measure that calculates the potential losses a portfolio could face during periods of financial stress.
- Incremental Risk Charge (IRC) is a capital charge that covers the risk of default and migration for unsecuritized credit products.
- Comprehensive Risk Measure (CRM) helps in managing and mitigating systemic risk by considering correlations between default risks of different instruments.
- Net Stable Funding Ratio (NSFR) and Liquidity Coverage Ratio (LCR) are liquidity measures introduced under Basel III to strengthen bank liquidity and decrease

leverage.

- Basel III was developed in response to the financial crisis of 2007-2008 to enhance the resilience of the banking sector.
-

Q.2743 Which of the following correctly describes the time horizon considered by the Liquidity Coverage Ratio (LCR) and the Net Stable Funding Ratio (NSFR)?

- A. LCR: Focuses on a 30-day period; NSFR: Focuses on a 2-year period.
- B. LCR: Focuses on a 1-year period; NSFR: Focuses on a 30-day period.
- C. LCR: Focuses on a 2-year period; NSFR: Focuses on a 30-day period.
- D. LCR: Focuses on a 30-day period; NSFR: Focuses on a 1-year period.

The correct answer is **D**.

The Liquidity Coverage Ratio (LCR) focuses on a 30-day period, while the Net Stable Funding Ratio (NSFR) focuses on a 1-year period. The LCR is designed to ensure that a bank has an adequate stock of unencumbered high-quality liquid assets (HQLAs) that can be converted into cash to meet its liquidity needs for a 30-day time horizon under a significantly severe liquidity stress scenario. On the other hand, the NSFR is a longer-term structural ratio designed to address liquidity mismatches and provide incentives for banks to use stable sources of funding. It aims to measure the amount of longer-term, stable sources of funding employed by an institution relative to the liquidity profiles of the assets funded and the potential for contingent calls on funding liquidity arising from off-balance sheet commitments and obligations, over a one-year horizon.

Choice A is incorrect because while it correctly identifies the time horizon for the LCR as a 30-day period, it incorrectly states that the NSFR focuses on a 2-year period. The NSFR actually focuses on a 1-year period. The NSFR is designed to ensure that long-term assets are funded with at least a minimum amount of stable liabilities in relation to their liquidity risk profiles over a one-year period.

Choice B is incorrect because it incorrectly identifies the time horizons for both the LCR and the NSFR. The LCR focuses on a 30-day period, not a 1-year period as stated in this choice. Similarly, the NSFR focuses on a 1-year period, not a 30-day period. The LCR is designed to ensure that a bank has an adequate stock of unencumbered high-quality liquid assets (HQLAs) that can be converted into cash to meet its liquidity needs for a 30-day time horizon. The NSFR, on the other hand, is designed to ensure that long-term assets are funded with at least a minimum amount of stable liabilities in relation to their liquidity risk profiles over a one-year period.

Choice C is incorrect because it incorrectly identifies the time horizons for both the LCR and the NSFR. The LCR focuses on a 30-day period, not a 2-year period as stated in this choice.

Similarly, the NSFR focuses on a 1-year period, not a 30-day period. The LCR is designed to ensure that a bank has an adequate stock of unencumbered high-quality liquid assets (HQLAs) that can be converted into cash to meet its liquidity needs for a 30-day time horizon. The NSFR, on the other hand, is designed to ensure that long-term assets are funded with at least a minimum amount of stable liabilities in relation to their liquidity risk profiles over a one-year period.

Things to Remember

- The Liquidity Coverage Ratio (LCR) is part of the Basel III regulatory framework and is aimed at promoting short-term resilience of a bank's liquidity risk profile.
 - The Net Stable Funding Ratio (NSFR) complements the LCR by promoting more medium-to long-term structural liquidity by ensuring that banks have stable funding sources to support their activities over a one-year horizon.
 - HQLAs (High-Quality Liquid Assets) are a key component in calculating the LCR and are assets that can be easily and quickly converted into cash with little to no loss in value.
 - The LCR and NSFR are important tools for regulators to monitor and ensure the stability and resilience of the banking system, especially during times of financial stress.
 - Both ratios are designed to prevent liquidity crises and promote a more stable and sustainable banking sector.
-

Q.3237 Exim Bank estimates its stable funding to be \$100 million. Further, net cash outflows over the coming 30 days are estimated to hit \$155 million. Exim bank has capital of \$10 million and its total exposure stands at \$150 million. The bank's high-quality liquid assets are valued at \$140 million.

Determine the bank's liquidity coverage ratio (LCR) as stipulated in Basel III.

A. 0.9032

B. 0.875

C. 1.1

D. 1.4

The correct answer is **A**.

According to Basel III rules, the bank needs a minimum liquidity coverage ratio (LCR) of 100%. The LCR focuses on the bank's ability to see it through a 30-day period of disrupted liquidity. The LCR formula is as follows:

$$\text{LCR} = \frac{\text{high-quality liquid assets}}{\text{net cash outflows in a 30-day period}}$$

In this case,

$$\text{LCR} = \frac{\$140 \text{ million}}{\$155 \text{ million}} = 0.9032 = 90.3\%$$

It's evident that Exim bank has not met the minimum 100% requirement and is in violation of the rule.

Q.3242 A bank has a previous-period stressed VAR of \$20 million, a multiplication factor (M) of 4, and a stressed VAR average over the previous 60 trading days of \$7 million. Which of the following values is the correct stressed VAR amount for this bank?

- A. \$28 million
- B. \$20 million
- C. \$48 million
- D. \$8 million

The correct answer is **A**.

The calculation of SVAR is defined as follows:

$$\begin{aligned} & \max(\text{previous SVAR}, M \times \text{average SVAR}) \\ & = \max[\$20 \text{ million or } \$28 \text{ million } (4 \times \$7 \text{ million})] \end{aligned}$$

Therefore, the max amount is \$28 million.

Q.3245 Question: Bank ABC is subject to Basel III regulations. The bank has the following balance sheet information (in millions):

- Tier 1 Capital: \$250
- Tier 2 Capital: \$150
- Total Assets: \$2,000
- On-balance-sheet derivative exposures: \$100
- Off-balance-sheet derivative exposures: \$150
- Repurchase agreement exposures: \$200

Under Basel III, what is the leverage ratio for Bank ABC?

- A. 5.77%
- B. 10.20%
- C. 13.08%

D. 9.10%

The correct answer is **B**.

Under Basel III, the leverage ratio is calculated as Tier 1 Capital divided by the exposure measure. The exposure measure includes total assets, off-balance-sheet items, and derivative exposures.

First, calculate the exposure measure:

Exposure Measure = Total Assets + On-balance-sheet Derivative Exposures + Off-balance-sheet Derivative Exposures + Repurchase Agreement Exposures

Exposure Measure = \$2,000 + \$100 + \$150 + \$200 = \$2,450 million

Next, calculate the leverage ratio:

Leverage Ratio = (Tier 1 Capital / Exposure Measure) × 100

Leverage Ratio = (\$250 / \$2,450) × 100

Leverage Ratio = 0.10204 × 100

Leverage Ratio = 10.204% ≈ 10.20% (rounded to two decimal places)

Things to Remember

- The leverage ratio is a key regulatory requirement under Basel III to ensure that banks maintain an adequate level of capital to support their activities.
- Tier 1 Capital is considered the highest quality capital and includes common equity tier 1 (CET1) capital and additional tier 1 capital instruments.
- Tier 2 Capital includes items such as subordinated debt, hybrid instruments, and other capital instruments that provide additional loss-absorbing capacity.
- Total Assets represent the sum of all assets on the bank's balance sheet, including both on-balance-sheet and off-balance-sheet items.
- Derivative exposures, both on-balance-sheet and off-balance-sheet, are included in the exposure measure calculation to capture the potential risks associated with these financial instruments.
- Repurchase agreement exposures are also considered in the exposure measure

calculation as they represent off-balance-sheet commitments that could pose risks to the bank.

- The leverage ratio is expressed as a percentage and is calculated by dividing Tier 1 Capital by the exposure measure and multiplying by 100.
-

Q.3246 Consider a bank balance sheet with

1. Common stock of \$730,000,000;
2. Unrealized long-term marketable equity securities gain: \$33,000,000;
3. Five -year subordinated debt: \$28,000,000;
4. Goodwill: \$92,000,000.

Based Basel III capital requirements and solely on the above information, the tier 1 and tier 2 capital numbers are, respectively:

- A. \$730,000,000 and \$0
- B. \$730,000,000 and \$61,000,000
- C. \$822,000,000 and \$33,000,000
- D. \$671,000,000 and \$28,000,000

The correct answer is **D**.

Tier 1 capital consists of equity plus unrealized gains/losses less goodwill = $730 + 33 - 92 = \$671$ million.

Tier 2 capital includes five year subordinated debt of \$28 million.

Things to Remember

- Basel III Capital Requirements are regulatory standards introduced after the 2008 financial crisis to strengthen the regulation, supervision, and risk management within the banking sector.
- Tier 1 Capital is the core measure of a bank's financial strength from a regulator's point of view. It includes common equity tier 1 capital and additional tier 1 capital.
- Tier 2 Capital is supplementary capital that includes items such as subordinated debt, hybrid instruments, and collective impairment provisions.
- Goodwill represents the premium paid for a company over its book value and is considered a non-core banking asset.
- Unrealized gains or losses on marketable securities are included in the calculation of regulatory capital to reflect the potential volatility in the value of these assets.

Q.3250 Steve Warne is an advisor at a local Bank which is attempting to transition to the new Basel III standards. Specifically, they are wondering if their liquidity and funding ratios meet the updated requirements as specified by the Basel Committee. Given the following information, what is the bank's current liquidity coverage ratio?

- High-quality liquid assets = \$236
- Marketable securities = \$107
- Required amount of stable funding (RSF) = \$320
- Cash inflows over the next 30 days = \$214
- Cash outflows over the next 30 days = \$487
- Long-term economic capital = \$640
- Available amount of stable funding (ASF) = \$305

A. 48.46%

B. 86.45%

C. 206.3%

D. -115.67%

The correct answer is **B**.

The 30-day liquidity coverage ratio (LCR) = $\frac{\text{the stock of high-quality liquid assets}}{\text{the net cash outflow over a 30-day period.}}$

Under Basel III, this ratio must equal or exceed 100%.

$$\text{Net cash outflow} = 214 - 487 = -\$273$$

$$\text{Bank's liquidity coverage ratio} = \frac{\$236}{\$273} = 0.8645 = 86.45\%.$$

Note that a negative cash outflow implies a net cash inflow.

Q.4285 After the global financial crisis, it was realized that the minimum capital charges under

the market risk amendment were not sufficient to address trading book risks. Which of the following is one of the significant changes implemented in 2011 to address these trading book risks, which was later known as Basel 2.5?

- A. VaR computation was tailored to include a stressed VaR component
- B. A portion of operational risk was required on top of credit and market risk
- C. The risk weights in credit risk formulas were to be based on modern credit risk and banks' internal measures
- D. It was ruled out that the Tier 1 capital was necessary for the preservation of maintenance, while Tier 2 capital was to be used for the recapitalization of a financial institution in resolution and decrease the level of failures on the depositors

The correct answer is **A**.

The global financial crisis of 2007-2009 exposed the inadequacy of the minimum capital charges under the market risk amendment in addressing the underlying trading-book risks. In response to this, the Basel Committee introduced several changes, one of which was the inclusion of a stressed VaR component in the VaR computations. This change was aimed at ensuring that the VaR calculations took into account extreme market conditions, thereby providing a more accurate measure of the potential losses that could be incurred in such scenarios. This change was part of a broader set of reforms known as Basel 2.5, which were designed to strengthen the resilience of the banking sector and reduce the likelihood of future financial crises.

Choice B is incorrect. Operational risk, while important, was not the primary focus of Basel 2.5 amendments. The changes were primarily aimed at addressing trading book risks and did not mandate additional operational risk coverage on top of credit and market risk.

Choice C is incorrect. While modernizing credit risk measures and aligning them with banks' internal measures is a crucial aspect of overall risk management, it was not a specific change implemented under Basel 2.5 to address trading book risks.

Choice D is incorrect. The classification of Tier 1 and Tier 2 capital for preservation of maintenance or recapitalization purposes does not directly relate to the changes made in Basel 2.5 for addressing trading book risks.

Things to Remember

- **Basel Accords:** The Basel Accords are a set of international banking regulations developed by the Basel Committee on Banking Supervision to ensure financial institutions maintain adequate capital reserves to cover risks.
- **Market Risk:** Market risk is the risk of losses in on and off-balance sheet positions arising from adverse movements in market prices.

- Value at Risk (VaR): VaR is a statistical measure used to quantify the level of financial risk within a firm or portfolio over a specific time frame.
 - Operational Risk: Operational risk is the risk of loss resulting from inadequate or failed internal processes, people, and systems, or from external events.
 - Tier 1 and Tier 2 Capital: Tier 1 capital is the core measure of a bank's financial strength, while Tier 2 capital is supplementary capital that includes items such as undisclosed reserves, revaluation reserves, and hybrid instruments.
-

Q.4286 According to Basel regulations (Basel 2.5) on stressed VaR, how is the stressed period identified?

- A. Banks must use the most recent one-year period of historical data to calculate stressed VaR.
- B. Banks must identify the most stressful one-year (250 day) period from the most recent seven years.
- C. Banks can use any one-year period from their historical data that shows the highest volatility.
- D. Banks must use a fixed one-year period from the 2007-2008 financial crisis for all portfolios.

The correct answer is **B**.

According to Basel 2.5, Stressed VaR is calculated by identifying a one-year period (equivalent to 250 trading days) from the most recent seven years that was most stressful for a bank's current portfolios. This period is identified based on the bank's current risk profile and the historical market conditions during that year. The Stressed VaR is then calculated using the risk factors and market data from this identified stressful period. This approach ensures that the bank's capital requirement is sufficient to cover potential losses under extreme market conditions similar to those experienced during the identified stressful year.

Choice A is incorrect. Using only the most recent one-year period would defeat the purpose of stressed VaR, which is to counter the tendency of VaR to decline during periods of low volatility.

Choice C is incorrect. Banks cannot choose any historical period; they are specifically limited to looking at the most recent seven years of data.

Choice D is incorrect. The stressed period is portfolio-specific and must be the most stressful period for the bank's current portfolio, not a fixed period that applies to all banks regardless of

their holdings.

Things to Remember

- Stressed VaR is a risk measure used in Basel 2.5 to assess a bank's capital requirement under extreme market conditions.
 - It is important to select a one-year period from the most recent seven years that exhibited stress in the bank's current portfolio to calculate Stressed VaR.
 - The historical market conditions during the identified stressful year are crucial in determining the Stressed VaR.
 - Stressed VaR helps ensure that banks have enough capital to cover potential losses during severe market downturns.
-

Q.4287 The 99% 10-day VaR for ABC Bank is \$800. The average 99% VaR for the recent 60 days is \$360. Over the past seven years, the most stressful 10-day 99% VaR is \$950 and the most stressful 60-day average 99% VaR is \$370. The multiplier on the average 99% VaR for the recent 60 days is 3.0, and that of the most stressful average 99% VaR for the recent 60 days over the past seven years is 3.5. What is the estimated market risk capital charge for this bank under Basel 2.5?

- A. \$1,850
- B. \$1,160
- C. \$2,375
- D. \$2,460

The correct answer is **C**.

As per Basel 2.5, the market risk is given by:

$$MR_{2.5} = \max(VaR_{t-1}, m_r VaR_{avg}) + \max(SVaR_{t-1}, m_s SVaR_{avg})$$

Where:

- VaR_{t-1} is the traditional 10-day, 99% VaR from the previous day.

- VaR_{avg} is the 99% average VaR of the most recent days.
- SVaR_{t-1} is the most stressful 10-day 99% VaR over the past seven years.
- SVaR_{avg} is the most stressful 60-day average 99% VaR over a period of seven years.
- $m_r = 3$ and $m_s = 3.5$ are the respective multipliers.

Given:

- $\text{VaR}_{t-1} = \$800$
- $\text{VaR}_{\text{avg}} = \360
- $\text{SVaR}_{t-1} = \$950$
- $\text{SVaR}_{\text{avg}} = \370

The market risk capital charge is calculated as:

$$\begin{aligned}\text{MR}_{2.5} &= \max(\$800, 3 \times \$360) + \max(\$950, 3.5 \times \$370) \\ &= \max(\$800, \$1080) + \max(\$950, \$1295) \\ &= \$1080 + \$1295 = \$2375\end{aligned}$$

Therefore, the estimated market risk capital charge for ABC Bank under Basel 2.5 is \$2,375.

Things to Remember

- Basel 2.5 is an international regulatory framework that sets out the minimum capital requirements for market risk that banks need to hold.
- Value at Risk (VaR) is a statistical measure used to quantify the level of financial risk within a firm or portfolio over a specific time frame.
- Multiplier is used to adjust the VaR based on historical stress levels to ensure that the capital charge adequately covers potential losses.
- Market risk capital charge is a key component of a bank's overall capital requirement and is designed to cover potential losses from adverse market movements.
- Understanding the different components of market risk, such as VaR and stress VaR, is crucial for calculating the capital charge accurately.

Q.4288 Which of the following is one of the variants of calculating incremental default risk charge (IDRC) as proposed by Basel 2.5?

- A. A standardized approach similar to that of Basel I
- B. Current exposure similar to that of Basel I
- C. Internal rating-based (IRB) based on a one-year time horizon
- D. All of the above

The correct answer is **D**.

The Basel Committee proposed adding IDRC to specific risk which through two forms:

- An internal model of default risk tailored to 99.9th percentile at one-year time horizon similar to the IRB approach
- When the internal model is unavailable, either standardized or current exposure approach of calculating specific risk similar to that of Basel I.

Things to Remember

- Incremental Default Risk Charge (IDRC) is a measure introduced by Basel 2.5 to capture the incremental default risk of specific risk in the trading book.
 - Basel II.5 aimed to enhance the capital requirements for specific risk in the trading book by introducing IDRC.
 - The IDRC is calculated based on a one-year time horizon and is tailored to the 99.9th percentile of the distribution of potential losses.
 - Internal Rating-Based (IRB) approach allows banks to use their internal models to calculate IDRC based on their specific portfolios and risk profiles.
 - The standardized approach and current exposure approach are alternatives provided for calculating IDRC when internal models are not available.
-

Q.4289 Which of the following are the components of Tier 1 capital in the context of Basel III capital definition?

- A. Common equity and Retained earnings
- B. General loan loss reserves
- C. Subordinated debt
- D. Goodwill and other intangibles

The correct answer is **A**.

Tier 1 capital is divided into:

- Tier 1 Equity Capital, also called Core Tier 1 capital.
- Additional Tier 1 Capital.

Tier 1 Equity Capital consisted of the following:

- Common equity.
- Retained earnings.
- Limited amount of minority interest and unrealized gains and losses.

Additional Tier 1 Capital includes:

- Unguaranteed, unsecured, non-cumulative perpetual preferred equity instruments subordinated to depositors and subordinated debt callable after five or more years.
- Debt with suitable factors leads to conversion to equity or write-downs.
- Approved minority interest excluded in Core Tier 1 capital.

A and B are incorrect. These are components of Tier 2 capital.

D is incorrect. Goodwill and other intangibles are actually deducted when calculating Tier 1 capital.

Things to Remember

- Basel III is a global regulatory framework on bank capital adequacy, stress testing, and market liquidity risk agreed upon by the members of the Basel Committee on Banking

Supervision.

- Tier 1 capital is the core measure of a bank's financial strength from a regulator's point of view.
 - Common equity and retained earnings are the most important components of Tier 1 capital as they represent the highest quality of capital.
 - General loan loss reserves are part of Tier 2 capital, which is less loss-absorbing than Tier 1 capital.
 - Subordinated debt is included in Additional Tier 1 Capital, which is a supplementary measure of a bank's financial strength.
 - Goodwill and other intangibles are deducted from Tier 1 capital as they do not represent tangible assets that can absorb losses in times of financial distress.
-

Q.4290 Assume that a bank has common equity of \$100 million, retained earnings of \$80 million, minority interest and unreleased gains and losses of \$20 million, and goodwill and other intangibles of \$5 million. What is the value of Tier 1 equity capital in the context of the Basel III accord?

- A. \$190 million
- B. \$195 million
- C. \$205 million
- D. \$100 million

The correct answer is **B**.

In the context of Basel III, Tier 1 equity capital consists of common equity, retained earnings, and a limited amount of minority interest and unrealized gains and losses less goodwill and other intangibles. So, in this case:

Tier 1 equity capital = $100 + 80 + 20 - 5 = \$195$

Things to Remember

- Common Equity
 - Retained Earnings
 - Minority Interest
 - Unrealized Gains and Losses
 - Goodwill and Other Intangibles
 - Tier 1 Capital Ratio
 - Basel III Accord
-

Q.4291 The estimated risk-weighted assets of a bank is \$200 million. In the context of Basel III, the Core Tier 1 (Tier 1 Equity Capital) of the bank is at least:

- A. \$10 million
- B. \$4.5 million
- C. \$9 million
- D. \$12 million

The correct answer is **C**.

Basel III changed the minimum capital requirements such that the Core Tier 1 capital must be at least 4.5% of the risk-weighted assets (RWA). So, in this case, the Core Tier 1 must be at least:

$$4.5\% \times \$200 \text{ million} = \$9 \text{ million}$$

Q.4292 The estimated risk-weighted assets of a bank stand at \$400 million. In the context of Basel III, what is the bank's minimum Tier 1 capital?

- A. \$18 million
- B. \$12 million
- C. \$16 million
- D. \$24 million

The correct answer is **D**.

The minimum Tier 1 capital increases from 4% in Basel II to 6%, applicable in 2015, over RWAs. This 6% is composed of 4.5% of CET1, plus an extra 1.5% of Additional Tier 1 (AT1).

So in this case, Tier I capital must be at least \$24 million:

$$0.06\% \times \$400 \text{ million} = \$24 \text{ million}$$

Q.4293 In the context of Basel III, the Tier 2 capital is designed to address the losses after failure and thus protects the depositors and other creditors of the bank. Which of the following is a component of Tier 2 capital?

- A. Common equity
- B. General loan loss reserves
- C. Retained earnings
- D. Approved minority interest not included in Core Tier 1

The correct answer is **B**.

Components of Tier 2 Capital include subordinated debt and general loan loss reserves are components. Subordinated debt refers to unsecured, unguaranteed debt instruments that are subordinated to depositors and other creditors. These instruments must have a maturity of five or more years and can only be called after five or more years. On the other hand, general loan loss reserves are not allocated to absorb losses on specific positions. They include capital limited at 1.25% of standardized approach Risk-Weighted Assets (RWAs) or 0.6% of Internal Ratings-Based (IRB) RWAs.

A and C are incorrect. These are components of Tier 1 Capital.

D and is incorrect. This is a component of Additional Tier 1 Capital

Things to Remember

- Basel III is a global regulatory framework on bank capital adequacy, stress testing, and market liquidity risk agreed upon by the members of the Basel Committee on Banking Supervision.
- Tier 1 capital is the core measure of a bank's financial strength from a regulator's point of view, consisting of common equity and disclosed reserves.
- Tier 2 capital is supplementary capital that includes items such as subordinated debt, general loan loss reserves, hybrid instruments, and other undisclosed reserves.
- General loan loss reserves are provisions set aside by banks to cover potential losses from non-performing loans and are part of Tier 2 capital.
- Approved minority interest not included in Core Tier 1 is a component of Additional Tier 1 capital, not Tier 2 capital.

Q.4295 Which of the following statements correctly describes Systemically Important Financial Institutions (SIFIs)?

- A. They are the entities subject to less supervision and regulation
- B. They are entities whose failure or distress will affect the whole market or the whole economy.
- C. They are the entities whose failure affects only its stakeholder but not the broader market system or the economy
- D. They are the market entities whose failure can be reversed by government financing without affecting its stakeholders

The correct answer is **B**.

Systemically Important Financial Institutions (SIFIs) are indeed entities whose failure or distress can have a significant impact on the entire market or economy. This is because SIFIs are typically large, interconnected financial institutions that play a critical role in the functioning of the financial system. Their operations are so integral to the financial system that their failure could trigger a cascade of failures among other financial institutions, leading to a systemic crisis. This is why SIFIs are subject to additional regulatory scrutiny and are required to have robust risk management systems in place to prevent their failure.

Choice A is incorrect. Systemically Important Financial Institutions (SIFIs) are not subject to less supervision and regulation. In fact, due to their significant role in the financial system and potential impact on the economy, they are often subject to more stringent regulations and oversight compared to other financial institutions.

Choice C is incorrect. The failure of a SIFI does not only affect its stakeholders but also has far-reaching implications for the broader market system or economy. This is because SIFIs play a crucial role in maintaining financial stability, and their distress or failure can lead to systemic risks.

Choice D is incorrect. While government financing can sometimes be used as a measure to prevent the failure of a SIFI from causing widespread economic disruption, it does not mean that such interventions would have no effect on its stakeholders. Stakeholders may still suffer losses or face other negative consequences as a result of such interventions.

Things to Remember

- **Systemic Risk:** SIFIs are key contributors to systemic risk, which is the risk of a widespread financial collapse or crisis that can have severe adverse effects on the economy as a whole.
- **Regulatory Capital Requirements:** SIFIs are often required to hold higher levels of regulatory capital to ensure their stability and ability to withstand financial shocks.

- Resolution Plans: SIFIs are typically required to develop detailed resolution plans, also known as living wills, to outline how they could be resolved in an orderly manner in the event of financial distress or failure.
 - Global Systemically Important Banks (G-SIBs): Some SIFIs are designated as G-SIBs by international regulatory bodies like the Financial Stability Board, indicating their global systemic importance.
-

Q.4296 The liquidity coverage ratio (LCR) of a bank is approximated to be 1.30. Under Basel III liquidity requirements, does the bank fulfill the required LCR?

- A. No, because $LCR > 1$
- B. Yes, because $LCR > 1$
- C. No, because $LCR < 2$
- D. Yes, because $LCR < 2$

The correct answer is **B**.

The Basel III regulatory framework was introduced by the Basel Committee on Banking Supervision to strengthen the regulation, supervision, and risk management within the banking sector following the 2008 financial crisis. One of the key measures introduced under Basel III is the Liquidity Coverage Ratio (LCR). The LCR is designed to ensure that banks have an adequate stock of unencumbered high-quality liquid assets (HQLA) that can be converted into cash to meet their liquidity needs for a 30-day time horizon under a significantly severe liquidity stress scenario. The LCR is defined as the ratio of the stock of HQLA to total net cash outflows over the next 30 calendar days. A bank's LCR must be greater than or equal to 1. This means that the bank's stock of high-quality liquid assets should be at least equal to its total net cash outflows over the next 30 days. In this case, the bank's LCR is 1.30, which is greater than 1. Therefore, the bank is in compliance with the Basel III liquidity requirements.

Choice A is incorrect. The statement that the bank is not in compliance because $LCR > 1$ is incorrect. According to Basel III regulations, a bank's LCR should be greater than or equal to 1. This means that the bank has enough high-quality liquid assets to meet its net cash outflows for a 30-day period.

Choice C is incorrect. The assertion that the bank isn't compliant because its $LCR < 2$ isn't correct either. There's no requirement under Basel III for a bank's LCR to be less than or equal to 2.

Choice D is incorrect. The claim that the bank complies with Basel III liquidity requirements because its $LCR < 2$ doesn't hold true as well. As per Basel III, there's no upper limit on how

high an institution's LCR can be; it just needs to be at least one.

Things to Remember

- Basel III regulatory framework was introduced to strengthen the regulation, supervision, and risk management within the banking sector after the 2008 financial crisis.
 - Liquidity Coverage Ratio (LCR) is a key measure under Basel III to ensure banks have enough high-quality liquid assets to meet their liquidity needs for a 30-day period under stress scenarios.
 - LCR is calculated as the ratio of the stock of High-Quality Liquid Assets (HQLA) to total net cash outflows over the next 30 days.
 - A bank's LCR must be greater than or equal to 1 to comply with Basel III requirements.
 - There is no upper limit on how high an institution's LCR can be under Basel III; it just needs to be at least 1.
-

Reading 125: High-level Summary of Basel III Reforms

Q.3092 The following are motivations for revising the Basel III framework EXCEPT:

- A. To align definitions with the internal ratings-based approach (IRB) by introducing a new definition for default.
- B. To expand banks' borrowing powers to enable them mitigate market risk in periods of stress.
- C. To improve liquidity by requiring banks to hold liquid assets sufficient to run the bank for 30 days during times of stress.
- D. To limit procyclicality by requiring banks to hold sufficient retained earnings that can be drawn down during periods of economic stress.

The correct answer is **B**.

The motivation to expand banks' borrowing powers to enable them to mitigate market risk in periods of stress was not a reason for revising the Basel III framework. In fact, the opposite is true. The revised Basel III framework aimed to limit the use of leverage by banks. This was done in response to market analysis that revealed that banks often borrowed excessively, which only exacerbated financial pressure during times of stress. Therefore, the revised requirements sought to further restrict the use of debt among banks, rather than expand their borrowing powers.

Choice A is incorrect. The Basel III framework did indeed introduce a new definition for default to align definitions with the internal ratings-based approach (IRB). This was done to ensure consistency in risk measurement and management across banks.

Choice C is incorrect. Improving liquidity by requiring banks to hold liquid assets sufficient to run the bank for 30 days during times of stress was one of the key motivations behind the revisions in Basel III framework. This requirement, known as Liquidity Coverage Ratio (LCR), ensures that banks have an adequate stock of unencumbered high-quality liquid assets that can be converted into cash easily and immediately in private markets.

Choice D is incorrect. The aim to limit procyclicality by requiring banks to hold sufficient retained earnings that can be drawn down during periods of economic stress was indeed one of the motivations behind Basel III's revisions. This measure helps ensure that banks have a buffer during downturns, thereby reducing their vulnerability and enhancing their resilience.

Things to Remember

- Basel III framework: A set of international banking regulations developed by the Basel Committee on Banking Supervision to strengthen regulation, supervision, and risk management within the banking sector.

- Internal Ratings-Based Approach (IRB): A risk-sensitive approach used by banks to calculate regulatory capital requirements based on their internal risk assessments.
 - Liquidity Coverage Ratio (LCR): A regulatory requirement that banks hold enough high-quality liquid assets to cover net cash outflows for a 30-day stress period.
 - Procyclicality: The tendency of the financial system to magnify economic fluctuations, where risk-taking behavior by banks and other financial institutions exacerbates economic cycles.
 - Retained Earnings: The portion of a company's net income that is retained by the company rather than distributed to shareholders as dividends.
-

Q.3094 Which of the following is **not** an approach for calculating credit risk capital?

- A. Standardized approach
- B. Internal ratings based approach – Foundation
- C. Internal ratings based approach – Advanced
- D. Standardized approach- advanced

The correct answer is **D**.

The 'Standardized approach- advanced' is not a recognized method for calculating credit risk capital. The Basel II framework, which provides guidelines for banking supervision, does not include an 'advanced' version of the Standardized approach. The Standardized approach is a simpler method that uses external credit assessments for determining risk weights. It does not have an 'advanced' version like the Internal Ratings Based (IRB) approach. The IRB approach, on the other hand, allows banks to use their internal estimates of risk parameters, and it has two versions: Foundation and Advanced.

Choice A is incorrect. The Standardized Approach is a recognized method for calculating credit risk capital. In this approach, banks use external credit ratings to quantify the risk of their credit exposures.

Choice B is incorrect. The Internal Ratings Based (IRB) – Foundation approach is also a recognized method for calculating credit risk capital. Under this approach, banks use their own internal estimates of probability of default (PD) while other parameters like loss given default (LGD) and exposure at default (EAD) are provided by the regulator.

Choice C is incorrect. The Internal Ratings Based (IRB) – Advanced approach allows banks to use their own internal estimates for all parameters including PD, LGD and EAD in the calculation of risk-weighted assets and hence, the regulatory capital.

Things to Remember

- Basel II framework provides guidelines for banking supervision
 - Standardized approach uses external credit assessments for determining risk weights
 - Internal Ratings Based (IRB) approach allows banks to use their internal estimates of risk parameters
 - IRB approach has two versions: Foundation and Advanced
-

Q.3095 Capital Bank, a hypothetical a global systematically important bank (G-SIB) based in Europe, is subject to a 5% risk-weighted higher-loss absorbency requirement. In line with Basel III reforms, the bank would be subject to a leverage ratio buffer requirement of:

- A. 5%
- B. 10%
- C. 2.5%
- D. Zero: the bank has already surpassed the required 3% risk-weighted higher-loss absorbency requirement

The correct answer is C.

The leverage ratio buffer requirement for a G-SIB is set at 50% of its risk-weighted higher-loss absorbency requirement. This is a measure put in place to mitigate the potential ripple effects associated with the failure of G-SIBs. Therefore, if a G-SIB has a 5% risk-weighted higher-loss absorbency requirement, it would be subject to a leverage ratio buffer of 2.5%. This requirement is part of the Basel III reforms, which aim to improve the banking sector's ability to absorb shocks arising from financial and economic stress, improve risk management and governance, and strengthen banks' transparency and disclosures.

Things to Remember

- Risk-weighted higher-loss absorbency requirement is a measure of the capital that banks need to hold to cover potential losses on their assets.
 - G-SIBs (Global Systemically Important Banks) are banks whose distress or failure could cause significant disruption to the global financial system and economy.
 - The leverage ratio is a measure of a bank's capital to its exposure, providing a simple measure of capital adequacy without adjusting for risk.
 - The Basel III reforms aim to strengthen the regulation, supervision, and risk management within the banking sector to promote financial stability.
 - Capital conservation buffer is an additional capital requirement that banks must hold to ensure they have enough capital to continue lending during periods of financial stress.
-

Q.3096 Prime Bank's risk-weighted assets stood at \$200 million as of December 2018. What is this bank's common equity requirement plus the capital conservation buffer, according to Basel III?

- A. \$9,000,000
- B. \$14,000,000
- C. \$12,000,000
- D. \$16,000,000

The correct answer is **B**.

Under Basel III, Common Equity Tier I (CETI) risk-weighted requirements consist of a capital ratio of 4.5% plus an additional capital conservation buffer of 2.5%, making up a CET ratio of 7%. With risk-weighted assets of \$200 million, therefore, the bank's CETI requirement will be \$14 million ($= 7\% \times \200m)

Things to Remember

- Basel III is a global regulatory framework on bank capital adequacy, stress testing, and market liquidity risk agreed upon by the members of the Basel Committee on Banking Supervision.
- Common Equity Tier 1 (CET1) capital is the highest quality of capital under Basel III regulations, consisting of common shares and retained earnings.
- The Capital Conservation Buffer is an additional requirement under Basel III that aims to ensure that banks build up capital buffers outside periods of stress.
- Risk-weighted assets are a measure of the total assets of a bank, adjusted for risk. Different types of assets have different risk weights assigned to them.
- Capital ratios, such as the CET1 ratio, are important indicators of a bank's financial health and ability to absorb losses.

Q.3097 The Basel III reforms announced in 2017 require banks to calculate Credit Value adjustment risk using all of the following methods EXCEPT:

- A. The internal modeled approach

- B. The standardized approach
- C. The simpler basic approach
- D. All of the above

The correct answer is **A**.

The internal modeled approach is not included in the Basel III reforms announced in 2017 for calculating Credit Value Adjustment (CVA) risk. The Basel III reforms were designed to strengthen the regulation, supervision, and risk management of banks. As part of these reforms, banks are required to calculate CVA risk using specific methods. However, the internal modeled approach is not one of these methods. The reforms instead emphasize the use of two main methods: the standardized approach (SA-CVA) and the simpler basic approach (BA-CVA). The internal modeled approach was removed from the guidelines, indicating a shift away from internal models towards more standardized and simpler methods for calculating CVA risk. This change was made to increase the comparability and consistency of CVA risk calculations across banks and to reduce the reliance on banks' internal models, which can vary significantly in their design and assumptions.

Choice B is incorrect. The standardized approach is indeed a part of the Basel III reforms for calculating CVA risk. This approach uses a set of predefined risk weights and exposure amounts to calculate the CVA risk, which provides consistency across different banks.

Choice C is incorrect. The simpler basic approach is also included in the Basel III reforms for calculating CVA risk. This method simplifies the calculation process by using a fixed percentage for certain types of exposures, making it easier for smaller banks to comply with the regulations.

Choice D is incorrect. All of the above cannot be correct as both The Standardized Approach and The Simpler Basic Approach are included in Basel III reforms for calculating Credit Value Adjustment (CVA) Risk.

Things to Remember

- Basel III reforms aim to strengthen the regulation, supervision, and risk management of banks.
- Credit Value Adjustment (CVA) risk is the risk of loss due to changes in the credit risk of counterparties.
- The Standardized Approach (SA-CVA) and the Simpler Basic Approach (BA-CVA) are the main methods recommended for calculating CVA risk under Basel III reforms.
- Internal modeled approach, while not included in Basel III reforms for CVA risk calculation, was previously used by banks to calculate CVA risk based on their internal models.

- The shift towards standardized and simpler methods for CVA risk calculation aims to increase comparability and consistency across banks and reduce reliance on potentially varied internal models.
-

Q.3098 The new standardized approach for determining a bank's operational risk capital requirements assumes that:

- I. Operational risk increases at a decreasing rate with a bank's income
 - II. Banks which have experienced greater operational risk losses historically are more likely to experience operational risk losses in the future
- A. I only
 - B. II only
 - C. Both I and II
 - D. Neither I nor II

The correct answer is **B**.

The new standardized approach for determining a bank's operational risk capital requirements is based on two components: a measure of a bank's income and a measure of a bank's historical losses. It assumes that banks which have a history of operational risk losses are more likely to experience operational risk losses in the future. This is because past performance is often a good indicator of future performance, especially when it comes to operational risk. Banks that have experienced operational risk losses in the past have demonstrated that they have vulnerabilities in their operations that can lead to losses. These vulnerabilities may be due to a variety of factors, such as inadequate internal controls, poor risk management practices, or a lack of effective oversight. Unless these issues are addressed, the bank is likely to continue experiencing operational risk losses in the future. Therefore, the assumption that banks with a history of operational risk losses are more likely to experience such losses in the future is a key component of the new standardized approach for determining operational risk capital requirements.

Choice A is incorrect. The new standardized approach for calculating a bank's operational risk capital requirements does not assume that the operational risk escalates at a diminishing rate as the bank's income increases. Instead, it assumes that the operational risk increases proportionally with the increase in income.

Choice C is incorrect. As explained above, assumption I is not correct under the new standardized approach for calculating a bank's operational risk capital requirements. Therefore, both assumptions I and II cannot be correct.

Choice D is incorrect. Assumption II is indeed correct under this approach as banks with a history of higher operational risk losses are considered more prone to future operational risk losses.

Things to Remember

- **Operational Risk:** The risk of loss resulting from inadequate or failed internal processes, people, and systems, or from external events. It includes legal risk but excludes strategic and reputational risk.
 - **Standardized Approach:** A method for calculating regulatory capital requirements that is prescribed by regulators and is based on standardized formulas and parameters.
 - **Income Measure:** A component used in the standardized approach to determine a bank's operational risk capital requirements, reflecting the size and complexity of the bank's operations.
 - **Historical Losses:** Past instances of operational risk losses experienced by a bank, which are considered in assessing the likelihood of future operational risk losses.
 - **Internal Controls:** Policies and procedures implemented by a bank to ensure the reliability of financial reporting, compliance with laws and regulations, and the effectiveness and efficiency of operations.
-

Q.3099 An American bank has the following exposure:

Business line	BI coefficient	Business line relevant indicator
A	8%	48
B	10%	44
C	12%	20

The bank's supervisor has set an internal loss multiplier of 1. The capital requirement for operational risk for the bank, using the standardized approach, is equal to:

- A. 10.0
- B. 10.64
- C. 5.76
- D. 12.0

The correct answer is **B**.

The operational risk capital requirement can be summarized as follows:

$$\text{Operational risk capital} = \text{BIC} \times \text{ILM}$$

where:

$$\text{Business Indicator Component (BIC)} = \sum(\alpha_i \times \text{BI}_i)$$

α_i is the BI coefficient for business line i , and BI_i is the business line indicator

ILM = internal loss multiplier = 1

Thus, value of the capital requirement = $(48 \times 0.08) + (44 \times 0.10) + (20 \times 0.12) = 10.64$

Q.3101 A hypothetical a global systematically important bank (G-SIB) based in Europe, is subject to a \$200 million risk-weighted higher-loss absorbency requirement. In line with Basel III reforms, the bank would be subject to a leverage ratio buffer requirement of:

- A. \$100 million
- B. \$50 million
- C. \$200 million
- D. \$400 million

The correct answer is **A**.

The leverage ratio among G_SIBS is set at 50% of the bank's riskweighted higher-loss absorbency requirement. Therefore, a G-SIB with a \$200m risk-weighted higher-loss absorbency requirement would be subject to a leverage ratio buffer of \$100m.

Things to Remember

- Global Systemically Important Banks (G-SIBs) are financial institutions whose distress or failure could cause significant disruption to the global financial system.
 - Risk-weighted higher-loss absorbency requirements are part of Basel III reforms aimed at strengthening the resilience of the banking sector by ensuring that banks hold adequate capital to cover potential losses.
 - The leverage ratio is a measure of a bank's capital to its exposure and is designed to act as a backstop to the risk-based capital requirements.
 - The leverage ratio buffer is a requirement that G-SIBs must hold additional capital above the minimum leverage ratio to further enhance their resilience.
-

Q.3102 Bank A has \$200 million in tier 1 capital and \$100 million in tier 2 capital. Bank A loaned \$50 million to XYZ Corporation, which has 30% riskiness, and \$100 million to Brighter World, Inc., which has 50% riskiness. The bank's capital adequacy ratio is equal to:

- A. 3.52
- B. 1.51
- C. 2.2
- D. 4.61

The correct answer is **D**.

$$\text{Capital adequacy ratio} = \frac{\text{Tier I capital} + \text{Tier II capital}}{\text{Risk weighted assets}}$$

Bank A has risk-weighted assets of \$65 million (\$50 million $\times$ 0.3 + \$100 million $\times$ 0.50).

It also has capital of \$300 million, (\$200 million + \$100 million).

Its resulting capital adequacy ratio is 4.61 ($\frac{\$300 \text{ million}}{\$65 \text{ million}}$).

Q.3103 Four European banks, A, B, C, and D have the following capital amounts and risk weighted assets (in \$m):

Bank	A	B	C	D
Tier I capital	5	8	15	25
Tier II capital	3	3	5	10
Risk-weighted assets	30	40	240	230

Which of the four banks is in violation of the capital adequacy requirements as set out in the Basel III reforms announced in 2017?

- A. Bank A
- B. Bank B
- C. Bank C
- D. Bank D

The correct answer is **C**.

According to the revised Basel III guidelines, the minimum capital adequacy ratio, including the capital conservation buffer, is 10.5%. As can be seen from the calculations below, only bank C has failed to attain the minimum ratio.

$$\text{Capital adequacy ratio} = \frac{\text{Tier I capital} + \text{Tier II capital}}{\text{Risk weighted assets}}$$

Bank	A	B	C	D
Tier I capital	5	8	15	25
Tier II capital	3	3	5	10
Risk-weighted assets	30	40	240	230
CAR (Capital adequacy ratio)	26.7%	27.5%	8.3%	15.2%

Q.3104 A French bank has the following exposure:

Business line	BI coefficient	Business line relevant indicator
Consumer banking	12%	100
Global banking	15%	200
Wealth management	18%	50

The bank's supervisor has set an internal loss multiplier of 1. The capital requirement for operational risk for the bank, using the standardized approach, is equal to:

- A. 67
- B. 80
- C. 51
- D. 45

The correct answer is C.

The operational risk capital requirement can be summarized as follows:

$$\text{Operational risk capital} = \text{BIC} \times \text{ILM}$$

where:

$$\text{Business Indicator Component (BIC)} = \sum(\alpha_i \times \text{BI}_i)$$

α_i is the BI coefficient for business line i , and BI_i is the business line indicator

ILM = internal loss multiplier = 1

Thus, value of the capital requirement = $(100 \times 0.12) + (200 \times 0.15) + (50 \times 0.18) = 51$

Q.3105 In the most recent global financial crisis (2007/2008), banks suffered huge losses resulting from CVA risk – losses related to the deterioration of a counterparty's creditworthiness in derivative contracts. In the aftermath of the crisis, the Basel Committee has enhanced the CVA framework with a view to:

- A. Keep losses associated with CVA risk at less than 10% of the total value of the derivatives.

- B. Totally eliminate CVA losses by conducting due diligence on all counterparties before a contract comes into force.
- C. Enhance the risk sensitivity of the framework by recognizing more risk drivers.
- D. Limit derivative contracts at not more than 20% of the total capital for a bank.

The correct answer is **C**.

The enhanced CVA framework has 3 main objectives:

- **To enhance risk sensitivity**

The revised CVA framework takes into account the exposure component of CVA risk as well as the risk of associated hedges.

- **To enhance robustness of the CVA framework**

The updated guidelines remove the use of an internally modeled approach and instead emphasize the use of two main methods: (I) the standardized approach (SA-CVA), and (II), the simpler basic approach (BA-CVA). In addition, banks with minimal engagement activities in derivative transactions can use their credit counterparty risk (CCR) capital requirements as a proxy for their CVA charge.

- **To improve consistency of the CVA framework**

The standardized and basic approaches of the revised CVA framework have been revised to be consistent with the approaches used in the revised market risk framework.

Things to Remember

- **Counterparty Credit Risk (CCR):** CCR is the risk that a counterparty to a financial contract will default before the contract is settled.
- **Credit Valuation Adjustment (CVA):** CVA is the adjustment made to the price of a derivative instrument to account for the credit risk of the counterparty.
- **Basel Committee on Banking Supervision:** The Basel Committee is an international body that provides recommendations on banking regulations and capital requirements to enhance financial stability.

- Standardized Approach (SA-CVA): SA-CVA is a method under the enhanced CVA framework that provides a standardized approach for calculating CVA charges.
 - Basic Approach (BA-CVA): BA-CVA is a simpler method under the enhanced CVA framework that allows banks with minimal derivative activities to use their credit counterparty risk capital requirements as a proxy for their CVA charge.
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Q.3106 The Basel Committee has agreed on various additional enhancements to the IRB approaches to further reduce unwarranted RWA variability. Which of the following correctly outlines a measure that has been put forth for adoption by banks?

- A. Secured exposures: increasing the haircuts that apply to the collateral; Unsecured exposures: reducing the LGD parameter from 45% to 40% for exposures to non-financial corporates.
- B. Secured exposures: reducing the LGD parameters; Unsecured exposures: reducing the LGD parameter from 25% to 20% for exposures to non-financial corporates.
- C. Secured exposures: decreasing the haircuts that apply to the collateral; Unsecured exposures: reducing the LGD parameter from 45% to 40% for exposures to non-financial corporates.
- D. Secured exposures: increasing the LGD parameters; Unsecured exposures: increasing the LGD parameter from 40% to 50% for exposures to non-financial corporates.

The correct answer is **A**.

The Basel Committee has indeed proposed adjustments to the supervisory specified parameters in the Foundation – Internal ratings based approach (F-IRB). For exposures that are secured by non-financial collateral, the committee has suggested increasing the haircuts that apply to the collateral. A 'haircut' in this context refers to a reduction applied to the value of an asset that is being used as collateral for a loan. The purpose of this haircut is to provide a cushion for the lender in case the value of the collateral falls. In addition to this, for exposures that are unsecured, the committee has proposed reducing the Loss Given Default (LGD) parameter from 45% to 40% for exposures to non-financial corporates. LGD is a measure of the potential loss to a lender or investor in the event of default by a borrower. By reducing the LGD parameter, the committee aims to reflect a lower potential loss on unsecured exposures to non-financial corporates.

Choice B is incorrect because it incorrectly states that the Basel Committee has proposed reducing the LGD parameters for secured exposures. In fact, the committee has proposed increasing the haircuts that apply to the collateral for secured exposures. Furthermore, the committee has not proposed reducing the LGD parameter from 25% to 20% for exposures to non-financial corporates. The correct proposal is to reduce the LGD parameter from 45% to 40% for

such exposures.

Choice C is incorrect because it incorrectly states that the Basel Committee has proposed decreasing the haircuts that apply to the collateral for secured exposures. In fact, the committee has proposed increasing these haircuts. Furthermore, the committee has not proposed reducing the LGD parameter from 45% to 40% for exposures to non-financial corporates.

Choice D is incorrect because it incorrectly states that the Basel Committee has proposed increasing the LGD parameters for secured exposures and increasing the LGD parameter from 40% to 50% for exposures to non-financial corporates. In fact, the committee has proposed increasing the haircuts that apply to the collateral for secured exposures and reducing the LGD parameter from 45% to 40% for unsecured exposures to non-financial corporates.

Things to Remember

- Basel Committee on Banking Supervision (BCBS) is an international committee that provides a forum for cooperation on banking supervisory matters.
 - Internal Ratings-Based (IRB) approach is a method of calculating regulatory capital requirements for credit risk based on a bank's own internal risk assessment models.
 - Loss Given Default (LGD) is a key parameter in credit risk modeling that represents the proportion of an exposure that is expected to be lost in the event of a default.
 - Haircuts are adjustments made to the value of collateral to account for potential fluctuations in its value and to protect the lender in case of default.
 - Reducing LGD parameters for certain exposures can lead to lower capital requirements for banks, as it reflects a lower potential loss in case of default.
-

Q.3107 The initial phase of the Basel III framework focused, in part, on increasing the quality of bank regulatory capital to cover unexpected losses. As such, the Minimum Tier I capital:

- A. Rose from 4% to 6%.
- B. Rose from 5% to 6%.
- C. Rose from 5% to 7%.
- D. Rose from 4% to 7%.

The correct answer is **A**.

The Basel III framework was introduced in response to the deficiencies in financial regulation revealed by the financial crisis of 2007-08. It aimed to strengthen the banking sector's ability to deal with financial and economic stress, improve risk management and governance, and strengthen banks' transparency and disclosures. One of the key aspects of this framework was the focus on improving the quality of bank regulatory capital to cover unexpected losses. As such, the Minimum Tier I capital requirement was increased from 4% to 6%. This increase was intended to ensure that banks have a sufficient buffer of high-quality capital that can absorb losses during periods of financial and economic stress.

Choice B is incorrect. The Basel III framework did not increase the Minimum Tier I capital requirement from 5% to 6%. The initial phase of Basel III increased the requirement from 4% to 6%, not from 5%.

Choice C is incorrect. This choice incorrectly states that the Minimum Tier I capital requirement rose from 5% to 7%. In reality, under Basel III, it was increased from a lower percentage of 4%, not from a higher percentage of 5%.

Choice D is incorrect. While this option correctly identifies that the final target for Minimum Tier I capital requirement under Basel III was indeed set at a higher level of around 7%, it incorrectly suggests that this increase was implemented in the initial phase itself. In fact, during its initial phase, Basel III only raised this requirement to an intermediate level of about 6%. The further increase up to around approximately 7% was planned for later stages.

Things to Remember

- Basel III framework was developed by the Basel Committee on Banking Supervision to strengthen the regulation, supervision, and risk management of the banking sector.
 - Tier I capital is the core measure of a bank's financial strength from a regulator's point of view and consists of common equity tier 1 and additional tier 1 capital.
 - Common Equity Tier 1 (CET1) capital is the highest quality of capital and includes common shares and retained earnings.
 - Additional Tier 1 (AT1) capital includes instruments that are subordinated to all other claims in winding up a bank but rank above Tier 2 capital.
 - Minimum capital requirements are set to ensure that banks have enough capital to absorb unexpected losses and continue operating in times of financial stress.
-

Q.3109 When were the Basel III reforms announced?

- A. In 2010
- B. In 2014
- C. In 2015
- D. In 2017

The correct answer is **D**.

The Basel III reforms were officially announced in 2017. The Basel III reforms are a comprehensive set of reform measures designed to improve the regulation, supervision, and risk management within the banking sector. These reforms were developed in response to the financial crisis of 2007-2008. They are intended to strengthen bank capital requirements and introduce new regulatory requirements on bank liquidity and bank leverage. The announcement of these reforms in 2017 marked a significant milestone in banking regulation, with the aim of reducing the risk of future banking crises.

Choice A is incorrect. The Basel III reforms were not announced in 2010. This year marked the publication of the initial version of Basel III, but it was not officially announced as a reform until later.

Choice B is incorrect. The Basel III reforms were not announced in 2014 either. While there were discussions and proposals around banking regulations during this time, the official announcement of the Basel III reforms did not occur in this year.

Choice C is incorrect. Similarly, 2015 is also an incorrect answer as it does not correspond to the official announcement year for Basel III reforms.

Things to Remember

- Basel III reforms aim to improve regulation, supervision, and risk management in the banking sector
- These reforms were developed in response to the financial crisis of 2007-2008
- Basel III reforms focus on strengthening bank capital requirements, liquidity, and leverage
- The reforms are designed to reduce the risk of future banking crises
- Basel III reforms have a global impact on banking regulations

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Q.3111 A Bank holding company based in Germany has two subsidiaries, A and B. The business indicator values of each are given in the table below:

Bank	A	B
BI	€800 million	€1.2 billion

In light of this information, which of the following statements is correct?

- A. Bank A would be expected to calculate operational risk capital based on the Advanced Measurement Approach while Bank B would employ the standardized measurement approach.
- B. Only Bank B would be expected to set aside capital for operational risk.
- C. Bank B would be expected to use loss experience in the standardized approach calculations.
- D. Neither Bank A nor Bank B would be expected to set aside some capital for operational risk.

The correct answer is **C**.

Bank B would be expected to use loss experience in the standardized approach calculations. This is because, according to the Basel III reforms announced in 2017, all banks are required to use the standardized approach in operational risk capital calculations. For banks with Business Indicator (BI) values of more than €1 billion (falling into bucket 2-3), internal loss experience must be taken into account while calculating operational risk capital. In this case, Bank B, with a BI value of €1.2 billion, falls into this category and therefore, would be expected to incorporate its loss experience into its calculations.

Choice A is incorrect. The Advanced Measurement Approach (AMA) and the Standardized Measurement Approach (SMA) are not determined by the Business Indicator (BI) value of a bank. Both Bank A and Bank B can use either approach depending on their operational risk management capabilities, regulatory approval, and other factors.

Choice B is incorrect. Both banks would be expected to set aside capital for operational risk as it is a requirement under Basel III regulations for all banks, regardless of their BI values.

Choice D is incorrect. As explained above, both Bank A and Bank B would be expected to set aside some capital for operational risk as per Basel III regulations.

Q.3115 The following are verified historical loss data for an international bank over a 10-year period (in billions of Euros)

[3.8, 2.9, 2.8, 2.8, 0.6, 0.4, 0.1, 0.2, 0.1, 0.2]

Determine the bank's Internal Loss Multiplier as computed under the Standardized Measurement Approach (The bank's Business Indicator Component is €18 billion)

A. 1.39

B. 0.9288

C. 1.0

D. 1.0449

The correct answer is **D**.

$$ILM = \ln[\exp(1) - 1 + (\frac{LC}{BIC})^{0.8}]$$

Where $LC = 15$ times a bank's average historical losses over the preceding 10 years.

Thus,

$$LC = (\frac{3.8 + 2.9 + 2.8 + 2.8 + \dots + 0.1 + 0.2}{10}) \times 15 = 20.85$$

$$ILM = \ln[\exp(1) - 1 + (\frac{20.85}{18})^{0.8}] = 1.0449$$

Q.3116 The following are verified historical loss data for a large established bank over a 10-year period (in billions of Euros)

[0.8, 0.9, 0.7, 0.8, 0.06, 0.04, 0.10, 0.09, 0.03, 0.0]

The bank has a Business indicator of €960 million.

Determine the bank's operational risk capital, ORC, as computed under the Standardized Measurement Approach

A. €115 million

B. €3.52 million

C. €361 million

D. €100 million

The correct answer is **A**.

Recall that

$$\text{ORC} = \text{BIC} \times \text{ILM}$$

Where

$$\text{BIC} = \sum (\alpha_i \times \text{BI}_i)$$

And,

$$\text{ILM} = \ln[\exp(1) - 1 + \left(\frac{\text{LC}}{\text{BIC}}\right)^{0.8}]$$

Also recall that for firms with BI levels less than €1bn, the ILM is set to 1, and therefore internal loss data does not affect the capital calculation.

Thus, the operational risk capital in this case is a function of the business Indicator Component only.

With a BI of €960 million, the bank falls under bucket 1 of the Basel guidelines and therefore the relevant BI coefficient is 0.12.

$$\text{BIC} = 0.12 \times 960 = \text{€}115.2 \text{ million}$$

So,

$$\text{ORC} = 115.2 \times 1 = 115.2$$

Q.3119 An international lender based in Dubai has a Business Indicator of €34.5 billion. Determine the Business Indicator Component for the bank.

A. €0.12bn.

B. €4.35bn.

C. €35bn.

D. €5.28bn.

The correct answer is **D**.

To answer the question, it's important to have the BI bucket divisions and corresponding marginal coefficients as outlined in Basel III reforms.

BI Bucket	1	2	3
BI Range	$\leq 1 \text{ bn}$	$\text{€}1 \text{ bn} < \text{BI} \leq \text{€}30 \text{ bn}$	$\text{€}30 \text{ bn}$
Marginal BI Coefficient	0.12	0.15	0.18
BI of € 40	$\text{€}1\text{bn} \times 12\%$ $= \text{€}0.12\text{bn}$	$\text{€} = (30 - 1) \times 15\%$ $= \text{€}4.35\text{bn}$	$= \text{€}(34.5 - 30) \times 18\%$ $= \text{€}0.81\text{bn}$
BIC=sum of Buckets 1-3 = €5.28bn			

BIC = Sum of buckets 1 to 3 = €5.28bn

Q.3120 A hypothetical European Bank has a business indicator (BI) of EUR 40 billion. The bank's loss component is EUR 1.2 billion. Using the information in the following table, calculate the bank's operational risk capital.

BI Bucket	1	2	3
BI Range	$\leq 1 \text{ bn}$	$1 \text{ bn} < \text{BI} \leq 30 \text{ bn}$	30 bn
Marginal BI Coefficient	12%	15%	18%

- A. €0.63 billion
- B. €0.55billion
- C. €4.30 billion
- D. €4.5 billion

The correct answer is **C**.

The operational risk capital requirement (ORC) can be calculated as follows:

$$\text{ORC} = \text{BIC} \times \text{ILM}$$

Where the Business indicator component (BIC) is given by:

$$\text{BIC} = \sum (\alpha_i \times \text{BI}_i)$$

$$\text{BIC} = (12\% \times \text{€}1) + (15\% \times (\text{€}30 - \text{€}1)) + (18\% \times (\text{€}40 - \text{€}30)) = \text{€}6.27$$

And, the Internal losses multiplier (ILM) is expressed as:

$$\text{ILM} = \ln[\exp(1) - 1 + \left(\frac{\text{LC}}{\text{BIC}}\right)^{0.8}]$$

$$\text{ILM} = \ln[\exp(1) - 1 + \left(\frac{1.2}{6.27}\right)^{0.8}] = \ln 1.98 = 0.6855$$

Finally,

$$\text{ORC} = \text{€}6.27 \times 0.6855 = \text{€}4.30$$